ADVANCED DIGITAL CONTROL TECHNIQUES FOR BRUSH-LESS DC (BLDC) MOTOR DRIVES

BY

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Submitted in partial fulfillment of the
requirements for the degree of
Doctorate of Philosophy in Electrical Engineering
in the Graduate College of the
Illinois Institute of Technology

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Chicago, Illinois
December 2006

ACKNOWLEDGMENT

I would like to thank my academic advisor, Dr. Ali Emadi, for his support and encouragement during my undergraduate and graduate studies. Also, the financial support received from the Diversifying Higher Education Faculty in Illinois (DFI) institution greatly facilitated my academic career. Programs similar to DFI are a great resource for minority students to find financial and moral support. I sincerely thank the DFI founders and the current governing personal for their contributions to this great organization.

For the past eight and a half years, family and friends were a source of strength and encouragement. In particular, I would like to acknowledge my mother, Rosa, who year in year out unselfishly provided everything from financial support to a hot meal during exam time. She has shaped and groomed me into the person that I am today and for that I am forever indebted to her. My two sisters, Rocío and Mariana, and my stepfather, Leonardo, were always quick to provide a word of wisdom and encouragement at the first sign of frustration on my part. I thank them and hope that I can return their generosity as I start a new facet of my life.

Aside from support from my family, I was privileged to have the companionship of my girlfriend, Olivia, during my academic life. The life of a graduate student can be very trying due to the long hours and demanding tasks. The love, care, and support that Olivia provided fueled me to push forward even at the most difficult of times. I sincerely thank her and look forward to starting a new chapter in our lives as one through marriage.

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Symbol	Definition
b	viscous friction constant
$\vec{B}$	magnetic field density
D	duty cycle
emf	electromotive force
E_f	motor terminal back emf
ϕ	magnetic flux
$\vec{F}$	vector force
f_s	switching frequency
i	instantaneous current
I	average current
I_H	State 2 average current
I_L	State 1 average current
$I_{\max}$	maximum band gap current
$I_{\min}$	minimum band gap current
J	rotor moment of inertia
k_d	derivative constant
k_e	back-emf constant
k_I	integral constant
k_p	proportional constant

Symbol	Definition
k_t	torque constant
$\vec{l}$	conductor length and direction
L	motor terminal inductance
M_p	overshoot
N	number of conductor turns
P	power
R	motor terminal resistance
T	torque
T_c	conduction interval duration
T_d	developed torque
T_L	load torque
τ_m	mechanical time constant
T_N	net torque
t_{off}	switch off-time
t_{on}	switch on-time
t_p	peak time
t_r	rise time
t_s	settling time
T_s	switching period
θ_1	conduction angle beginning

Symbol	Definition
θ_2	conduction angle ending
$\theta_{2,\min}$	minimum conduction ending
θ_r	angular position
ω	angular velocity
ω_H	State 2 rotor speed
ω_L	State 1 rotor speed
ω_n	natural frequency
ω_{ss}	steady state angular velocity
$u(t)$	step function
V_{dc}	dc voltage source
ζ	damping ratio

ABSTRACT

Development of advanced electric motor drives has yielded increases in efficiency and reliability. Residential and commercial appliances such as refrigerators and air conditioning systems use conventional motor drive technology. The machines found in these appliances are single phase induction or brushed DC machines which are characterized by low efficiency and high maintenance, respectively. A brush-less DC (BLDC) motor drive (an advanced motor drive system) is characterized by higher efficiency, lower maintenance, and higher cost.

In a market driven by profit margins, the appliance industry is reluctant to replace the conventional motor drives with the advanced motor drives due to their higher cost. In this Ph.D. proposal, a new digital control concept will be introduced for BLDC motor drives. The new digital controller treats the motor like a digital system that may only operate at a few predefined states. Those states establish machine operating conditions which produce distinct constant motor speeds. Speed regulation is achieved by alternating states during operation, which makes the concept of the controller extremely simple for integrated-circuit (IC) development.

A design methodology for this controller, applicable to any BLDC machine, is presented. The design procedure is derived from Newton's 2nd Law and the characteristic equations of a BLDC machine. The design procedure takes into account the acceptable output speed ripple. Also, is demonstrated that through the use of an observer, disturbance rejection is achieved. Computer simulation results are presented along with experimental proof-of-concept.

CHAPTER 1

INTRODUCTION AND OVERVIEW

1.1 Introduction

Brushless-dc (BLDC) motor drives are continually gaining popularity in motion control applications, major avenues being consumer appliances, industrial machinery, aerospace applications, and automotive applications. A major challenge holding back the complete acceptance of BLDC motor drives by industry and appliance manufactures is the high cost associated with BLDC motor drives. A large portion of the cost is attributed to the controller. Therefore, it is necessary to have a low cost, but effective BLDC motor speed controller.

In this thesis, conventional BLDC motor drive control methods such as proportional and proportional-integral controllers are reviewed. Speed regulation design via classic control systems theory is outlined. As an alternative to classic control theory, a novel digital controller is presented. The digital controller deals with the BLDC motor as a digital system. The motor operates at a few predefined states that result in predefined motor speeds. Speed regulation is achieved by switching from one state to another during operation. That makes the concept of the controller extremely simple, which should facilitate its realization onto an integrated circuit (IC) development. The digital controller is based on nothing more than a couple of "IF and THEN" statements. The controller can be implemented either in a low-cost application specific integrated circuit (ASIC) or as an embedded controller in a digital signal processor (DSP).

1.2 Overview of Chapters

Chapter 2 introduces the basic principles of electric machines and electric motor drives. Fundamental physic theories behind the operation of electric machines are presented, namely Faraday's law, Lenz's law, and Lorentz's law. Machine construction with regards to permanent magnet placement is discussed. Also, the power electronic converter used to drive a BLDC machine is presented and analyzed. The concept of PWM voltage control is explained, followed by a generalized method of generating a PWM signal. Also, Hysteresis current control is presented as an indirect voltage controller. Lastly, the six-step switching sequence necessary to operate the motor is presented.

Chapter 3 focuses on the development of a precise BLDC machine model to be used in computer simulations. The BLDC machine and power electronic converter are mathematically modeled and converter into the S-domain in order to realize a system model in the Matlab/Simulink software environment. Throughout the system model development, special notice is given to modeling non-linearities that characterize a BLDC machine. Basic simulation results are provided which validate the modeling technique with regards to the machine and power electronic converter. The last section of this chapter details the development of a robust experimental setup that was used to validate the novel digital controller presented in this thesis.

Chapter 4 introduces the classic control theory behind feedback closed-loop systems. Basic sample equations and their transfer functions for dynamic modeling in the s-domain are presented. The significance of poles and zeros on the complex s-plane is related to the dynamic performance of a system. The concepts of proportional and proportional-integral control for speed regulation are introduced. A brief description of the design tool, Root Locus Design, is covered. Also, the control concepts are used to find the feedback transfer function for a BLDC motor in terms of the motor parameters. Classic control techniques are applied to a BLDC machine and simulation results are discussed. Classic control techniques can be implemented digitally in the z-domain. The last section in this chapter discusses the use of classic digital techniques applicable to motor drives.

Chapter 5 introduces the novel BLCD electric motor drive digital controller. The concept of the digital controller is presented in detail. The digital control concept is realized in two ways: namely Conduction Angle Digital Control and Current Mode Control. A design method is presented that is derived from Newton's second law of motion applied to rotary bodies. The design method is generalized so as to make it applicable to any BLDC motor under the assumption that a constant load torque is applied and that the load magnitude is known. The proposed digital controller is realized in Matlab/Simulink in order to validate the derived design equations. Experimental implementation of the proposed digital control for Conduction Angle Digital Control was carried out.

Chapter 6 further develops the proposed digital control for applications were the load torque is not constant and unknown. Requirements for the controller sample frequency are addressed and a design method for calculating the required sample rate is derived. Simulations are carried out to validate the theoretical calculations for non-constant and unknown load conditions. Experimental results are provided to solidify the validity of the proposed controller.

Chapter 7 summarizes the thesis accomplishments and explores various possible applications for the newly developed digital controller. New research avenues resulting from the work presented in this thesis are discussed.

1.3 Research Goals & Accomplishments

The goal of this thesis was to develop a design strategy for the proposed digital control concept. The design strategy was to be applicable to any BLDC motor, while maintaining its ease of design. This thesis lays out the groundwork for the development of a new low-cost IC for control of BLDC motors. As a base for comparison, when using conventional control schemes with a digital signal processor (DSP), the cost may be as much as twenty dollars. The price reduces to about four dollars is an analog IC is utilized. Preliminary IC development research estimates the in mass production, the new control concept can be implemented for less than half of a conventional analog controller which will make BLDC motors more affordable for manufacturers to use in their products. The low cost efficient (70-99%) motor drives will slowly replace the inefficient (40-60%) single phase induction motors found in most homes. Consumers will save money on their energy bills and the environment will be spared from tons of harmful pollutants.

The digital motor drive system was successfully simulated in the Matlab/Simulink environment. Also, a robust experimental set-up was created with real-time interface and data acquisition capabilities. Simulation results and experimental results are presented and analyzed. Based on those results, the novel digital control is established as a valid control method for BLDC machines.

CHAPTER 2

ELECTRIC MOTOR DRIVES FOR BRUSH-LESS DC (BLDC) MACHINES

2.1 Introduction

Electric machines play a tremendous role in providing comfort to everyday life. Air conditioners (A/C), automobiles, fans, and power tools are just a few examples of systems that take advantage of the electric machine. In automobiles, power windows, power seats, and windshield wipers use electric machines to make for a more pleasant driving experience. Cooling fans in computers are essential to maintain the temperature of today's powerful digital signal processors (DSP's) within safe limits. Air conditioning units would be highly ineffective without the help of an electric machine powered blower to circulate the cool or warm are throughout a living space. It is hard to imagine a time without these commodities since electric machines are found all around us. Advancements in power electronics have allowed for the development of advanced electric machines that can achieve higher efficiencies and a wider range of operation with low maintenance requirements and high durability. The drawback to these advanced machines is most noticeable in the added complexity due to the required power electronic converters that drive them. This thesis strives to develop an inexpensive novel digital controller for a BLDC machine. However, before introduction of the novel control technique, fundamental operation principles of a BLDC must be thoroughly understood. Only then, can a suitable controller be developed.

2.2 Electric Motor Drives Overview

Over the years, electric motor drives have evolved from inefficient complex systems into drives that are easier to design and more versatile. Just a few decades ago, speed control was achieved by means of crude methods. Such methods include speed control via current limiting resistors introduced with mechanical or magnetic switches. Also, there were limitations in the selection process of a motor. If an ac motor was to be used in any application, it would have to have direct access to an alternative current (ac) source. Similarly, direct current (dc) motors were used mainly where access to a dc source was available. For many industrial applications, a dc motor was required with only the availability of an ac source. In such a case, the complex series configuration of an induction motor, dc generator, and dc motor was required. Needless to say, there are major expenses and inefficiencies with that set-up when compared to today's motor drives. A modern electric motor drive system offers greater design flexibility and better overall system efficiency. Figure 2.1 is a block diagram representation of the major components that make up a modern electric motor drive system.

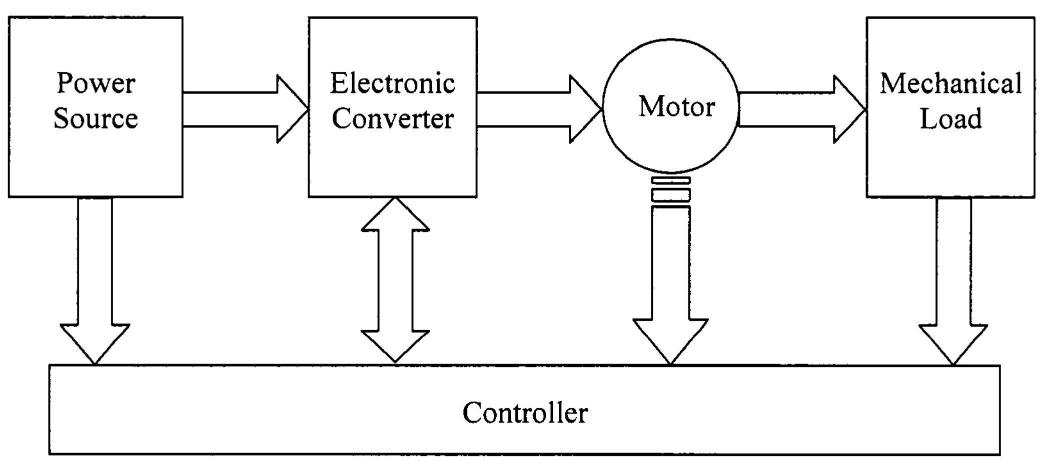


Figure 2.1 Electric Motor Drive Block Diagram

The major components are the power source, electronic converter, electric motor, mechanical load, and the controller. The power source is determined by the type of electricity that is available at the site for the electric motor drive system. For residential applications, the power source is one hundred and twenty volts alternating current. For automobile applications, a fourteen volt direct current bus is available. The electronic converter has the task of converting the power source electricity type into that which is suitable to drive the electric motor. Electric motors are chosen so as to meet the torque and speed requirements of the mechanical load. The controller is design to ensure that the dynamic and steady state behavior of the motor is sufficient to match the performance demands required by the load.

Loads exhibit a wide range of torque characteristics. Some loads exhibit increased torque with increases in speed, while other loads may have the opposite behavior. In addition, the torque to speed relationship is often times non-linear. An example of a non-linear load that is highly dependant on speed is a fan. Another load type is when the torque is constant independent of its speed, such as a motor driving a water pump. The power consumed by any load is given by the product of the torque and the speed. Some examples of non-linear loads are blowers and water pumps [18]. Equation 1.1 relates power to the torque and speed for any rotation body.

$$P = T\omega \tag{1.1}$$

where, P = power

T = torque

ω = angular velocity

A variety of electric motors are available to match the performance requirements of any load. It is the task of the design engineer to choose the motor type that is best suited for the load. Figure 2.2 summarizes the speed-torque characteristics of electric machines, [10]. Induction motors are characterized by a low start-up torque, relative to its maximum torque, T_{max}. After the transient turn-on phase, an induction motor operates in the steady state region specified by the top portion of the curve. That operating region is mostly linear and very similar to that of a dc shunt and dc separately excited motor. Synchronous motors exhibit a constant speed under any torque condition. The speed is determined by its ac source and by the motor construction. In practice, exceeding the rated torque of a synchronous motor would certainly lower its speed and possibly damage the motor.

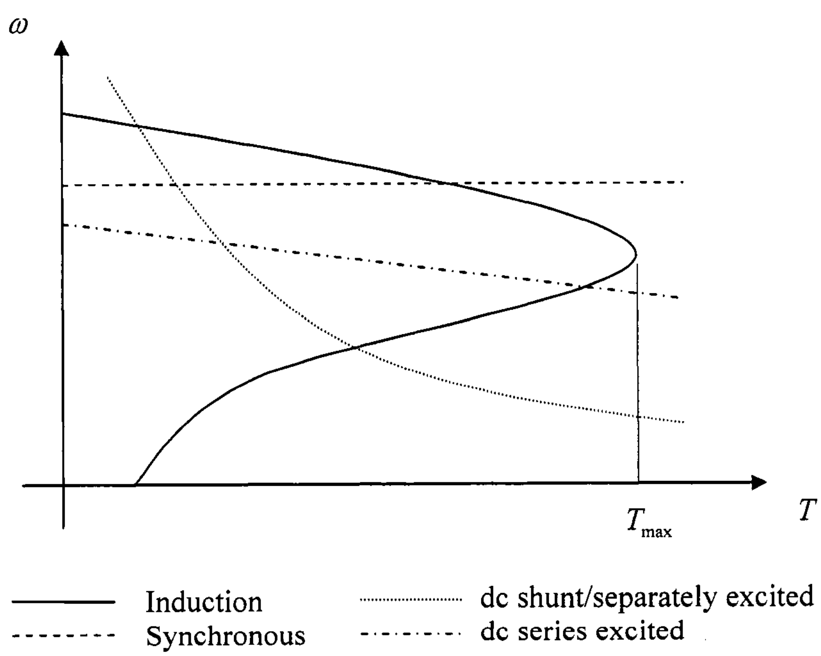


Figure 2.2 Speed vs. Torque Characteristic of Various Motors

A dc motor, connected in shunt or separately excited configuration, is characterized by small linear reductions in speed as the torque is increased. The speed-torque curve of a dc series motor is very similar to that of an automobile transmission. At low speeds the series motor has the capability of providing a very high torque, while at high speeds the motor can only provide a small torque. The characteristics of a BLDC motor are essentially the same as that of a dc separately excited motor. Even though the motor construction is different, the operating principles are the same. The fundamental difference lies in that the current is commutated electronically by solid state switches rather than mechanically via rotor contacts. The type of contacts normally used are wired brushes, however solid metallic spring loaded cubes may also be used to achieve mechanically controlled current commutation.

Brush technology for dc machines has been thoroughly investigated. In [16] silver-graphite brushes arranged in parallel achieved safe current conduction of up to twenty kiloamps. Though, that is a great feat allowing excellent machine performance, there is a unavoidable drawbacks to brushes. Current arcing translates to losses which result in two problems. First, losses generating due to arcing raise the temperature of the brush-commutator system resulting in increased wear rate. Second, active machine cooling may be required to offset the temperature increase which is costly and decreases system efficiency, [17]. Wearing of brushes may also increase the machine resistance (resulting in a significant plant variation), which can alter the performance characteristics for a motor drive system.

The power sources available for motor drives applications will be either an ac or dc source. In the United States, residential buildings have single phase ac electricity at sixty hertz. Some industrial buildings may have three phase ac sources, also at sixty hertz. These three phase power sources are necessary in cases were high power consumption is the characteristic of a loads. In Europe, single and three phase ac sources are also available. However, their standard operating frequency is fifty hertz. The other type of source, a dc source, is most commonly found in a vehicle. A conventional vehicle's electronic system operates from a fourteen volt dc bus.

The function of a power electronic converter is to transform the available power source type into whatever power source is necessary to drive the motor. Converters add tremendous design flexibility when it comes to choosing an electric motor. For example, if a load is best suited to be driven by a dc motor and only an ac power source is available, then an ac to dc converter is used to link the to objects. The most common type of converters are shown in block diagram form, see Figure 2.3.

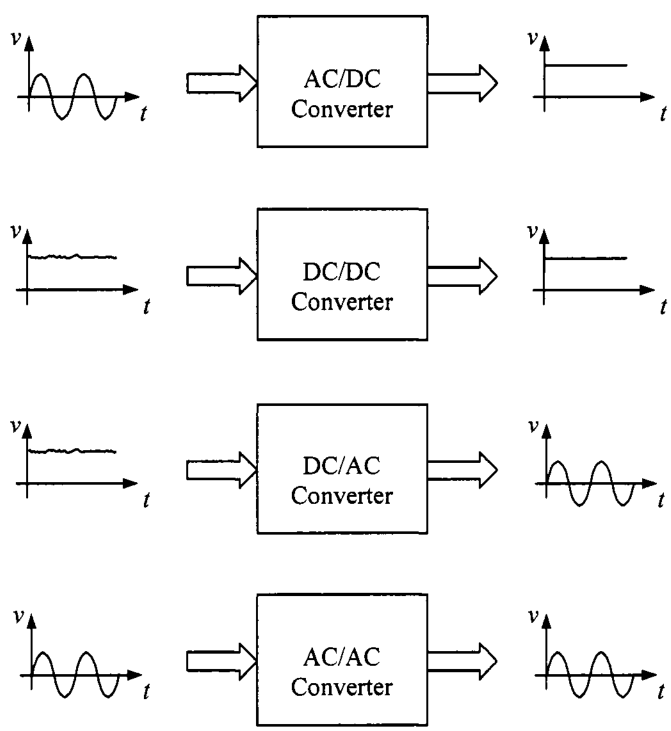


Figure 2.3 Common Power-Electronic Converters

At first glance, ac to ac converters may seem to be redundant systems, but these converters are required in some cases. For example, if the operating frequency of the motor must be something other than sixty hertz, then an ac to ac converter is required. Also, sometimes ac to ac converters are used to improve the quality of an ac source and to change the voltage amplitude. Similarly, dc to dc converters improve the quality of the voltage and have the ability to decrease or increase the voltage magnitude.

2.3 Electric Machine Physics

The operation of BLDC machines is considerably more complicated in comparison to conventional dc machines. A dc machine will rotate by simply applying a dc voltage to its winding terminals, while a BLDC machine requires a power electronic converter that applies an ac trapezoidal voltage whose frequency is dependant on the rotor angular velocity. Despite that difference, dc and BLDC machines operate under the same physics principals, which is why a separately excited dc machine can be used to introduce how a BLDC machine operates. The fundamental physics principles that describe the functionality of a dc machine are summarized by Faraday's law, Lenz's law, and Lorentz's law.

In 1831 Michael Faraday was successful in attaining experimental proof that magnetism can induce a current. He wound two separate windings on a magnetic core and while applying a voltage to one of the windings, a momentary current was observed in the second winding. The momentary current was only observed the instance when a voltage was applied to the winding or removed from the winding. As it is well known, his simple experiment was proof that a time-varying magnetic field will generate an electromotive force (emf), which is nothing more than a voltage. The emf has the ability to induce a current in a suitable closed circuit, which may consist of a conductor, a capacitor, or any imaginary linear path trough space [12]. Therefore, an emf is a voltage that is created when a varying magnetic field is applied to a conductor or when a conductor undergoes relative motion with respect to a steady-state magnetic field. The aforementioned physical phenomena is known as Faraday's law as defined by Equation 2.1. When discussing dc or BLDC machine operation, the term back-emf is often mentioned. The prefix "back" is necessary to indicate the polarity of the emf induced in the machine conductors. The negative sign in Equation 2.1 corresponds to the word "back" to indicate that the additional induced flux produced by the emf is in such a way that it apposes the flux which generated that emf. The minus sign was introduced by Heri Lenz and is known as Lenz's law, published in 1834.

$$emf = -N\frac{d\varphi}{dt} \tag{2.1}$$

where,
emf = electromotive force in volts
φ = magnetic flux
N = number of conductor turns

The third law, fundamental to understanding the operation of a dc machine, is described by Lorentz force equation, see Equation 2.2. Lorentz equation indicates that placing a current carrying conductor in a magnetic field produces force on that conductor as describe by Equation 2.2. The produced force has a magnitude whose orientation in three-dimensional space can be found by application of the right-hand-rule to the current vector "crossed" with the magnetic field vector. Figure 2.4 is an illustrative interpretation of Lorentz's law, the magnetic field is going into the page, normal to the length and force vectors [4].

$$\vec{F} \equiv i\,\vec{l} \times \vec{B} \tag{2.2}$$

where, $\vec{F}$ = force vector

i = current magnitude in the conductor

$\vec{l}$ = conductor length

$\vec{B}$ = magnetic field density

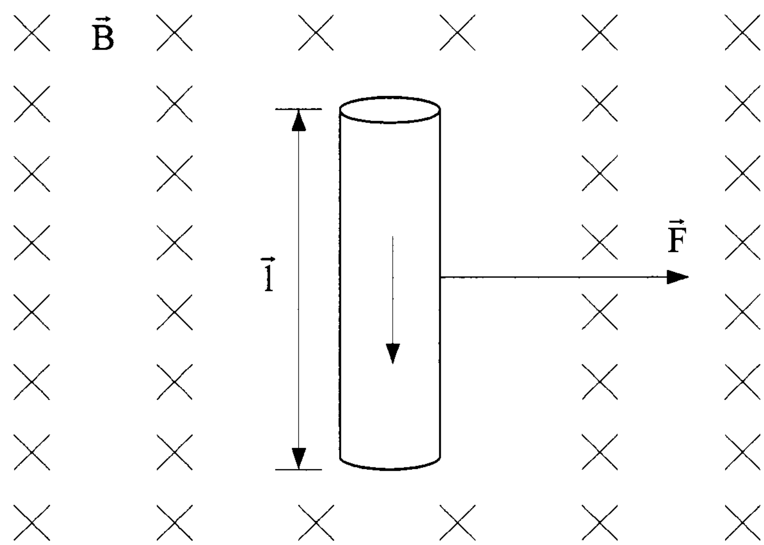


Figure 2.4 Illustrative representation of Lorentz force equation.

Equations 2.1 and 2.2 served to explain the physics behind the operation of a dc machine and allows for a more illustrative relation between the equations and the operation of the machine. Figure 2.5 is that of a cross section of a separately excited dc machine (note: a one pole machine is depicted for simplicity). It consists of coil windings in the stator which produces a steady magnetic field as depicted by the flux lines. The current conducting coils on the rotor produce a force as indicated by the Lorentz force equation. That force causes the rotor to rotate, which results in a conductor undergoing relative motion with respect to a steady magnetic field. That relative motion gives rise to Faraday's law inducing a back-emf. The orientation of the emf is such that it apposes the stator magnetic field as shown in the Figure 2.2.

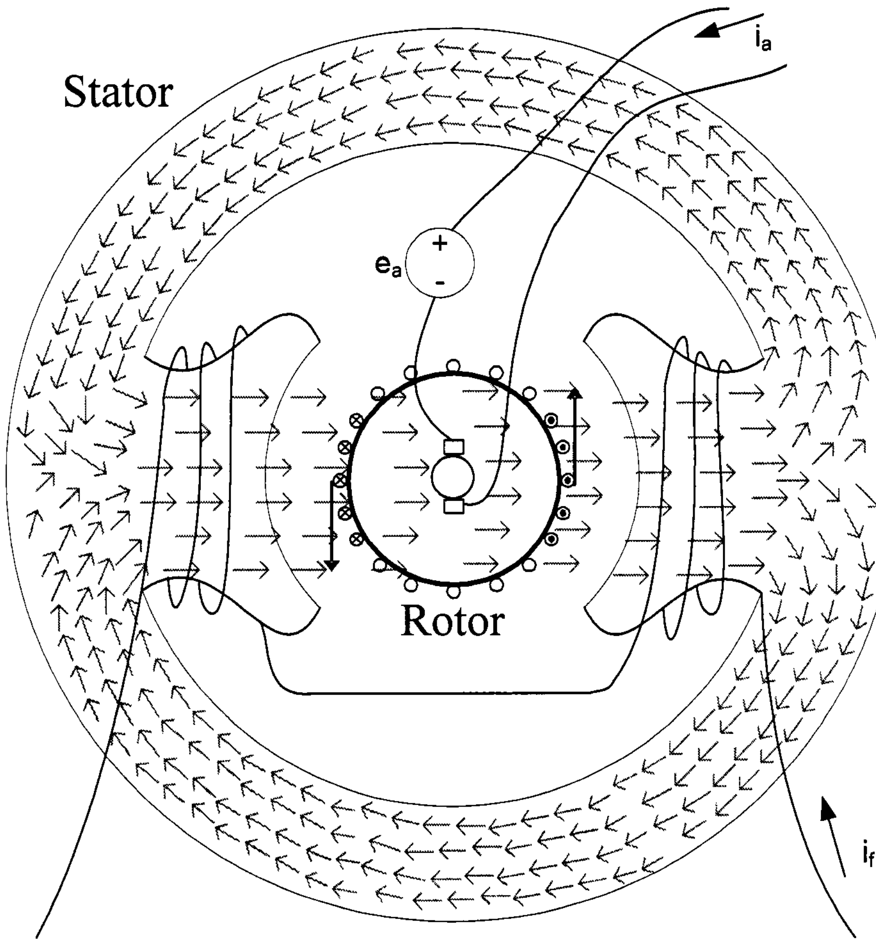


Figure 2.5 Single pole separately exited dc machine cross-section

Figure 2.5 was used to create and equivalent circuit model of a dc machine as shown in Figure 2.6. The stator windings, also known as the field windings, have finite impedance hence the need for R_f and L_f in the circuit model. In the rotor, also known as the armature, there is also an impedance represented by R_a and L_a. Aside from the impedance, the armature contains the back-emf modeled as a voltage (e_a) that is directly proportional to the rotor angular velocity (ω). The electromagnetic torque (T_{em}) produced is directly proportional to the armature current. The T_{em} is developed by the moment created by the force vector as indicating by Lorentz force equation.

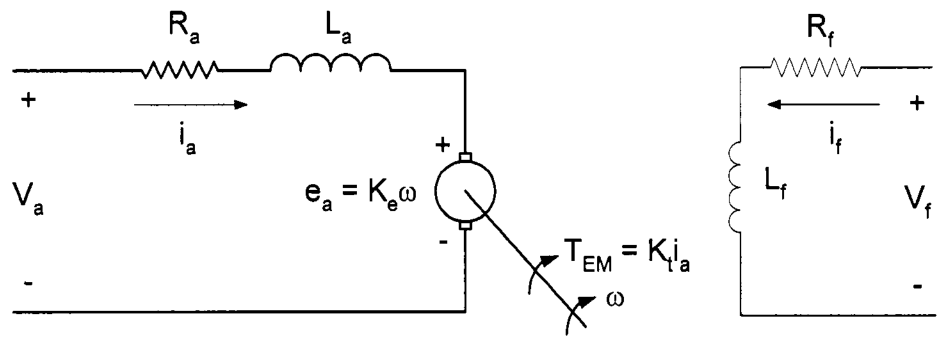


Figure 2.6 DC machine schematic.

An equivalent circuit modeling the electromechanical behavior of a dc machine allows for detailed dynamics and steady state analyses of the machine parameters (i.e. armature current, rotor angular velocity, rotor angular position, etc.). The equivalent circuit set of coupled differential equations, which form the machines characteristic equations. Equations 2.3 through 2.6 are first order coupled linear differential equations that for all practical purposes accurately model any dc machine.

$$v_f(t) = i_f(t)R_f + L_f \frac{di_f(t)}{dt} \tag{2.3}$$

where, v_f = field voltage

i_f = field current

R_f = field resistance

L_f = field inductance

$$v_a(t) = i_a(t)R_a + L_a \frac{di_a(t)}{dt} + e_a(t) \tag{2.4}$$

where, v_a = armature voltage

i_a = armature current

R_a = armature resistance

L_a = armature inductance

$$T_{em} = k_t\, i_a = \omega(t)b + J\frac{d\omega(t)}{dt} + T_L \tag{2.5}$$

where, T_{em} = electromagnetic torque

T_L = Load torque

ω = rotor angular velocity

b = viscous damping constant

J = rotor inertia

$$\omega(t) = \frac{d\theta(t)}{dt} \tag{2.6}$$

ω = rotor angular velocity

θ = rotor angular position

$$k_t = k_e = k\varphi_f = k' i_f \tag{2.7}$$

A separately excited dc machine operating with a constant dc voltage applied to its field windings behaves very similar to a BLDC machine. The constant field voltage simplifies Equation 2.3 from a differential equation to an algebraic equation. That results in the elimination of the dynamics associated with the field current i_f. Therefore, the back-emf gain and the torque sensitivy gain, k_e and k_t respectively, are considered to be constants.

2.4 Permanent Magnet (PM) BLDC Machine Construction and Characteristics

A PM BLDC machine cannot be classified as a conventional dc or ac machine, since it is partially both. The definition proposed by the National Electric Manufacturers Association (NEMA) is as follows: "A brush-less dc motor is a rotating self-synchronous machine with a permanent magnet rotor and with known rotor shaft position for electronic commutation. A motor meets this definition whether the drive electronics are integrated with the motor or separate from it," [27]. There is no mention pertaining to the number of armature phases, magnet pole sequence on the rotor, or method of position sensing. Theoretically, a brushless-dc machine may have any number of phase windings. A higher number of phase windings serves to improve the electromagnetic torque quality. For a ripple-less torque, an infinite number of phase windings is required. Of coarse, this is not possible in practice, instead the torque ripple is significantly reduced when three phases are employed with a six-pulse current commutation sequence [18].

The performance characteristics of a brushless-dc motor are very similar to that of a separately excited or shunt dc motor. In fact, a brushless-dc motor is essentially an inverted separately excited dc motor. In a dc motor, the stator windings provide a stationary magnetic field which can be replaced by permanent magnets as shown in Figure 2.7. The rotor contains slots in which additional windings are placed to provide a changing magnetic field with respect to the rotor. The windings provide a current path, so as to produce the force described by Lorentz's law. Once that force results in angular rotation, the polarity of the current in the rotor windings must be reversed to continue with the rotation of the rotor. To reverse the current polarity, metal brushes and a mechanical commutator are required.

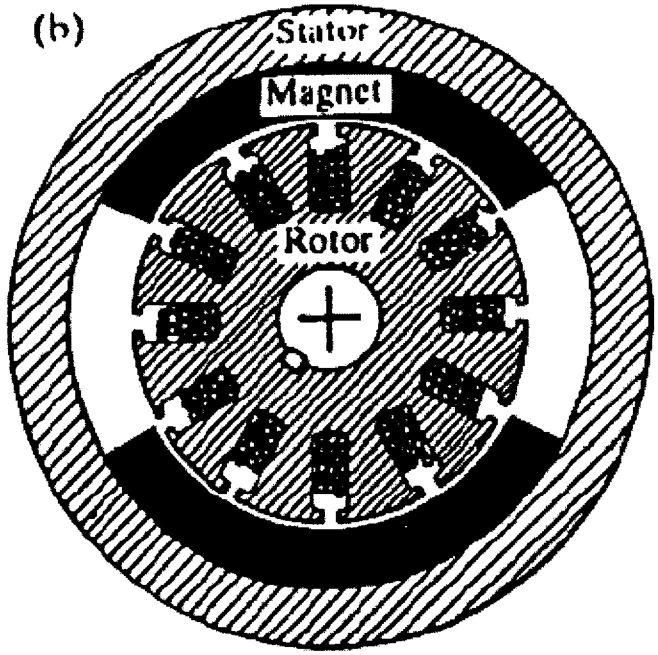


Figure 2.7 DC Motor Cross Section

In a brushless-dc motor, see Figure 2.8, the permanent magnets are transferred from the stator onto the rotor. The magnets provide a constant stationary magnetic field with respect to the rotor. The winding slots are built into the stator and the changing magnetic field is provided by the current polarity changes in the slot windings. Current polarity must change in accordance to the rotor magnetic field, which requires position sensing of the rotor. Low–resolution sensors, such as hall-effect sensors, are fixed on the stator to provide this information. Current commutation in brushless-dc motors is done via solid state switches, which eliminates the need for brushes, as the name suggests. The benefits of removing the brushes are lower maintenance, no sparks, no dust, no RF-noise, lower friction, no commutator, and better cooling conditions. Some drawbacks to brushless-dc motors are the added costs associated with permanent magnets and the need for positions sensing, making speed control more complex compared to a dc motor.

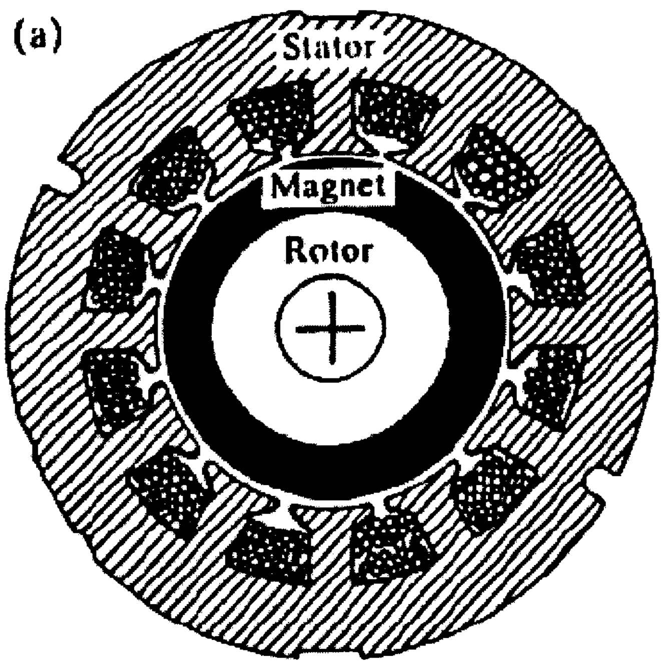


Figure 2.8 Brushless-DC Motor Cross Section

Permanent magnet brushless-dc motors can be classified according to the permanent magnet type, the shape of the back-emf waveform, the form of mounting the permanent magnets, or by its winding configuration. However, in this text only the back-emf is used to classify the motor, as it is common practice. There are two types of back-emf that a permanent magnet brushless-dc motor can generate, namely trapezoidal and sinusoidal waveforms, [13]. Therefore, the two types of permanent magnet brushless-dc motors are sinusoidal brushless-dc and trapezoidal brushless-dc. The latter is commonly referred to as a BLDC motor, and will be called as such from here on in. The back-emf waveform is determined by the manner in which windings are placed in the stator, either concentrated or distributed windings. Concentrated windings produce a trapezoidal back-emf while distributed windings result in a sinusoidal back-emf. In either case, the motor cross section will take the form of that illustrated in Figures 2.9 and 2.10.

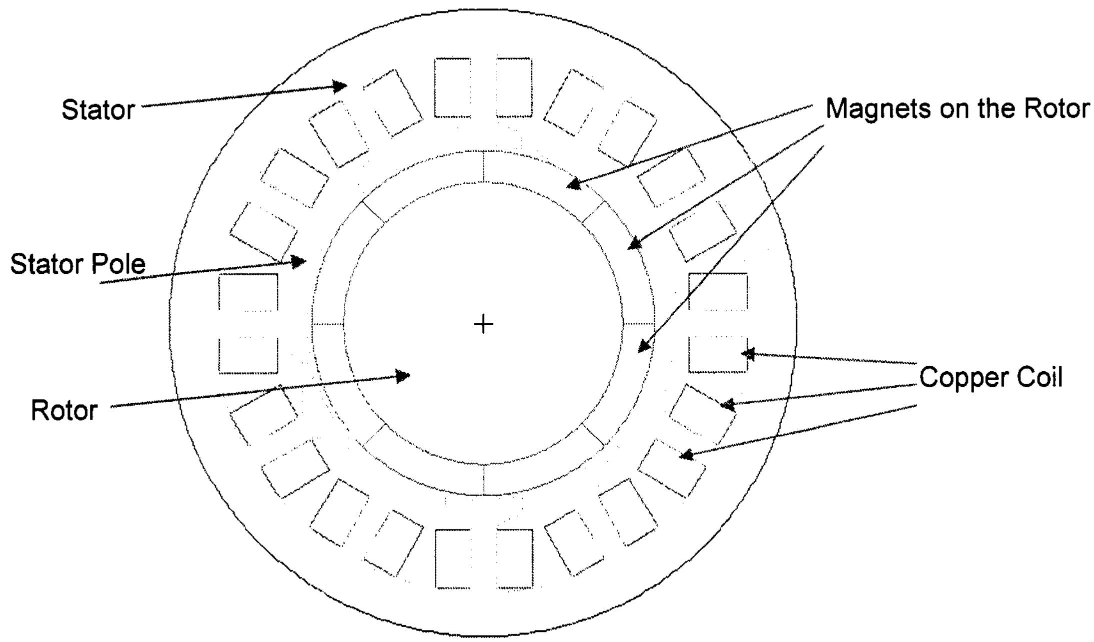


Figure 2.9 BLDC Motor Cross-Section for Surface Mounted PMs.

The fundamental difference in the two cross sections is in the placement of the permanent magnets. In Figure 2.9, the permanent magnets are adhered onto the surface of the rotor. Placing the magnets on the rotor surface cuts back on manufacturing time, but during high speed operation, there is the danger adhesive failure that can result in the magnets "flying" off the rotor. In Figure 2.10, the magnets are inserted beneath the surface of the rotor. This practice requires additional machining of the rotor to create the slots for the magnets. Manufacturing time is increased, resulting in added cost. The added benefit is that the motor can be operated at very high speed without the danger of failure due to the magnets.

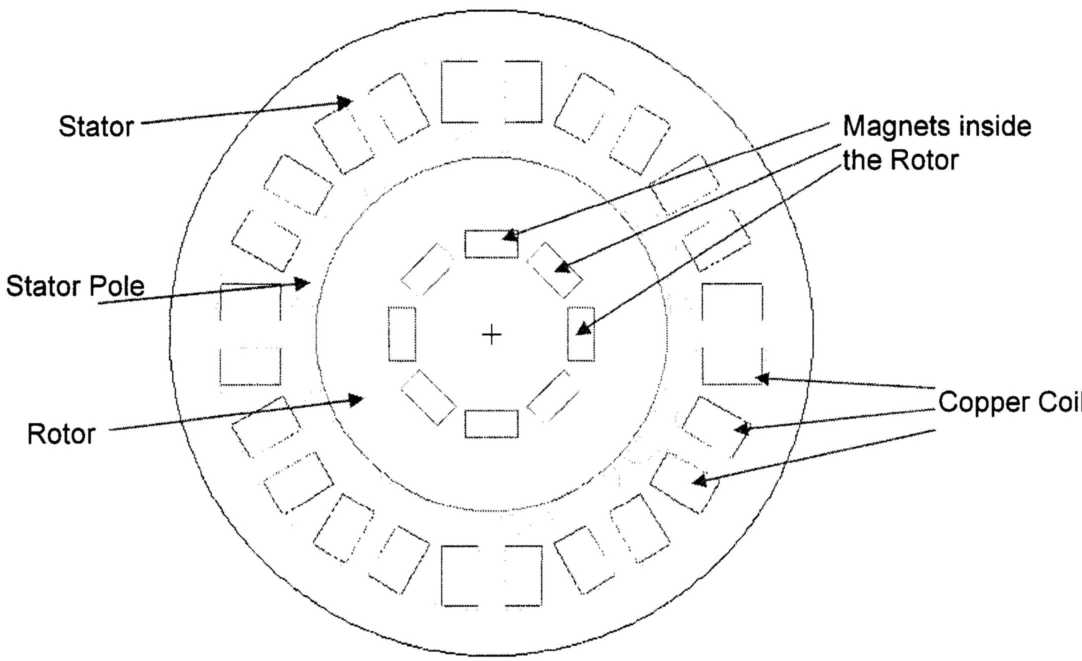


Figure 2.10 BLDC Motor Cross-Section for Interior Mounted PM's

The physical behavior of the BLDC machine has been explained through its similarities to a separately excited dc machine. BLDC machines require a dc/ac inverter along with rotor position information in order to operate. The dynamics of the machine are described by a set of mathematical differential equations similar to those defined for a dc machine. To attain the electrical equations for a BLDC machine, basic circuit analysis was used to find the per-phase voltage, Equation 2.8. The equation is only shown for phase-a to neutral since the equations for phase b and c only differ in the subscript notation.

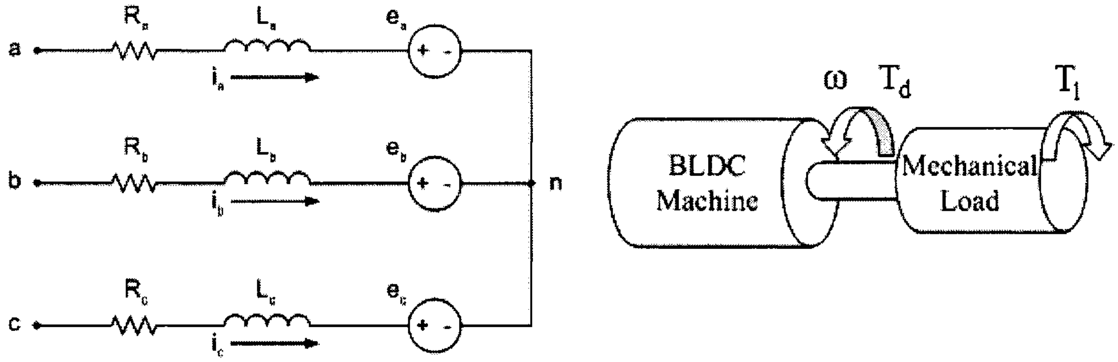


Figure 2.11 Three-phase BLDC machine equivalent circuit and mechanical model.

$$v_{an}(t) = i_a R_a + L_a \frac{di_a(t)}{dt} + e_a(t) \qquad (2.8)$$

where, $v_{an}(t)$ = per phase voltage

$i_a(t)$ = phase current

$e_a(t)$ = per phase voltage back-emf

R_a = per phase resistance

L_a = per phase inductance

Equation 2.9 is the mechanical equation that relates the machine's angular velocity to the developed electromagnetic torque, load torque, and motor parameters.

$$T_{em}(t) = \omega(t)b + J\frac{d\omega}{dt} + T_L(t) \qquad (2.9)$$

where, $T_{em}(t)$ = developled elecrtromagnetic torque

$\omega(t)$ = rotor angular velocity

b = viscous fricton canstant

J = rotor moment of inertia

T_L = load torque

Equations 2.8 and 2.9 are coupled through the developed electromagnetic torque (T_{em}) and the back-emf (e_{phase}) which are described by Equations 2.10 and 2.11 respectively.

$$T_d = k_{t\text{-}a}\, i_a + k_{t\text{-}b}\, i_b + k_{t\text{-}c}\, i_c \tag{2.10}$$

$$e_a = k_e \omega(t) \tag{2.11}$$

where, $k_{t\text{-}x}$ = per-phase torque sensitivity

$k_{e\text{-}x}$ = per-phase back-emf

In conclusion the dynamics of a BLDC machine are described by the five first order non-linear coupled differential equations. Those equations are summarized in Figure 2.12.

BLDC Machine Characteristic Equations

$$\left(\begin{array}{l} v_{an}(t) = i_a R_a + L_a \dfrac{di_a(t)}{dt} + e_a(t) \\[2ex] v_{bn}(t) = i_b R_b + L_b \dfrac{di_b(t)}{dt} + e_b(t) \\[2ex] v_{cn}(t) = i_c R_c + L_c \dfrac{di_c(t)}{dt} + e_c(t) \\[2ex] T_d(t) = \omega(t) b + J \dfrac{d\omega}{dt} + T_L(t) \\[2ex] \omega(t) = \dfrac{d\theta(t)}{dt} \end{array}\right)$$

where, $v_{xn}(t) = f(\theta, i_x)$

$e_x(t) = f(\theta, \omega)$

Figure 2.12 BLDC machine characteristic equations.

2.5 Power Electronic Driver Circuitry

Brushless-dc machines have multi-phase windings in the stator, however the most popular motor construction is that of a three-phase motor, namely phase a, b, and c. To achieve motor rotation, a waveform matching the back-emf must be applied to the three phases. For BLDC machines, the back-emf should be of trapezoidal shape as illustrated in Figure 2.13. The back-emf is a function of the rotor position θ_r, which can be derived from the hall-effect sensors. The ideal current waveform relative to rotor angular position is also given for each phase in Figure 2.13, [23].

To produce the trapezoidal excitation voltage, a three-phase dc to ac inverter is required. For lower power drive systems, MosFets are used as shown in Figure 2.14, but for higher power systems IGBTs are the switches of choice, [24]. The BLDC motor is connected as shown below and the MosFet switching is summarized by the truth table derived from the hall-effect position sensors, see Table 2.1. The switching must follow the six step sequence, however the first step was chosen arbitrarily.

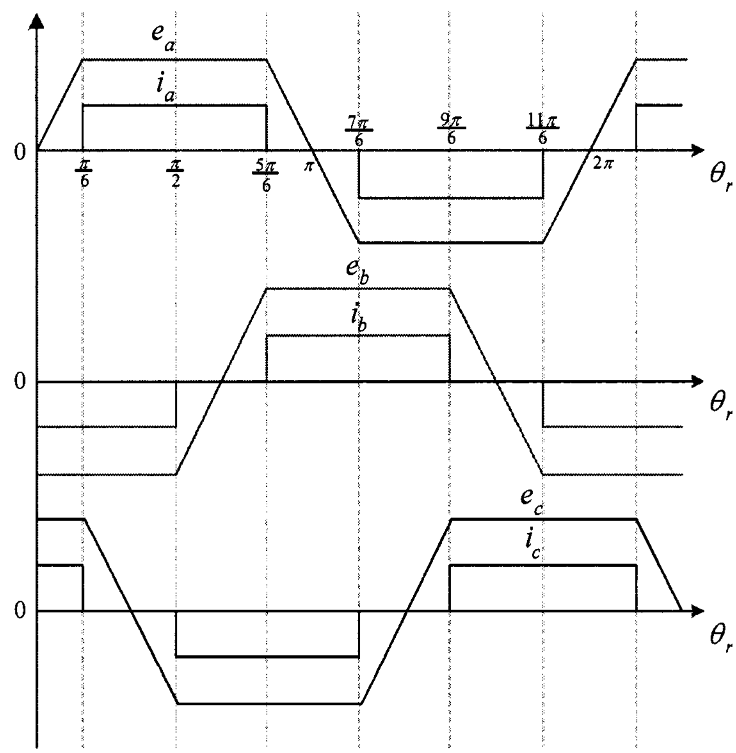


Figure 2.13 Trapezoidal BLDC Motor Ideal Back-EMF and Current.

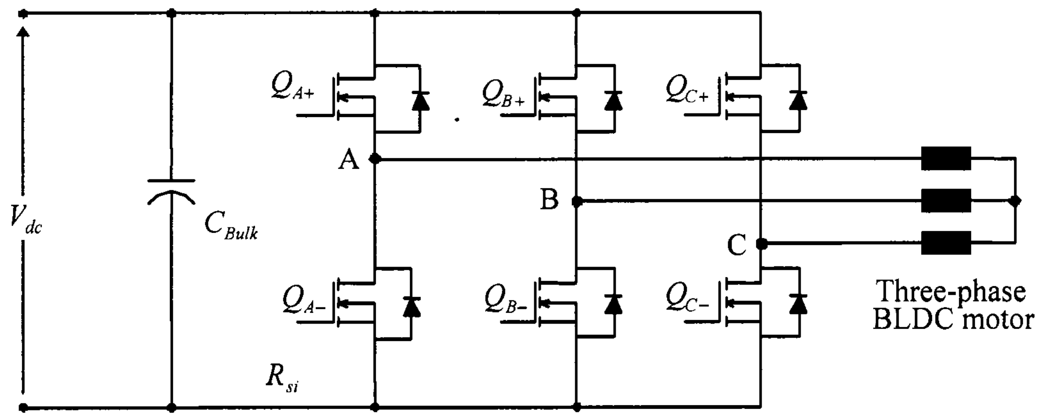


Figure 2.14 Three-Phase DC to AC Inverter

Table 2.1 Hall-Effect Truth Table for Switching

Step	Q_{A+}	Q_{B+}	Q_{C+}	Q_{A-}	Q_{B-}	Q_{C-}
1	ON	OFF	OFF	OFF	OFF	ON
2	ON	OFF	OFF	OFF	ON	OFF
3	OFF	OFF	ON	OFF	ON	OFF
4	OFF	OFF	ON	ON	OFF	OFF
5	OFF	ON	OFF	ON	OFF	OFF
6	OFF	ON	OFF	OFF	OFF	ON

The power source in Figure 2.14 is typically a dc supply. Therefore the purpose of the power electronic converter is to periodically apply a positive, negative, or zero dc supply voltage to the motor phase terminals so as to regulate the current injected into the motor phases. The power electronic converter performs hard switching on the current which requires anti-parallel diodes due to the inductive nature of the load, namely the BLDC machine.

To demonstrate the current control capabilities during positive and negative conduction of the phase currents, the switching scenario is outlined and explained. When positive current is desired in phase A, top switch A and bottom switch B (bottom switch C will provide the same result) are closed. Figure 2.15 shows that a positive voltage results across the A-B motor terminals. Current is positive and increasing in magnitude until top switch A is opened. Due to the inductive nature of the motor windings, current must be continuous therefore bottom diode in leg A will conduct, see Figure 2.16, resulting in zero volts applied across the motor A-B terminals. The current will remain positive but will decrease until top switch A is turned on once again. Pulse width modulation or Hysteresis current control may be used for the switching scheme of the top A switch. For the model described in this chapter Hysteresis current control is utilized due to it direct control capability of the phase currents.

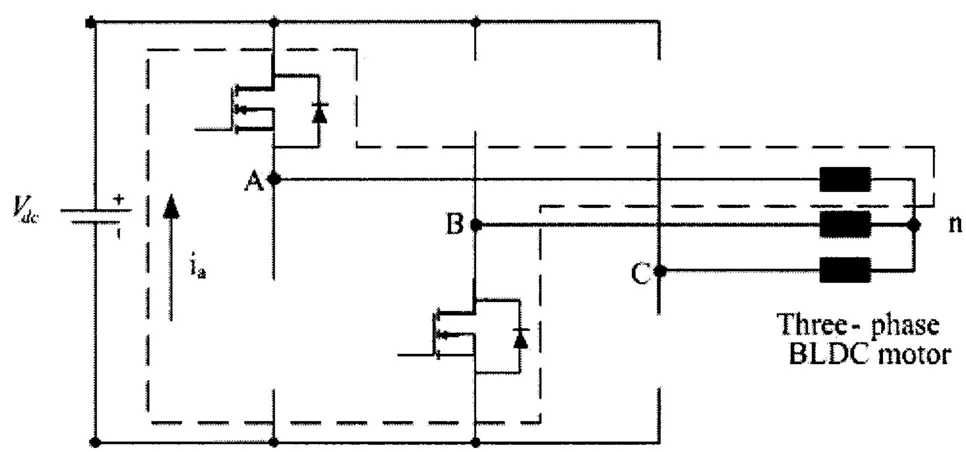


Figure 2.15 Phase A positive voltage excitation.

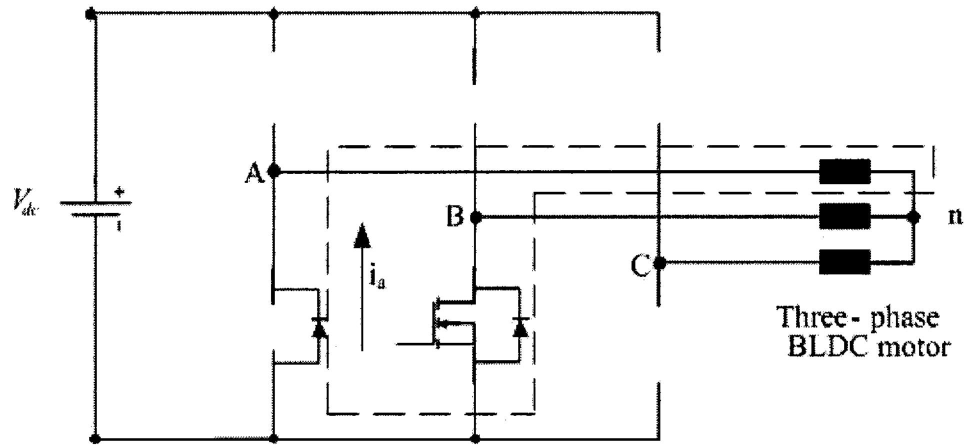


Figure 2.16 Free-wheeling phase current during positive conduction.

In the case where negative current is required in phase A, bottom switch A and top switch B (top switch C will also produce the same current in phase A) are turn on. That results in the negative battery voltage applied to the A-B motor terminals. As with positive current conduction, freewheeling mode is achieved by turning off top switch B. Doing so results in a zero voltage applied to the A-B motor terminals which will serve to decrease the magnitude of the phase A current. Current conduction has been outlined for positive and negative current injections, however there are periods of zero current conduction as indicated by Figure 2.13. During those time intervals, both top switch A and bottom switch B remain in the off state. Therefore current conduction is only allowed through the anti-parallel diodes of legs A and B. The top and bottom diodes of the opposing inverter legs will alternate so as to maintain the current near zero. That only applies during the ideal zero conduction portions of the phase currents.

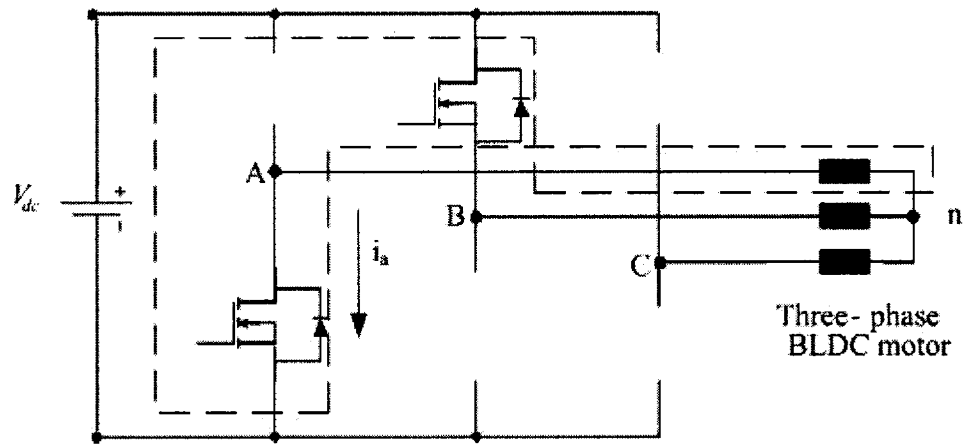


Figure 2.17 Phase A negative voltage excitation.

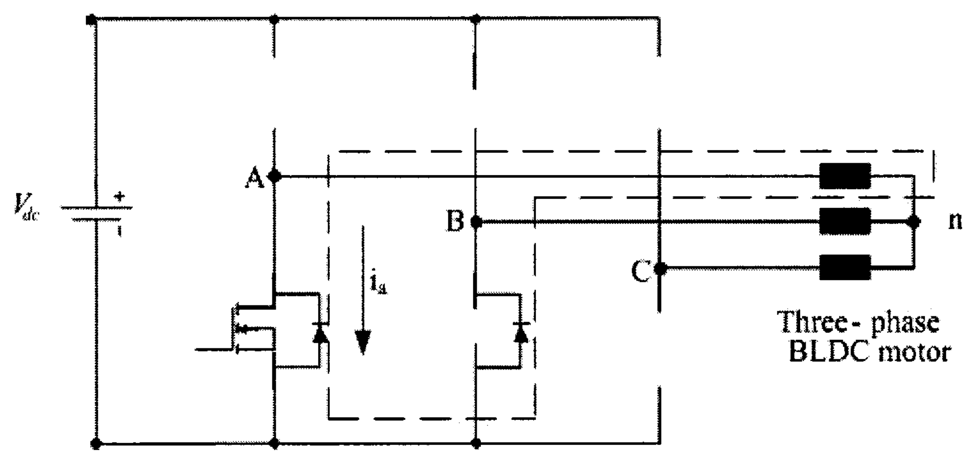


Figure 2.18 Free-wheeling phase current during negative conduction.

2.5.1 Voltage Control. It has been shown that the speed of a BLDC machine is directly proportional to the voltage applied to its terminals. To achieve speed regulation of a BLDC motor, some type of variable voltage source is required. One possible means of producing a variable voltage dc supply is to place a buck converter between the dc voltage supply and the three phase dc to ac inverter as shown in Figure 2.19.

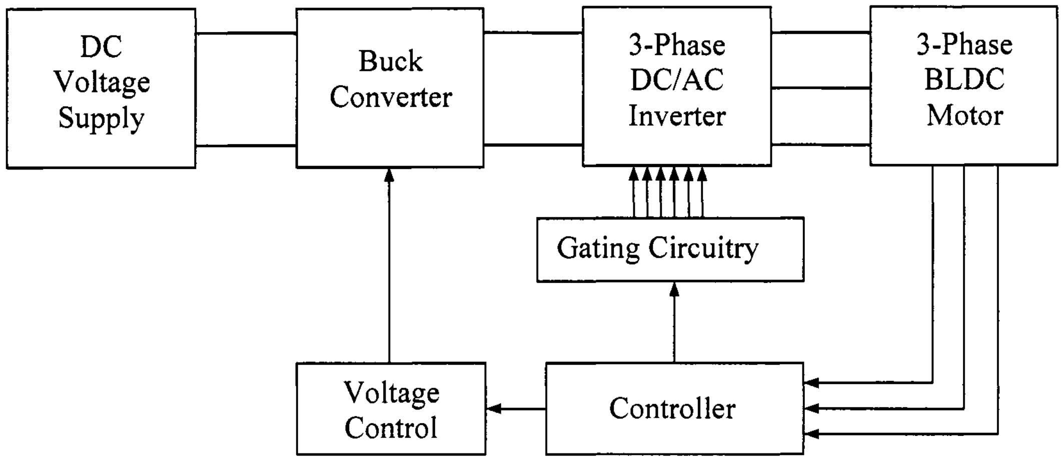


Figure 2.19 Voltage Control via Buck Converter

The addition of the buck converter to the driver circuit topology provides the means to vary the dc voltage supply. However, introducing the additional component also adds cost and complexity to the overall system, making it an unattractive solution for practical implementation. Instead, it is desired to include the voltage control characteristic of a buck converter into the three phase dc to ac inverter, see Figure 2.20.

There are two methods for implementing voltage control into the three phase dc to ac inverter, namely PWM voltage control and Hysteresis current control. PWM voltage control is easy to realize in a practical system, but often times requires some kind of over-current protection to ensure the safety of the motor. On the other hand, Hysteresis current control possesses an inherent over-current protection feature, but is considered to be more complex than PWM control.

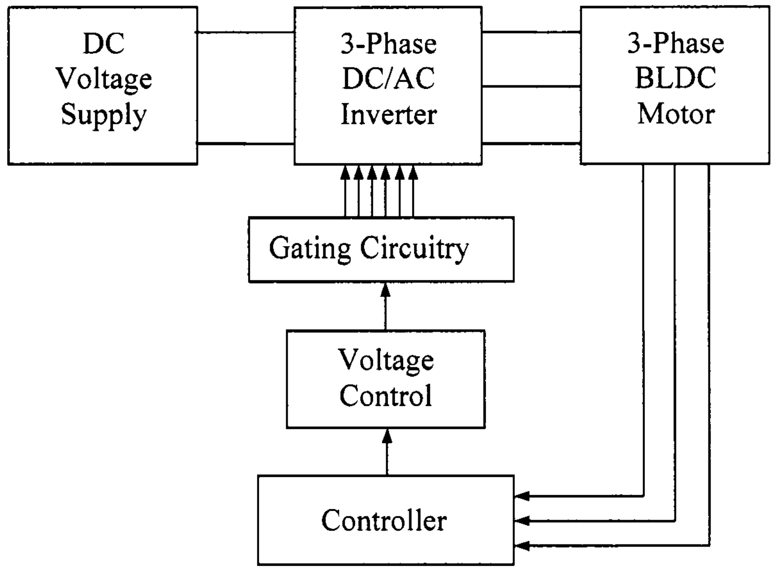


Figure 2.20 Voltage Control via DC/AC Inverter

2.5.2 PWM Control. Pulse width modulation (PWM) serves to control the average output voltage given a fixed input voltage. That is accomplished by using power switches to vary the time in which the dc input is applied to the load and the time that it is not applied, t_{on} and t_{off} respectively. The PWM control concept is better understood when applied to a basic switch-mode dc to dc converter as shown in Figure 2.21.

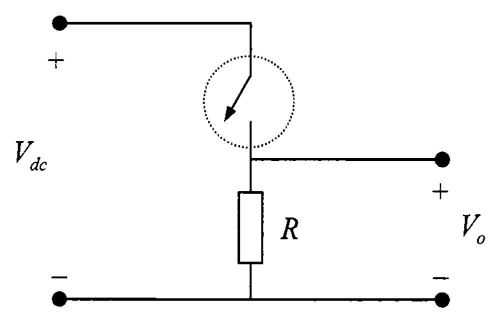


Figure 2.21 Basic Switch-Mode DC to DC Converter

The on-time (t_{on}) is defined as the time in which the encircled switch is closed, allowing the input dc voltage (V_{dc}) to be applied to the resistive load. Off-time (t_{off}) is when the encircled switch is opened resulting in no voltage applied to the load. Therefore, during the on-time $V_o = V_{dc}$ and during the off-time $V_o = 0$.

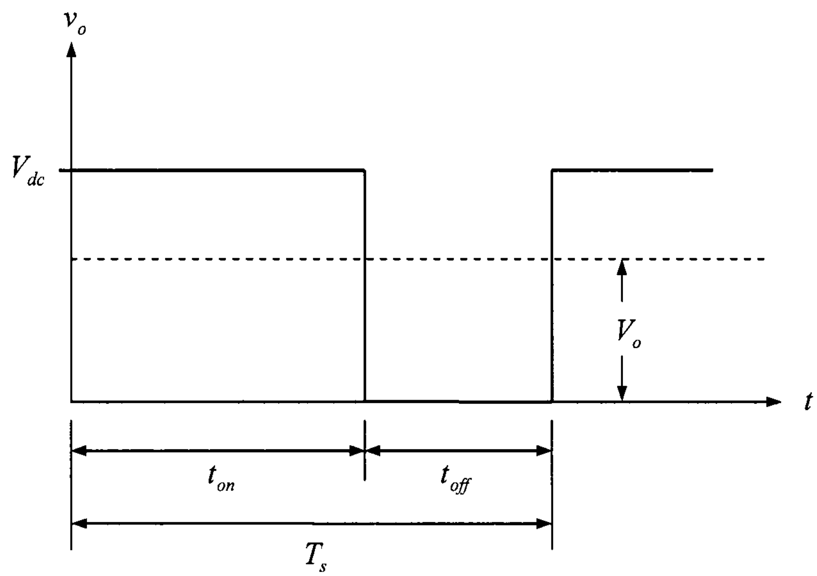


Figure 2.22 Switch-Mode DC/DC Converter Voltage Wave-Form

The output voltage wave-form to a switch-mode dc to dc converter is illustrated in Figure 2.22. As shown, the switching nature of the converter results in a variable average voltage at the output. The switching time period (T_S) is simply the sum of the on-time and off-time. Further classification of PWM control is made with respect to the nature of the switching time period. If the switching time period is variable, then it is a variable frequency PWM control. On the other hand, when the switching time period is constant, it is called constant frequency PWM control.

A common problem with having variable frequency switching under any circumstances is in the difficulty of filtering the harmonics produced at a wide range of frequencies. With constant frequency PWM control, the harmonics produced are at the switching frequency and its multiples, making filtering much easier. Therefore, only constant frequency PWM is considered in this section.

It is important to derive a relation between the input voltage and output voltage as a function of the on-time and off-time. First, a parameter called the duty ratio (D) is defined as the on-time duration to the switching time period, see Equation 2.12.

$$D = \frac{t_{on}}{T_s} \tag{2.12}$$

Then, taking the integral of the output voltage over one time period and dividing by the time period yields the input to output relationship.

$$\begin{aligned} V_o &= \frac{1}{T_s}\int_0^{T_s} v_o(t)dt \\ &= \frac{1}{T_s}\left(\int_0^{t_{on}} V_{dc}dt \int_{t_{on}}^{T_s} 0dt\right) \\ &= \frac{t_{on}}{T_s}V_{dc} \end{aligned} \tag{2.13}$$

Substitution of the duty ratio as defined by Equation 2.12 yields

$$V_o = DV_{dc} \tag{2.14}$$

A nice feature of PWM control is that there is a linear relationship between the input and the output, just like a linear amplifier. This type of voltage control is well suited for use in linear control systems as presented Chapter 4.

The basic switch-mode dc to dc converter in Figure 2.22 may only function as drawn for a purely resistive load. In practice, the load will be resistive-inductive, as is the case with any electric machine. Even resistive loads will always contain some stray inductance. An inductive load means that damage to the switches may take place if the current through the inductors is forced towards zero in a very short time. The forcing of the current occurs every time the switch is turned off or on, which resulting in very high voltage spikes as defined by Equation 2.15.

$$V_L = L\frac{di_L}{dt} \tag{2.15}$$

where, V_L = inductor voltage

i_L = inductor current

L = inductance

The high voltage spikes can become large enough and exceed the switch voltage rating despite over design engineering measures. One way to eliminate the high voltage spikes during turn-off time is by allowing the current to go down towards zero at a slower rate. To accomplish this, a diode is introduced into the circuit to allow a current path during the turn off time. The diode is labeled a free wheeling diode since it provides a free wheeling path for the current to naturally die down towards zero. For a three phase dc to ac inverter, the free wheeling path is provided by the diodes in anti-parallel configuration with respect to a switch, see Figure 2.23.

Now that the concept of PWM control is well understood, a means of generating the PWM signal must be presented. The PWM signal must be composed of On and Off pulses that are fed to the switches during operation. Figure 2.24 illustrates a common approach to producing the PWM signal.

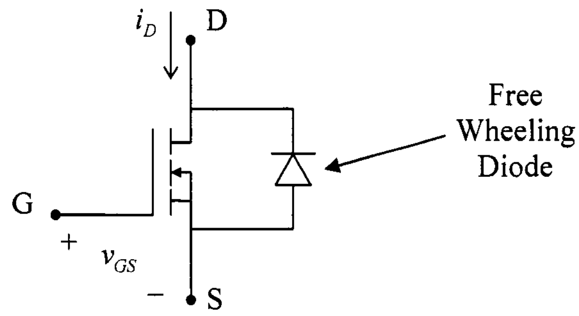


Figure 2.23 Switch with Anti-Parallel Diode

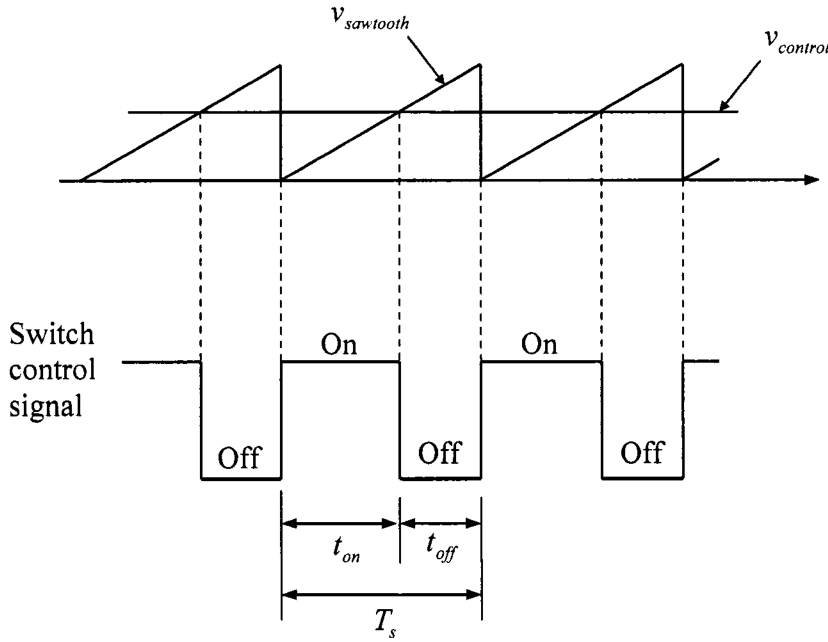


Figure 2.24 PWM Signal Generation

The voltage control signal ($v_{control}$) is compared with a saw-tooth waveform to produce the PWM signal. If the voltage control signal is greater than the saw-tooth signal, then the switch is turned on for current conduction. If the voltage signal is less than the saw-tooth signal, then the switch is turned off. A block diagram representation of the above figure is presented in Figure 2.25. The repetitive waveform can also take the form of a triangular wave-form, similar results are attained to that of a saw-tooth wave-form. An attractive feature of PWM voltage control is that harmonics generated from turning the switches on and off a are at a fixed frequency. Therefore, filters can be easily design to suppress the propagation of harmonics throughout the motor drive control circuitry.

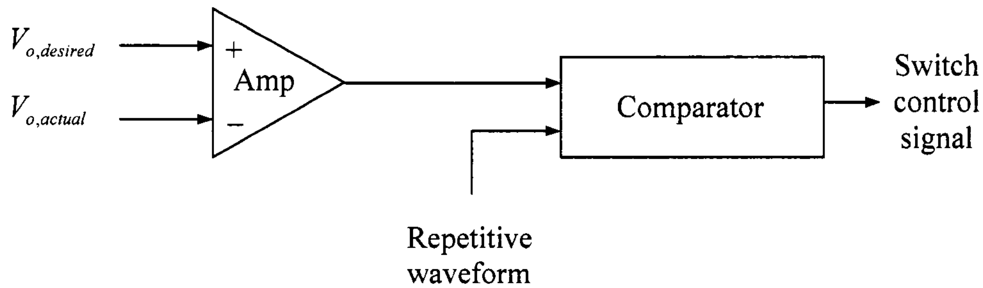


Figure 2.25 PWM Block Diagram

2.5.3 Hysteresis Current Control. Hysteresis current control, achieves voltage control indirectly by monitoring the current through the load and forcing it to stay within a predefined band-gap, [3]. For example, if the load is inductive-resistive, forcing the current to remain within a band-gap will result in an average output voltage as illustrated in Figure 2.26.

The on-time and off-time applied to the power switch are not fixed values. They vary depending on the behavior of the current, resulting in a variable switching frequency. The variable switching frequency poses the problem of switching harmonics over a wide frequency range making harmonic filtration difficult. However due to its nature of operation, Hysteresis current control has inherent over-current protection since it is always being monitored. This is a very attractive feature for high performance motor drive systems. Control of the current magnitude ensures that the electric machine will not fail due to dramatic load changes. Such a load change may occur when an object may prohibit the rotation of the rotor causing a locked rotor situation. Under locked rotor conditions, PWM control would command a duty ratio equal to one. When that occurs, there is essentially a short circuit which will certainly damage the motor if no additional protective measures are taken.

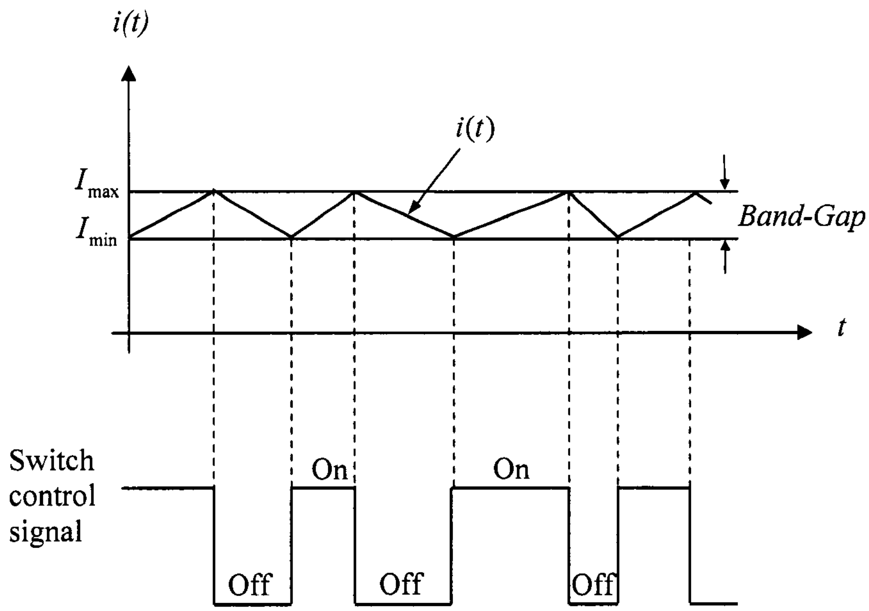


Figure 2.26 Hysteresis Current Control.

To implement Hysteresis current control, comparators and logic gates are used to determine the off and on times of the switch. The current is monitored in order to determine the possible states that the instantaneous current can take with respect to the band-gap. Table 2.2 below explicitly defines the three states in which the instantaneous current may exists.

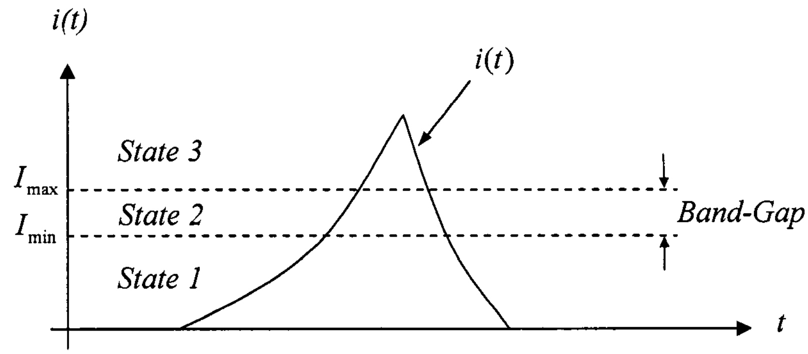


Figure 2.27 Hysteresis Current Control States

Table 2.2 Hysteresis Current State Definitions.

State 1	State 2	State 3
$i(t) < I_{max}$	$i(t) < I_{max}$	$i(t) > I_{max}$
$i(t) < I_{min}$	$i(t) > I_{min}$	$i(t) > I_{min}$

Logic gates are used to manipulate the defined states. Therefore, logic values are given by the following two bit notation. The choice of 10, 00, and 10 (for states one, two, and three) were made to accommodate the logic devices that are used to produce the switch control signals.

State 1 => 10

State 2 => 00

State 3 => 10

From observing Figure 2.23, it is determined that if the instantaneous current is in State 1, then the switch control signal should correspond to the on status of the switch. Allowing the switch to conduct will force the current to become more positive. When the current is in Stage 3, the switch should be turned off, forcing the current to decrease. State 2 requires memory of the previous State to determine whether the switch should be turned on or off. If the previous state was State 1, the switch should remain on. However, if the previous state was State 3, then the switch is to remain off.

An edge triggered set-reset flip-flop is memory capable device well suited to perform the required task of deciding the on or off state of the switch. The truth table of an edge triggered set-reset flip-flop is in Table 2.3. The non-inverted output (Q), is the switch control signal that determines the status of the switch. Also included in the table is the resulting state that corresponds to the flip-flop output Q.

Table 2.3 Edge Triggered Set-Reset Flip-Flop Truth Table

S	R	Q	Qn	State
0	0	no change		State 2
X	0 => 1	0	1	State 3
0 => 1	X	1	0	State 1

2.6 Position Sensors

Brushless-dc machine operation requires of rotor position information to allow for appropriate solid state switch firing. Three leading technologies are commonly used to fulfill the position information requirement. Those technologies are hall-effect sensors, resolvers, and optical encoders. The most commonly used sensor type is a hall-effect sensor set. They are low cost and provide position resolution to within thirty electrical degrees, which is sufficient to operate a BLDC machine. If precise speed regulation is required, a higher resolution position sensor is needed. Both optical encoders and resolvers offer much higher position resolution. The difference in the two sensors is most evident in their robustness under harsh mechanical environments. In general, optical encoders are fragile in comparison to and encoder. Resolvers can easily survive in automotive propulsion applications where high temperatures and extreme vibration is common. The position sensor type will always depend on the particular application. Often times, redundant position sensors are used to increase the survivability of the motor drive system. Sensor-less position techniques are also available as back up position detecting methods.

2.7 Conclusion

Chapter 2 covered the basic operating principles of a BLDC motor drive system. The BLDC machine characteristic equations and the need for a power electronic converter were discussed. Hysteresis current control and PWM voltage control were presented as two possible speed regulation methods. The next step was to develop a system model to be used in a computer simulation package. The model is to mimic the machine and power electronics so as to serve as a test bed for classic and new control methods. Chapter 3 focuses on the development of a simulation model followed by a description of the experimental test bed developed to validate the digital control that will be presented in Chapters 5 and 6.

CHAPTER 3

BLDC MACHINE MODEL AND EXPERIMENTAL SETUP

3.1 Introduction

To evaluate the functionality of any controller, a precise model for the BLDC machine must be created and implemented in a computer simulation package. This chapter presents the steps taken towards the development of a precise machine model for a BLDC machine. Also, the power electronic converter was modeled in order to explicitly illustrate the current dynamics. A general model was created in Matlab/Simulink so as to model any BLDC machine given the motor parameters (i.e. phase resistance/inductance, rotor inertia, torque sensitivity constant, etc.) Figure 3.1 illustrates the block diagram for typical motor drive setup.

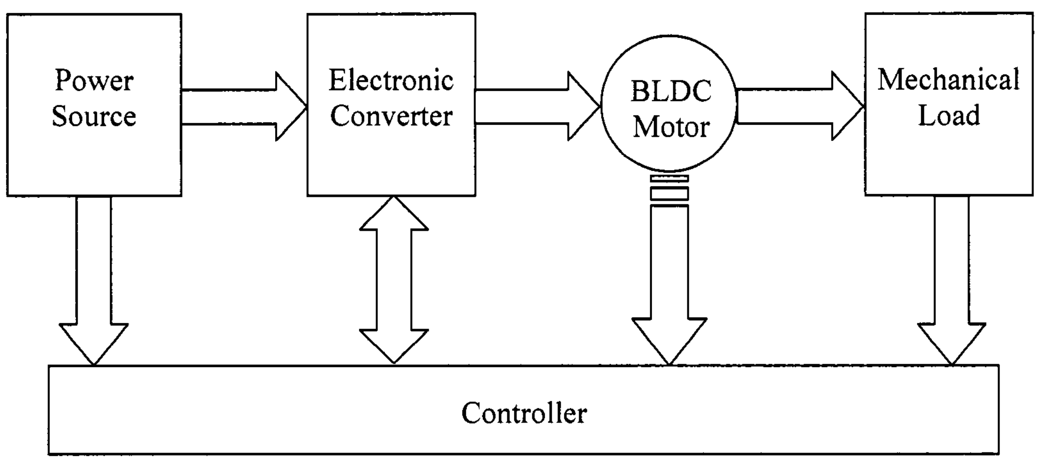


Figure 3.1 BLDC motor drive block diagram.

The computer model must include the illustrated components, all of which are assumed to be ideal. Switching losses in the power electronic converter were neglected and all sensors (current, speed, and voltage) are assumed to be ideal. Therefore no signal attenuation or delay was attributed to the feedback information.

3.2 Mathematical System description

The first step towards the development of any machine model, is to derive the characteristic equations that describe its dynamic and steady state behavior. To attain the electrical equations for a BLDC machine, basic circuit analysis was used to find the per-phase voltage, Equation 3.1. The equation is only shown for phase-a to neutral since the equations for phase b and c only differ in the subscript notation.

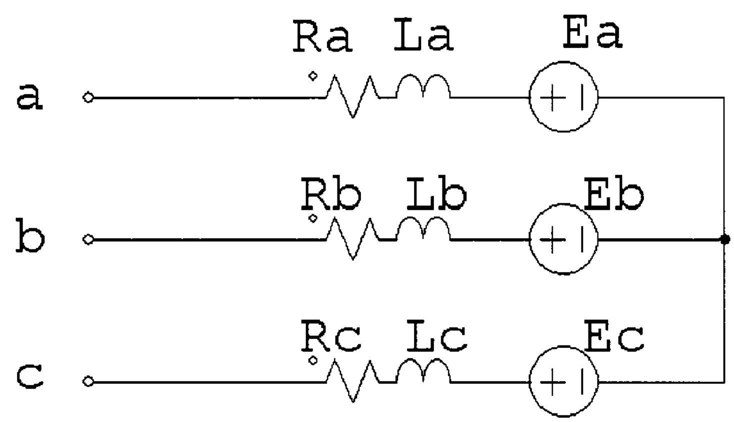


Figure 3.2 BLDC machine equivalent circuit.

$$v_{an}(t) = i_a R_a + L_a \frac{di_a(t)}{dt} + e_a(t) \tag{3.1}$$

where, $v_{an}(t)$ = per phase voltage

$e_a(t)$ = per phase back-emf

$i_a(t)$ = phase current

R_a = per phase resistance

L_a = per phase inductance

In addition to the electrical equation, a mechanical equation is derived in order to describe the machine's angular motion, see Equation 3.2.

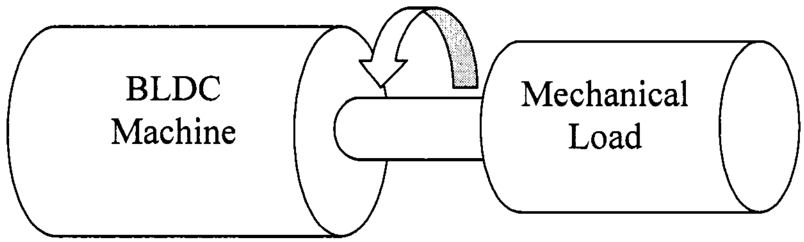


Figure 3.3 BLDC mechanical coupling.

$$T_d(t) = \omega(t)b + J\frac{d\omega}{dt} + T_L(t) \tag{3.2}$$

where, $T_d(t)$ = developled torque

$\omega(t)$ = rotor angular velocity

b = viscous fricton canstant

J = rotor moment of inertia

T_L = load torque

In order to couple the two equations, the developed electromagnetic torque (T_d) and the back-emf (e_{phase}) are rewritten in terms of the current and speed, respectively.

$$T_d = k_{t\text{-}a}\, i_a + k_{t\text{-}b}\, i_b + k_{t\text{-}c}\, i_c \tag{3.3}$$

$$e_a = k_e\omega(t) \tag{3.4}$$

where, $k_{t\text{-}x}$ = per-phase torque sensitivity

$k_{e\text{-}x}$ = Per-phase back-emf constant

Equations 3.5, 3.6, 3.7 and 3.6 are the Laplace transforms of Equation 3.1 and 3.2 which include the relations for e_a and T_d as outlined above. The equations were utilized to construct the block diagram shown in Figure 6.4. Two non-linearity's are explicitly shown; namely the back-emf and the torque sensitivity. The back-emf is a function of rotor angular position and rotor angular speed, while the torque sensitivity is a function of rotor position and phase current.

$$V_{an}(s) = I_a(s)\,R_a + I_a\,(s)s\,L_a + k_{e,a}\,\omega(s) \tag{3.5}$$

$$V_{bn}(s) = I_b(s)\,R_b + I_b\,(s)s\,L_b + k_{e,b}\,\omega(s) \tag{3.6}$$

$$V_{cn}(s) = I_c(s)\,R_c + I_c\,(s)s\,L_c + k_{e,c}\,\omega(s) \tag{3.7}$$

$$k_{t,a}I_a(s) + k_{t,b}I_b(s) + k_{t,c}I(s) = \omega(s)b + \omega(s)sJ + T_L(s) \tag{3.8}$$

3.3 System Model

The mathematical system description was used to develop a non-linear state space block diagram model of the BLDC machine. The equations were transformed into the S-domain so as to represent the time domain differential equations into algebraic S-domain equations. Block outputs were included to explicitly show the equation realization at the summing junctions.

The developed block diagram is in fact an electro-mechanical system, as is the case with any electric motor drive system. Rotor angular motion is described by Newton's second law of motion applied to a rotating body. [32]. The mechanical dynamics and static behavior are described by the mechanical equation as illustrated in Figure 3.4. Current commutation in each motor phase winding embodies the electrical characteristics of the BLDC machine. The three electrical equations as shown in Figure 3.4 describe the electric dynamic and static behavior.

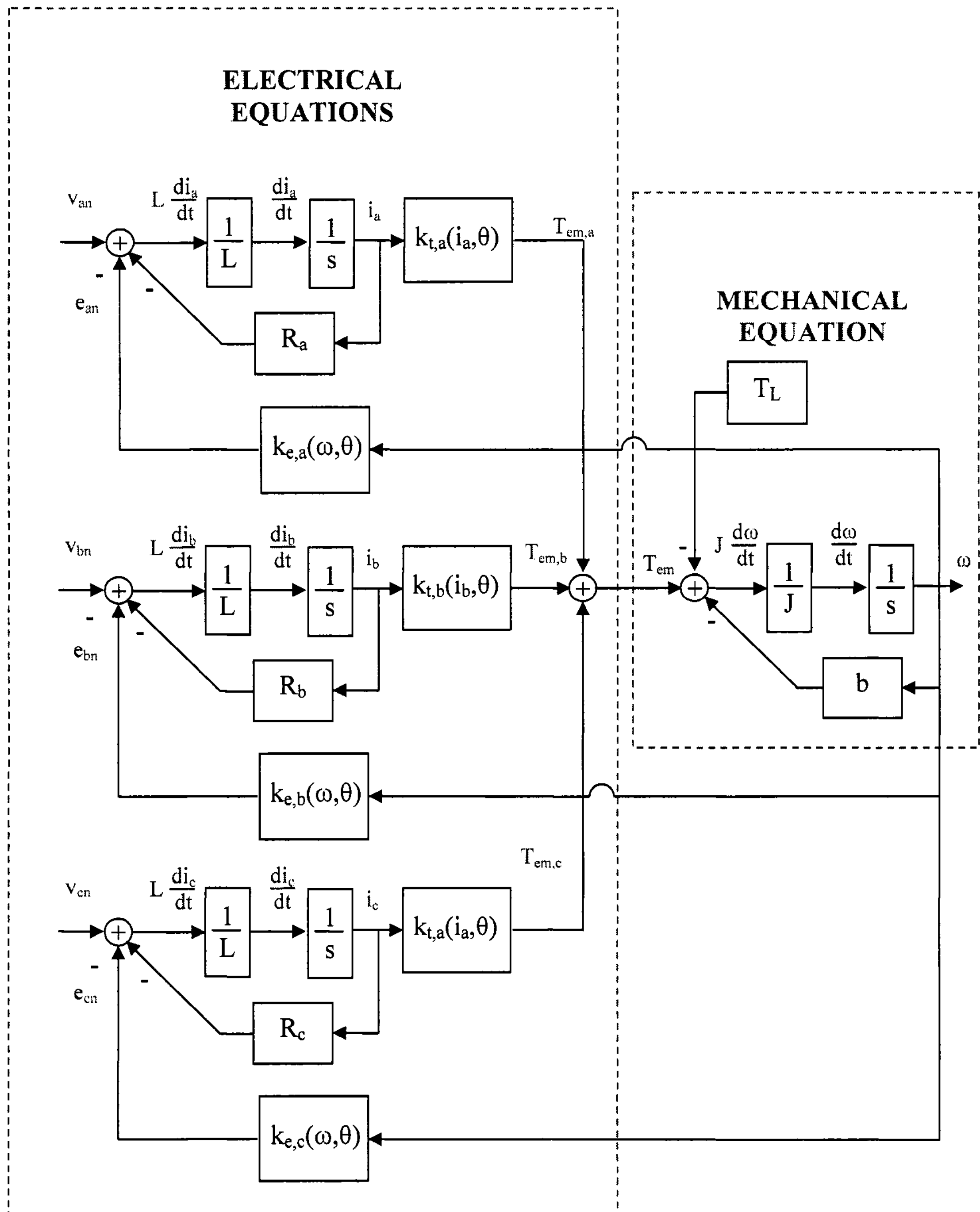


Figure 3.4 BLDC machine model block diagram.

3.3.1 Trapezoidal Back-emf Waveform Construction. A BLDC machine is characterized by a non-linear back-emf waveform (see Figure 3.5), therefore it must be explicitly defined via look-up table or a conditioned base scheme. Examination of the ideal waveform reveals that the shape of back-emf is a function of the rotor position. Also, the amplitude of the back-emf is linearly related to the rotor angular velocity (ω).

The first task in modeling the back-emf was to create a normalized trapezoidal waveform as described in Figure 3.5. Therefore, a Simulink block was created to produce a normalized back-emf waveform given the rotor position. Given the rotor position, the ideal back-emf waveform was used as a guide to create a piece-wise normalized trapezoidal function. The piece-wise function is composed of functions defined for specific areas within one wave-cycle. Table 3.1 summarizes the piece-wise function definitions for the normalized trapezoidal waveform created for phase A of the BLDC machine model.

The same procedure was followed to generate the trapezoidal waveforms for phase B and phase C with their appropriate one hundred and twenty degrees offset. As previously mentioned, the normalized trapezoidal waveform described in Table 3.1 can also be realized via a look up table. The simulation package must have interpolation capabilities in order to effectively realize adequate waveform resolution. The look-up table will reduce simulation time since no calculation time is required to produce an output. Though, which ever method is utilized, the global simulation sampling time is of great importance towards the accuracy of the simulation results.

Table 3.1 Back-emf function definitions for phase A.

Rotor Position	Piece-wise Functions
$0 \le \theta_r < \frac{\pi}{6}$	$y = \theta_r^{'}$
$\frac{\pi}{6} \le \theta_r < \frac{5\pi}{6}$	$y = \frac{\pi}{6}$
$\frac{5\pi}{6} \le \theta_r < \frac{7\pi}{6}$	$y = -\theta_r^{'} + \pi$
$\frac{7\pi}{6} \le \theta_r^{'} < \frac{11\pi}{6}$	$y = -\frac{\pi}{6}$
$\frac{11\pi}{6} \le \theta_r^{'} < \frac{12\pi}{6}$	$y = \theta_r^{'} - 2\pi$

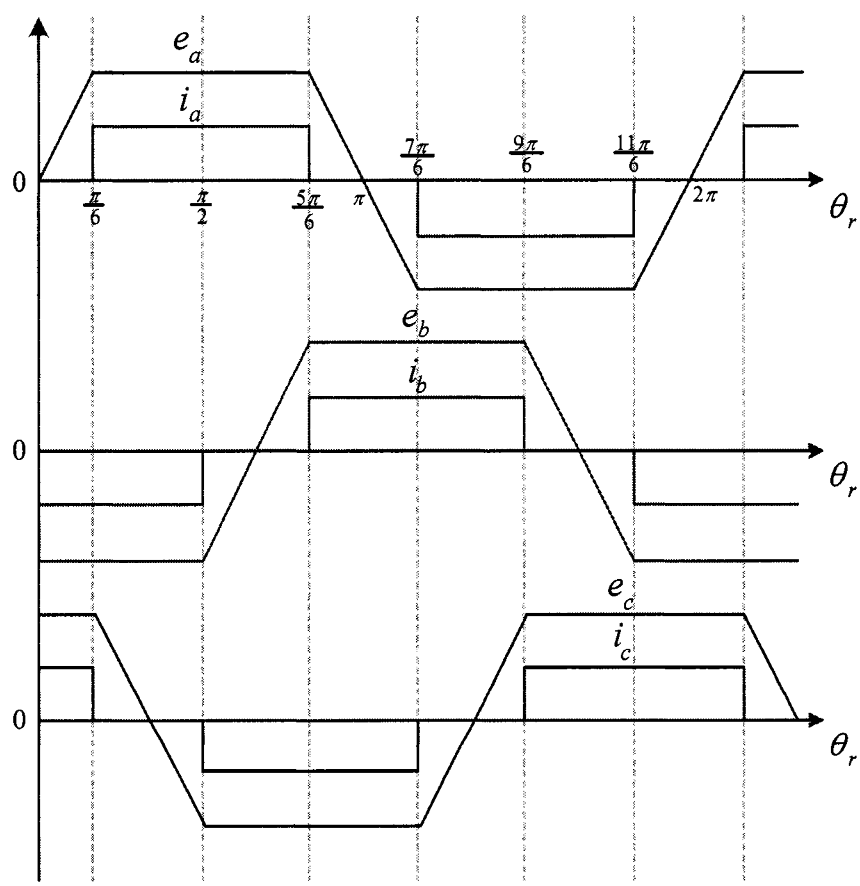


Figure 3.5 Ideal back-emf and current waveforms.

Production of the back-emf waveform is contingent upon rotor position information. The back-emf was defined for $0 \le \theta_r < 2\pi$, therefore the position output was constrained to those limits with the use Equation 3.9. In the equation, the variable "n" is an integer value representing the number of rotations undergone by the rotor referenced to its initial position.

$$\theta_r = | \theta_{norm} - n |*2\pi \tag{3.9}$$

Equation 3.9 resets the position to zero whenever the position exceeds a multiple of 2π. Taking the absolute value ensures that the function works for negative rotor positions. Figure 3.6 illustrates how the output of Equation 3.9 (θ_r) is produced, given the normalized rotor position (θ_{norm}). The normalized rotor position is defined as the rotor position measured in revolutions as apposed to radians. The BLDC machine block diagram shown in Figure 3.4 generates speed information in radians per second which when integrated produces rotor position in radians. That value is divided by a factor of 2π in order to attain rotor position in revolutions. Figure 3.7 demonstrates the normalized position output in comparison to the actual rotor position.

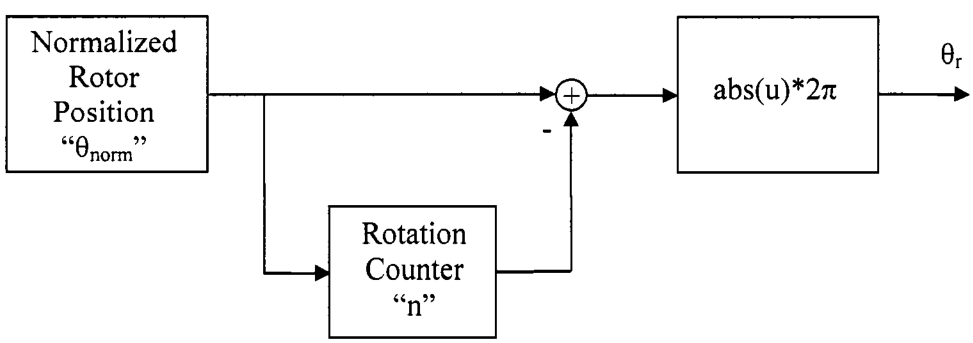


Figure 3.6 Rotor position decoding block diagram.

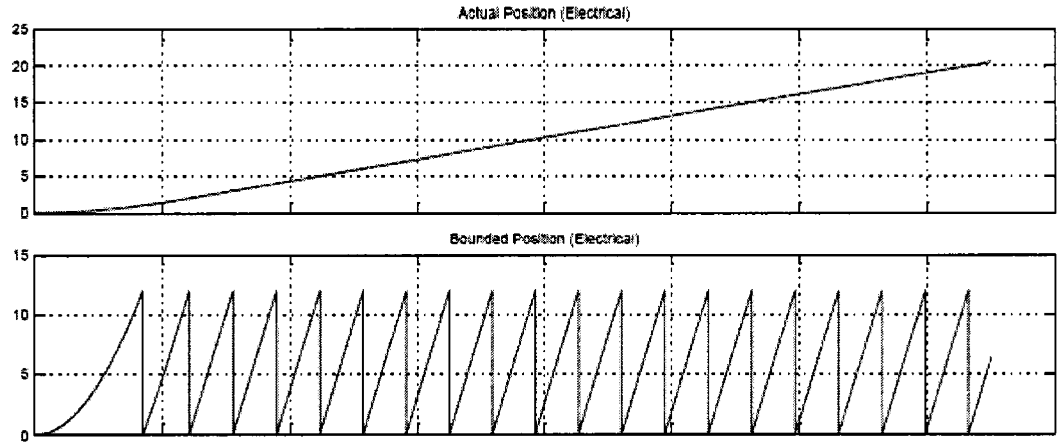


Figure 3.7 Rotor position decoding simulation.

The normalized back-emf waveform was generated, see Figure 3.8, however its dependency on rotor speed must be factored in since it directly effects the amplitude of the trapezoidal waveform. The back-emf constant ($k_{e,phase}$) is normally used to indicate the proportionality between the rotor angular velocity and back-emf peak to peak value. For modeling accuracy purposes this information is critical. The back-emf constant can readily be obtained from the BLDC machines manufacturer's datasheet. In the case were that information is not available, the peak to peak value can be experimentally found by back driving the machine to a known speed and measuring the open circuit phase to neutral voltage. Then, simply divide the measured voltage by the rotor speed which results in the back-emf constant in volts per radian per second.

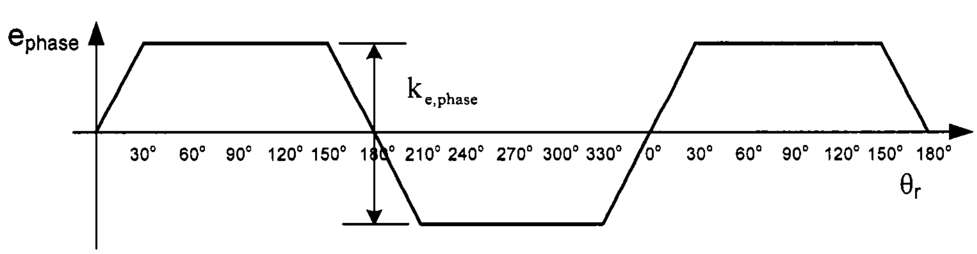


Figure 3.8 Phase to neutral back-emf waveform.

3.3.2 Torque Sensitivity Waveform Construction. The function definitions in Table 3.1 were used to generate a modeling blocks that generates the per-phase trapezoidal waveforms of the back-emf. Another motor parameter that has the same waveform with respect to rotor position is the torque sensitivity ($k_{t,phase}$). In fact, if the torque sensitivity and the back-emf are measured in SI units, then there values are identical. The difference lies in that the torque produced per-phase is the product of the torque sensitivity (Nm per amps) and the per-phase current injection, see Figure 3.9.

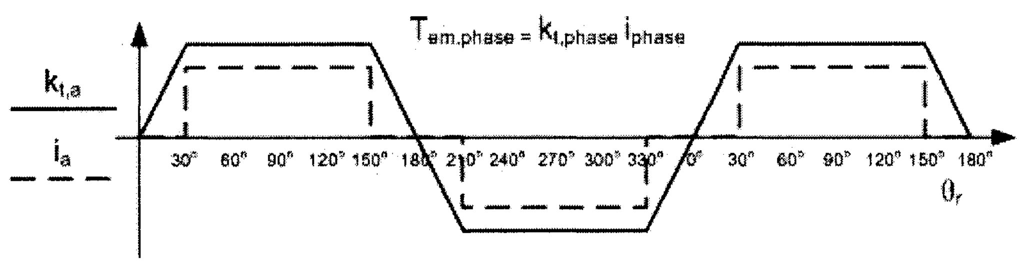


Figure 3.9 Ideal Back-emf and current waveforms.

3.4 Three Phase DC/AC Inverter Model.

Aside from the BLDC machine model, the power electronic converter necessary to operate the machine must also be modeled. The converter is a three phase DC to AC converter. It consists of six solid state semiconductor switches. MosFets and IGBT's are the most common types of switches used; in lower power applications MosFets are preferred over IGBT's. For the purpose of modeling, the switches are assumed to be ideal. It is assumed that no parasitic capacitance, inductance, or resistances are present during switching or conduction.

The power electronic model must be capable of applying positive, negative, and zero voltage across the motor phase terminals. Doing so, manipulates the current injecting to the motor. Due to its direct electromagnetic torque control capability, a Hysteresis current scheme was modeled as apposed to a fixed frequency PWM scheme.

3.5 Matlab/Simulink BLDC Machine and Converter Model

The aforementioned modeling methods were combined to create the Matlab/Simulink model shown in Figure 3.10. The model was created for any BLDC machine, the necessary inputs are the machine parameters which are readily found in the manufacturer's data sheet or through experimental testing. Once the machine parameters are known, it is only necessary to input them into to the machine parameter window illustrated in Figure 3.11.

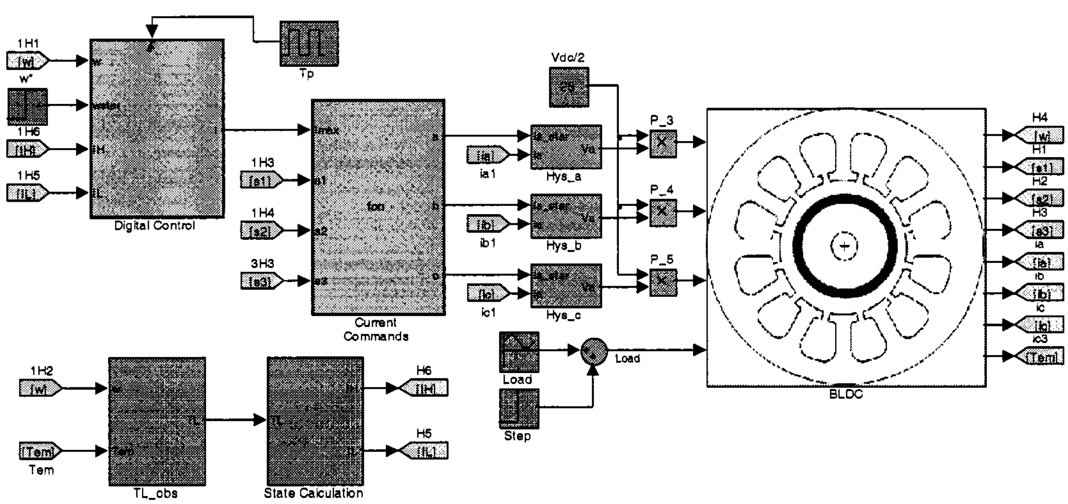


Figure 3.10 Matlab/Simulink BLDC Machine Model.

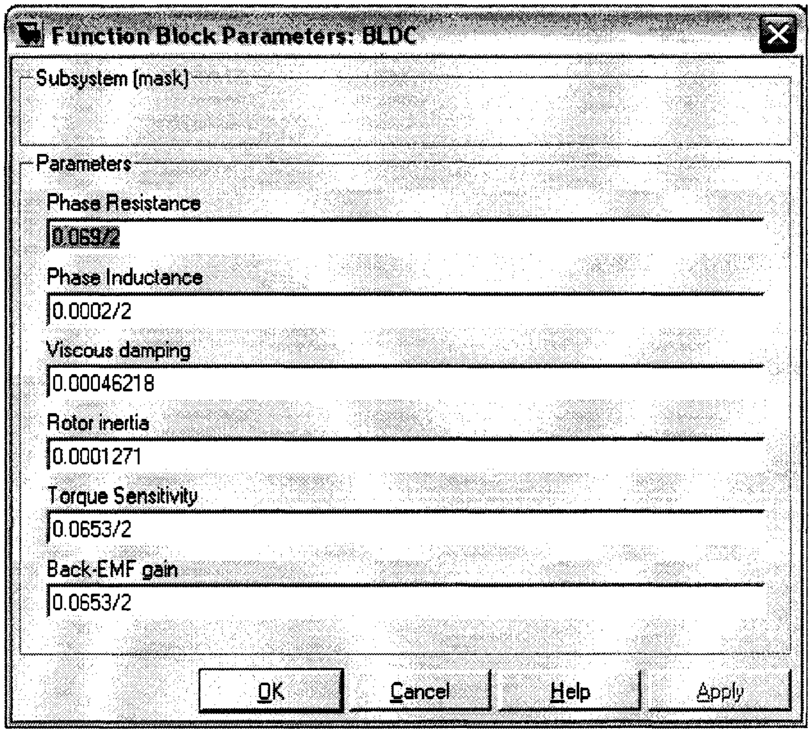


Figure 3.11 Parameter input window for BLDC machine model.

3.5.1 BLDC Machine Model. The BLDC machine model as described in Section 3.2 was realized as illustrated in Figure 3.12. Inputs to the machine model are that of the three phase voltages and the load torque. Outputs were created so as to generate the same position information available when hall-effect sensors are utilized, namely a three-bit digital word: S1, S2, and S3. Aside from position information, an ideal speed sensor is used to output the rotor speed which is then integrated to achieve precise rotor position information. The relationship between the mechanical rotor position and electrical rotor position was also modeled. Two common pole pair values for BLDC machines are two and four. In the case of two pole pairs, for every completer rotor revolution two electrical cycles occur. That is to say that two current cycle will occur and in the case of four pole pairs, then four current cycles will occur. For the most part, the number of pole pairs is crucial to the inherent torque ripple which characterized BLDC machines. More specifically, the torque ripple frequency is directly affected. The higher the torque ripple, the higher the torque ripple frequency given ideal phase current injections.

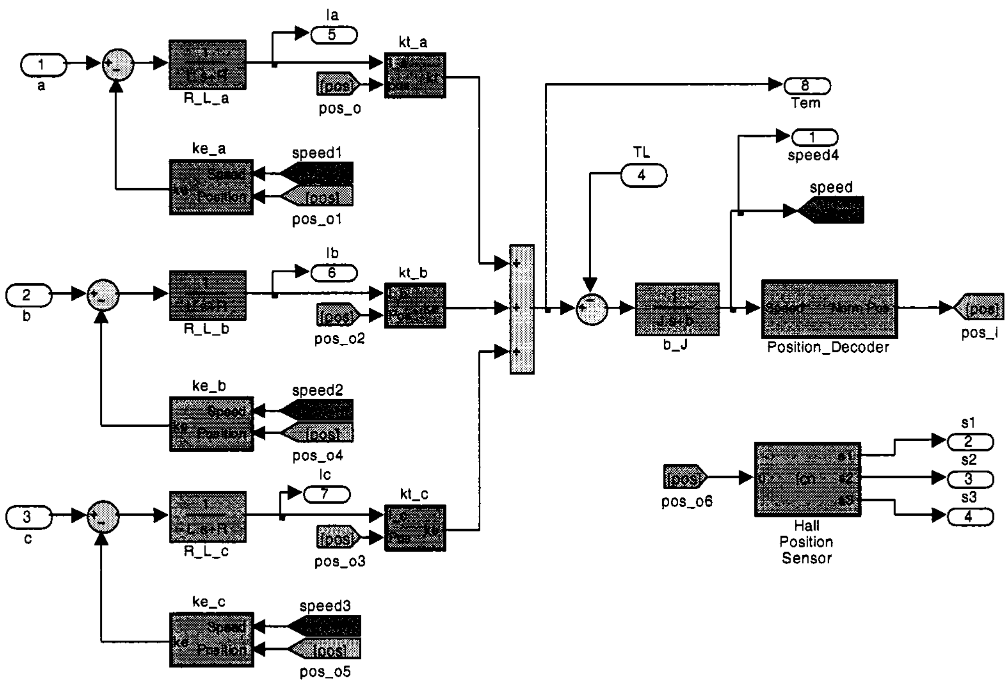


Figure 3.12 BLDC Machine Model.

3.5.2 Hysteresis Current Regulator. Modeling of Hysteresis current regulation was accomplished through the use of a function block and an SR flip-flop, see Figure 3.13. The role of the function block was to compare feedback phase currents to the predefined Hysteresis band. Its output was then fed into the SR flip-flop which serves to appropriately pulse the top inverter switches during positive conduction intervals in the motor phases. Figure 3.14 illustrates the three motor phase currents which result from using the hysteretic current controller on the BLDC machine model. It is apparent that the currents follow closely that of the ideal current injection.

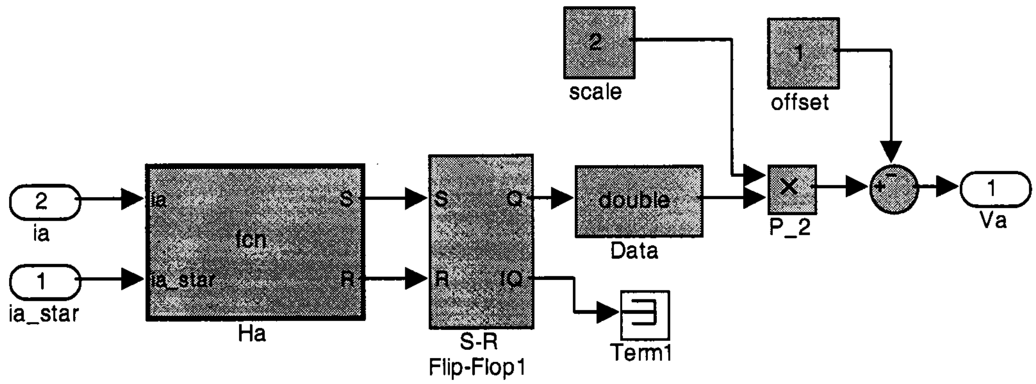


Figure 3.13 Hysteresis current regulator model.

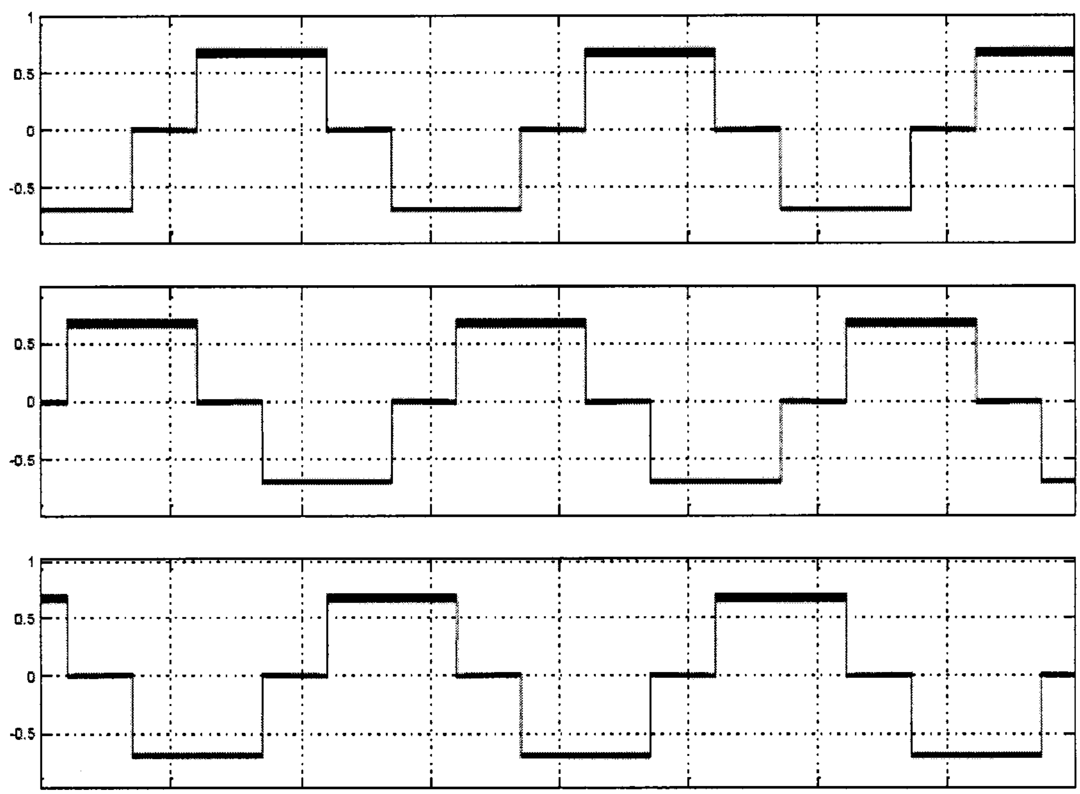


Figure 3.14 Phase currents generated with Hysteresis current controller (5ms/div).

The time scale used in Figure 3.14 was five milliseconds per division, therefore it is difficult explicitly view the hysteretic current switching between the Hysteresis band. Figure 3.15 offers a zoomed in view for one of the phases shown in Figure 3.14. It can be seen that the phase current remains within the Hysteresis band of seven amps and six and a half amps. The gating signal to the solid state switch coordinates with the current bang-bang nature as shown in the top plot of Figure 3.15.

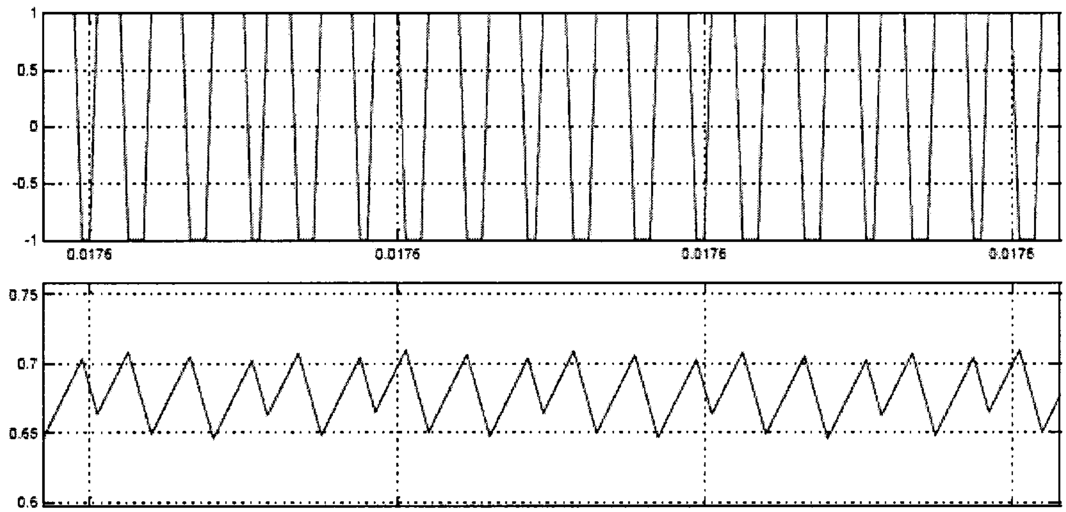


Figure 3.15 Gate switching for hysteretic current control.

3.5.3 Position Decoding From Hall Sensors. Ideal current for a BLDC machine was outlined in Figure 3.5, observation of the ideal waveforms reveal that from the three-bit digital word the top and bottom switch commands are derived. The function block in Simulink was created to implement the truth table in Table 2.1. From the function, the commanded square wave currents were created and fed into the Hysteresis current regulator described in Figure 3.13. A sample simulation which shows the three hall sensor signals (S1, S2, and S3) along with the normalized current commands are displayed in Figure 3.16. With the normalized current commands in hand, it was only necessary to amplify those commands by the appropriate gain determined by any controller implemented to drive the BLDC machine. In practice, Hysteretic switching should only be applied during positive current conduction.

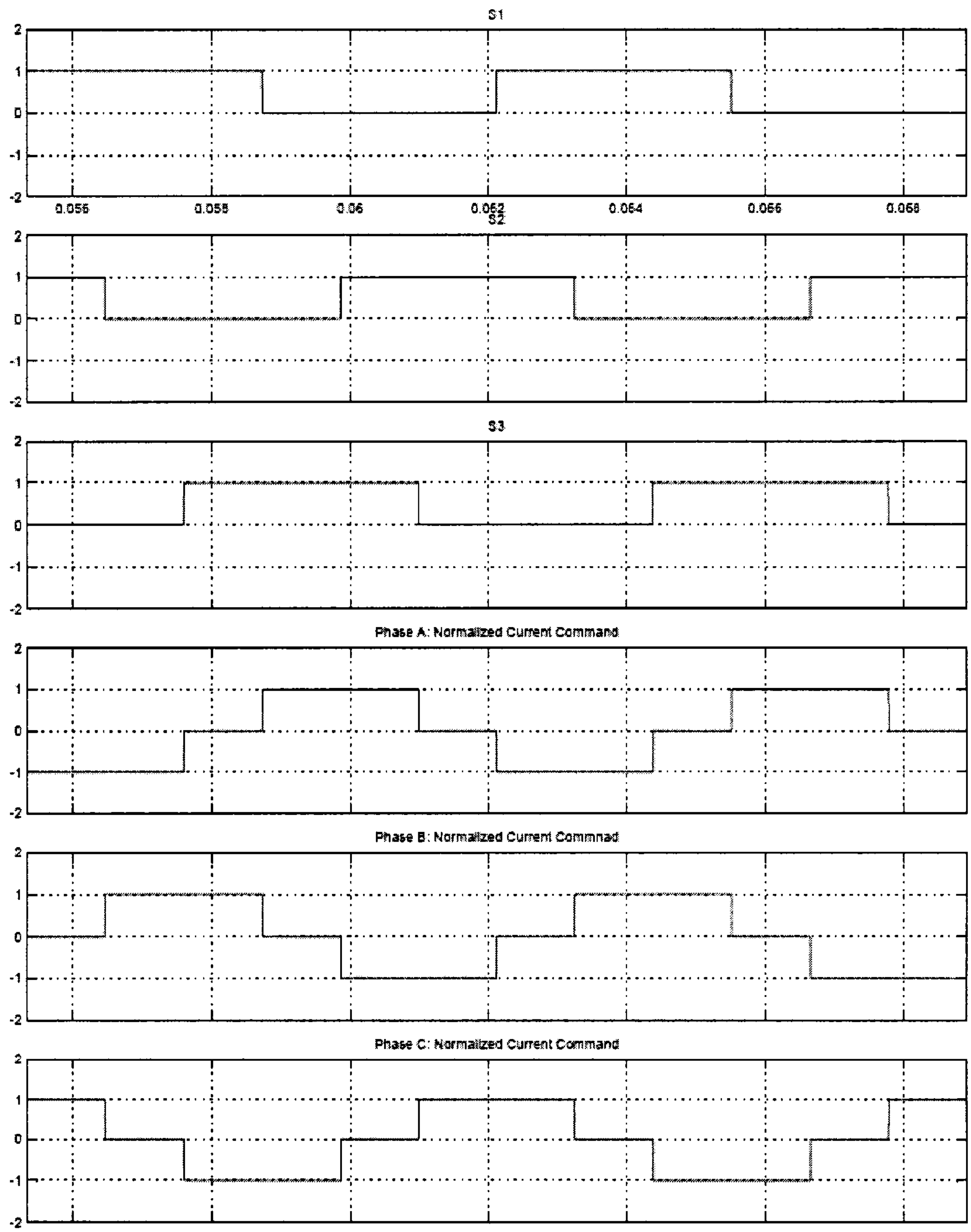


Figure 3.16 Modeled hall sensors and resulting current commands.

3.5.4 System Modeling Conclusion. Thus far, the operating principles of a BLDC machine were outlined in Chapter II. Chapter III focused on the development of a model for any BLDC machine. It was only necessary to attain the machine parameters in order to model the machine accurately. Those parameters include:

- R_{phase} Phase Resistance
- L_{phase} Phase Inductance
- J Rotor Inertia
- b Viscous damping constant
- $K_{e,phase}$ Back-emf waveform and amplitude
- $K_{t,phase}$ Torque sensitivity waveform and amplitude

All the above motor parameters are readily available from the BLDC machine's datasheet. Incase the data sheet is not available, the parameters can be found through various experimental techniques.

3.6 Experimental Setup

A computer model suitable to test any BLDC machine was created. The next step was to construct and experimental setup capable of implementing simulations carried out with the derived machine model. Some of the experimental setup features must include Hysteresis current control, accurate speed and position information, a secure three phase DC/AC inverter, digital signal processor, and an appropriately sized BLDC machine and load. The dc bus voltage should be a "stiff" voltage and was attained with a high capacity lead acid battery coupled with electrolytic (low frequency filter) and metal-film (high frequency filter) capacitors. Figure 3.17 illustrates a block diagram of the experimental setup. The block diagram shows all the major components including sensors required to run any BLDC machine with a closed loop controller. The remainder of this section will focus on the individual blocks and the details of their construction.

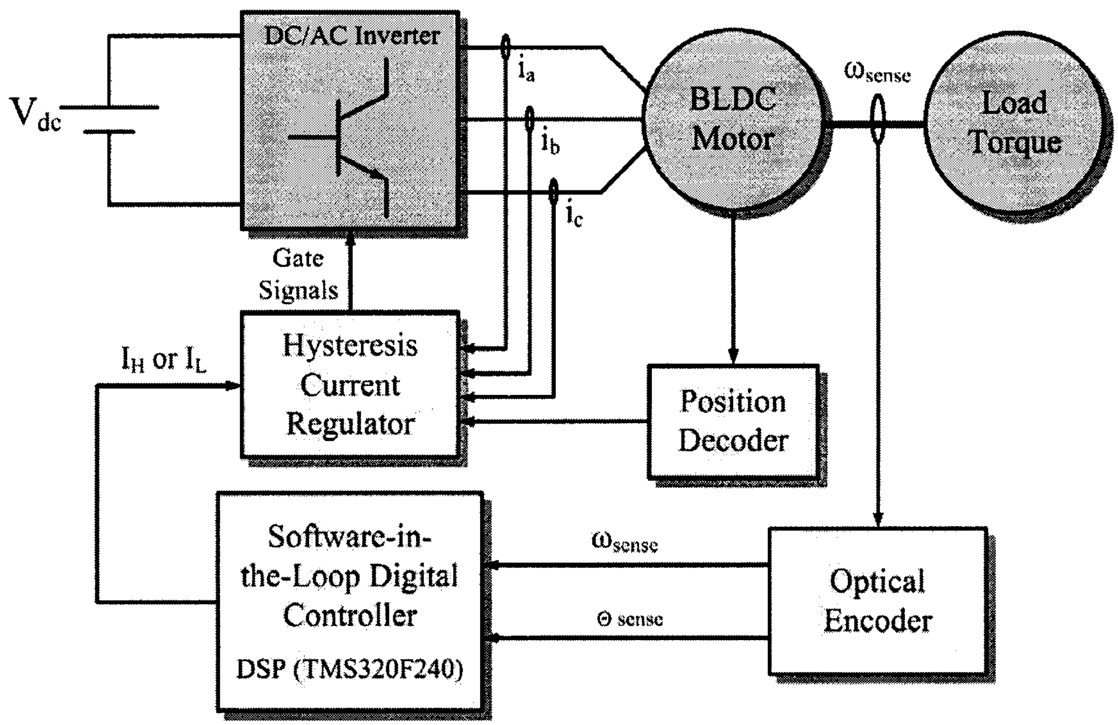


Figure 3.17 Experimental setup block diagram.

3.6.1 Three-Phase DC/AC Inverter. A standard three-phase dc to ac inverter was constructed. The inverter consisted of power MOSFETs rated for one hundred volts and thirty amps, well over the power rating of the BLDC machine. The MosFet are assumed to be ideal switches with a small turn on resistance. However to manage the switches ON or OFF status, fifteen volts must be applied from source to gate in order turn ON the device and zero volts will turn OFF the MosFet. The control signal is a logic five volt low power signal which must be amplified to the required fifteen volts. Aside from the signal amplification, it is necessary to maintain electrical isolation between the digital signal command and the amplified gate to source voltage. An digital opto-coupler isolator IC was utilized to maintain electrical isolation. The purpose of the opto-isolator is two-fold. First, the isolation offers protection to the digital signal processor, which holds the control, safety, and diagnostic features of the motor drive system. Second, electrical noise produced from the high frequency switching is suppressed and not significantly reflected onto the control side of the isolation. The block diagram in Figure 3.18 illustrates the components necessary to amplify a digital ON/OFF command to the gate of a solid state switch.

The isolation is signified by the distinct ground symbols labeled Digital Ground and Power Ground. The Schmitt Trigger Buffer serves to suppress unwanted turning OFF or ON of the switch. It accomplished the suppression by allowing for a threshold in which the state of the switch is not altered. For example, if the control device signal is five volts and noise causes the signal to spike down towards zero momentarily, then the output of the Schmitt Trigger will not switch states. State will only change is the spike is large enough to reach a lower threshold. The same idea applies to spikes towards five volts when a zero volt signal is outputted by the device control. Figure 3.19 illustrates the experimental setup signals, captured with and oscilloscope, which show the device signal and the amplified voltage applied to the switch. The device signal is five volts and the output voltage is fifteen volts as expected.

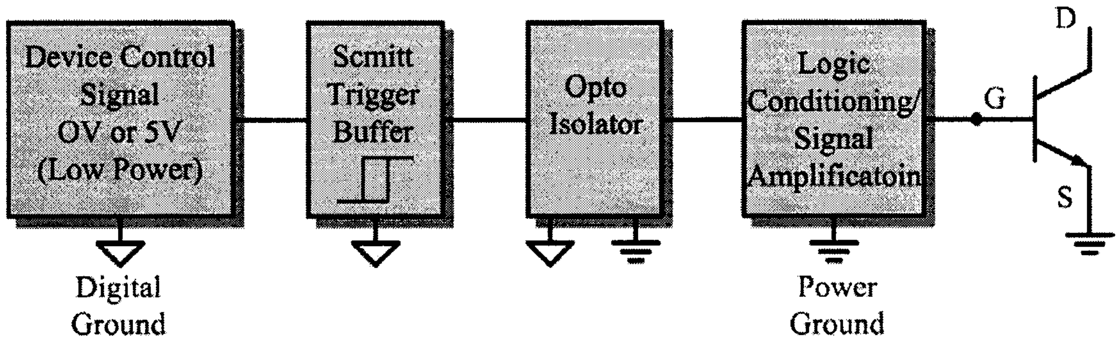


Figure 3.18 Three phase DC/AC MosFet Inverter.

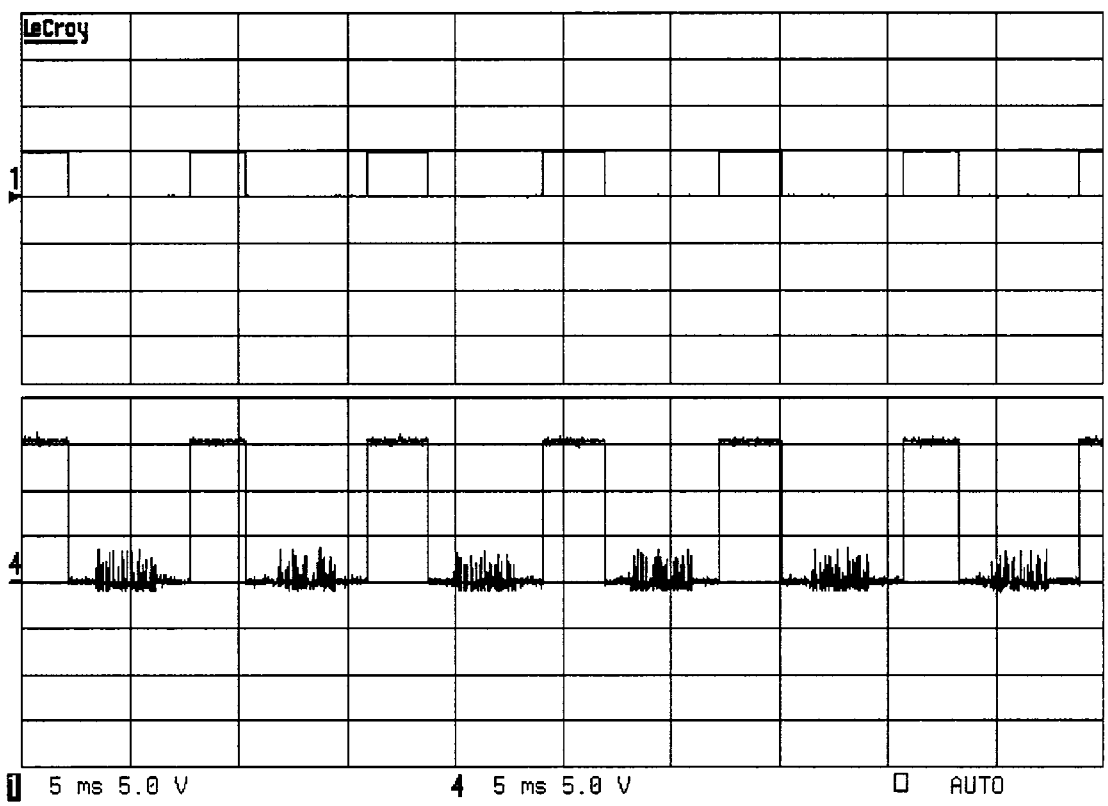


Figure 3.19 Device control signal and resulting amplified gate voltage.

Another practical consideration concerning the turning ON/OFF of a switch is the need for a snubber circuit to suppress voltage spikes from drain to source due to the parasitic inductance of the switch and the inductive nature of the load. Turning OFF the switch results in a large change in current which causes high voltage spikes or "ringing" as illustrated in Figure 3.20. A similar situation occurs at turn ON. The spikes are usually severe enough to damage the switch on account to voltage stress from drain to source. Application of a snubber circuit which is nothing more than an RC filter is enough to suppress the ringing to a level that will protect the switch. Figure 3.21 shows a snubber across the switch terminals as it was implemented in the experimental setup.

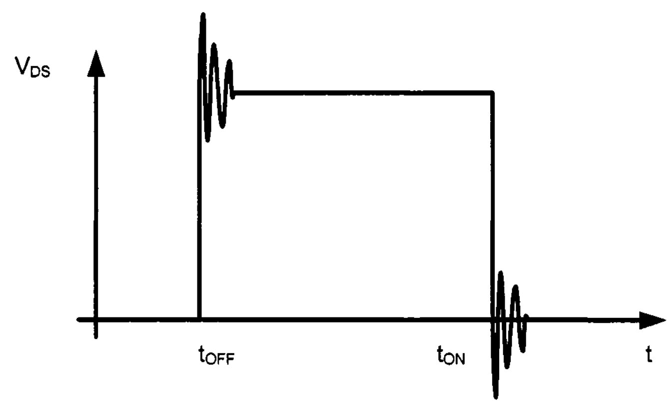


Figure 3.20 Practical drain to source voltage.

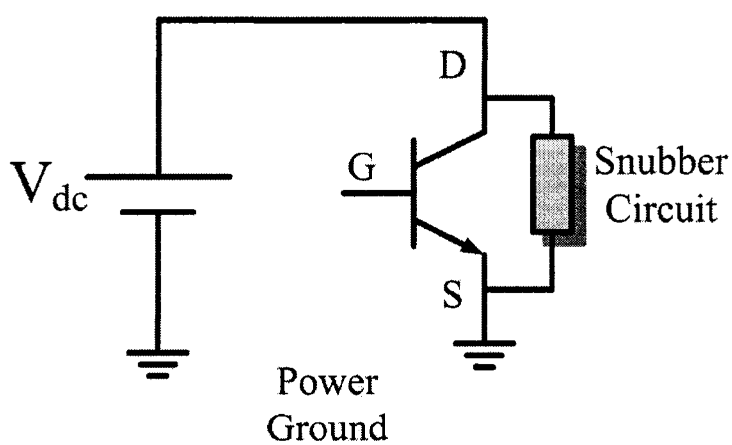


Figure 3.21 Snubber circuit voltage spike suppression.

The drain to source voltage across one switch from the DC/AC inverter was captured and is illustrated in Figure 3.22. As expected, voltage spikes during turn on and turn off are suppressed by the snubber circuit. The filtering bandwidth was tuned for a maximum switching frequency of forty kilohertz.

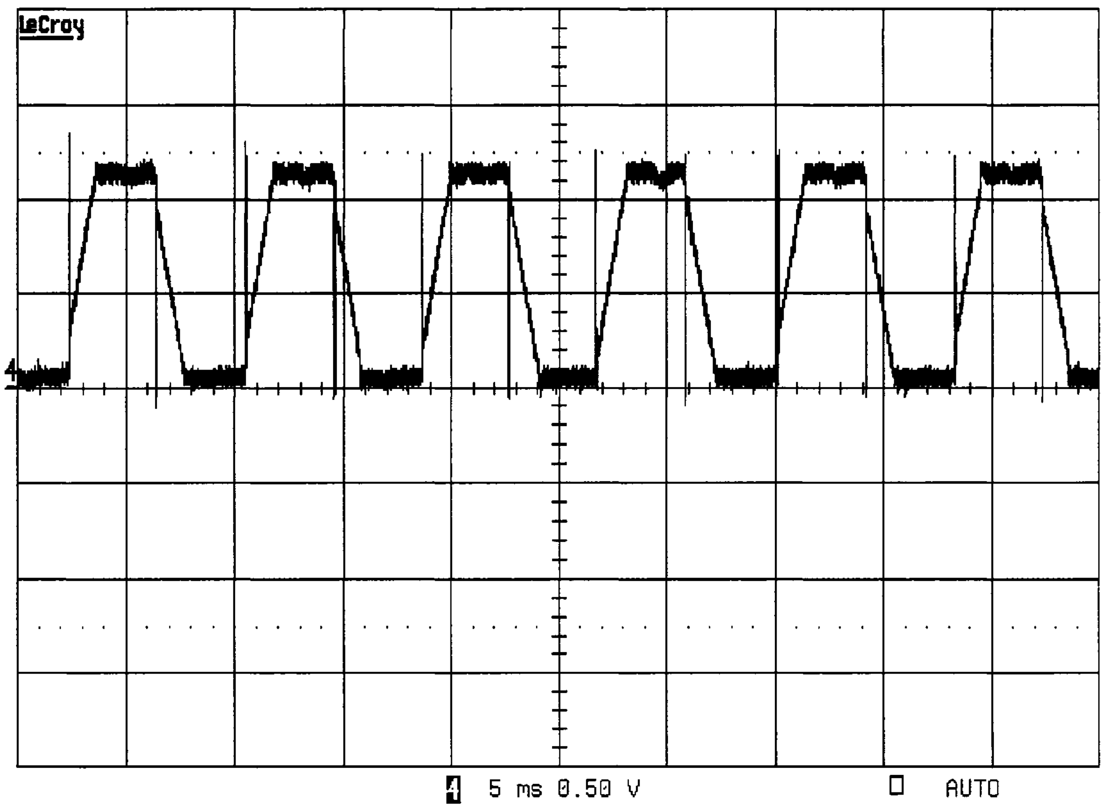


Figure 3.22 Gate to source voltage with snubber circuit (1/20 scaling).

3.6.2 Hysteresis Current Regulator. The experimental setup requires hysteresis current control for all three motor phase currents. As described in Section 3.5.2, the current was to stay within the hysteresis band as defined by the controller. By using hysteresis current control the ideal current waveform is attained to a high degree of accuracy as illustrated by the captured current waveforms in Figure 3.23. A zoomed in view showing the hysteretic switching during the positive conduction phase is shown inF Figure 3.24.

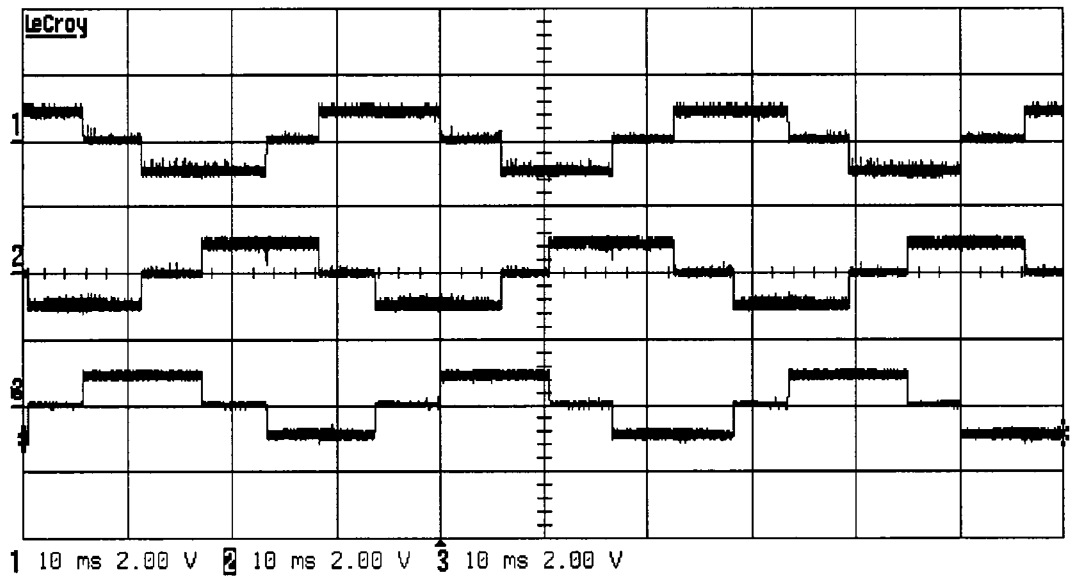


Figure 3.23 Phase current injections via Hysteresis current control.

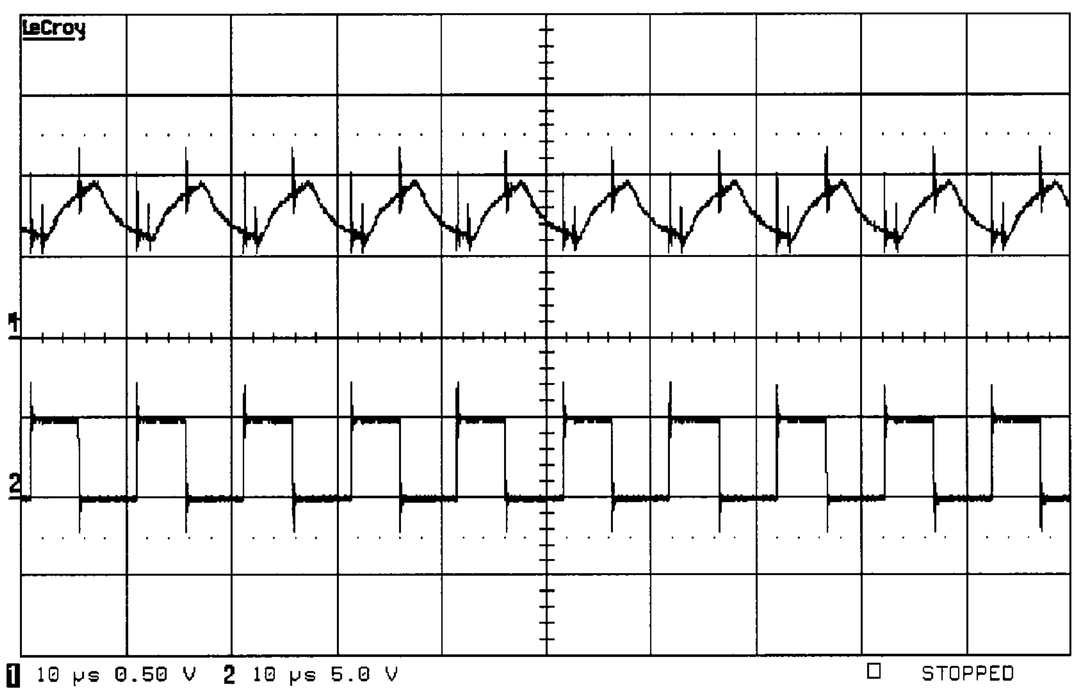


Figure 3.24 Hysteretic switching command and corresponding phase current.

3.6.3 Current Sensing Hardware. To make use of the Hysteresis hardware, monitoring of the instantaneous current is required. For each phase, closed loop current transducers were used. The transducers produce a current signal proportional to the current flow through each motor phase. Low-pass RC filters were used to reduce electro-magnetic interference. Figure 3.25 shows the circuit model implemented for each current sensor used in the experimental setup.

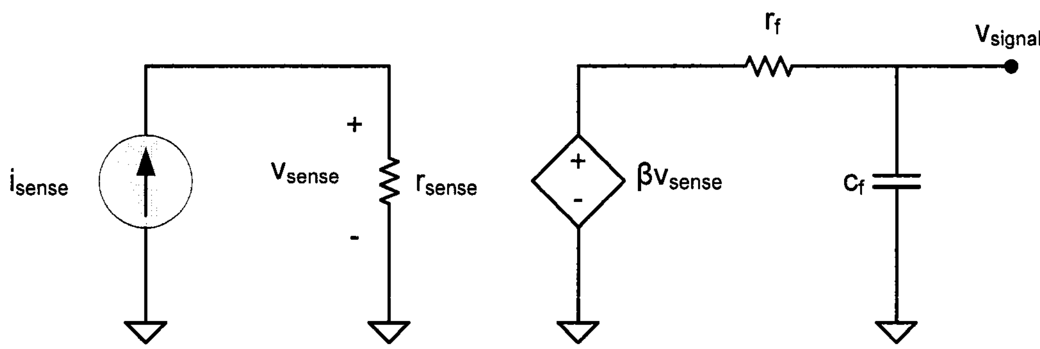


Figure 3.25 Current sensing circuit model.

3.6.4 Position Decoding. Position information is necessary for the correct current commutation as described by Section 3.5.3. The truth table in Table 2.1 was implemented with quad NAND gate IC's. The position decoding logic produces the intervals in which the phase currents should conduct. Figure 3.26 illustrates the ideal current conduction derived from the hall position sensor signals (Note: only S1 and S3 are necessary to determine the phase A ideal current waveform). The conduction signal for the other two phases follow the same logic with the appropriate phase offset. Phase B current conduction is determined from hall sensor signals S1 and S2 while phase C is determined from S2 and S3.

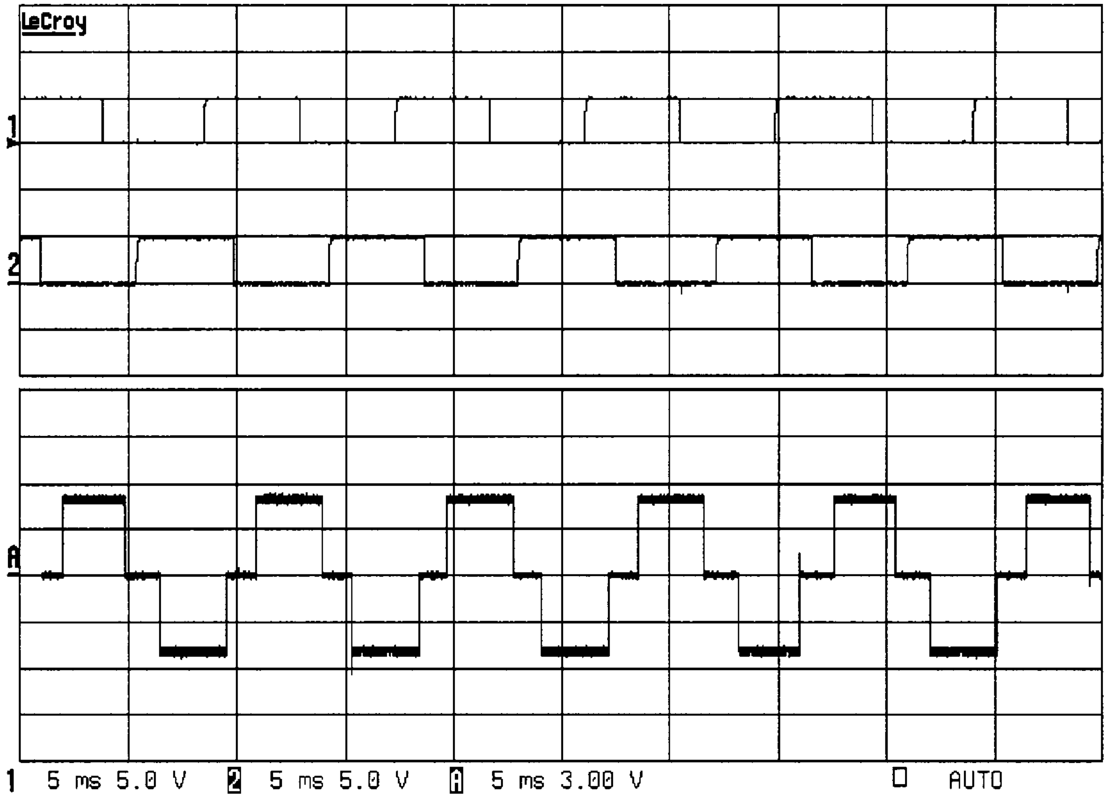


Figure 3.26 Position decoding form hall effect sensors.

3.6.5 Optical Encoder. Speed regulation quality is largely dependent upon the speed feedback which is attained by some kind of speed sensor. The constructed experimental setup was equipped with an optical encoder in order to determine the rotor speed. An optical encoder consists of a rotating disk, a light source, and a photodetector (light sensor). The disk, which is mounted on the rotating shaft, has coded patterns of solid and transparent sectors. As the disk rotates, these patterns interrupt the light emitted onto the photodetector, generating a digital or pulse signal output. The generated pulses include two at the same frequency and offset by ninety degrees. The offset is utilized to determine rotor direction. A third pulse, which is at a much lower frequency, is called the index which serves to reset the pulse count on the other two signals. A sample of the signal attained form an optical encoder are is illustrated in Figure 3.27.

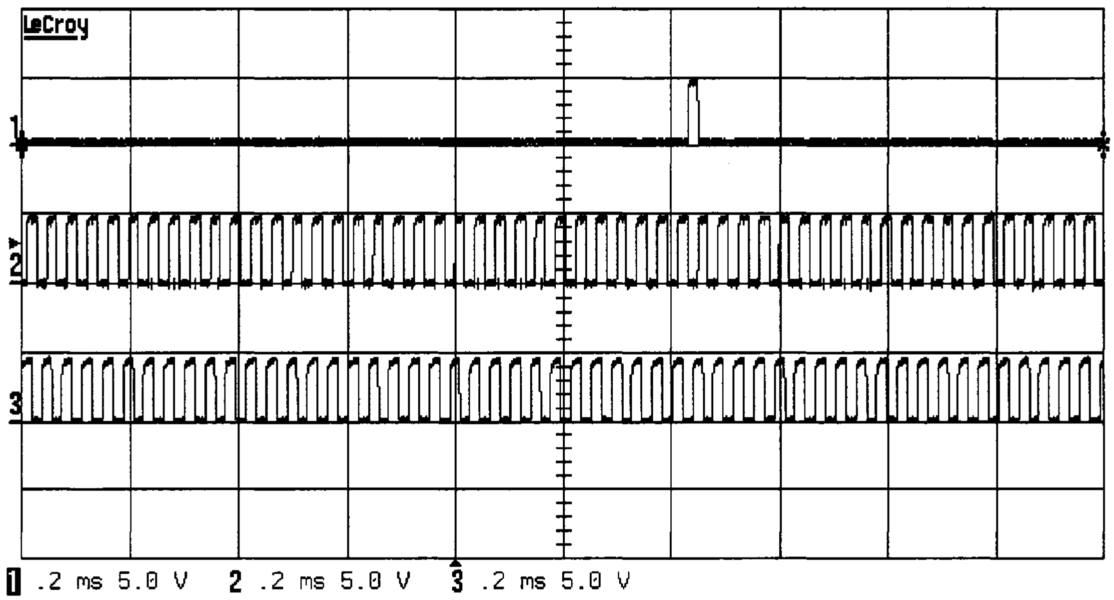


Figure 3.27 Optical encoder signas (1: IDX, 2:Ph0, 3: Ph90)

3.6.6 Load and Motor. The experimental setup was constructed to operate any BLDC machine rated for less than one hundred volts at three kilowatts. The BLDC machine must poses hall sensors in order to derive current commutation. Also, the machine should be furnished with an optical encoder so as to attain rotor speed information. Experimental testing will be carried out on two machines that meet all the above requirements, namely BN42-EU-02 (see Figurer 3.28) and BN34-45EU-01 both acquired from Poly-Scientific. Both machines are three-phase wye connected with trapezoidal back-emf. Machine ratings and all other parameter information is provided in Table 3.2 and Table 3.3. A permanent magnet dc machine was used to act as a passive load for the BLDC machine. Power resistors where paralleled to the motor terminals in order to apply a load torque. A picture of the completed experimental setup is provided in Figure 3.29.

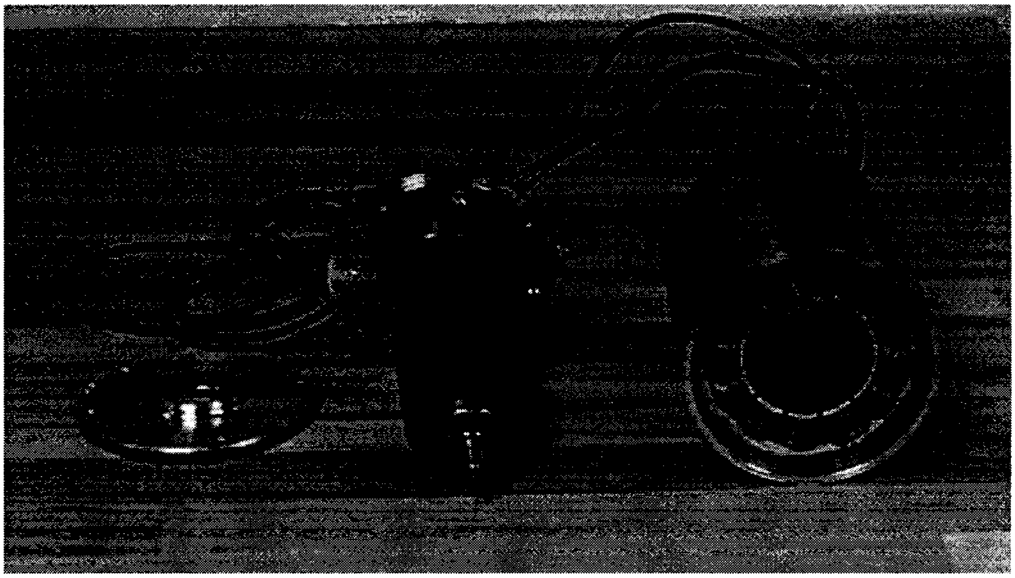


Figure 3.28 BLDC machine: model # BN42-EU-02.

Table 3.2 Data Sheet for BLDC Motor from Poly-Scientific (BN42-EU-02)

Parameter	Units	Value
Terminal Voltage	volts DC	50
Rated Speed	rpm	2820
Rated Torque	Nm	2.92
Rated Current	Amps	20.2
Rated Power	Watts	861
Torque Sensitivity	Nm/amp	0.164
Back-emf	volts/krpm	17.1
Terminal Resistance	ohms	0.106
Terminal Inductance	mH	0.428
Motor Constant	Nm/sq.rt.watt	0.516
Rotor Inertia	g-cm^2	4943
Number of Poles		4

Table 3.3 Data Sheet for BLDC Motor from Poly-Scientific (BN34-45EU-01)

Parameter	Units	Value
Terminal Voltage	volts DC	24
Rated Speed	rpm	3300
Rated Torque	Nm	1.33
Rated Current	Amps	20.2
Rated Power	Watts	459
Torque Sensitivity	Nm/amp	0.0653
Back-emf	volts/krpm	6.83
Terminal Resistance	ohms	0.069
Terminal Inductance	mH	0.200
Motor Constant	Nm/sq.rt.watt	0.275
Rotor Inertia	g-cm^2	1271
Number of Poles		8

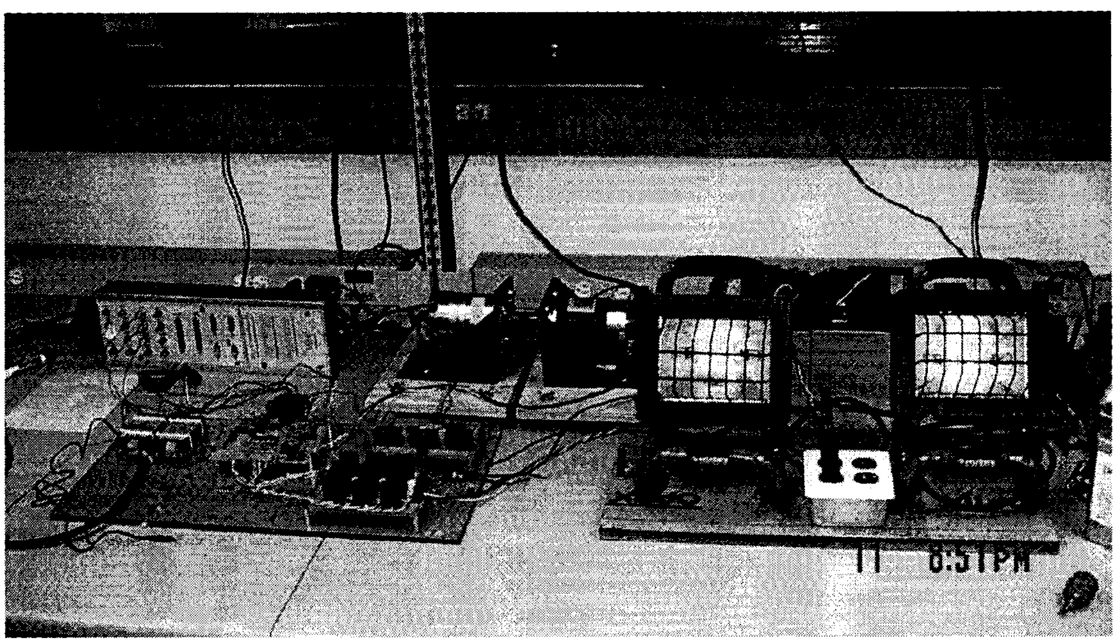


Figure 3.29 BLDC Machine experimental Setup.

3.7 Conclusion

A Matlab/Simulink model was created to precisely predict the steady state and dynamic behavior of a BLDC machine. Features such as Hysteresis current behavior and position decoding from model hall sensors were included to increase the accuracy of any simulation results. As a tool to validate any simulations, an experimental setup was construct so as to have similar features to those modeled in Matlab/Simulink. Many precautionary steps were taken to ensure the safe operation of the experimental setup and safety to the operator. From here on in, the developed machine model and experimental set up are used to model and validate the novel digital control technique, which is the focus of this thesis.

CHAPTER 4

CLASSIC CONTROL TECHNIQUES

4.1 Introduction to Control Systems

There are various forms of controls that may be used to manage a dynamic system. Such systems include manual control, which requires a human operator. If only a machine is used to control a system, as is the case of a thermostat in temperature control, then it is an automatic control. Automatic control systems try to maintain a constant output in spite of disturbances. Therefore, they may sometimes be referred to as regulators or tracking/servo systems. A further classification of automatic control depends on the method of regulating the output. If there is no dependence or measure of the output in its regulation, then it is called an open-loop control. When the output is measured and used for control calculations, then it is referred to as a feedback-control. Feedback-control is necessary for the speed regulation of any electric machine as in the case with speed regulation of BLDC motors.

An example of feedback control can be found in the cruise control feature of an automobile. Figure 4.1 illustrates a basic block diagram of a cruise control system. When the driver is going to cruise on the express-way, he or she may choose to activate the cruise control function in order to maintain the automobile's speed constant. Once the driver does that, an automatic feedback system is given the task of maintaining the vehicles speed constant in spite of disturbances such as wind gusts or steep inclines. The block diagram for the automatic feedback cruise control system is shown in Figure 4.1. The major components include the controller, the car engine (actuator), auto body (plant), and the speedometer (feedback), [11].

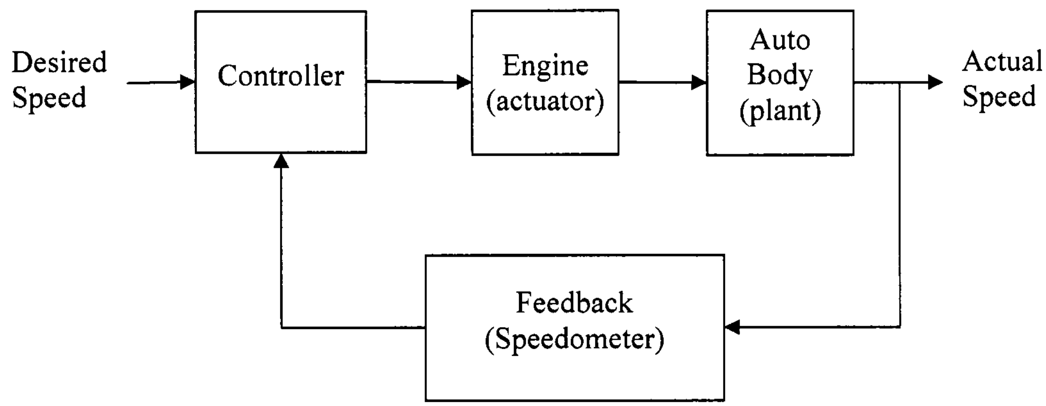


Figure 4.1 Block Diagram of Cruise Control Feedback System

The function of the controller is to compare the actual speed to the desired speed in order to produce a corrective signal. If the actual speed is less than the desired speed, then the corrective signal serves to increase the actual speed. On the other hand, if the actual speed is greater than the desired speed, then the corrective signal decreases the actual speed. Finally, if the actual speed is equal to the desired speed, then no corrective action is taken. For the cruise control example, the corrective signal may correspond to changes in the throttle angle of the engine. Increasing the throttle angle will cause an increase in propulsion force, which would be needed if the car comes across a disturbance. Disturbances may come in the form of wind gusts or when the car begins ascending an inclined road.

Changes in the throttle angle cannot increase or decrease the vehicles speed directly, therefore there is a need for an actuator, the vehicle engine, to translate changes in the corrective signal to changes in speed of the vehicle. The engine receives the corrective signal from the controller and applies it to the car by varying the propulsion force in accordance to the controller signal. The propulsion force acts upon the vehicle, the plant, which is the object being controlled.

Lastly, some type of sensor is required to provide feedback of the output to the controller. In this case, the feedback is attained via the speedometer. The speedometer provides a signal proportional to the auto's speed so that the controller can compare it to the desired speed. Feedback will allow the controller to recognize the presence of a disturbance allowing it take the corrective action necessary to increase the speed back to the desired speed.

Taking a closer look at Figure 4.1, the main components of a feedback control system are the controller, actuator, plant, and feedback. The parameters of the actuator and plant are determined by the physical properties of the object being controlled. In the case of electric motor control, the electrical and mechanical time constants will determine the plant model. The feedback is a result of the control signal being applied to the plant. The controller parameters are specified by the control-system designer and different parameter definitions will result in different system performance. These differences are especially noticeable under dynamic operations. A good design will have the output of the plant track the input reference under dynamic conditions and under different types of disturbances. That in fact is the purpose of feedback automatic systems, to have the plant output track the reference input in spite of disturbances. A summary of an automatic feedback (closed-loop) system is shown in Figure 4.2.

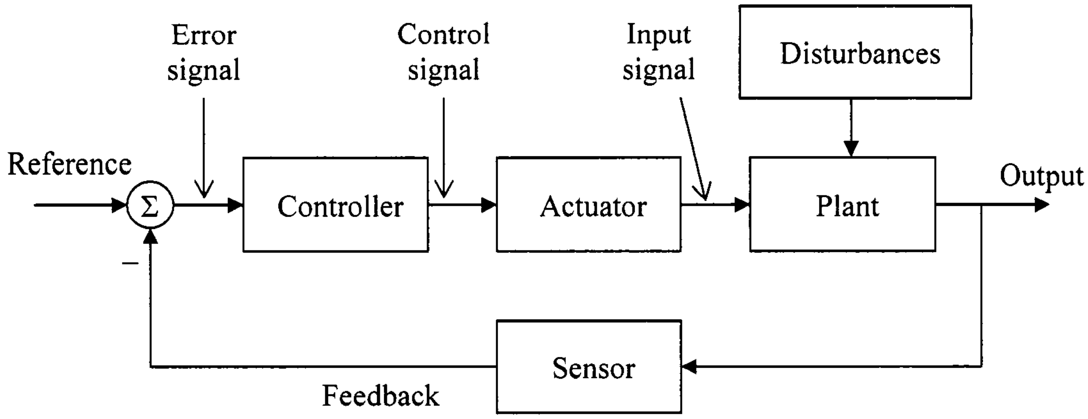


Figure 4.2 Automatic Feedback System Block Diagram

4.2 S-Domain Plant Models

To design the controller, a mathematical model that describes the dynamic behavior of the actuator and plant must first be defined. When working with complex systems, higher order differential equations may result making mathematical analysis quite troublesome. It is convenient to carry out design in a domain other than time, so as to reduce the model equations to simpler algebraic equations. Three domains are available for analysis of dynamic behavior, namely the s-plane, the frequency domain, and state space. All modeling and design will be carried out in the s-domain, therefore it is necessary to highlight the basic conversion tables of the Laplace Transform. Table 4.1 and Table 4.2 list useful Laplace transforms and properties for common functions. The Laplace transform will be used to convert mathematical models from the time domain to the s-domain, which results in the conversion of the time domain differential equation to an algebraic equation in the s-domain.

Table 4.1 Properties of Laplace Transforms

Laplace Transform	Time Function	Comment
$F(s)$	$f(t)$	Transform pair
$\alpha F_1(s)+\beta F_2(s)$	$\alpha f_1(t)+\beta f_2(t)$	Superposition
$F(s)e^{-s\lambda}$	$f(t-\lambda)$	Time delay $(\lambda \geq 0)$
$\frac{1}{a}F(\frac{s}{a})$	$f(at)$	Time scaling
$F(s+a)$	$e^{-at}f(t)$	Shift in frequency
$s^m F(s)-s^{m-1}f(0)$ $-s^{m-2}f^{(1)}(0)-...-f^{(m-1)}(0)$	$f^{(m)}(t)$	Differentiation
$\frac{1}{s}F(s)$	$\int f(\rho)d\rho$	Integration
$F_1(s)F_2(s)$	$f_1(t)*f_2(t)$	Convolution
$\lim\_s \to \infty\_sF(s)$	$f(0^+)$	Initial Value Theorem
$\lim\_s \to 0\_sF(s)$	$\lim\_t \to \infty\_f(t)$	Final Value Theorem

Table 4.2 Table of Laplace Transforms

$F(s)$	$f(t),\ t \ge 0$
1	$\delta(t)$
$\frac{1}{s}$	$1(t)$
$\frac{1}{s^2}$	t
$\frac{2!}{s^3}$	t^2
$\frac{m!}{s^{m+1}}$	t^m
$\frac{1}{s+a}$	e^{-at}
$\frac{1}{(s+a)^m}$	$\frac{1}{(m-1)!}t^{m-1}e^{-at}$
$\frac{a}{s(s+a)}$	$1-e^{-at}$
$\frac{a}{s^2(s+a)}$	$\frac{1}{a}\left(at-1+e^{-at}\right)$
$\frac{a}{s^2+a^2}$	$\sin at$
$\frac{s}{s^2+a^2}$	$\cos at$
$\frac{s+a}{(s+a)^2+b^2}$	$e^{-at}\cos bt$
$\frac{b}{(s+a)^2+b^2}$	$e^{-at}\sin bt$

4.3 Transfer Function for Feedback Control System

Once a model is attained for the plant and actuator in the s-domain, it is necessary to obtain a transfer function that describes the input {Y(s)} to output { R(s)} relationship of the closed-loop feedback system. Each block in Figure 4.2 is represented by a separate s-function. One function will describe the controller while another describes the actuator and so on. Therefore, to derive an input out relation, $H(s) = \frac{R(s)}{Y(S)}$, rules of block diagram manipulation must be established. A series combination of two or more blocks is equivalent to the product of the functions in each block. A series combination can be reduced to a single block representation as shown in Figure 4.3.

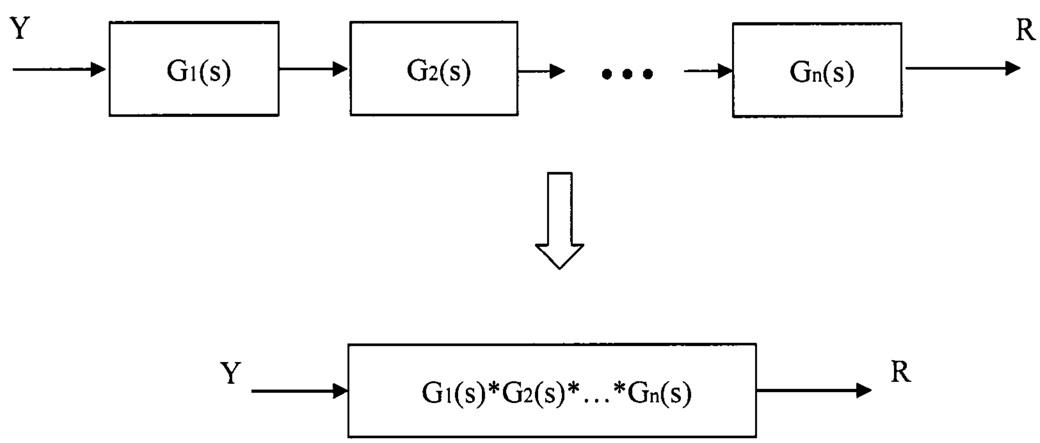


Figure 4.3 Series Combination Block Reduction

The parallel combination of two or more blocks is equivalent to the single block representation of the summation of the functions as shown in Figure 4.4.

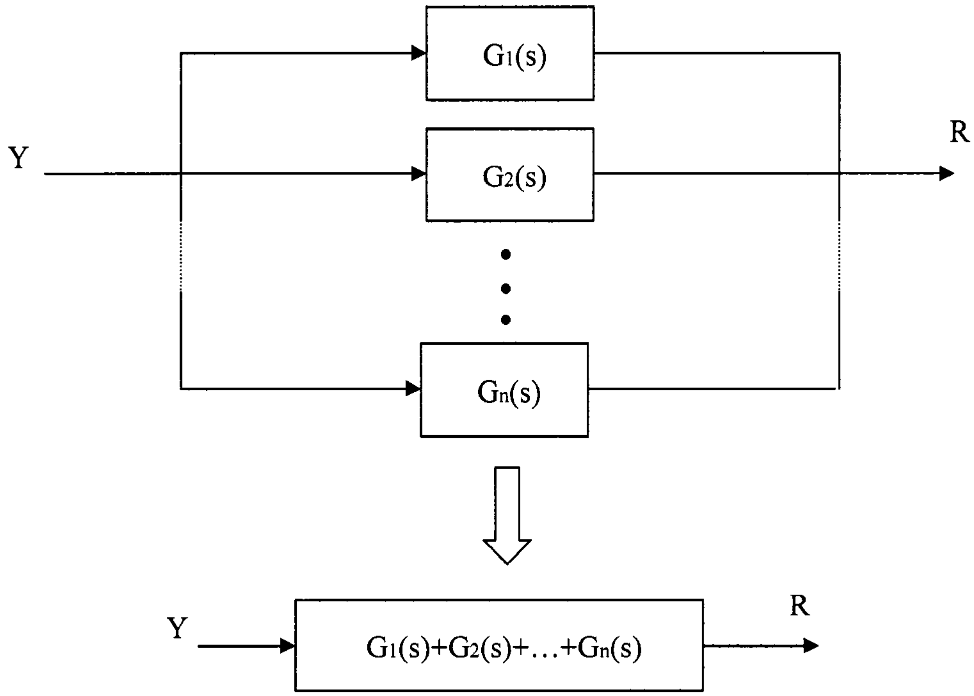


Figure 4.4 Parallel Combination Block Reduction.

In the case of a closed loop negative feedback system, the equivalent block reduction is possible in the following manner. The output to input relation is the represented by the forward path function, G1(s), divided by the sum of one and the product of the forward path function and the feedback path function, G2(s). The feedback block diagram is illustrated in Figure 4.5.

Having established the fundamental rules of block reduction, a simple two pole plant model is used to point out important characteristics of the closed loop feedback transfer function. Figure 4.6 represents a basic feedback system, where D(s) is the controller function, P(s) is the representation of the actuator and plant function, and F(s) is the feedback function. The model for the plant and actuator is represented by Equation 4.1, which is that of a two pole system.

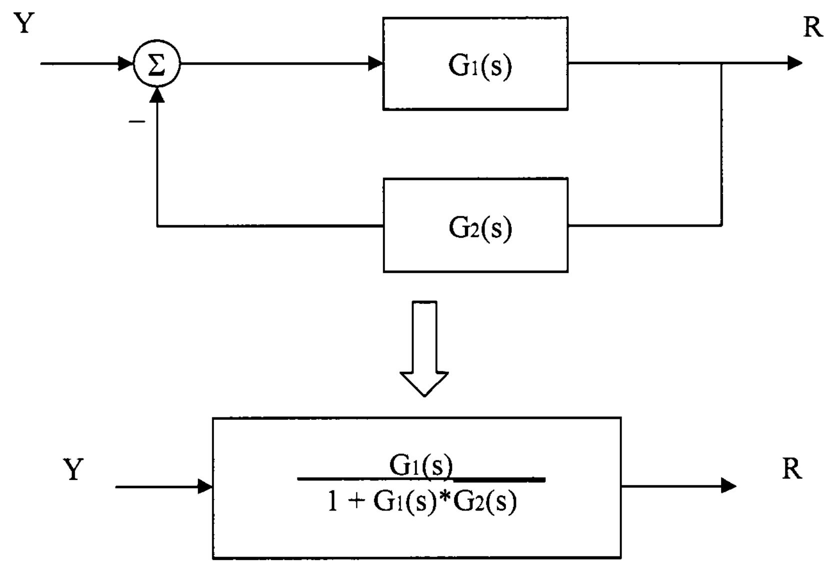


Figure 4.5 Negative Feedback Block Reduction.

$$P(s) = \frac{\omega_n^2}{s(s + 2\omega_n\zeta)} \tag{4.1}$$

where, ω_n = natural frequency

ζ = damping ratio

The natural frequency and damping ratio are parameters that will aid to establish the dynamic behavior of the plant. Setting the denominator of P(s) equal to zero and solving for s will reveal the position of the poles in the complex s-plane.

$$S_1 = -2\omega_n\zeta \qquad S_2 = 0$$

The poles correspond to the time constants of the output transient response to an input. Referring back to the Laplace transform table, the inverse Laplace of P(s) will take the form of Equation 4.2.

$$P(t) = A + Be^{-2\omega_n\zeta} \tag{4.2}$$

Where, A and B are constants that can be determined via partial fraction expansion of Equation 4.1. The pole values are predetermined by the physical properties of the plant, therefore there is no flexibility to altering the dynamic performance in the absence of a closed-loop feedback system. Figure 4.6 illustrates the plant in a closed loop system, the functions of for the controller, plant and actuator, and feedback are as follows;

$$D(s) = 1 \qquad P(s) = \frac{\omega_n^2}{s^2 + 2\omega_n\zeta s} \qquad F(s) = 1$$

The block diagram reduction rules are applied to reduce the output to input transfer function to Equation 4.3.

$$H(s) = \frac{R(s)}{Y(s)} = \frac{D(s)P(s)}{1 + D(s)P(s)F(s)} \tag{4.3}$$

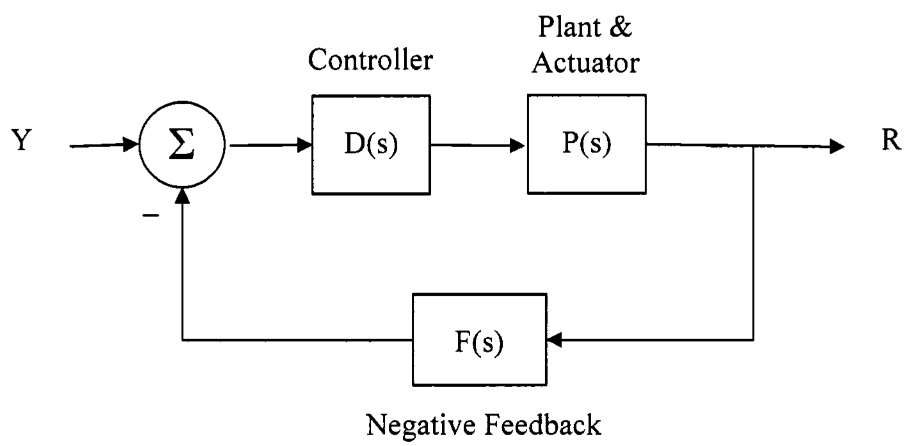


Figure 4.6 Block Diagram of Two-Pole Feedback System.

Substitution of the function values results in the transfer function given by Equation 4.4.

$$P(s) = \frac{R(s)}{Y(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{4.4}$$

The new input out relationship looks different compared to Equation 3.1 and along with the difference in appearance, the dynamic behavior of the closed loop system will also differ. To find the new pole values the denominator of H(s) is set equal to zero and solved for s. The equation below is the quadratic equation, which is used to find the solution for the poles.

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$Quadratic\_Equation = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where, a = coefficient of s^2

b = coefficient of s^1

c = coefficient of s^0

Therefore, the poles of H(s) are

$$s_{1,2} = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$= -\zeta\omega_n \pm \sqrt{(\zeta\omega_n)^2 - \omega_n^2}$$

Previously defined, ζ is the damping ratio which has a value between one and zero ($0 \leq \zeta < 1$), resulting in complex values for s_1 and s_2 when ζ is not equal to zero. To highlight the difference in the pole locations, two plots were created. Figure 4.7 depicts the pole location of the open-loop poles as defined by Equation 4.1. Figure 4.8 illustrates the location of the closed-loop conjugate poles pair.

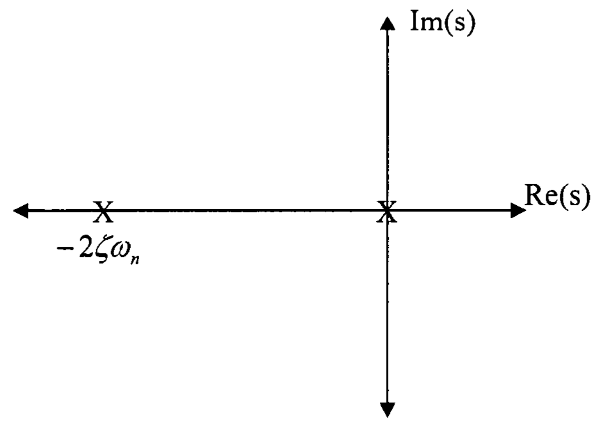


Figure 4.7 Open-Loop Pole Locations

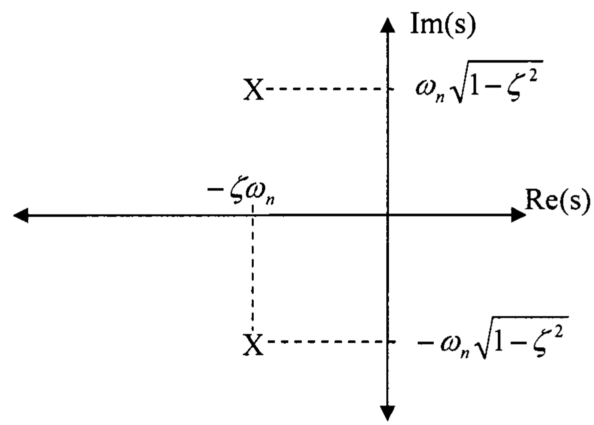


Figure 4.8 Closed Loop Poles Location

The plot of the transfer function poles on the s-plane is a great design tool. The graphical location reveals the dynamic behavior of the control system. A measure of dynamic behavior is taken when the input of a control system, Y(s), is set equal to the step function as defined by Equation 4.5.

$$\text{Step\_function} = u(t) = \begin{Bmatrix} 0, & t < 0 \\ 1, & t \geq 0 \end{Bmatrix} \tag{4.5}$$

The step change to the input shows how well the output will track the input via measurement of several parameters. Those key parameters are the rise time (t_r), settling time (t_s), overshoot (M_P), and the peak time (t_P), which are all a function of the natural frequency and damping ratio. Figure 4.9 is a plot of the output response to a step input of one at t = 0.

Rise time (t_r) is defined as the time, in seconds, required for the output to rise from ten percent to ninety percent of the desired value. The rise time is not heavily affected by variations in the damping ratio. Therefore, setting the damping ratio to a value of $\zeta = 0.5$, yields and approximation of the rise time to be a function of only the natural frequency as described by Equation 4.6

$$t_r \cong \frac{1.8}{\omega_n} \tag{4.6}$$

The overshoot (Mt_P) and the peak time (t_P) can be derived to formulate exact quantities. Overshoot is only a function of the damping ratio and the relation is non-linear as described by Equation 4.7. Peak time is a function of both the damping ratio and natural frequency and that exact value is given by Equation 4.8. A plot relating the overshoot to the damping ratio is provided in Figure 4.10.

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \tag{4.7}$$

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \tag{4.8}$$

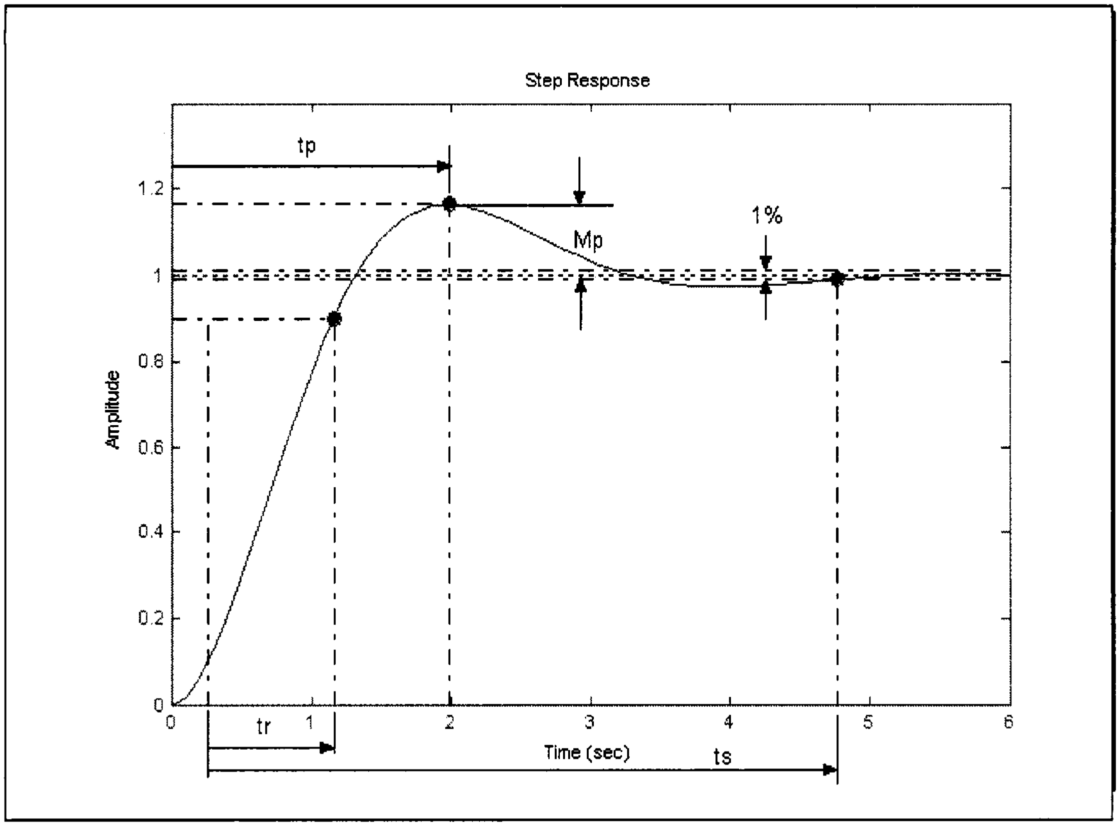


Figure 4.9 Step Response Characteristics

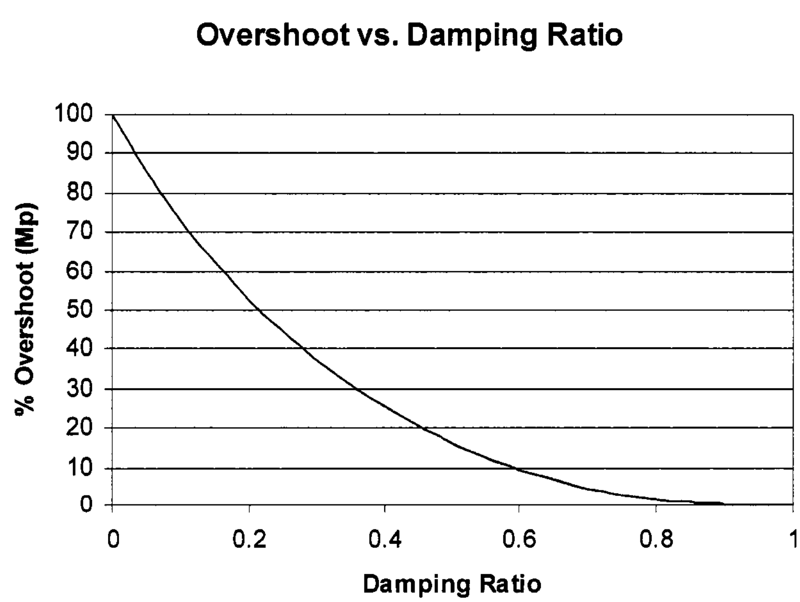


Figure 4.10 Percent Overshoot vs. Damping Ratio

The last measure of dynamic response is the settling time (t_s). The settling time is defined as the time it takes the output to track the step-input to within plus or minus a small percentage. In other words, it is a measure of the amount of time it takes for the transient exponential terms to decay leaving only the steady state response of the system. The complex pole pair of the closed loop system in the s-domain correspond to exponential terms in the time domain. The criteria for settling time is to have these exponential terms decay to within the small percentage of one percent of the steady state value. The exponential term can then be equated to 0.01 and solved for (t_s) to yield the settling time as described by Equation 4.9.

$$e^{-\zeta\omega_n t_s} = 0.01$$

$$-\zeta\omega_n t_s = \ln(0.01) = -4.6$$

$$t_s = \frac{4.6}{\zeta\omega_n} \tag{4.9}$$

4.4 Root Locus Design Method

The dynamic response of an open-loop or closed loop system can be determined by graphically measuring the output response to a step-input, as illustrated in Figure 4.4. In the complex-pole pair example, the controller and the feedback were set equal to unity, $D(s) = 1$ and $F(s) = 1$ respectively. Changes in the controller or feedback loop would result in altering the dynamic response characteristics, namely the rise time, overshoot, settling time, and peak time. The changes can be observed by varying the controller D(s) and graphically measuring the changes in the step response. That results in an inefficient trial and error design method. If a system is required to have a specific rise time constraint, then the designer must guess controller parameters until the constraint is met.

To avoid that inefficient design method, root-locus design is used to find the controller D(s) that meets the dynamic performance requirements. As an example of root locus design, the two pole system shown in Figure 4.6 will be analyzed once again, given a plant model of $P(s) = \frac{\omega_n^2}{s^2 + 2\omega_n \zeta s}$. The first step is to solve the denominator of the plant for s when set equal to zero. It was previously shown that the solution was $s_1 = -2\omega_n \zeta$ and $s_1 = 0$. Second, the pole locations are plotted on the s-plane with X's and the root locus diagram is drawn as shown in Figure 4.11 (If zeros were present, they would be denoted by "O" on the s-plane). Lastly, the root-locus diagram depicts the path of the poles as the controller changes its value. Therefore, the controller may be design to place the poles anywhere on the root locus diagram.

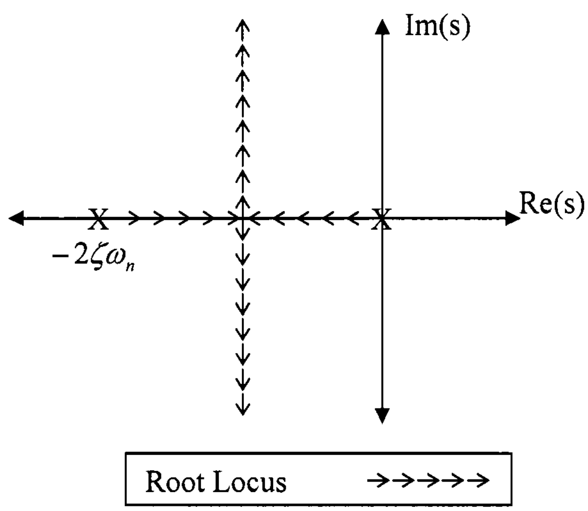


Figure 4.11 Two-Pole System Root-Locus Diagram

A more general approach to the root locus design method requires a closer look at the closed loop transfer function which was previously defined by Equation 4.3 as

$H(s) = \frac{R(s)}{Y(s)} = \frac{D(s)P(s)}{1 + D(s)P(s)F(s)}$. The roots of the denominator are of special interest when drawing the root-locus. The denominator can be rewritten as Equation 4.10.

$$1 + D(s)P(s)F(s) = 1 + KL(s) = 0 \tag{4.10}$$

The new s function L(s) is represented by its numerator $a(s)$ and its denominator $b(s)$. The roots of $a(s)$, correspond to the zeros (z_i) of the closed loop system, while the roots of $b(s)$ represent the poles (p_i), where m is the number of zeros and n is the number of poles. The constant K determines the root locus diagram as it is varied from zero to infinity.

$$a(s) = (s - z_1)(s - z_2)\ldots(s - z_m)$$

$$b(s) = (s - p_1)(s - p_2)\ldots(s - p_n)$$

In the case of the two pole example, with $P(s) = \frac{\omega_n^2}{s^2 + 2\omega_n\zeta s}$ and the controller set equal to unity, the closed-loop pole locations were shown to be $s_{1,2} = -\zeta\omega_n \pm \sqrt{(\zeta\omega_n)^2 - \omega_n^2}$. The root locus design method is useful when the final pole locations are know, but the controller value that provides those pole locations is not known. The root locus diagram is drawn and the controller value is determined graphically. Figure 4.12 shows the closed loop pairs, which are in fact on the root locus diagram created in Figure 4.11. Again, the determination of the controller is done graphically and is thoroughly explained in any control system textbook.

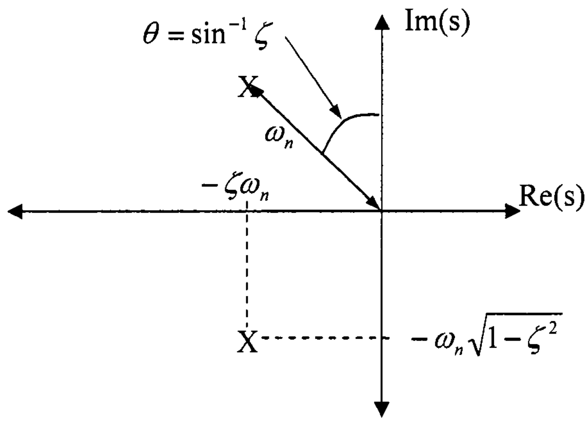


Figure 4.12 Plot of Pole Pairs on S-Plane

The root locus diagram has been created for the special case where the open loop transfer function contains only two poles. However, the open loop transfer function of a different plant model may contain two poles and one zero, three poles and no zeros, or four poles and three zeros, just to name a few examples (the number of poles must always be larger than the number of zeros for realizability). The rules that help determine the root locus diagram are listed below.

- If the number of poles and zeros to the right of a point on the Re(s) axis is odd, then that point is included in the root locus.
- If the number of poles and zeros to the right of a point on the Re(s) axis is even, then t hat point is included in the root locus.
- The number of poles minus the number of zeros, determine the angle in which the poles depart as the gain "K" goes to infinity.

4.5 Controller Types

There are three common controller functions used for closed-loop feedback systems. Those controller types are proportional control, integral control, and derivative control. Each affects the dynamic system performance differently. Proportional control is implemented when there is a need to speed up the system response. In other words, if the rise time is needed to be shorter, proportional control will do the trick. Integral control serves to improve the steady state performance of the closed-loop system. Derivative control adds stability by increasing the damping during transient conditions. That means that there is less overshoot for a step-input, however there is a trade off. Along with the added damping, some delay to the system is introduced, slowing down system performance. The mathematical models for all three controllers in the Laplace form are given by Equations 4.11, 4.12, and 4.13.

$$D_p(s) = K_p \tag{4.11}$$

$$D_I(s) = \frac{K_I}{s} \tag{4.12}$$

$$D_d(s) = sK_d \tag{4.13}$$

where, K_p = proportional constant

K_I = integral constant

K_d = derivative constant

4.6 Classic Control Applied to BLDC Machine

The machine parameters of a BLDC machine available at the Power Electronics Research Lab were used to verify the proper operation of the machine model. The machine parameters were attained from the manufacturer's datasheet, those parameters are summarized in Table 4.3. The rated voltage was applied to the motor model and the speed response, along with the phase (phase A) current were observed as shown in Figure 4.13 and 4.14.

Table 4.3 BLDC machine parameters.

Parameter	Units	Value
Terminal Voltage	volts DC	50
Rated Speed	rpm	2820
Rated Torque	Nm	2.92
Rated Current	Amps	20.2
Rated Power	Watts	861
Torque Sensitivity	Nm/amp	0.164
Back-emf	volts/krpm	17.1
Terminal Resistance	ohms	0.106
Terminal Inductance	mH	0.428
Motor Constant	Nm/sq.rt.watt	0.516
Rotor Inertia	g-cm^2	4943
Number of Poles		4

Figure 4.13 Open loop speed plot.

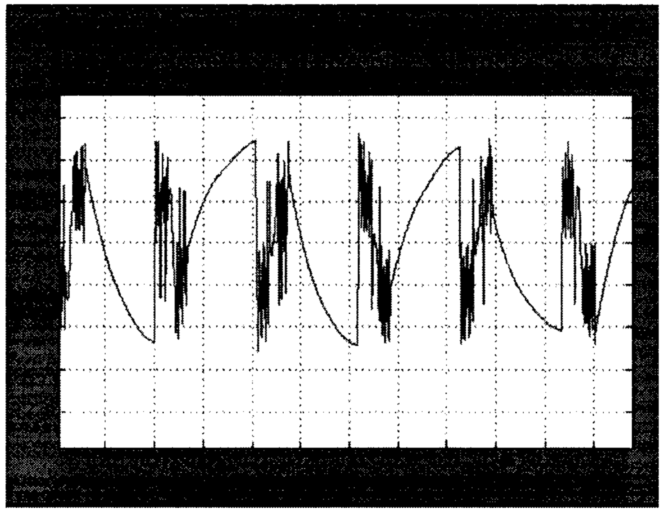


Figure 4.14 Open loop current plot.

The open loop speed behavior is completely determined by the machine dynamics. The current behavior also is determined by the machine electric characteristics and it is apparent that it largely deviates from the ideal square-wave shown in Figure 4.16. Therefore, it is necessary to introduce some form of speed and current regulators so as to control or alter the machine speed and current dynamics.

4.6.1 Current and Speed Regulation. Classic control system theory was used to introduce PI compensators with negative feedback to control the phase currents and the motor speed. First, the current loop was tuned to attain a bandwidth of $f_{bw,i} \approx 3kHz$. The current loop in Figure 4.15 is attained from Equations 4.22 and 4.23 when the rotor is assumed to be in the locked position.

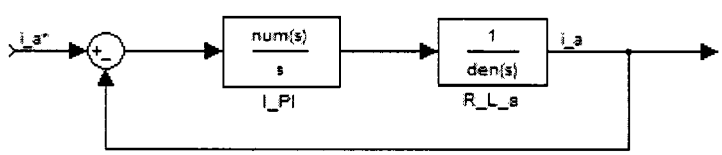


Figure 4.15 Current closed-loop block diagram, with locked rotor.

$$PI = K_P + \frac{K_I}{s} \qquad (4.21)$$

where, $K_p = proportional\ constant$

$K_I =$ integral constant

The PI block consists of a proportional gain summed with an integral gain multiplied by an integrator as given by Equation 4.21. In order to attain the desired bandwidth, the PI gains must be appropriately chosen as specified by Equations 4.22 and 4.23.

$$k_{P,i} = 2\pi f_{bw,i} R_{ph} \qquad (4.22)$$

$$k_{I,i} = 2\pi f_{bw,i} L_{ph} \qquad (4.23)$$

To ensure that the desired bandwidth was obtained, Matlab was used to generate bode plots of the resulting closed loop current transfer function. The resulting bode plots are shown in Figure 4.16. As expected the gain and phase start to role off at about 3kHz.

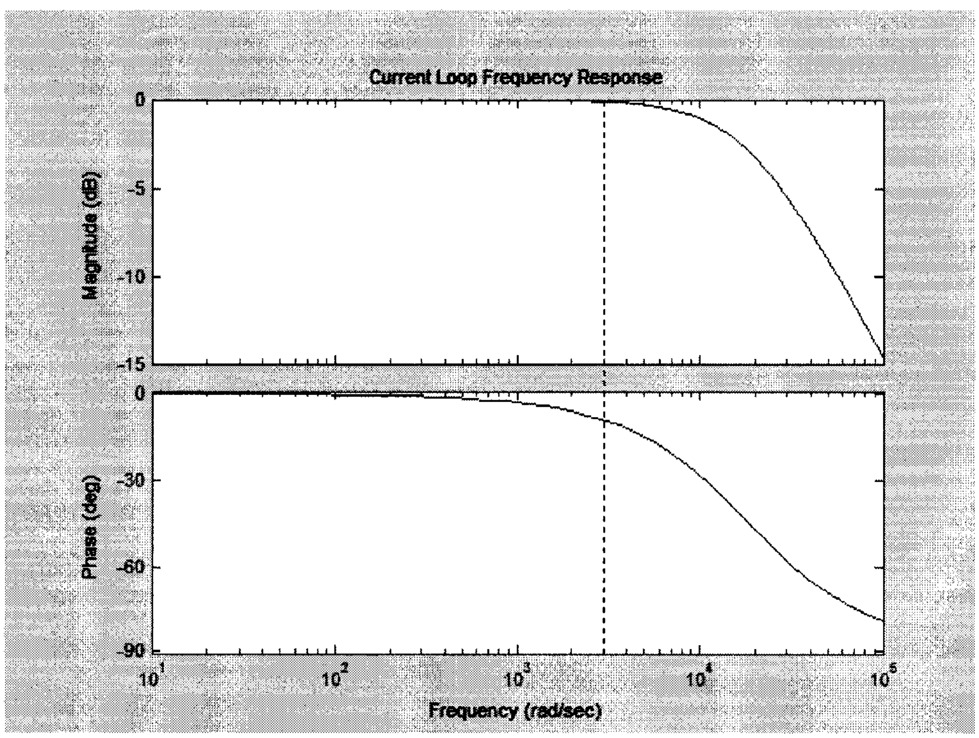


Figure 4.16 Current closed-loop frequency and phase response.

Next, the speed loop was tuned with the assumption that the current response was to be much faster than the speed response. The speed bandwidth was chosen to be $f_{bw,\omega} \approx 100Hz$, which is thirty times "slower" than the current bandwidth. Therefore, given the aforementioned assumption, the speed loop reduces to that shown in Figure 4.17. The unity gain block "T_gain" accounts for the assumption of the "fast" current loop to being infinitely fast compared to the speed loop.

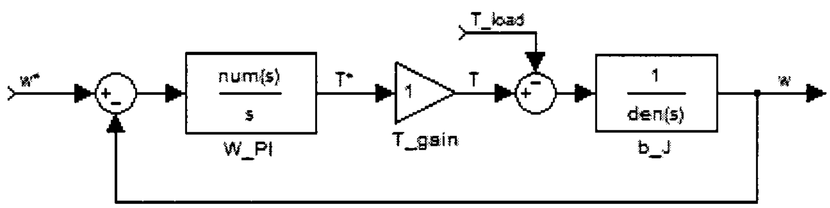


Figure 4.17 Speed closed-loop block diagram.

As with the current compensator, the speed compensator gains are calculated given the desired bandwidth and motor parameters. In order to attain the desired bandwidth, the PI gains must be appropriately chosen as specified by Equations 4.24 and 4.25.The resulting bode plots for the speed loop are illustrated in Figure 4.18.

$$k_{P,\omega} = 2\pi f_{bw,\omega} b \qquad (4.24)$$

$$k_{I,\omega} = 2\pi f_{bw,\omega} J \qquad (4.25)$$

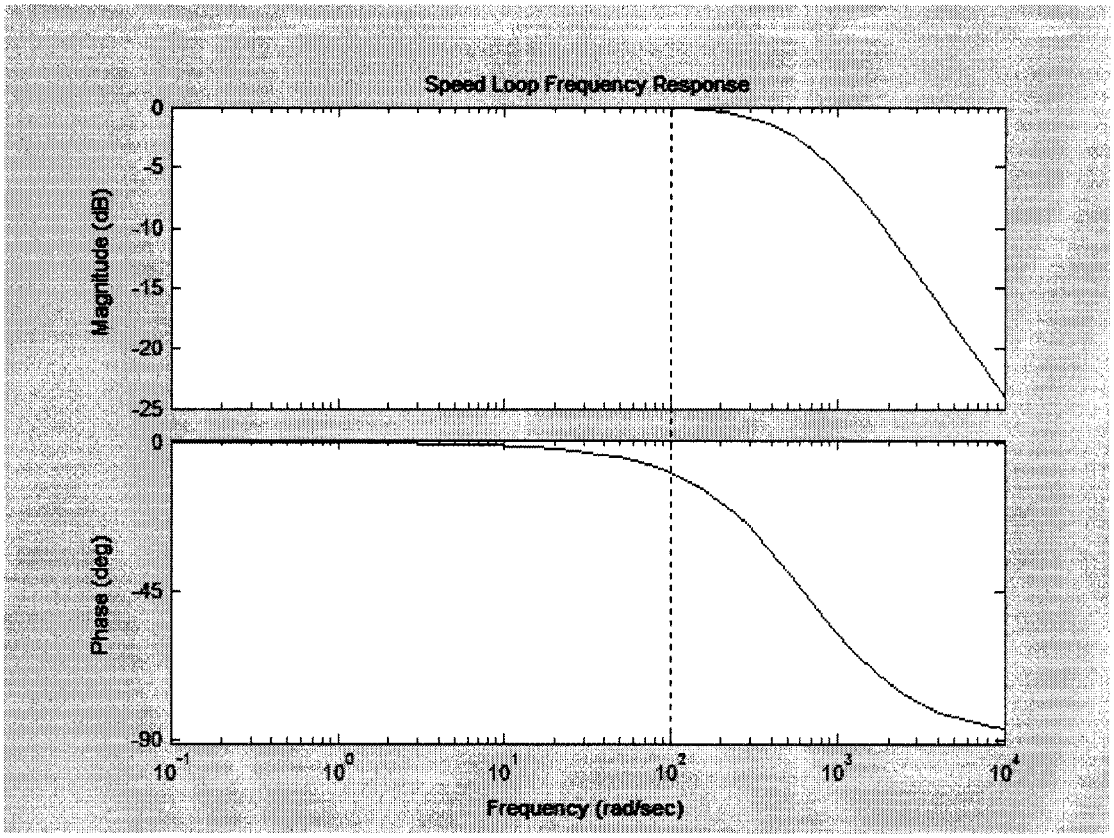


Figure 4.18 Speed closed-loop frequency and phase response.

Once the current and speed loops were appropriately tuned for the motor model parameters given in Table 4.4, the closed loop model was created in Simulink, see Figure 4.19. The machine model in Figure 3.12 has been converted into a subsystem (BLDC_Motor_Model) and current PI blocks have been added to each phase. Finally, the speed PI block is highlighted by the dashed box.

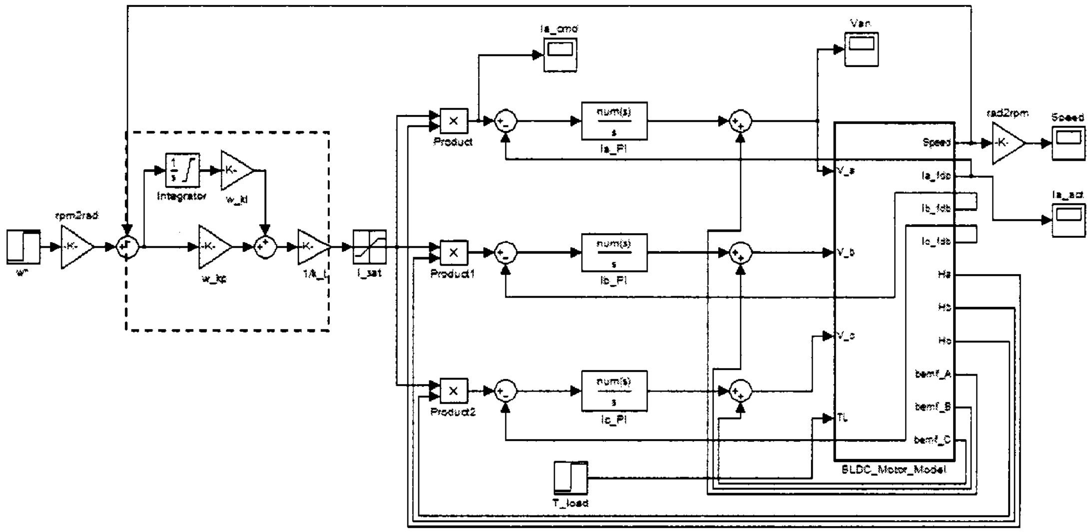


Figure 4.19 BLDC machine model with speed and current regulators.

4.6.2 Simulation Result. To verify the proper operation of the regulated BLDC machine model, an operation scenario was created. The scenario consisted of a step speed command of 2000 rpm (about 2/3 of the rated speed) and a step load command of 2 Nm (about 2/3 the rated load) applied 0.3 seconds after the speed command. The resulting phase (phase A) current and speed are shown in Figures 4.21 and 4.22. The current plot was as expected, it had a square-wave shape and the amplitude stepped up at the time of the load input. However, the speed response was not exactly as expected, the speed seemed to rise too slow and there was an overshoot that did not settle at 2000 rpm in the desired time corresponding to $f_{bw,\omega} \approx 100Hz$, 0.01seconds. Also, the applied load disturbance seemed to be corrected very slowly in comparison to the desired speed dynamics.

The slow rise time was due to the fact that a current limit was set for all three phase-currents. Therefore, Figure 4.23 shows that regardless of how "fast" the current loop is tuned, the motor response will be limited by the amount of current that can be injected into the motor (I_rated) without damaging the motor windings. The overshoot of the speed was due to the integrator build up of the error signal. The build up occurred because the speed loop was tuned to give a speed response that exceeded the physical capabilities of the machine.

Figure 4.24 illustrates the speed response when the load is applied at 0.3 seconds. The disturbance causes the speed to deviate from the commanded speed for over two seconds even though the speed loop was tuned to respond in about 0.01 seconds. The reason for the large deviation is due to the fact that the closed loop dynamics caused by load inputs differ from reference input dynamics. In tuning the speed loop, a pole zero cancellation was carried out in order to increase the speed loop bandwidth. However, the load to output path loses that pole zero cancellation, yielding a slower response corresponding to the natural mechanical time constant of the BLDC machine. Table 4.4 shows that the mechanical time constant for the machine modeled was about 0.5 seconds. That means that it will take roughly five time constants for a transient (load disturbance) to fade away. That, in fact, is what happened when the load disturbance was applied. Figure 4.25 shows that it takes about 2.5 seconds for a load disturbance to be corrected.

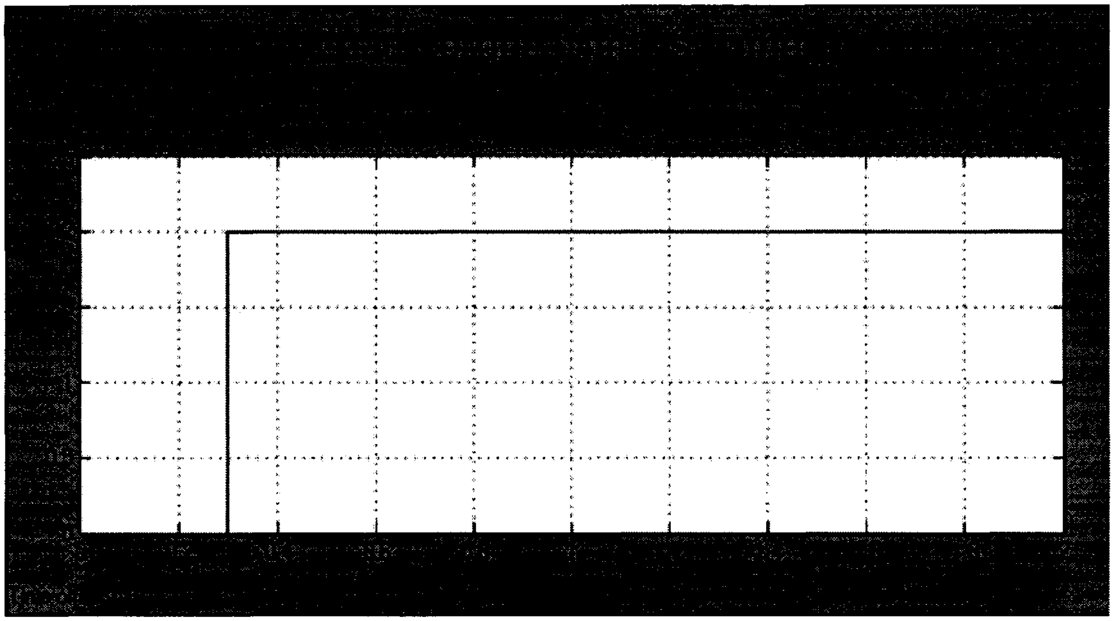


Figure 4.20 Applied torque vs. time.

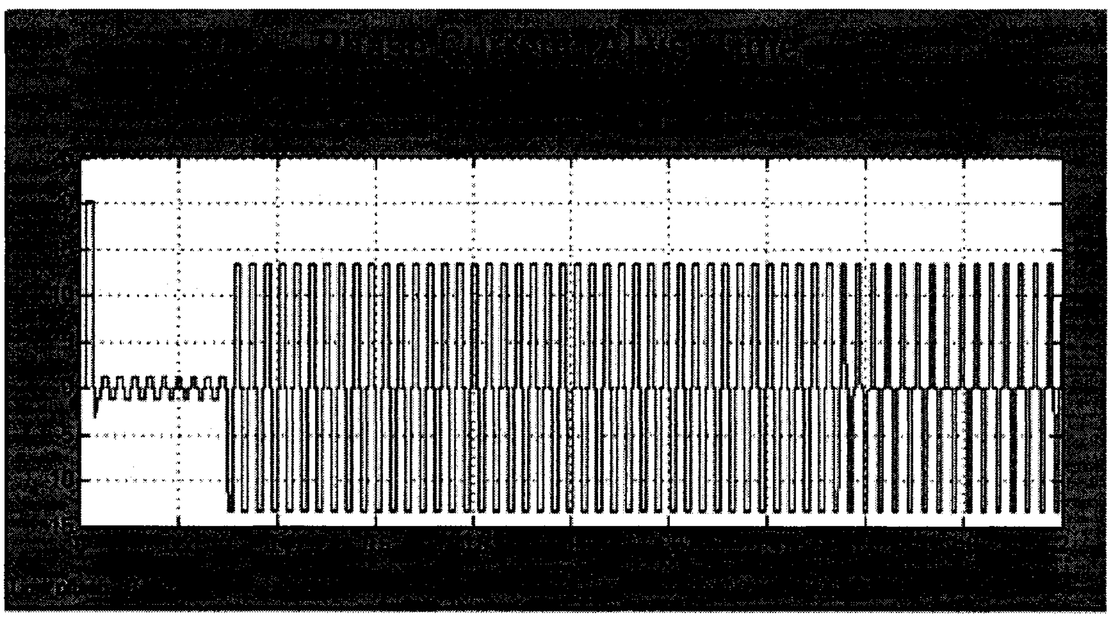


Figure 4.21 Phase current vs. time.

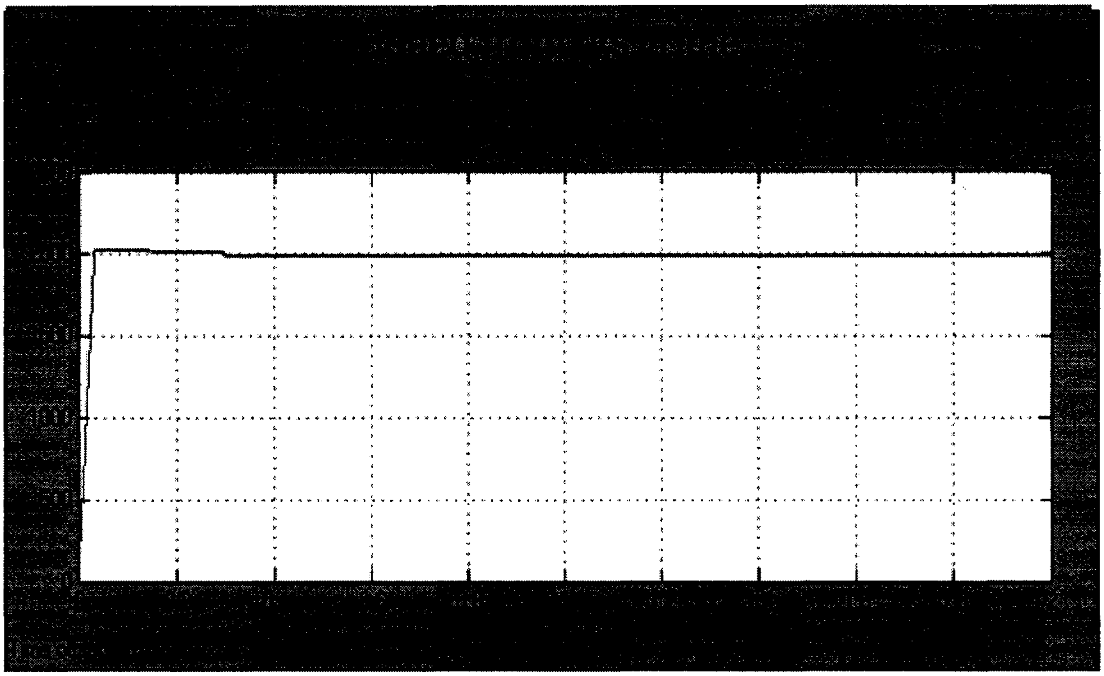


Figure 4.22 Speed vs. time.

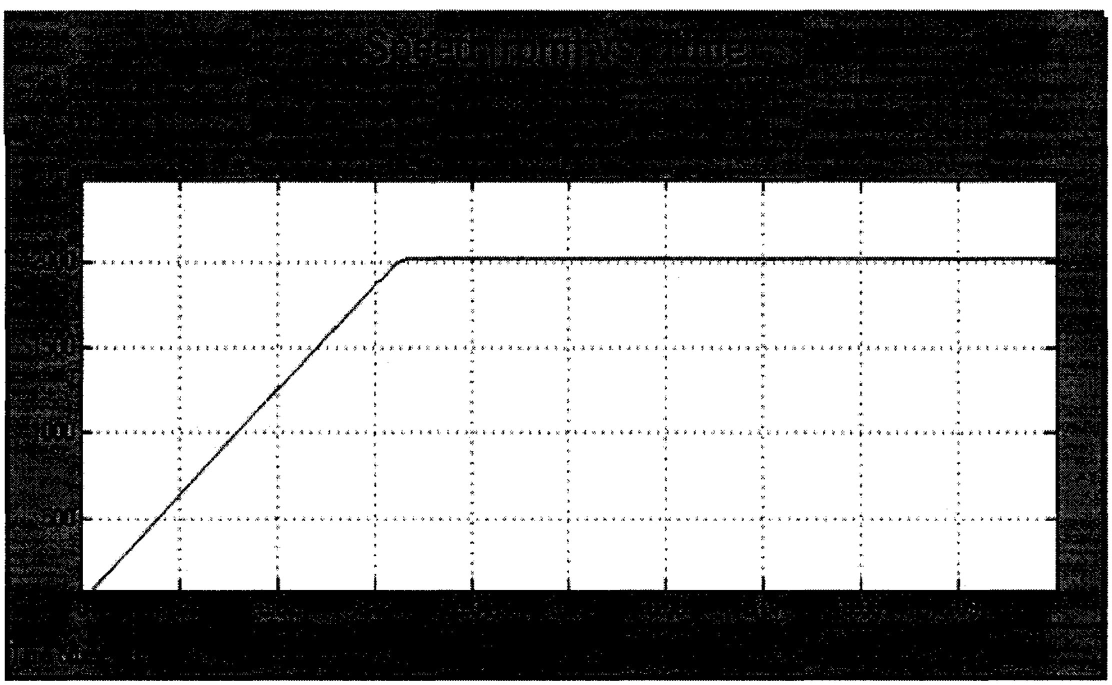


Figure 4.23 Speed vs. time (rise time view).

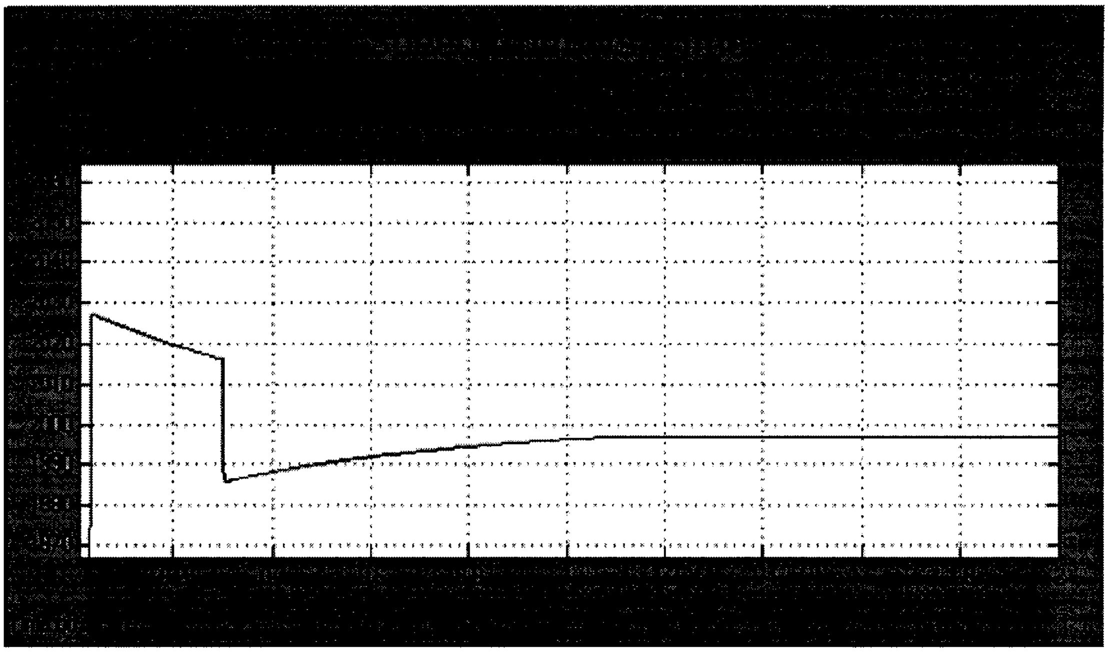


Figure 4.24 Speed vs. time (disturbance).

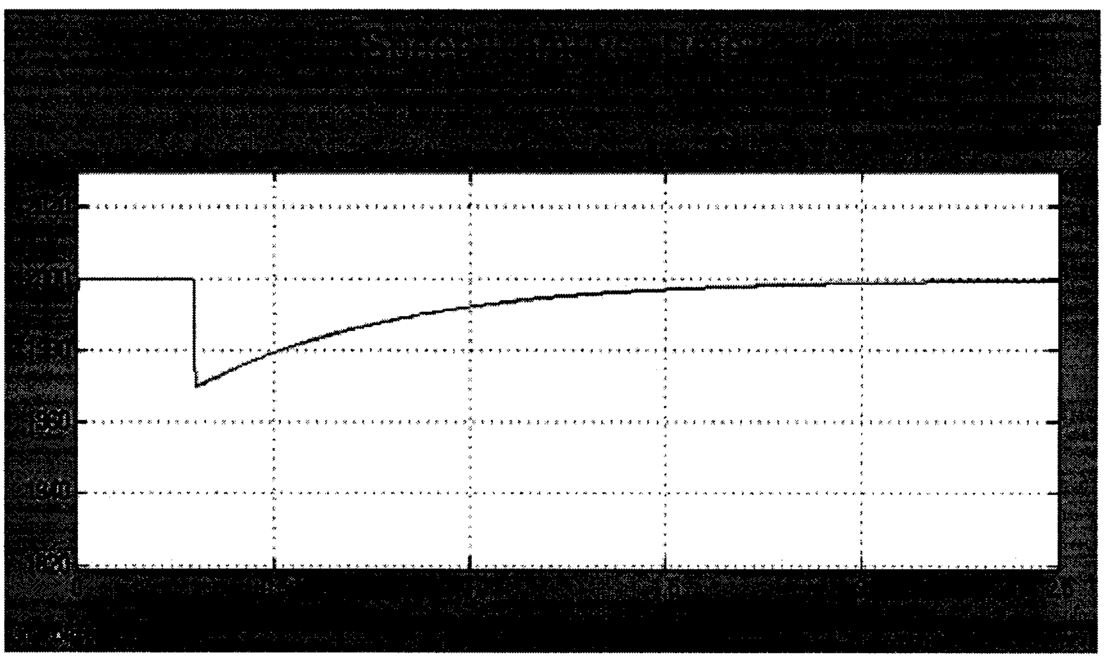


Figure 4.25 Ideal Speed (disturbance).

4.6.3 Conclusion. Simulink was used to successfully model an ideal BLDC machine as described in Chapter 3. Speed and torque regulators were introduced in order to simulate a high performance BLDC machine servo system. The tuning procedure for both regulators was outlined and closed loop response for the current and speed were analyzed. Upon completion of the BLDC machine model, it was noticed that the Simulink model that was created can be extended to model an ideal Permanent Magnet Synchronous Motor (PMSM) with a slight modification. By simply altering the back-emf wave-form from a trapezoidal shape to a sinusoidal shape, a PMSM model will result. That modification was done and the only difference in the performance compared to the BLDC machine was in the developed torque. The PMSM was free of any torque ripples were the BLDC machine contained torque spikes corresponding to the current commutation frequency. Figure 4.26 and 4.27 summarizes those findings.

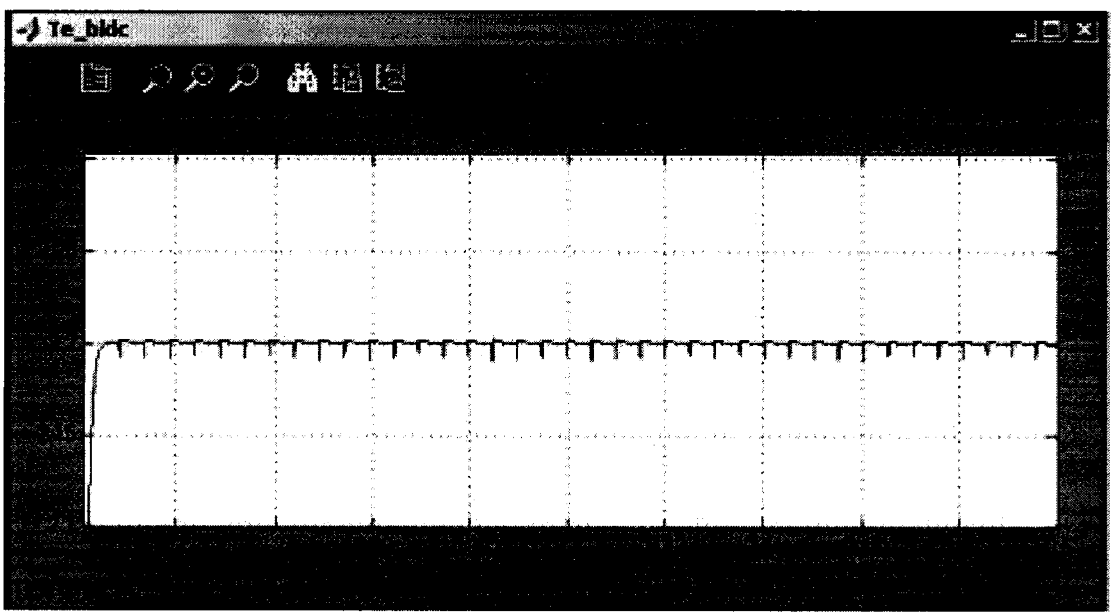


Figure 4.26 BLDC Machine Torque Plots

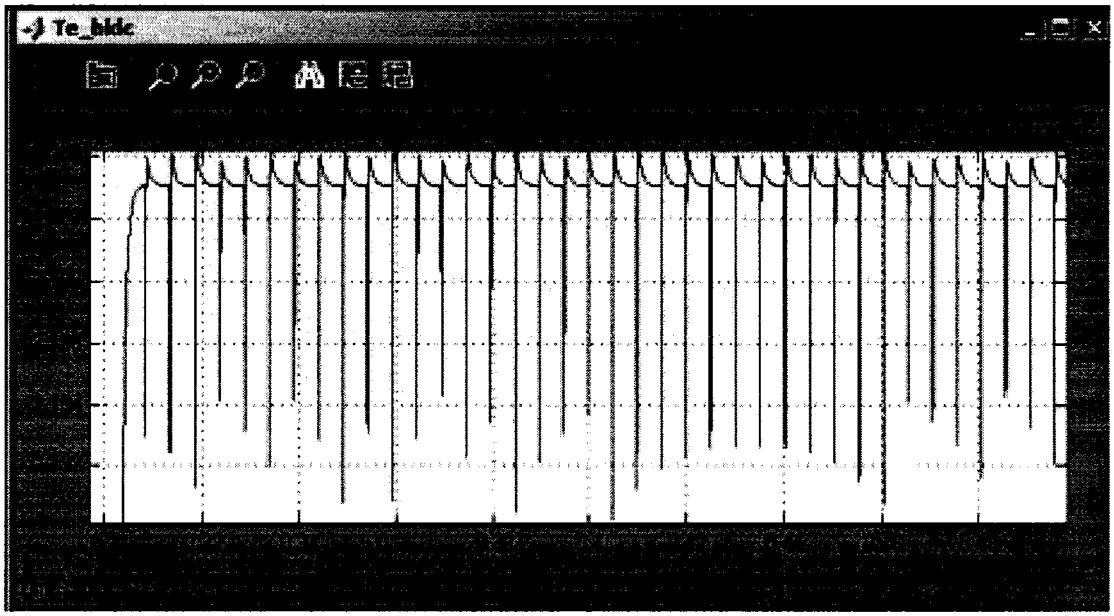


Figure 4.27 BLDC Machine Torque Plots

4.7 Classic Digital Control

The classic continuous control method applied to a BLDC machine as outlined in the previous section can be implemented in discrete time or the z-domain. The sampling time must be considered when deriving the gains to the control transfer functions. To demonstrate the methodology of digitally implementing a continuous time controller on a system, a temperature control design example is presented.

The objective at was to design a classic digital control for a temperature control system that was to meet specified transient and steady state requirements. The design method chosen was that of a digitally implemented PI compensator coupled with a lead compensator for performance enhancement. The block diagram that describes the temperature control system, see Figure 4.28, is composed of transfer functions representing the actuator and valve, the chemical heat process, and the temperature sensor. The controller will consist of the simplest controller that will meet the design requirements.

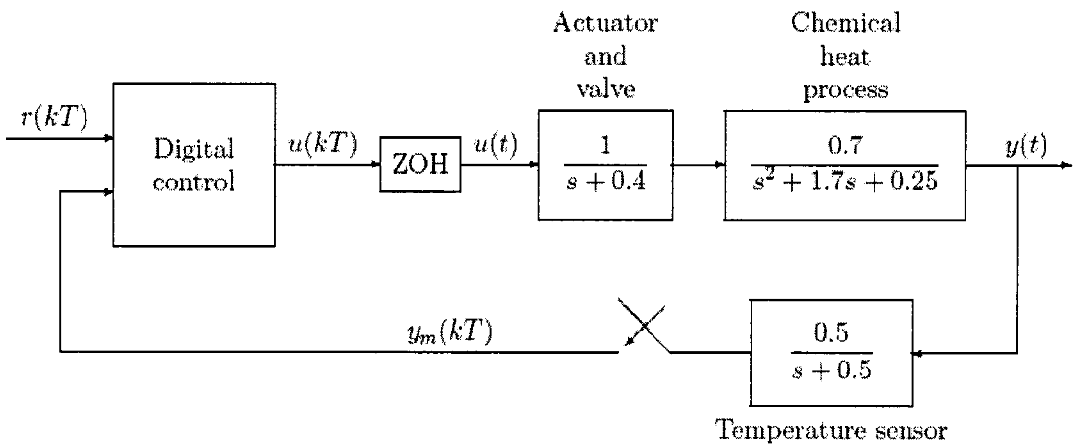


Figure 4.28 Classic digital control system block diagram.

4.7.1 Design Specifications. The temperature control unit is characterized by the mathematical model of its chemical heat process and actuator/valve, see Figure 4.29. The open loop response of this system to a step input is characterized by a rise time of about 15 seconds ($t_r \approx 15\sec$) and a significant steady state error ($e_{ss} \approx 700\%$) as shown in Figure 4.31. The Rise time (t_r) is defined as the time, in seconds, required for the output to rise from ten percent to ninety percent of the final value.

The desired closed loop performance of the heating system is to lower the rise time to about seven and a half seconds with an overshoot of less than five percent and zero steady state error to a step input. The performance requirements consist of a rise time of seven and a half seconds, an overshot les than five percent, and zero steady state error. From the specified requirements several other values can be approximated, such as the necessary sampling period (T_s), the damping ratio (ζ), and the natural frequency (ω_n).

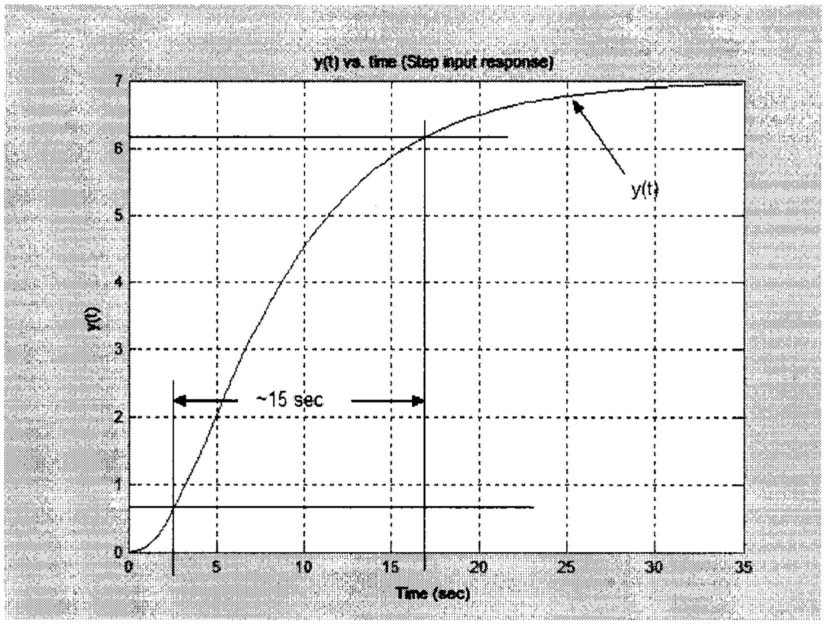


Figure 4.29 Open loop response of temperature control system.

The sampling period was approximated using the following "rule-of-thumb" for sampled systems. The sampling period should be roughly six to forty times faster than the rise time of the desired closed loop response. In this case, the sampling period was chosen as one twentieth of the desired rise time. The damping ratio was directly calculated with Equation 4.26, which relates the overshoot to the damping ratio. Finally, the desired natural frequency was approximated with Equation 4.27, which relates the rise time to the natural frequency. Those approximated and calculated values are consist of a sampling period of four tenths of a second, damping ration of less than seven tenths, and a natural frequency greater than twenty four hundredths.

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \tag{4.26}$$

$$t_r \cong \frac{1.8}{\omega_n} \tag{4.27}$$

4.7.2 Design Procedure. As mentioned in the introduction, it is desired to create a digital controller that will drive the temperature control system in Figure 4.30 with the desired transient and steady state characteristics. The first attempted controller was a discrete PI compensator; this required the calculation of the zero-order hold (zoh) of the actuator and plant. However, doubt arises on how to handle the non-unity feedback that results from the dynamics of the temperature sensor in the feedback path. Therefore, the temperature sensor was lumped into the plant model as shown in Figure 4.30. The design procedure was then carried out with the following assumption. If the output *ym(t)* meets the transient and steady state requirements, then the output *y(t)* will also meet those requirements. In fact the transient response of *y(t)* will be slightly better than that of *ym(t)*.

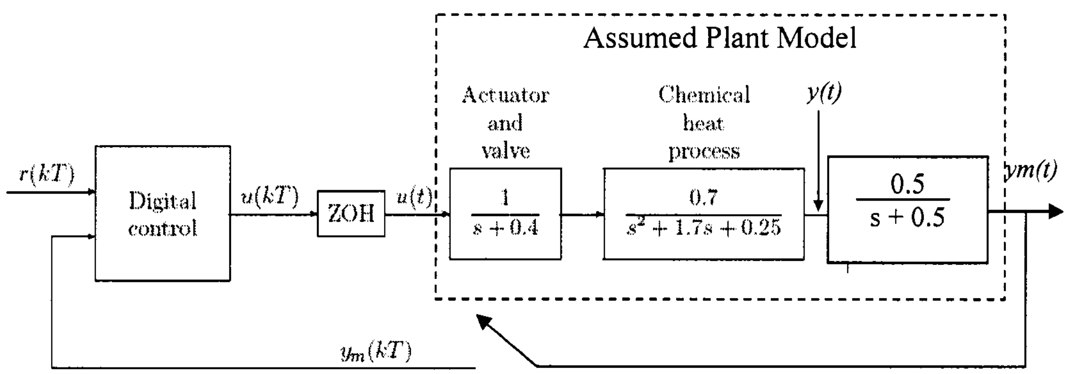


Figure 4.30 Modified plant model; includes temperature sensor.

The validity of the above assumption is shown in Figure 4.31. The figure shows a plot of the open loop response of the temperature control unit with and without the temperature sensor lumped together. The first curve is the uncompensated response of *y(t)* to a step input, while the second curve is the uncompensated response of *ym(t)*.

Examination of the curves reveals that the response of both curves will always be nearly identical with the exception of a phase delay. The steady state value of *ym(t)* will always equal that of *y(t)* since the gain of the temperature sensor at steady state is unity.

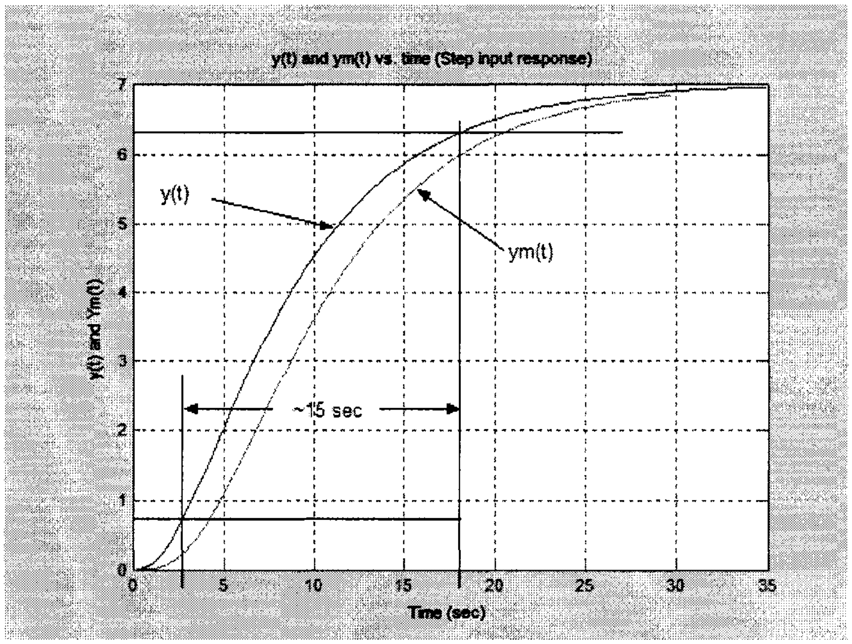
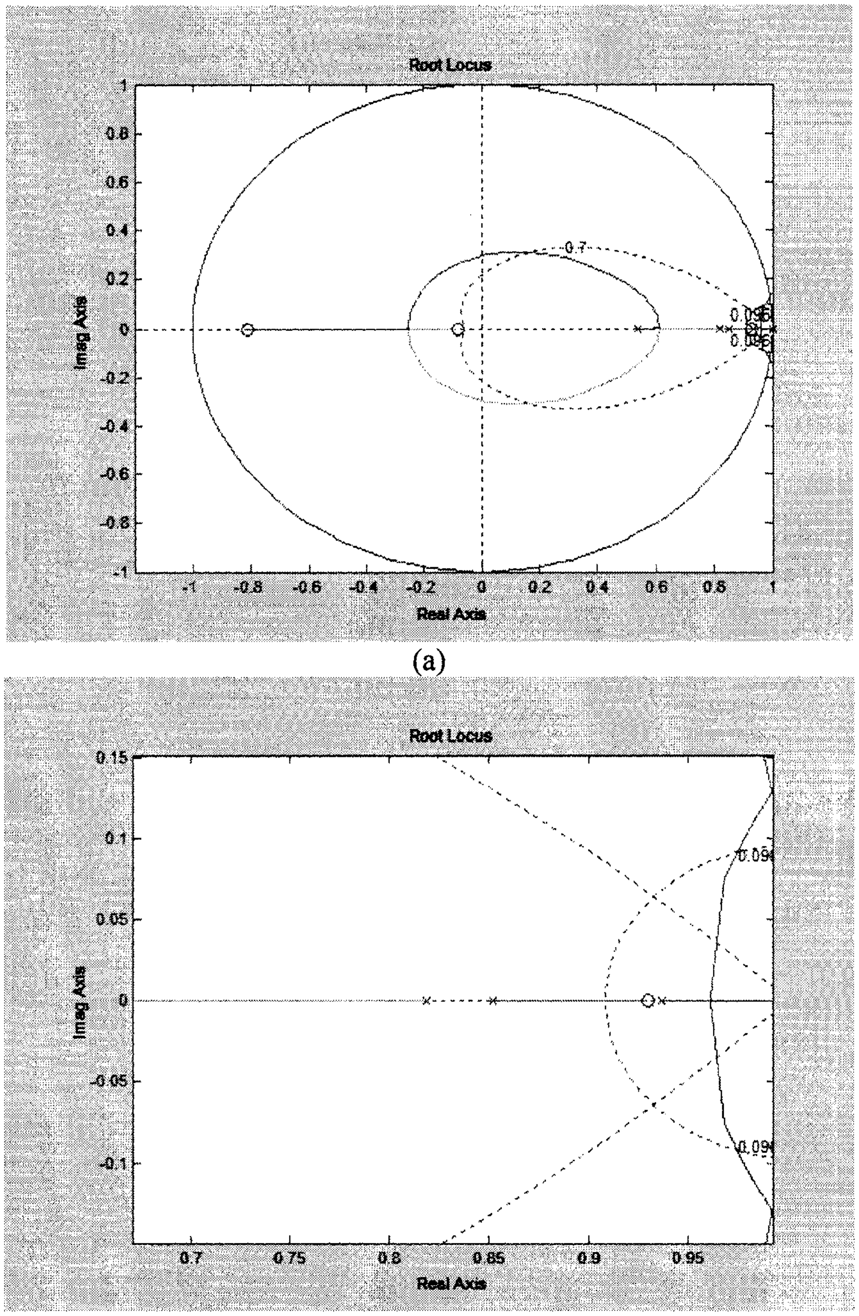


Figure 4.31 Open loop response output and sensor output.

Given the aforementioned assumption, the design procedure entailed the introduction of a digital PI compensator. Proportional-integral control was used since part of the requirements called for zero steady state error, which translates to integral action within the compensator. The first step was to find the zoh equivalent transfer function of the assumed plant model in Figure 4.32. The sampling frequency was chosen as Ts=0.4. The idea behind the PI compensator zero placement was to locate the zero very close to the slowest pole of the discrete plant model. Then, the root locus diagram was generated in Matlab, see Figure 4.32. The natural frequency and damping ratio constraint were also plotted and it was determined that the PI compensator alone would not be able to drive the plant so as to meet the requirements.

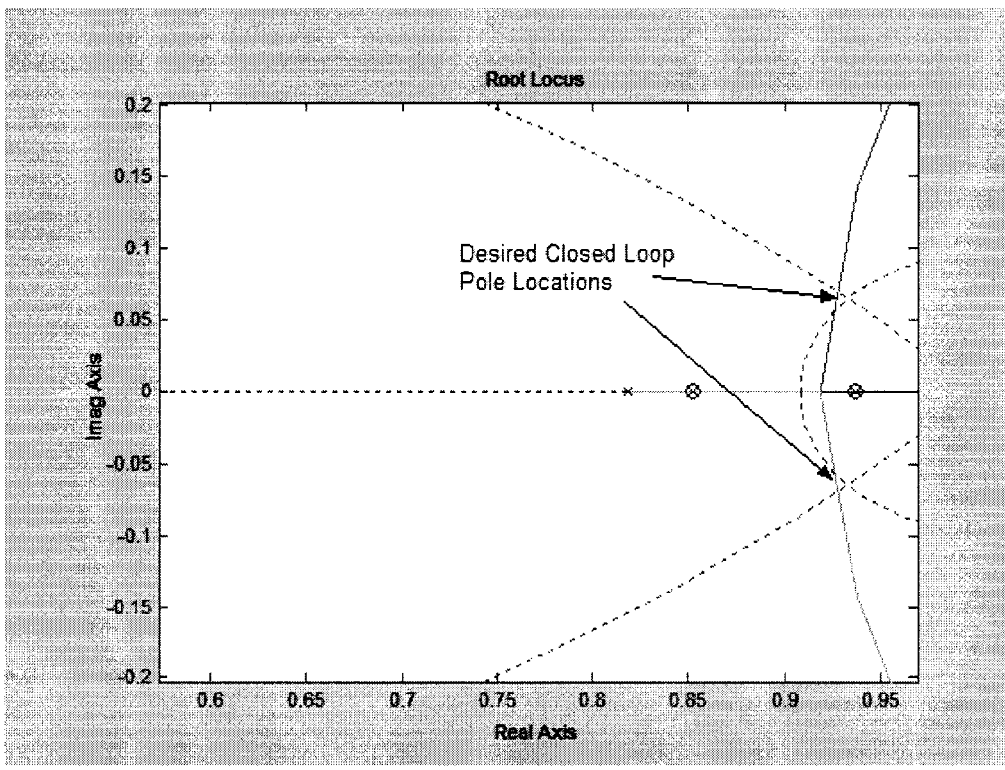


(a)

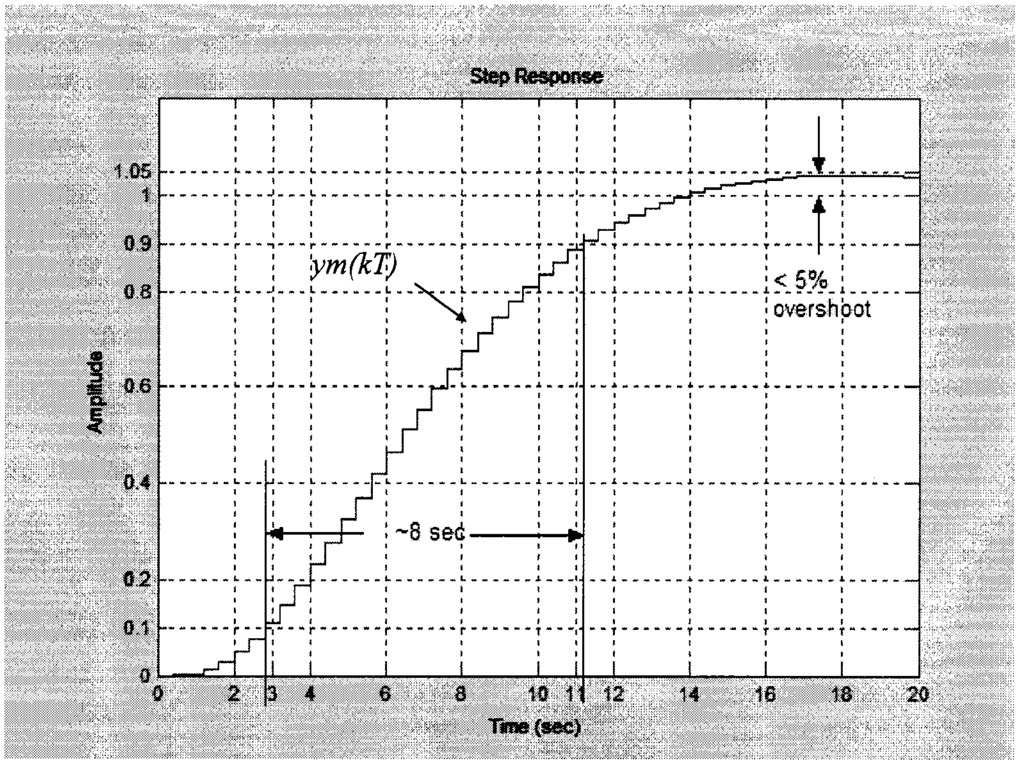
(b)

Figure 4.32 Root locus diagram of plant and PI compensator.

The second design idea was to add a lead compensator to once again try to cancel out a slow plant pole. The lead compensator zero was placed close to the second slowest plant pole. The lead compensator pole was placed near the origin (z_{p_lead}=0.1), the pole was placed as close as possible to z =1 without affecting the system performance. Figure 4.33a shows the root locus with the addition of the lead compensator, the closed loop poles can be place within the range that will meet the dynamic and steady state requirements. The exact position of the closed loop poles was tuned to produce the step response of *ym(t)* as illustrated in Figure 4.33b.



(a)



(b)

Figure 4.33 Final closed loop poles and output step response.

4.7.4 Analysis of Results. Figure 4.33b shows a rise time reduction of about half (~8 seconds) compared to the uncompensated system. Also, the overshoot is kept to within five percent and the integrator in the compensator ensure a Type 1 system. Therefore, there is a steady state error of zero. As mentioned earlier, the compensator was designed for *ym(t)* to meet the performance requirements. However *y(t)* will also meet the requirement with slightly better performance than *ym(t)*. The final design in block diagram form is illustrated in Figure 4.34. The gain K = 0.835 was determined via the "rlocfind" command in Matlab. The final closed loop pole values resulting with the above are summarized as follows:

- P1,2 = 0.9253 + 0.0669i
- P3 = 0.9344
- P4 = 0.8490

- P5 = 0.5128
- P6 = 0.1012

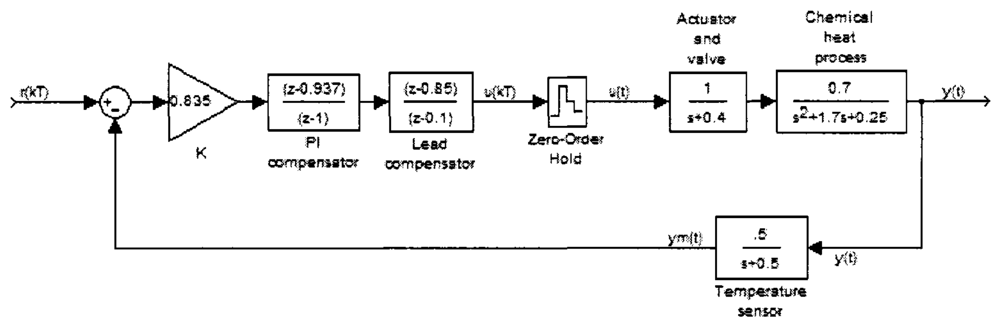


Figure 4.34 Final plant and controller design block diagram.

4.7.5 Input Magnitude and Output Response Relation. In the design of control systems for practical applications, it is important to take consider the capabilities of the actuator. In the temperature control system a compensator was designed to meet the specified requirements. Additional control technique could have been implemented to force the plant to perform far better or faster than the requirements. However, driving a system too fast (while maintaining stability) may prove to be very costly.

The input to the actuator for the temperature control problem is most likely some form of fuel. Fast compensation by the PI & lead compensator may dump a large amount of fuel to speed up the out response just by a few seconds. Therefore, when designing a control system, whether digital or analog, it is vital to take the practical capabilities of the plant into consideration. Since, in most cases, system performance is limited by practical limitations which come in the form of noise or the fact that finite energy sources are used to drive systems.

4.7.6 Conclusion. A series combination of a PI and lead compensator were used to achieve the specified requirements. Since the output *ym(t)* only differed from *y(t)* by a slight phase delay, it was lumped into the plant model in order to make the design procedure somewhat easier. Also, it was noted that it is not always better to go far and beyond in exceeding the performance requirements. This is due to the fact that the input magnitude of the actuator is often directly related to the cost of driving the system. For the temperature control system, a higher input magnitude to the actuator results in more fuel being used and for motor drives, the same problem results when too much electricity is applied to the motor driver.

CHAPTER 5

NOVEL DIGITAL CONTROL TECHNIQUE FOR BRUSH-LESS DC MOTOR DRIVES: STEADY STATE ANALYSIS

5.1 Introduction

Permanent magnet motors with trapezoidal back-emf and sinusoidal back-emf have several advantages over other motor types. Most notably, (compared to dc motors) they are lower maintenance due to the elimination of the mechanical commutator and they have a high power density which makes them ideal for high torque-to weight ratio applications [20]. Compared to induction machines, they have lower inertia allowing for faster dynamic response to reference commands. Also, they are more efficient due to the permanent magnets which results in virtually zero rotor losses [29]. The major disadvantage with permanent magnet motors is their higher cost and relatively higher complexity introduced by the power electronic converter used to drive them. The added complexity is evident in the development of a torque/speed regulator. Using the d-q transformation to ease the complexity of analyzing three phase machines may serve to design an adequate controller. However, development of a controller based on the transformation of the a-b-c equations to the d-q variables is only advantages for permanent magnet motors with sinusoidal back-emf. Applying the d-q transformation to a trapezoidal back-emf motor does not eliminate the angle dependant phase inductances [28]. The author in [20] applied the d-q transformation to a BLDC motor by fixing the synchronous reference frame to the instantaneous rotor flux linkage instead to the rotor geometric axis. However, this method is cumbersome since the instantaneous rotor flux linkage must be found experimentally and programmed into a DSP. Sliding mode control techniques have proved to be computationally extensive when adaptive parameter estimation is used to estimate load parameters [19], [33]. Hysteresis current control and pulse-width-modulation (PWM) control coupled with continuous control theory produce the most widely used BLDC motor control techniques [7]. Hysteresis current control is essential towards achieving adequate servo performance [6], namely instantaneously torque response yielding faster speed response compared to PWM control. For most applications, a proportional-integral (PI) current and speed compensators are sufficient to establish a well-regulated speed/torque controller. In other cases, state feedback control is needed to achieve more precise control of the BLDC motor. Classic control theory and linear system theory are well understood, but are highly complex and require extensive control systems knowledge to develop a well designed controller. Discrete control theory allows for such controllers to be digitally implemented with micro-controllers, microprocessors, or digital signal processors (DSPs). Digitizing analog controllers serves to add complexity to the overall design procedure.

It is important to note that digital implementation of a continuous control technique does not produce a pure digital controller. Instead, what results is a digitally implemented non-digital control technique. This thesis introduces a novel digital controller that treats the BLDC motor drive like a digital system. The BLDC system may only operate at a few predefined states that produce constant predefined motor speeds. Speed regulation is achieved by alternating states during operation, which makes the concept of the controller extremely simple for design and implementation purposes. This novel concept will help reduce the cost and complexity of motor control hardware. That, in turn, can boost the acceptance level of BLDC motors for commercial mass production applications, successfully fulfilling the promises of energy savings associated with adjustable speed drives.

Newton's 2nd Law and the characteristic equations of a BLDC machine were used to derive a design procedure for the novel controller. It will be shown that the design procedure involves a simple first order non-homogeneous differential equation. During steady state operation, the design procedure is reduced to a few simple algebraic equations. Computer simulations were used for proof-of-concept and implementation of the novel digital controller was carried out with the experimental setup described in Chapter 3.

5.2 Novel Digital Control

The proposed digital controller treats the BLDC motor like a digital system, which may only operate in two predefined states. Operation of the motor in State-1 will result in a motor speed of Low Omega (ω_L). Operation in State-2 results in a motor speed of High Omega (ω_H), where speed ω_H is greater than ω_L. If the commanded speed is ω^*, where $\omega_L \leq \omega^* \leq \omega_H$, the digital controller will achieve speed regulation by appropriately alternating states. The rules that the digital controller follows are extremely simple:

- If the actual motor speed is less than the commanded speed, then switch or stay at State-2 (ω_H).
- If the actual motor speed is greater than the commanded speed, then switch or stay at State-1 (ω_L).

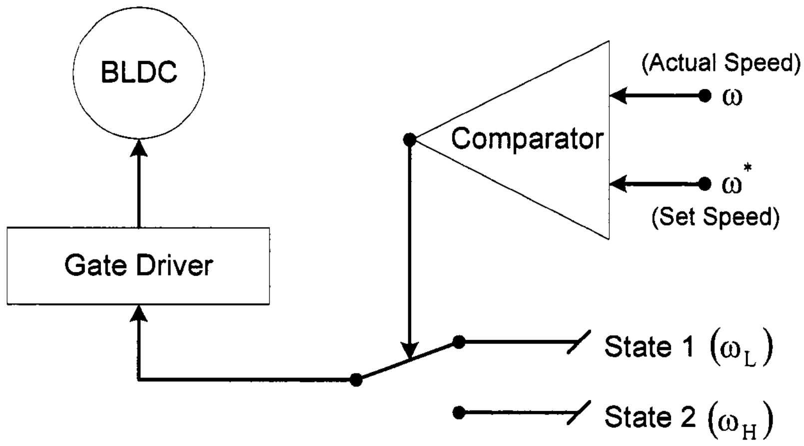


Figure 5.1 Proposed digital control.

The task of the digital controller is ideally suited for a DSP because the rules are nothing more than two IF THEN statements. Though, the main goal is not to use a DSP. Instead, the proposed method can be implemented in an application-specific-integrated-circuit (ASIC). The simplicity results in a very low cost IC. Figure 5.1 shows an illustrative description of the digital controller. The digital controller can be implemented several ways, the distinction lies in the method of attaining the two operating states.

The novelty of the control lies within the means of attaining the actuation signal which drives the BLDC motor. Control systems design allows for very large bandwidth BLDC motor drives systems. However, that bandwidth is always limited by practical constraints. Some of those constraints include the use of a finite power source and maximum current allowable in the stator windings before fusing occurs. In other words, a one kilowatt 20kHz bandwidth BLDC motor drive system, designed with classic control theory, may require hundreds of amps to achieve such a wide bandwidth. However, such a high current demand for a one kilowatt motor will certainly damage the motor. The proposed controller combines control system knowledge with practical knowledge of

BLDC motor drives. Figure 5.2 illustrates an actuation signal u(t) that results from applying a step speed changes at time t1 and t2. A classic controller reacts exponentially to the increase in speed error in order to track the reference speed.

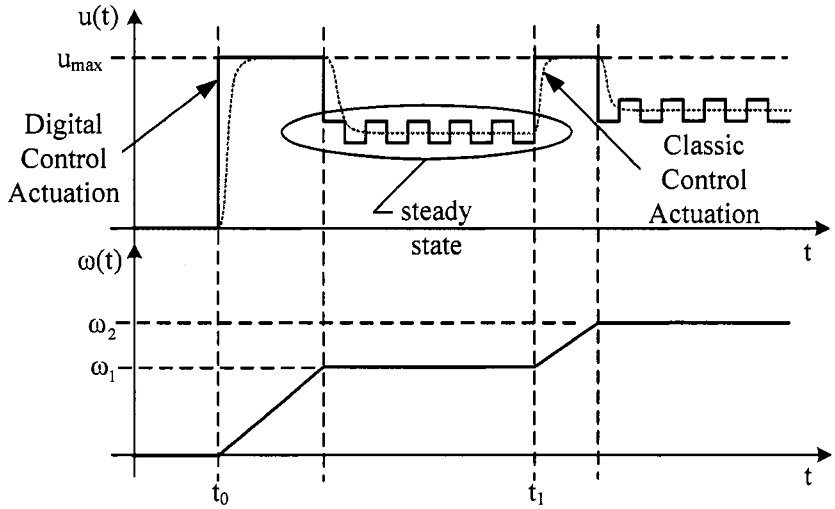


Figure 5.2 Digital control actuation signal.

The actuation signal shape depends on the controller gains which may result in an overshoot or a slow response. On the other hand, the digital control immediately applies the full actuation signal (rated current of the motor) immediately until the commanded speed is achieved. That will achieve the fastest speed response considering all practical limitations. Then, the two states are used to achieve speed regulation in steady state operation, which highlights the simplicity of this control algorithm. This chapter covers the design and experimental verification of steady state operation. Dynamic operation (namely disturbance rejection capability) will be developed and experimentally verified in Chapter 6. The following section will introduce two possible implementation methods for the proposed digital control, namely Conduction Angle Control and Current Mode Control. The design procedure for each method is derived and computer simulations are provided for each method. Experimental validation is provided for the Conduction Angle Control method.

5.3 Conduction-Angle Control

The name Conduction-Angle Control stems from the definitions of State-1 and State-2. Figure 5.3 shows that the ideal current for a BLDC motor is a square wave with a periodic conduction angle from θ_1 to θ_2. Modification of the conduction angle or conduction length produces ω_L and ω_H. At the beginning of each conduction-angle the controller decides the appropriate state so as to achieve speed regulation.

Conduction angle control is very similar to Hysteresis current control, it is necessary to define a current band as well as θ_1 and θ_2, see Figure 5.3 [31]. Once those parameters are appropriately defined for a given BLDC motor, the motor can be operated at a fixed speed. However, there is a need for speed control; therefore, two states are defined in order to produce ω_L and ω_H, as required by the digital controller. These states are produced by assigning two distinct values to θ_2, namely $\theta_{2,min}$ and $\theta_{2,max}$, while maintaining the current band and θ_1 constant, as shown in Figure 5.3.

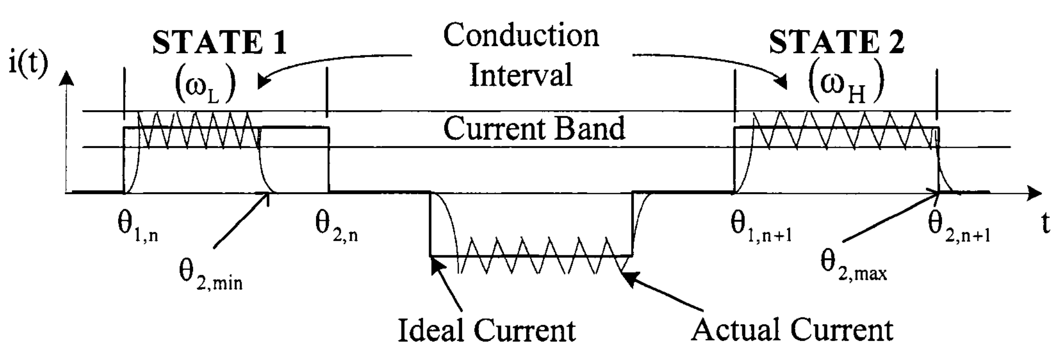


Figure 5.3 Conduction angle control state definitions.

When the motor is operated with $\theta_2 = \theta_{2,min}$, a fixed speed of ω_L results; current is injected into the motor for only part of the conduction interval. When $\theta_2 = \theta_{2,max}$, the fixed speed of ω_H results since current is injected for the entire conduction interval. When a commanded speed (ω^*) is given, where $\omega_L \leq \omega^* \leq \omega_H$, the digital controller will use the rules mention earlier to regulate the motor speed. In order to develop a simple design scheme for the digital controller, θ_1 is always set to the beginning of the conduction interval and $\theta_{2,max}$ is fixed to the ending of the conduction interval.

5.4 Current Mode Control

Current-mode control is also very similar to hysteresis current control. For this control technique, the three hysteresis current control parameters (current band, θ_1, and θ_2) are once again defined, see Figure 5.4. However, θ_1 and θ_2 are held constant while the current band (CB) is given two distinct values, namely CB_{min} and CB_{max} [30]. These values serve to produce the two necessary states in order to carry out the proposed digital control. Figure 5.4 illustrates the two states and as with the conduction-angle control, State 1 produces ω_L while State 2 yields ω_H. When a commanded speed (ω^*) is given, where $\omega_L \leq \omega^* \leq \omega_H$, the digital controller can regulate the motor's speed by following the rules mention earlier.

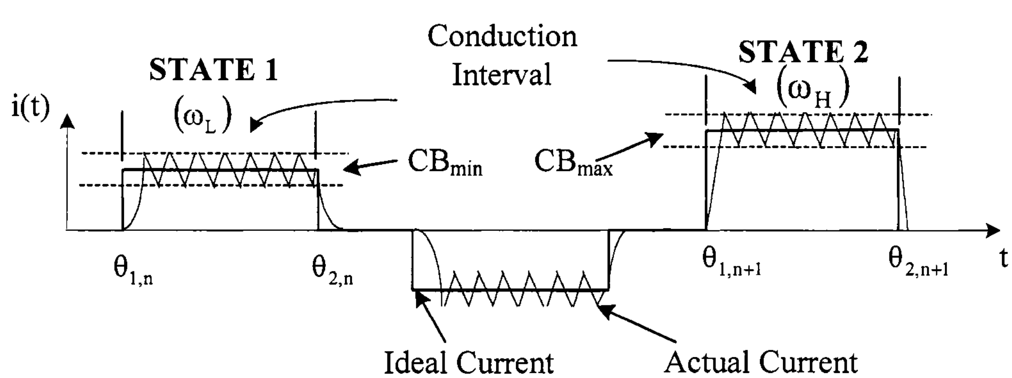


Figure 5.4 Current mode control state definitions.

5.5 Controller Design

The task at hand was to develop an efficient method of designing the digital control for any BLDC motor, so as to attain speed regulation for both implementation methods. Operating conditions of the motor and motor drive system must be defined so as to produce ω_L and ω_H. The first step was to find the equation for angular velocity as a function of the operating parameters under steady state conditions. To do so, Newton's 2nd law applied to rotary motion as shown in Equation 5.1 was used.

$$T_d = \omega(t)b + J\frac{d\omega(t)}{dt} + T_L \tag{5.1}$$

where, T_d = Developed Torque

$\omega(t)$ = rotor speed

b = Viscous Friction Constant

J = Rotor Moment of Inertial

T_L = Load Torque

The BLDC motor will only operates in two predefined States. Within those States and under steady-state conditions, the load torque and the developed torque will be constant. Then, Equation 5.1 is rewritten to form Equation 5.2, which is a first order non-homogeneous differential equation. The solution to the differential equation provides the instantaneous speed, which is a function of motor parameters and load conditions, see Equation 5.3.

$$\frac{d\omega(t)}{dt} + \frac{b}{J}\omega(t) = \frac{T_d - T_L}{J} \tag{5.2}$$

$$\omega(t) = \frac{T_d - T_L}{b} + \left(\omega(0) - \frac{T_d - T_L}{b}\right)e^{-(b/J)t} \tag{5.3}$$

The mechanical time constant (τ_m) is defined by the ratio of the moment of inertia to the viscous friction constant. Under steady-state conditions and with the substitution of the time constant into Equation 5.3 yields Equation 5.4, this is the equation for angular velocity as a function of the operating parameters in steady state.

$$\omega(t \rightarrow \infty) = \omega_{ss} = \frac{T_d - T_L}{J}\tau_m \tag{5.4}$$

Equation 5.4 defines the steady state angular velocity as a function of the motor mechanical parameters. However, rotor speed control is accomplished via applied voltage and the resulting current. The approximate developed torque is proportional to the peak current (I) as defined in Equation 5.5. The instantaneous torque sensitivity values k_{ti}, i= a, b, c are approximated by k_t which is the peak to peak value of k_{ti}. The current peak values are assumed to be constant since operation is in steady state.

$$T_d = k_{ta}i_a + k_{tb}i_b + k_{tc}i_c \approx k_t I \tag{5.5}$$

Equation 5.5 is substituted into Equation 5.4 and solved for the average current. The equation for the current is a function of the desired steady state rotor speed. It can be used to find the necessary current to produce ω_L and ω_H for a given load.

$$I(\omega_{ss}) = \frac{1}{k_t}\left(\frac{J}{\tau_m}\omega_{ss} + T_L\right) \tag{5.6}$$

The established relationship is between the rotor speed and the average current; this is sufficient to implement Current Mode Control since CB_{min} and CB_{max} can now be defined. However, to implement conduction angle control the relationship between the rotor speed and the conduction angle, θ_2, must also be established. The two possible values for θ_2 are $\theta_{2,max}$ and $\theta_{2,min}$. Theta-two max is always set equal to unity, allowing for current conduction during the entire conduction angle interval. Therefore, it is only necessary to find a relationship between the rotor speed and $\theta_{2,min}$. In other words, what $\theta_{2,min}$ value will result in a speed equal to ω_L. To accomplish this, let

$$I(\omega_{ss} = \omega_H) = I_H \tag{5.7}$$

and

$$I(\omega_{ss} = \omega_L) = I_L \,. \tag{5.8}$$

To attain ω_H, the hysterises current band (as defined in Figure 5.3) must have an average value equal to I_H. The peak current band remains the same for both states, but in State 1 the average current over the conduction interval is decreased to I_L. That is accomplished by ending the current conduction early, before the end of the conduction interval as shown in Figure 5.3. The average current in State 1 must be equal to the integral over the partial conduction angle as shown below. A direct calculation can now be used to determine $\theta_{2,min}$, as described by Equation 5.9. The period is normalized to unity so as to express $\theta_{2,min}$ as a ratio of the period.

$$I_L = \frac{1}{T}\left(\int_0^{\theta_{2,min}} I_H d\sigma + \int_{\theta_{2,min}}^{1} 0 d\sigma \right)$$

$$= I_H \Big|_0^{\theta_{2,min}} = I_H \theta_{2,min}$$

$$\theta_{2,min} = \frac{I_L}{I_H} \tag{5.9}$$

In summary, to implement the digital controller to any BLDC motor under a constant load torque, the following procedure should be followed.

1. Find the following motor parameters from the manufacturer's data sheet.

k_t = Torque sensitivity constant

b = Viscous Friction Constant

J = Rotor Moment of Inertia

τ_m = Mechanical time constant

2. Determine the desired operating speed and specify the load torque.
3. Choose ω_H and ω_L to cover the desired speed range.
4. Use Equation 5.6 to determine the values of I_H and I_L, *($I_H \rightarrow CB_{min}$ and $I_L \rightarrow CB_{max}$)*.
5. Use Equation 5.9 to determine the value of $\theta_{2,min}$, *(only necessary for Conduction Angle Control)*.

5.6 Simulations

To carry out simulations, it was necessary to develop a controller model that carried out both digital controller implementation methods. The first step was to define a detailed system-level block diagram which outlined the communication between the controller and the rest of the electric drive system [1]. The controller requires position information from the hall-effect sensors, as well as monitoring of the three phase currents. From that information, the controller must be able to determine the appropriate firing of the three-phase inverter, so as to implement the conduction-angle digital control. The specific tasks of the controller are presented below in detail:

- Determine the conduction interval for all three phases.
- Estimate the conduction interval duration (T_C).
- Maintain the phase currents within the current band during the conduction interval for all three phases.
- Determine when $\theta_{2,min}$ is reached during a conduction interval.

• Choose the appropriate state to apply to the next conduction interval.

The second task, estimate T_C, was accomplished with the use of resetable counters. For each phase, the hall sensors were used to activate the counters at the beginning of the negative conduction interval, see Figure 5.5. The counter stops at the end of that interval and stores the value as an estimate of the positive conduction interval. Once the positive interval ends, the counter is reset and the procedure repeats each period.

The third task was to maintain the phase currents within the current band gap during the conduction intervals. In other words, control the current via Hysteresis current control. For each phase, edge-triggered set reset flip-flops were used along with comparators that established the current band gap, I_{max} and I_{min}.

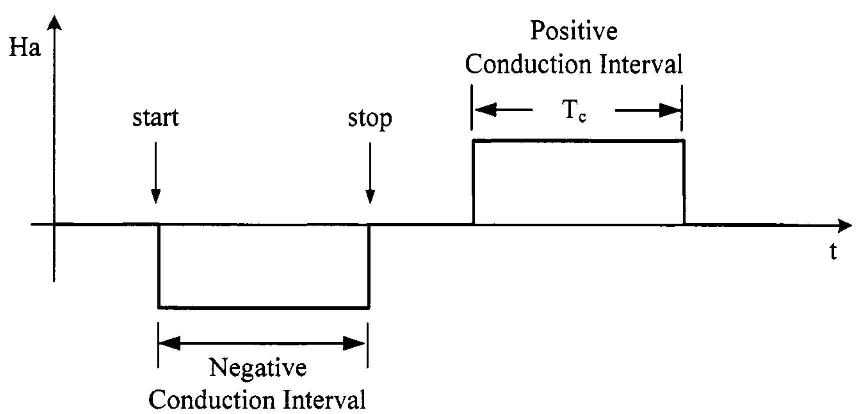


Figure 5.5 Counter stop and start indicators.

The last function that the controller must perform was to choose the appropriate state, given the actual speed of the motor. The motor speed can be derived in two ways. A speed sensor can be applied directly to the rotor or the speed can be calculated by measuring the frequency of the hall-effect sensor outputs. The controller chooses the state at the beginning of each positive conduction angle. That is done for each phase as shown in Figure 5.6.

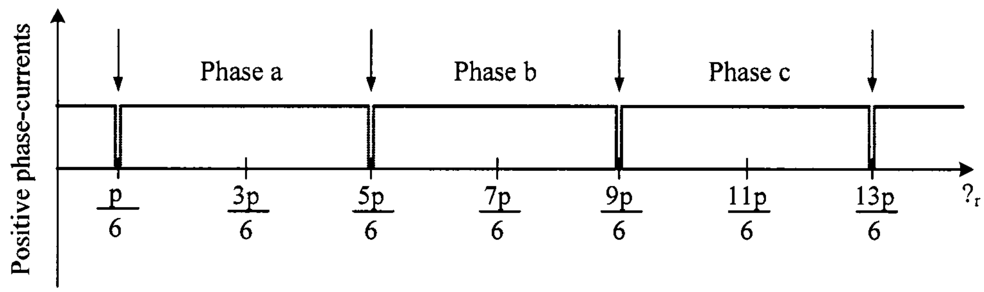


Figure 5.6 Instances of speed comparison for choosing State 1 or State 2.

Simulation models were developed to carry out all the aforementioned tasks. Figure 5.7 illustrates a block diagram of the simulation model created to simulate the proposed digital control methodologies. The BLDC motor model requires the input of various parameters to accurately carry out the simulations. These parameters were taken from an eight hundred and sixty watt BLDC motor, provided by *Poly-Scientific* (model: BN42-53EU-02LH). The BLDC motor was also used for experimental testing, which will be presented in the following section. The BLDC machine has a rated speed of about three thousand revolutions-per-minute, it can continuously source 2.92Nm of torque. The mechanical time constant is roughly half a second (τ_m= 0.516 sec), rotor inertia is J = 4.943 $Kg^{*}cm^2$, and all other motor parameters are found in the manufacturers' data sheet. The simulation goal was to accomplish speed regulation between ω_H and ω_L for a constant-torque load. In order to operate the motor in the predefined states, the current band-gap (CB_{min} and CB_{max}) and $\theta_{2,min}$ were found with the design method outlined in Section 5.5.

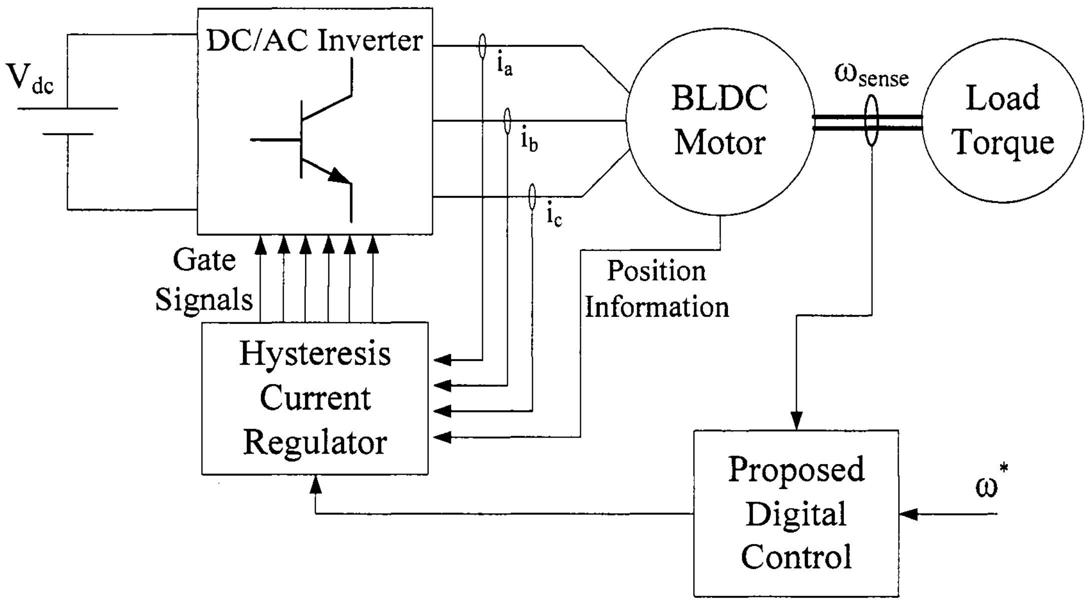


Figure 5.7 Block diagram of proposed digital control implemented in PSIM.

The design was carried out for $\omega_H \approx 1300$ rpm and $\omega_L \approx 800$ rpm. The constant-torque load was set to 0.3Nm. Speed regulation was tested at $\omega^* = 900$ rpm, 1000 rpm, and 1100 rpm. Table 5.1 outlines the operation conditions and designed parameters. Figures 5.8-5.10 and 5.11-5.13 display the motor speed and current waveforms for conduction angle control and current mode control respectively. The commanded speeds tested in the simulation were 900 rpm, 1000 rpm, and 1100 rpm. Table 5.2 and 5.3 summarize the simulation results, specifying the speed ripple associated with each commanded speed.

Table 5.1 Simulation parameters for 0.3 Nm load.

Given	Calculated Values	Desired Speeds
$T_L = 0.3Nm$	$I(\omega_{ss} = \omega_H) = I_H = 2.74A$	$\omega^* = 900$ rpm
$\omega_H = 1300rpm$	$I(\omega_{ss} = \omega_L) = I_L = 2.39A$	$\omega^* = 1000$ rpm
$\omega_L = 800rpm$	$\theta_{2,min} = I_L/I_H = 0.87$	$\omega^* = 1100$ rpm

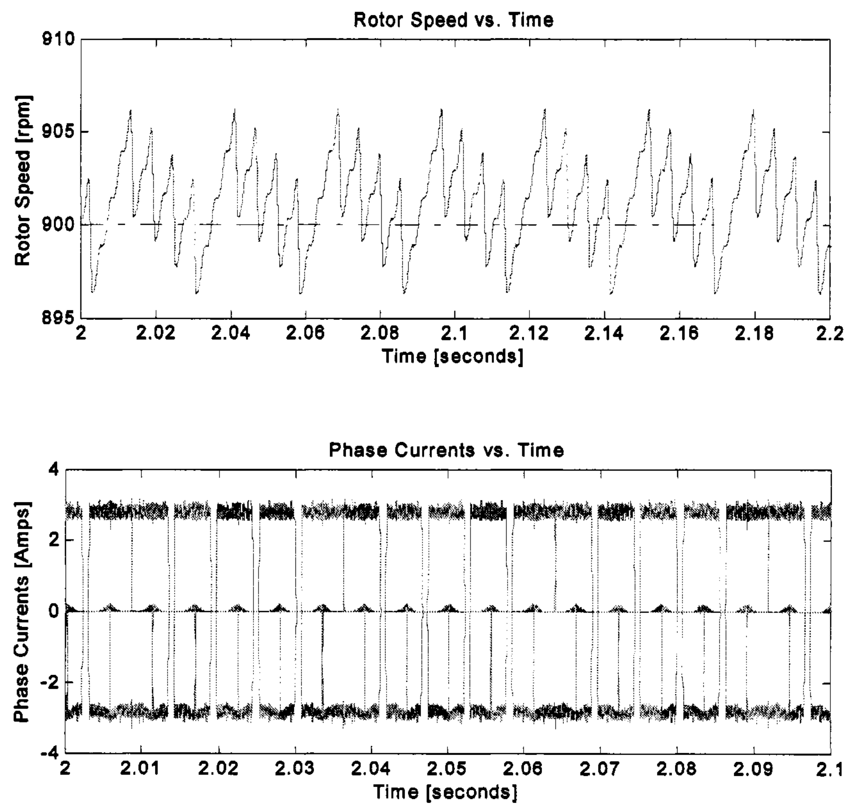


Figure 5.8 Speed and current results for $\omega^* = 900$ rpm with a 0.3Nm load.

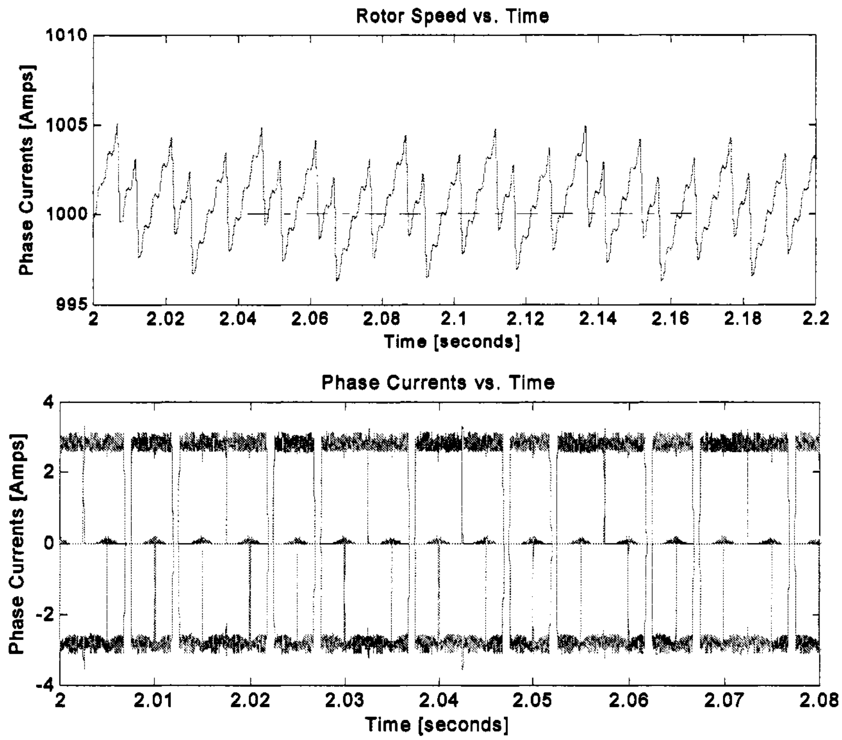


Figure 5.9 Speed and current results for ω* = 1000 rpm with a 0.3Nm load.

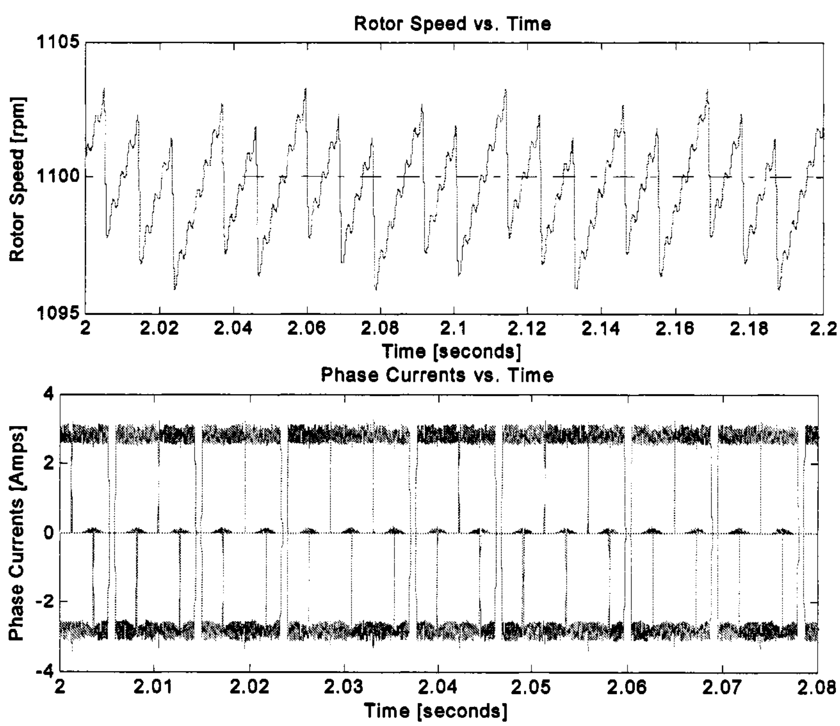


Figure 5.10 Speed and current results for ω* = 1100 rpm with a 0.3Nm load.

Table 5.2 Summary of simulation results for 0.3Nm load (Conduction angle control).

ω^*	$\omega_{average}$	$\omega_{\%Ripple}$
900 rpm	902 rpm	1.0 %
1000 rpm	1001 rpm	0.8 %
1100 rpm	1098 rpm	0.6 %

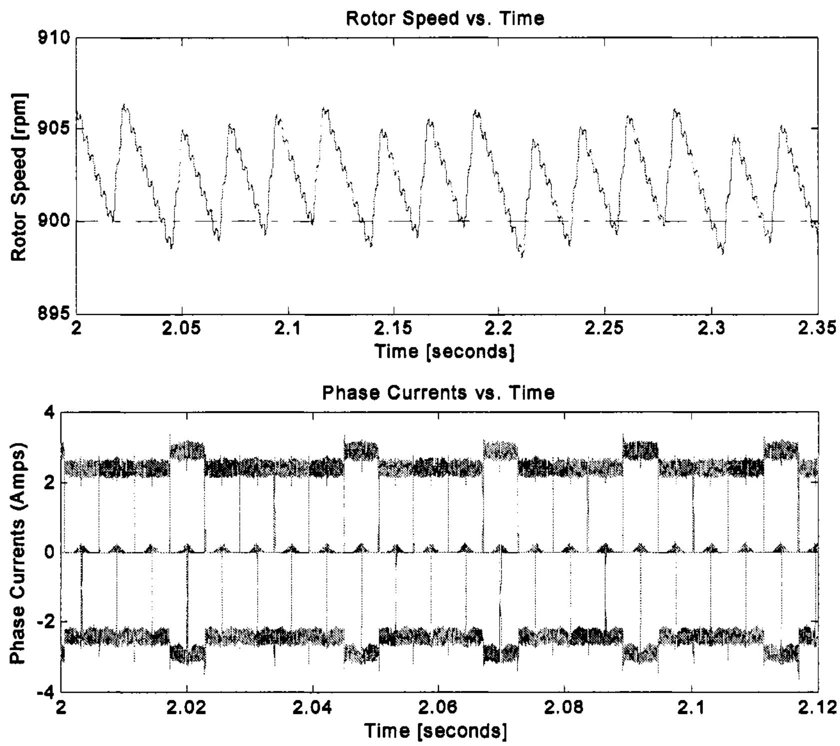


Figure 5.11 Speed and current results for ω^* = 900 rpm with a 0.3Nm load.

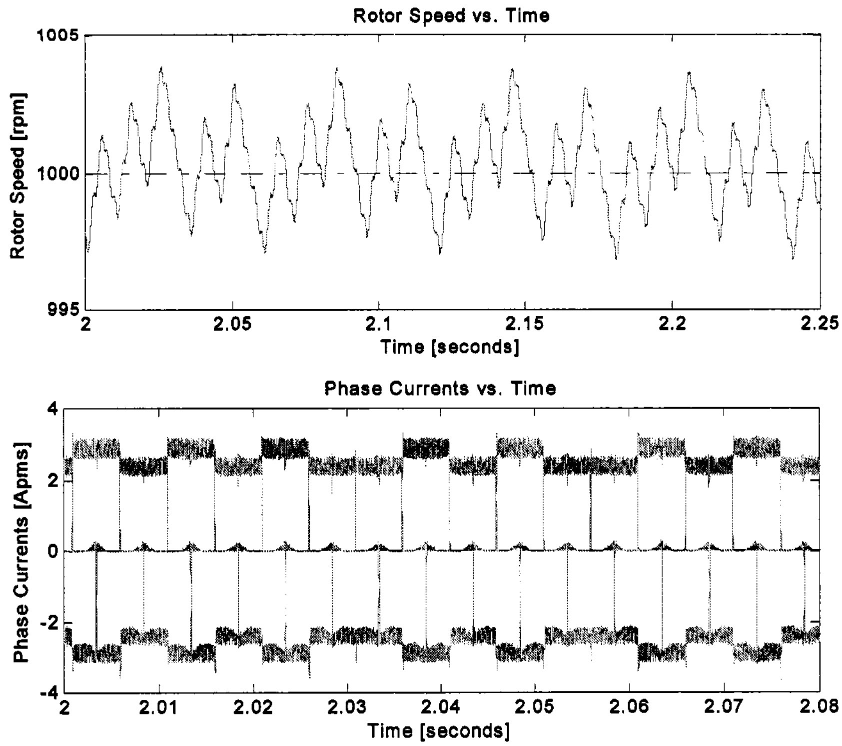


Figure 5.12 Speed and current results for ω* = 1000 rpm with a 0.3Nm load.

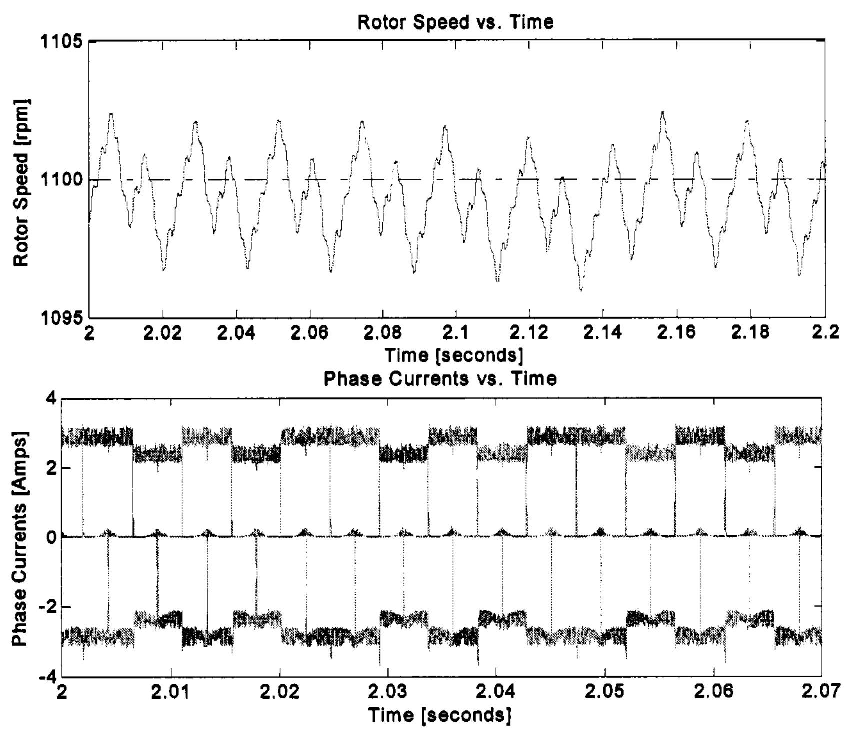


Figure 5.13 Speed and current results for ω* = 1100 rpm with a 0.3Nm load.

Table 5.3 Summary of simulation results for 0.3Nm load (Current mode control).

ω^*	$\omega_{average}$	$\omega_{\%Ripple}$
900 rpm	903 rpm	0.9 %
1000 rpm	1001 rpm	0.7 %
1100 rpm	1099 rpm	0.5 %

5.7 Experimental Results

The digital control design strategy was validated via computer simulation results. However, further proof-of-concept was desired. An experimental set-up was constructed in order implement and further validate the simulation results for the digital control, namely Conduction Angle Control. Conduction angle control displayed a slightly higher speed ripple compared to Current model control in simulations, therefore it was chosen as the method to be confirmed experimentally. All the components of conventional motor drive and the digital controller were individually designed and constructed as described in Chapter 3.

5.7.1 BLDC Motor and Mechanical Load. The BLDC motor used for experimental testing was acquired from Poly-Scientific (model: BN42-53EU-02LH). The three-phase wye connected motor has a trapezoidal back-emf. The ratings and all other motor information were taken from the manufacturer's data sheet. For the mechanical load, constant torque capability was required. Also, it was necessary for the load's moment of inertia to be much smaller than the motor's moment of inertia, since it was ignored during the controller design. Those requirements were met by the characteristics of magnetic brake. Finally, an adjustable motor mount was machined for motor-load coupling.

5.7.2 Three-Phase Inverter and Gate Driver. A standard three-phase dc to ac inverter was used. The switches were conservatively chosen MOSFETs rated for 100 volts and 30 amps, well over the power rating of the BLDC motor. To drive the switches, a high speed power MOSFET and IGBT driver IC was used (IR2133). The recommended connections given by the IC's data sheet were used to interface the inverter to the driver [15].

5.7.3 Current Control and Sensing Hardware. The digital control requires Hysteresis current control for the phase currents during conduction. Hysteresis hardware was constructed with comparators (MC3302) and edge-triggered set reset flip-flops. Dual-input nand-gates (SN7400) were used to construct the flip-flops. To make use of the Hysteresis hardware, monitoring of the instantaneous current is required. For each phase, closed loop current transducers (LA55-b) from *LEM Group* were used. The transducers produce a signal proportional to the current flowing through a wire. A low-pass filter was applied to the signal to reduce the emi effects on the control circuitry.

5.7.4 dSPACE and Simulink Model. The digital controller (conduction angle control) was implemented with the rapid prototyping and real-time interface system, dSPACE. The dSPACE system assists research and development by interfacing powerful engineering software, MATLAB/SIMULINK, to external hardware such as electric motors or power electronic converters [9]. Therefore, the controller was created in the SIMULINK platform. dSPACE was then used to control the operation of the electric drive system via its ControlDesk [8]. The Simulink model that was created to implement the controller, as described in Section 5.6, is shown in Figure 5.14. Once all the individual components were designed and assembled, they were interfaced to form the experimental set-up, see Figure 5.15.

The experimental set-up was used to verify the simulation results from Section 5.6. Testing was done for a constant load of 0.3Nm, (addition results are provided for a 0.1 Nm load in Appendix A). Presentation of the experimental results follows the same format as the simulation results. The operation conditions and designed parameters used were the same as those used for the simulations, see Table 5.4. Figures 5.16 through 5.20 display the motor speed and current waveforms for the commanded speeds of ω_H, 900 rpm, 1000 rpm, 1100 rpm, and ω_L respectively. Table 5.5 summarizes the experimental results, specifying the speed ripple associated with each commanded speed.

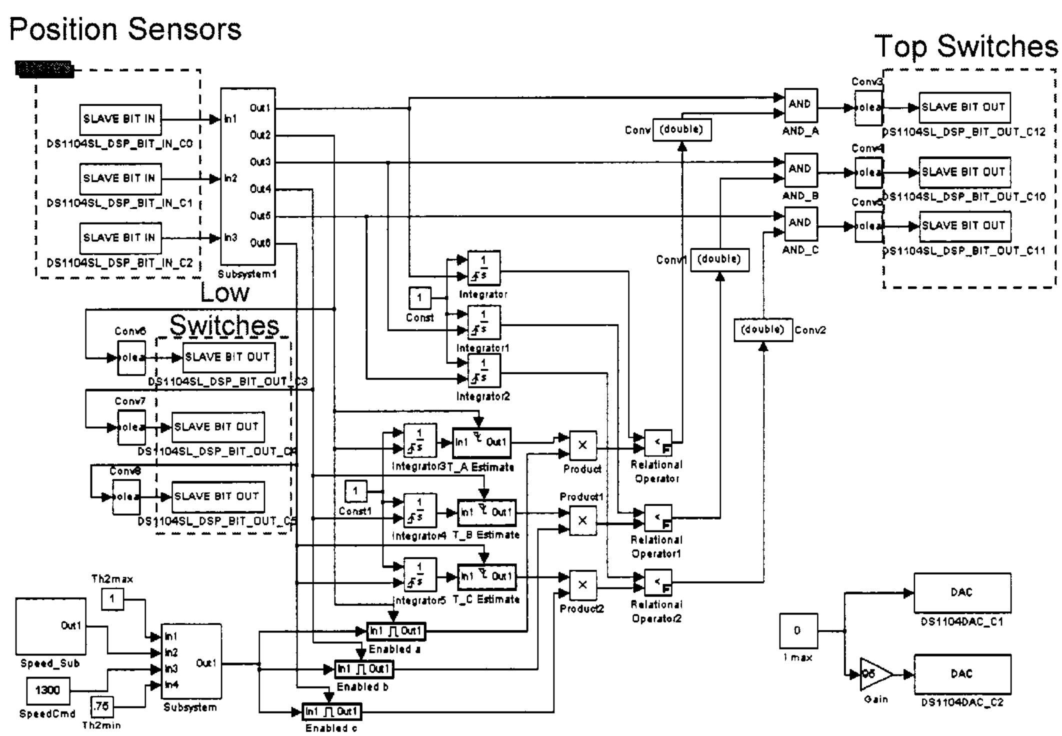


Figure 5.14 Digital control Simulink model.

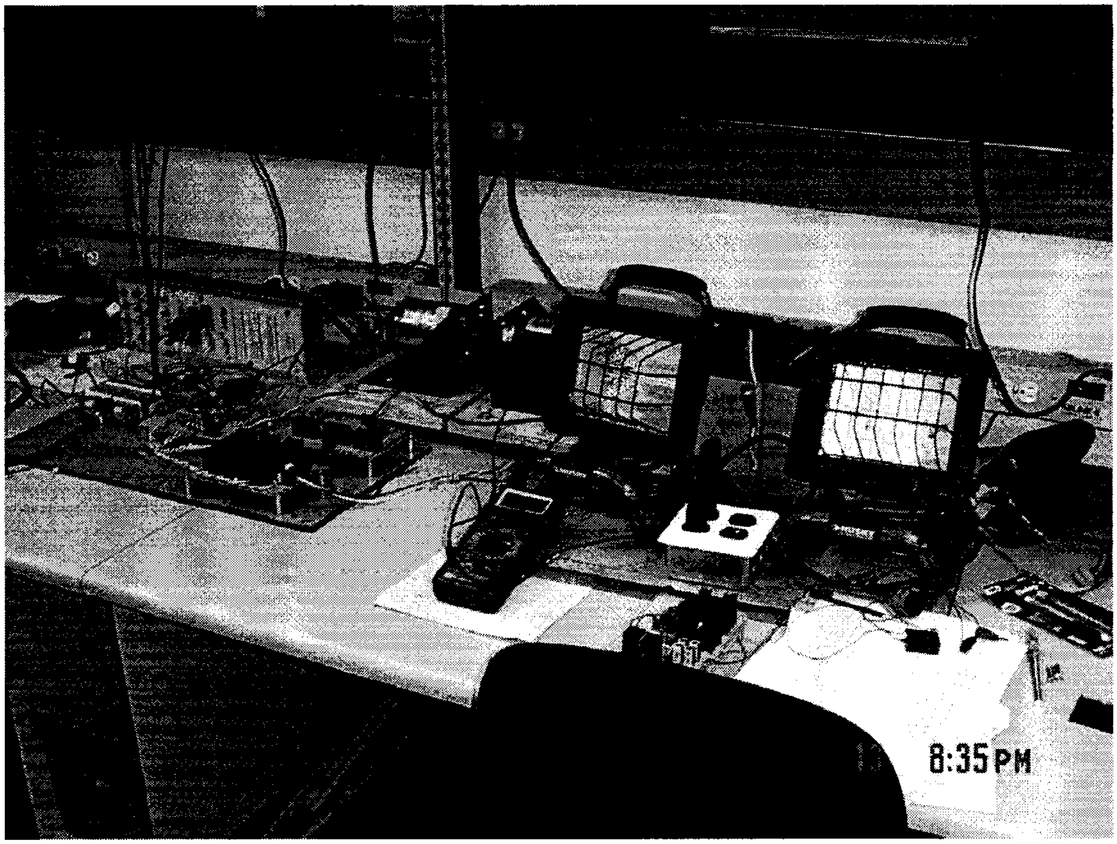


Figure 5.15 Experimental Setup for Conduction Angle Control

Table 5.4 Experimental Parameters for 0.3Nm Load

Given	Calculated Values	Desired Speeds
$T_L = 0.3Nm$	$I(\omega_{ss} = \omega_H) = I_H = 2.74A$	$\omega^* = 900\,rpm$
$\omega_H = 1300\,rpm$	$I(\omega_{ss} = \omega_L) = I_L = 2.39A$	$\omega^* = 1000\,rpm$
$\omega_L = 800\,rpm$	$\theta_{2,\min} = \frac{I_L}{I_H} = 0.87$	$\omega^* = 1100\,rpm$

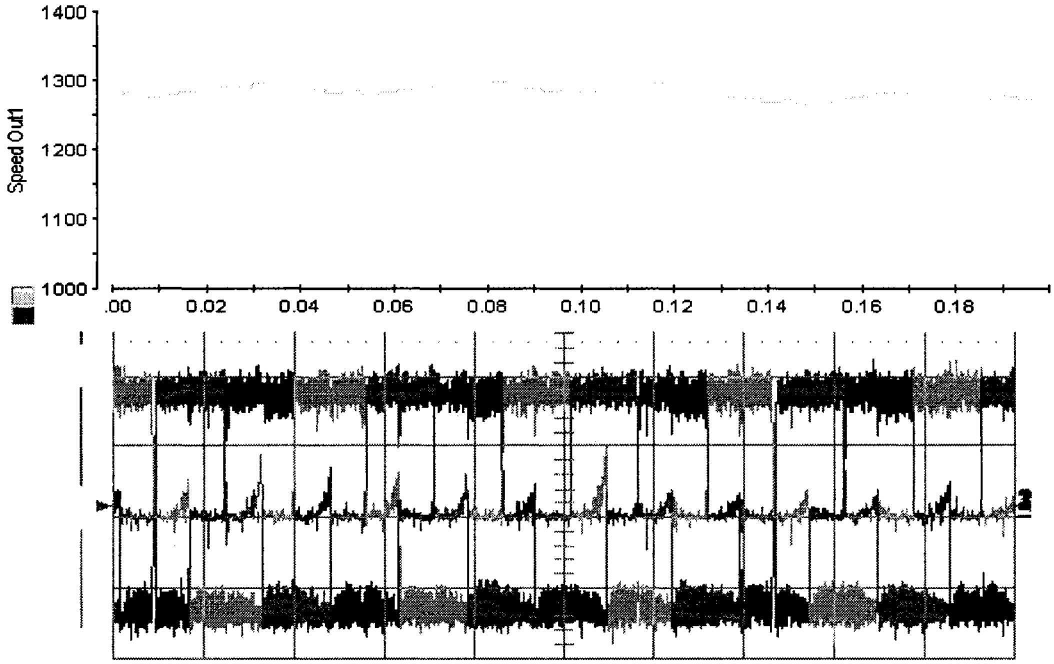


Figure 5.16 Speed & Current Results for ω_H with a 0.3Nm Load

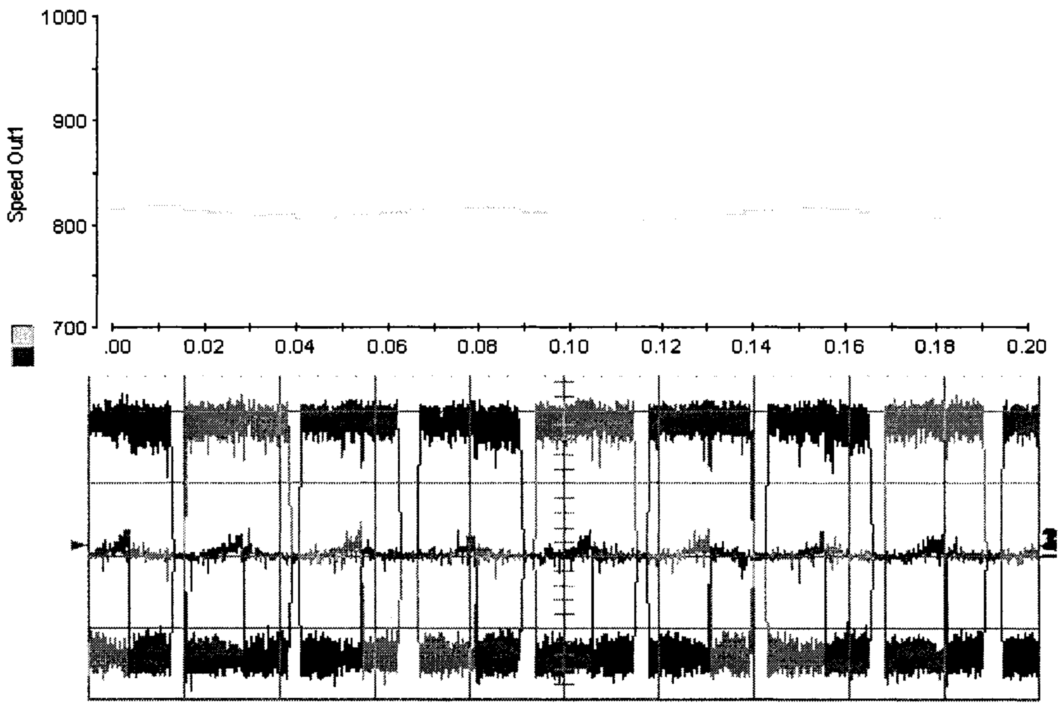


Figure 5.17 Speed & Current Results for ω_L with a 0.3Nm Load

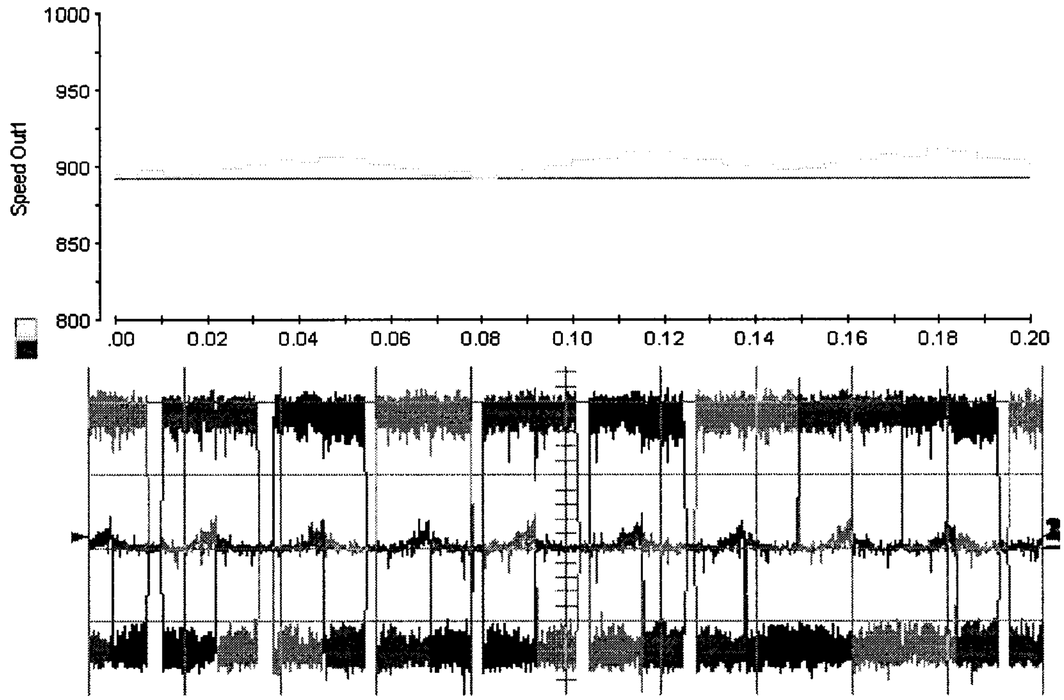


Figure 5.18 Speed & Current Results for ω* = 900 rpm with a 0.3Nm Load

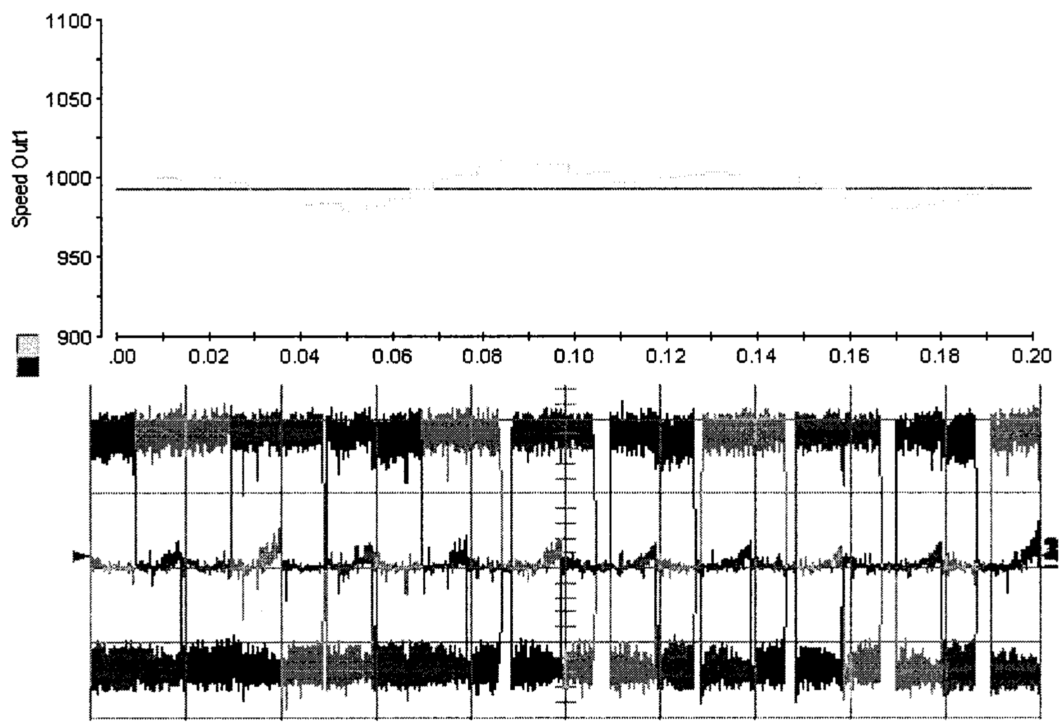


Figure5.19 Speed & Current Results for ω* = 1000 rpm with a 0.3Nm Load

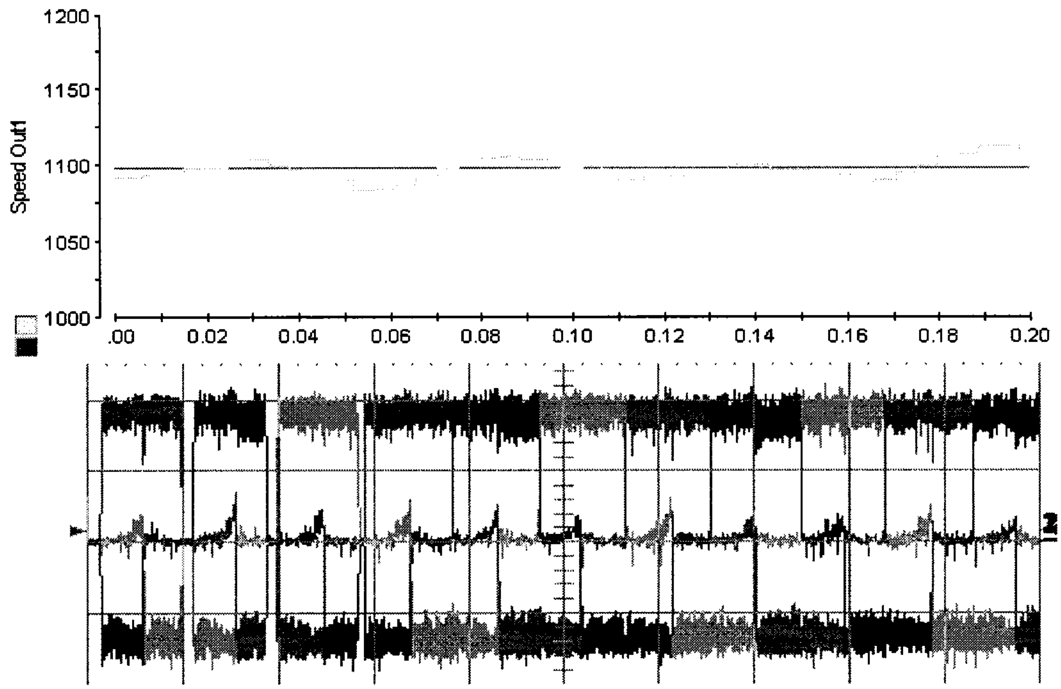


Figure 5.20 Speed & Current Results for ω* = 1100 rpm with a 0.3Nm Load

Table 5.5 Summary of Experimental Results for 0.3Nm Load

ω^*	$\omega_{average}$	$\omega_{\%Ripple}$
900 *rpm*	908 *rpm*	1.9 %
1000 *rpm*	1004 *rpm*	3.4 %
1100 *rpm*	1098 *rpm*	2.3 %

5.8 Conclusions

A new concept for digital control of BLDC motor has been introduced. The idea behind the new concept is to exploit the known fact of electromechanical systems. That fact is that the electrical time constants, in general, are at least an order of magnitude faster than those time constants associated with the mechanical parts. By quickly alternating the produced torque (which is proportional to the current) an average torque is produced resulting in an average speed. The digital controller was implemented via computer simulations in two ways, namely current mode control and conduction angle control. The latter was also verified experimentally.

The conduction angle control simulations and experimental results were in accordance with one another. Speed regulation for all the commanded speeds were well within acceptable limits. The highest speed ripple was recorded at 3.4% of the commanded speed in the experimental results. It is important to note that all the measured speed ripples included the inherent speed ripple associated with trapezoidal BLDC motors. The inherent speed ripple is largely due to the non-ideal trapezoidal back-emf, which does not have perfectly flat plateaus as assumed in the simulations.

The proposed digital control was able to achieve reasonable speed regulation within ω_H and ω_L. However, a general trend was observed while carrying out the simulations. At higher loads, the envelope speeds, ω_H and ω_L, were much more sensitive to the current band-gap values. Despite the problem of accurately attaining the desired envelope speeds at high loads, speed regulation was always achieved within ω_H and ω_L. Therefore, the proposed digital controller, as presented, is well suited for applications where speed ripple is not of significant importance. Such applications include air conditioning units whether they are a/c units or heat blowers to smaller units like ceiling fans.

In conclusion, this chapter has presented the initial investigation and proof-of-concept for a new way of looking at digital control for BLDC motors. Further controller development was carried out in the following chapter. A closed form calculation method is found to control the speed ripple under the entire motor speed range. Also, through the controller development thus far, the load torque has been assumed to be constant and known. However, in most applications the load torque is not constant and unknown. Therefore, Chapter 6 also focuses on enhancing the digital control with disturbance rejection capabilities.

CHAPTER 6

NOVEL DIGITAL CONTROL TECHNIQUE FOR BRUSH-LESS DC MOTOR DRIVES: STEADY STATE AND DYNAMICS

6.1 Introduction

This chapter will further development the new digital control technique for BLDC motor drives that was introduced in Chapter 5. In the previous chapter the controller was derived under the assumption only constant loads were applied to the BLDC machine. Also, the choice of ω_H and ω_L was somewhat arbitrary. The only requirement was for the commanded speed to lie within the high and low speed envelope. The instance at which the digital control chose one state over the other was dependant on the rotor angular velocity. The experimental results proofed that at lower speeds, the speed ripple would increase making well regulated low speed operation nearly impossible. The overall concept for the digital controller is further developed for Current Mode Control due to its continuous current injection of the conduction interval. I brief introduction on BLDC motor drives and leading control methodologies will be covered, followed by the derivation of the design equations for extended speed regulation under any load torque.

Just a few decades ago, high efficient motor drives were exclusive to high power industrial applications. The motor of choice for industries was either a three-phase induction motor or a three-phase synchronous motor. It was necessary to use a three phase motor instead of a single phase motor, since single phase motors have average efficiencies ranging from forty to sixty percent [22]. Three phase power is readily available to industries which helped to encourage the use and production of these three-phase ac motors. In residential buildings only single phase power is usually available.

Most residential motor drive applications are low power and employ single phase ac induction motors. The most common types of single phase induction machines are split-phase, capacitor start, and capacitor-start-and-run, all of which are designed to overcome inherent limitations such as poor starting torque and poor operating power factor [2]. The inefficiency of single phase motors is not a significant burden for individual residential consumers because the cost to power other electrical loads (i.e. lighting, electric heaters) overshadow the extra money spent on powering these inefficient motors. Advancement in motor drive technology along side with power electronics have allowed for high efficiency motor drives to find their way into many types of applications, industrial as well as residential whether it be high power or low power. One such advancement is the introduction of brushless dc (BLDC) machines with trapezoidal back-emf. These machines offer a high torque density, require little maintenance, and have much higher efficiencies compared to single phase induction motors [20]. Compared to induction machines, they have lower inertia allowing for faster dynamic response to reference commands. Also, they are more efficient due to the permanent magnets which results in virtually zero rotor losses [29]. The drawback with BLDC motors is that they require a power electronic converter to drive them, which adds system complexity and requires a sophisticated controller in order to operate.

Many technical papers describe control methods applied to BLDC motor drives. Hysteresis current control and pulse-width-modulation (PWM) control coupled with continuous control theory produce the most widely used BLDC motor control techniques [7]. Hysteresis current control is essential towards achieving adequate servo performance [6], namely instantaneously torque response yielding faster speed response compared to

PWM control. For most applications, proportional (P) or proportional-integral (PI) current and speed compensators are sufficient to establish a well-regulated speed/torque controller. Though great care must be taken in choosing the bandwidth of the motor drive system since too large of a bandwidth can result in an undesired wind up error due to practical limitations such as a finite power source. In other cases, linear control theory via state feedback using state observers and state tracking filters have been used to achieve more precise control of BLDC motors. Classic control theory and linear system theory are well understood, but are highly complex and require extensive control systems knowledge to develop well design controller. Discrete control theory allows for such controllers to be digitally implemented with micro-controllers, microprocessors, or digital signal processors (DSPs). Digitizing analog controllers serves to add complexity to the overall design procedure.

Sliding mode control (SMC) and variable structure systems (VSS) are other control methodologies. The aforementioned control techniques have certain benefits and drawbacks ranging from performance related to implementation cost. Ideally, SMC and VSS are said to exhibit excellent system performance, insensitivity to plant variations, and absolute disturbance rejection. However, in practice both methods exhibit the problem labeled the chattering phenomenon, which is essentially oscillation about an operating point [34]. Also, implementation of SMC techniques have proved costly in terms of DSP implementation due to computationally extensive operations when complemented by adaptive parameter estimation in an effort to reduce the chattering effect [19],[33].

6.2 Current Mode Digital Control

There are several methods of achieving the two states of operation. For example, in [31] and [30] two methods were introduced, namely Conduction Angle Control and Current Mode Control. In this paper, a form of Current Mode Control is employed which results in the actuation signals u_H and u_L taking the form of current signal. As stated in Chapter 5, current mode digital control treats the BLDC motor like a digital system, which may only operate in two predefined states, namely state-1 and state-2. Block diagram representation of the digital controller is illustrated in Figure 6.1. The figure shows the controller driving a BLDC motor under a constant torque load. The current mode digital control concept is best described by incorporating two open loop controllers as shown in Figure 6.1. The first open loop controller creates state-1 which generates the necessary electromagnetic torque to operate the motor at a speed of low omega (ω_L).Operation in state-2, which is attained with the second open loop controller, results in a motor speed of high omega (ω_H).

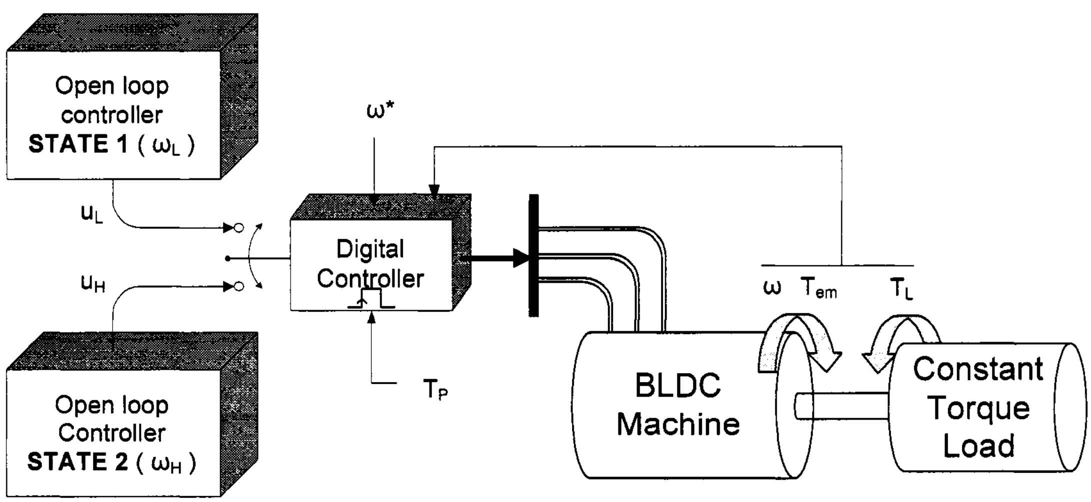


Figure 6.1 Proposed digital control.

Under the constraint that ω_H is greater than ω_L, when the commanded speed is ω^*, where $\omega_L < \omega^* < \omega_H$, the digital controller will achieve speed regulation by appropriately alternating states. In other words, the digital controller compares the motor speed to the commanded speed and determines which actuation signal, (either u_L or u_H) to apply to the BLDC machine.

6.3 Controller Design Derivation

Derivation of a design mythology begins with close examination of Figure 6.1 and explicitly pointing out the task of the digital controller. At every sampling instance, $t=kT_P$, the controller must decide which actuation signal (u_i, i = H or L) to apply to the BLDC machine. Therefore, it is necessary to define what the actuation signal is and how to determine the quantitative values that will produce the appropriate motor speeds. The design methodology should be general so as to be applicable to any BLDC machine. Operating conditions of the motor and motor drive system must be defined for ω_L and ω_H. The first step was to find the equation for angular velocity as a function of the operating parameters under steady state conditions. To do so, Newton's 2nd law applied to rotary motion was used, see Equatuion 6.1.

$$T_{em}(t) = \omega(t)b + J\frac{d\omega(t)}{dt} + T_L(t) \tag{6.1}$$

where, $T_{em}(t)$ = developed electromagnetic torque

$\omega(t)$ = rotor speed

b = viscous friction damping constant

J = rotor inertia

$T_L(t)$ = load torque

The BLDC motor will only operates in two predefined States. Within those states and under steady state conditions, the load torque and the developed torque will be constant, (non-constant loads will be considered in the following section). Even though alternating of states may occur at every sampling time $t=kT_P$, it is reasonable to assume that the developed mechanical torque will be in steady state. That assumption is made based on an inherent characteristic of most electromechanical systems. Electromechanical systems consist of an electrical and a mechanical subsystems whose dynamics are coupled in some manner.

In the case of a BLDC machine, the coupling occurs via the linear relation between the motor's active phase currents and the developed electromagnetic torque. The dynamics associated with the motor's phase current are at least an order of magnitude faster than that of the mechanical dynamics. Therefore, it is safe to assume that alternating states results in instantaneous alternating of developed electromagnetic torque relative to the mechanical dynamics [26]. Given the aforementioned, Equation 6.1 is written to form Equation 6.2, a first order non-homogeneous differential equation. Assuming an average electromagnetic torque and a constant load torque, the solution to the differential equation gives the instantaneous speed as a function of motor parameters and load conditions, see Equation 6.3:

$$\frac{d\omega(t)}{dt} + \frac{b}{J}\omega(t) = \frac{T_{em} - T_L}{J} \tag{6.2}$$

$$\omega(t) = \frac{T_{em} - T_L}{b} + \left(\omega(0) - \frac{T_{em} - T_L}{b}\right)e^{-(b/J)t} \tag{6.3}$$

The mechanical time constant (τ_m) is defined as the ratio of the moment of inertia to the viscous friction constant. Substitution of τ_m into Equation 6.3, along with setting time approaching infinity results in Equatoin 6.4. The steady state motor speed is described as a function of the rotor inertia and the mechanical time constant.

$$\omega(t \rightarrow \infty) = \omega_{\text{steady state}} = \frac{T_{em} - T_L}{J} \tau_m \qquad (6.4)$$

Equation 6.4 defines the steady state angular velocity as a function of the motor mechanical parameters. However, rotor speed control is accomplished via applied voltage and the resulting current. The induced torque is proportional to the instantaneous per phase current as defined in Equation 6.5. The current is assumed to be constant since operation is in steady state.

$$T_{em} = k_{ta}\, i_a + k_{tb}\, i_b + k_{tc}\, i_c \approx K_t\, I \qquad (6.5)$$

where, T_{em} = developed electromagnetic torque

k_{ti} = phase torque sensitivity (i = a, b, c)

i_i = phase currents (i = a, b, c)

K_t = peak value of torque sensitivity

I = approximate steady state current

Equation 6.5 is substituted into Equation 6.4 and solved for the average current. The equation for the current is a function of the desired steady state rotor speed, see Equation 6.6. It can be used to find the necessary current to produce ω_L and ω_H for a given load.

$$I(\omega_{ss}) = \frac{1}{k_t}\left(\frac{J}{\tau_m}\omega_{ss} + T_L\right) \qquad (6.6)$$

Equation 6.6 provides a closed form equation to determine the appropriate operating current values, which are in fact the actuation signals that produce the electromagnetic torque. Those two current are defined by 6.7 and 6.8:

$$I_H = \frac{1}{k_t}\left(\frac{J}{\tau_m}\omega_H + T_L\right) \tag{6.7}$$

$$I_L = \frac{1}{k_t}\left(\frac{J}{\tau_m}\omega_L + T_L\right) \tag{6.8}$$

The required currents are defined in terms of the motor parameters and the load torque. For a given BLDC machine, the required parameters are readily available from the manufacturer's data sheet. Load torque is assumed to be known (a method to obtain any dynamic load torque estimation will be outlined in the following sections). Therefore, what results is a mechanical system which draws parallels to a power electronic buck converter. Figure 6.2 depicts the electromagnetic torque produced by the digital controller. At every sampling instance, the controller will inject current into the motor so as to produce a high torque or a low torque period. Recall that motor operation in State-1 results in a motor speed of ω_L and state-2 produces ω_H. By exploiting the slow mechanical dynamics relative to the electrical dynamics, fast alternating of states results in an average speed that is equal to the command speed. Electromagnetic torque is undergoing hard switching, however like the L-C filter in a buck converter, the rotor and load inertia serve to filter the "noisy" torque into and average torque as depicted in Figure 6.2. Also shown in the figure is the behavior of the rotor speed during the hard switching instances.

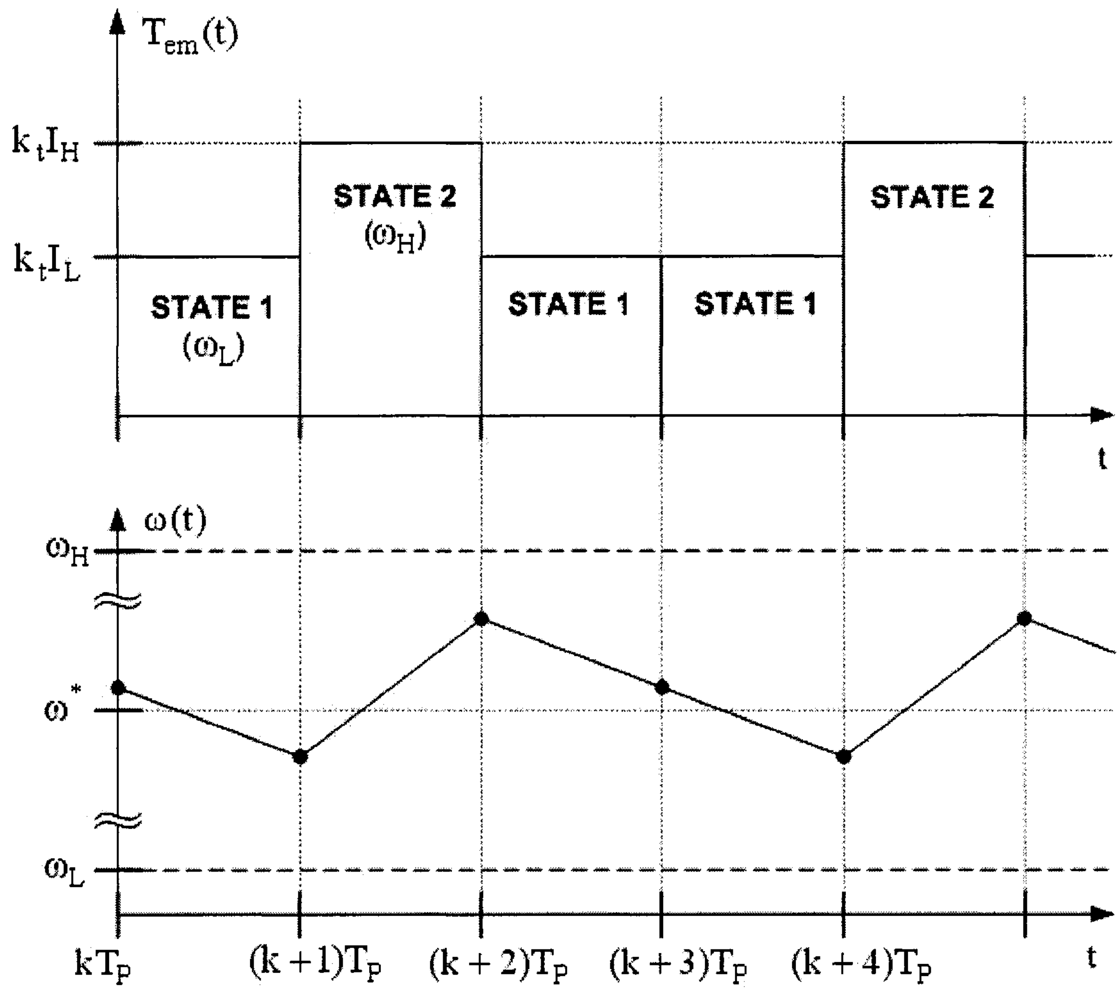


Figure 6.2 Instantaneous torque and speed behavior.

6.4 Sampling Time Calculation

Derivation of the design methodology was performed for an arbitrarily chosen period T_P. Motor speed ripple will largely depend on the choice of the period, therefore a relation between the ripple and the period must be established. To do so, Equation 6.3 is re-written with a couple of substitutions. The difference between the electromagnetic torque and the load torque is defined as the net torque ($T_N = T_{em} - T_L$). Also, the mechanical time constant (τ_m) is incorporated into the Equation 6.3 to form Equation 6.9.

$$\omega(t) = \frac{T_N}{J}\tau_m + \left(\omega(0) - \frac{T_N}{J}\tau_m\right)e^{-t/\tau_m} \tag{6.9}$$

where, T_N = net torque

τ_m= rotor speed

For an arbitrary initial condition, at the time equal to T_P the equation becomes:

$$\omega(t{=}T_P) = \omega(0) + \delta\omega = \frac{T_N}{J}\tau_m + \left(\omega(0) - \frac{T_N}{J}\tau_m\right)e^{-T_P/\tau_m} \tag{6.10}$$

Equation 6.10 is solved for the period by solving for the exponential term, which results in Equation 6.11.

$$e^{-T_P/\tau_m} = \frac{\omega(0) + \delta\omega - \frac{T_N}{J}\tau_m}{\omega(0) - \frac{T_N}{J}\tau_m} \tag{6.11}$$

The natural log of both sides of the equation is taken and the equality is solved for T_P to yield Equation 6.12.

$$T_P = -\tau_m \,\mathrm{Ln}\left(\frac{\omega(0) + \delta\omega - \frac{T_N}{J}\tau_m}{\omega(0) - \frac{T_N}{J}\tau_m}\right) \tag{6.12}$$

Equation 6.12 states that given a desired deviation in speed ($\delta\omega$) for any initial speed condition, the sampling or period can be directly calculated. However, the deviation in speed used to find the period is only for switching from one particular state to the other. There is speed deviation from the desired speed that is contributed from both states.

Therefore, to find the period necessary to limit the overall ripple $\Delta\omega$, speed deviations from both state must be considered.

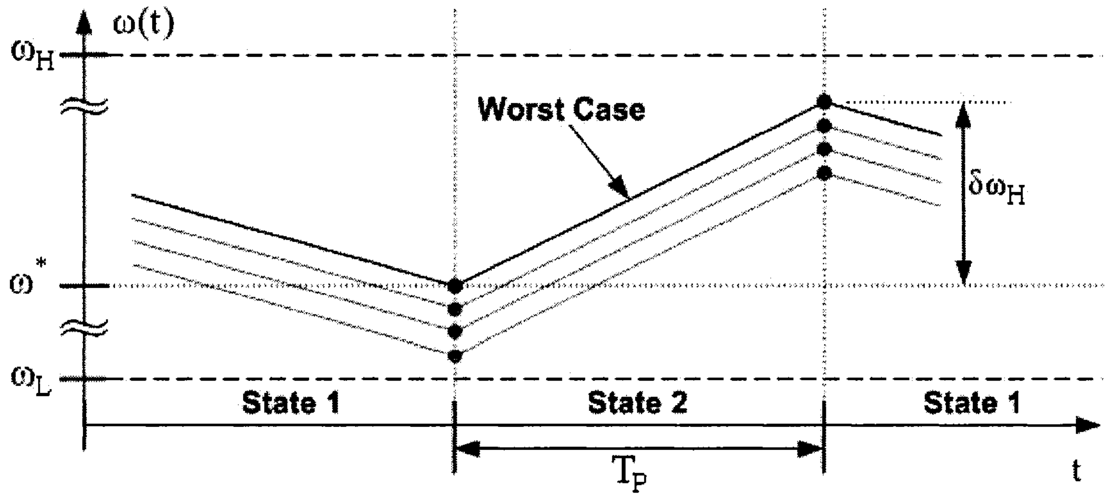


Figure 6.3 Speed ripple contribution from state 2.

Figure 6.3 along with Equation 6.12 are used to determine the necessary period that results in a desired overall speed ripple or change in speed ($\Delta\omega$). The figure reveals several instantaneous speed waveform possibilities. At the beginning of an arbitrary sampling time, when alternating from state-1 to state-2, the instantaneous speed will become less than the commanded speed. The controller will then switch states and the speed will begin to increase towards a fixed net value of $\delta\omega_H$. The largest speed deviation from the commanded speed results when switching of states occurs if $\omega(t) = \omega^* - \varepsilon$, where $\varepsilon \rightarrow 0$. The equation for that condition is given by Equation 6.13. A similar argument can be made to reveal the worst case speed deviation when alternating from State-2 to State-1, see Equation 6.14.

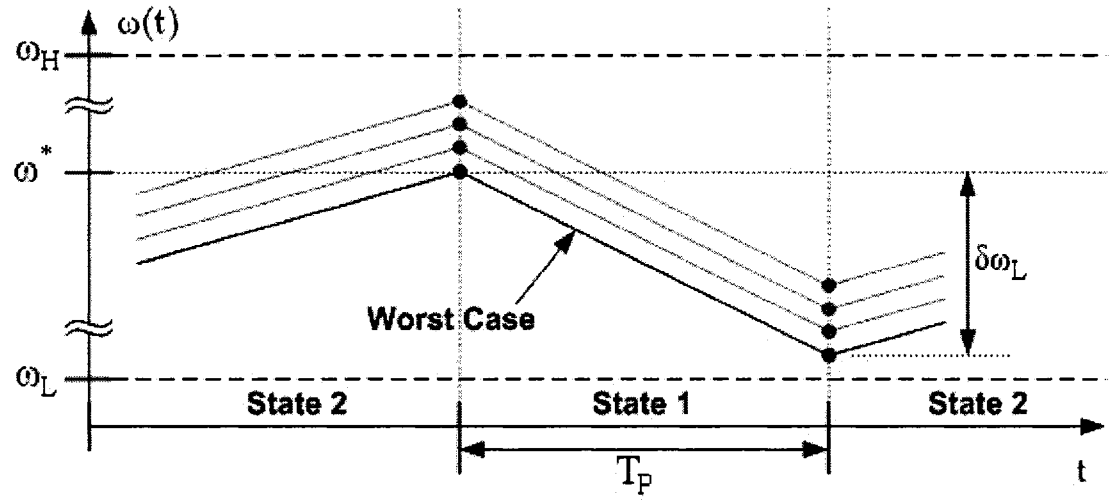


Figure 6.4 Speed ripple contribution from state 1.

Now that the speed deviation contributions resulting from switching have been found, the overall speed ripple is derived. Equation 6.13 is subtracted from Equation 6.14, which results in Equation 6.15.

$$\omega(t{=}T_P) = \omega^* + \delta\omega_H = \frac{K_t I_H - T_L}{J}\,\tau_m\left(1 - e^{-T_P/\tau_m}\right) + \omega^* e^{-T_P/\tau_m} \qquad (6.13)$$

$$\omega(t{=}T_P) = \omega^* - \delta\omega_L = \frac{K_t I_L - T_L}{J}\,\tau_m\left(1 - e^{-T_P/\tau_m}\right) + \omega^* e^{-T_P/\tau_m} \qquad (6.14)$$

where, $\delta\omega_L$ = speed deviation from ω^* during state-1

$\delta\omega_H$ = speed deviation from ω^* during state-2

$$\Delta\omega = \omega^* + \delta\omega_H - \omega^* + \delta\omega_L = \left(\frac{K_t I_H - T_L}{J}\,\tau_m - \frac{K_t I_L - T_L}{J}\,\tau_m\right)$$

$$\cdot\left(1 - e^{-T_P/\tau_m}\right) + \omega^* e^{-T_P/\tau_m} - \omega^* e^{-T_P/\tau_m} \qquad (6.15)$$

Simplification of Equation 6.15 yields the overall ripple as a function of the period (T_P), see Equation 6.16.

$$\Delta\omega = \left(\omega_H - \omega_L\right)\left(1 - e^{-T_P/\tau_m}\right) \tag{6.16}$$

If the period is forced towards zero ($T_P \rightarrow 0$) resulting in infinitely fast sampling, the speed ripple is reduced to zero as indicated by Equation 6.17.

$$\Delta\omega = \left(\omega_H - \omega_L\right)\left(1 - e^{-0/\tau_m}\right) = 0 \tag{6.17}$$

On the other hand, when the period is forced towards infinity ($T_P \rightarrow \infty$), the speed ripple reaches the maximum value, see Equation 6.18.

$$\Delta\omega = \left(\omega_H - \omega_L\right)\left(1 - e^{-\infty/\tau_m}\right) = \omega_H - \omega_L \tag{6.18}$$

Finally, Equation 6.16 is solved for the period to give a simplified relation to the speed ripple, see Equation 6.19. A relation between the overall speed ripple and the period has been established.

$$T_P = -\tau_m \operatorname{Ln}\left(1 - \frac{\Delta\omega}{\omega_H - \omega_L}\right), \ \Delta\omega < \omega_H - \omega_L \tag{6.19}$$

6.5 Disturbance Rejection

Derivation of the control parameters was carried out under the assumption that the load torque was constant and therefore under steady state operation. This section will demonstrate that even under a dynamic load, the proposed controller can regulate speed with the use of a state observer. The controller design procedure assumes that the load torque is known, however that is usually not the case. To resolve this issue, a state observer is employed to calculate an estimate of what the load torque is in an automated way. State observers are commonly used to provide state feedback information in cases where sensors such as a position sensor or a velocity sensor might be too costly or impractical for a particular application. To improve the accuracy of the observer, command feedforward (CFF) is used. Using CFF ensures a minimum phase lag between the actual states and the observed states [14]. Figure 6.5 provides a block diagram of the state observer used to calculate the load torque. The CFF signal is the estimated electromagnetic torque which is calculated as the sum of the phase currents multiplied by their respective torque sensitivity estimates.

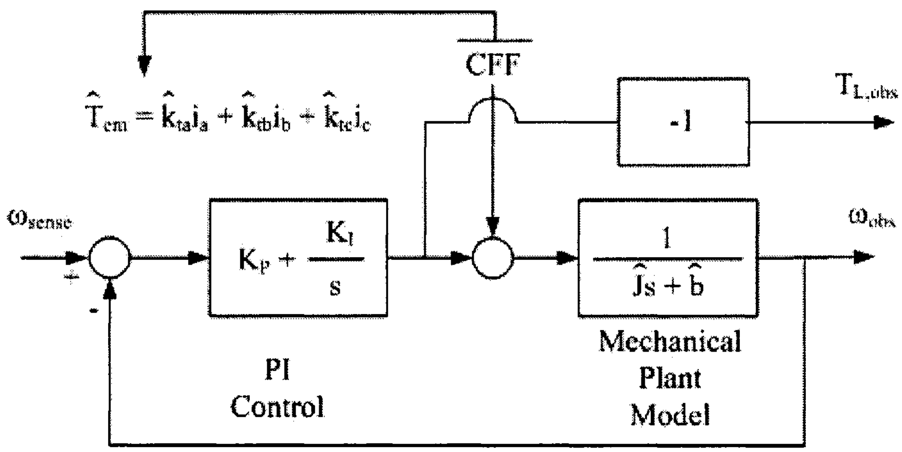


Figure 6.5 Observer block diagram with command feedforward.

The observer serves to mimic the mechanical plant in order to estimate the states of the system from the sensed state or system output [21],[5], in this case the sensed state is shaft speed. When the PI controller is properly tuned and the motor inertia and viscous friction constant are accurately modeled, the output of the PI must be equal to the negative of the load torque. Therefore, the observed load torque can be extracted from the observer as depicted in Figure 6.5. The first step towards tuning the PI controller gains is to chose the PI time constant, $\tau_{PI} = K_P/K_I$, so that it is equal to the estimated mechanical time constant $\tau_m = J/b$. This pole zero cancellation will be exact since the observer is realized as software in the absence of noise. Given the pole zero cancellation, a single pole at the origin remains with one degree of freedom in choosing the closed loop pole location through the choice of the integral gain K_I. The design equations resulting from the described tuning methodology are given in Equation 6.20 and Equation 6.21, where $f_{b\omega}$ is the desired frequency bandwidth.

$$K_P = 2\,\pi\, f_{b\omega}\, J \tag{6.20}$$

$$K_I = 2\,\pi\, f_{\,b\omega}\, b \tag{6.21}$$

6.6 Computer Simulations

To carry out simulations, it was necessary to develop a MATLAB/Simulink model that implemented the proposed controller and observer combinatoin. The first step was to define a detailed system-level block diagram which outlined the communication between the controller and the rest of the electric drive system [1], as depicted by Figure 6.6. Then, an ideal motor of a three-phase four pole BLDC motor was created as defined in Chapter 3. Last, the controller and auxiliary blocks were created with the following characteristics. The controller requires speed feedback as well as the observed torque extracted from the state observer. The control then produced the actuation signal, either I_H or I_L, in order to produce the appropriate operating state. The hysteresis current regulator takes the actuation signal along with rotor position information to fire the solid state switches accordingly.

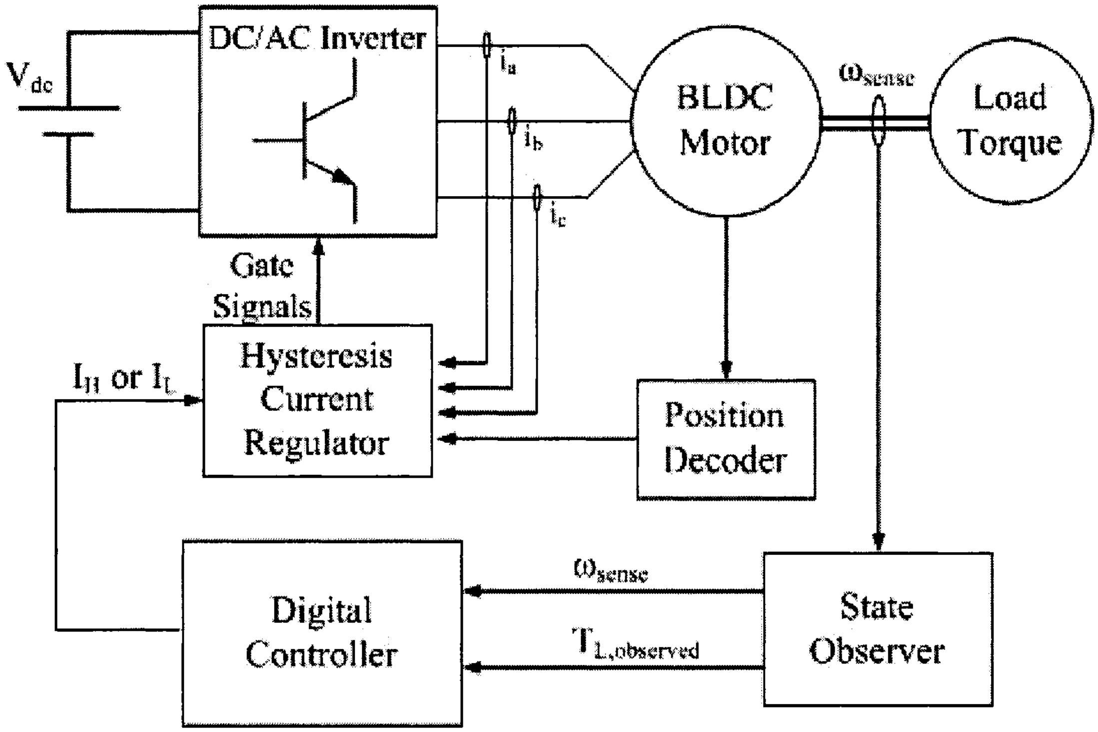


Figure 6.6 BLDC motor drive Simulink block diagram.

The developed Simulink model was used to test the performance of a BLDC machine. The BLDC motor modeled was a 459 Watt motor from *Polyscientific* whose parameters are readily available in the manufacturer's motor datasheet [25] and listed in Table 6.1.

With the motor parameters in hand, the derived design equations were used to calculate the controller parameters. Equation 6.19 was used to calculate the period for the desired speed ripple. Also, Equation 6.7 and Equation 6.8 were embedded in the Simulink block "Digital Control" in order to calculate the high and low currents with load torque information taken from state observer. The high and low speeds for the simulations were chosen as ω_H = rated speed and ω_L= zero speed. The speed ripple was desired to be minimal, therefore the period was calculated at eight hundred hertz using

Equation 6.19 so as to result in a speed ripple less than twenty revolutions per minute. The last design step was to tune the state observer with Equation 6.20 and Equation 6.21.

Table 6.1 Data Sheet for BLDC Motor from Poly-Scientific (BN34-45EU-01)

Parameter	Units	Value
Terminal Voltage	volts DC	24
Rated Speed	rpm	3300
Rated Torque	Nm	1.33
Rated Current	Amps	20.2
Rated Power	Watts	459
Torque Sensitivity	Nm/amp	0.0653
Back-emf	volts/krpm	6.83
Terminal Resistance	ohms	0.069
Terminal Inductance	mH	0.200
Motor Constant	Nm/sq.rt.watt	0.275
Rotor Inertia	g-cm^2	1271
Number of Poles		4

The bandwidth chosen for the observer was that of 10kHz. At first glance this bandwidth seems to be an unrealistic range. However since the observer is realized completely in software and immune to noise, the only limiting factor for the bandwidth of this observer is the resolution of the DSP realizing the observer. The proportional and integral gains calculated to achieve a 10kHz bandwidth are $K_I = 29.04$ and $K_P = 7.99$.

The calculated design parameters were incorporated into the SimuLink model and a time varying load was applied to test how well the controller regulates speed in the presence of disturbances. The applied load is defined by Equation 6.22 and consists of a sinusoidal load disturbance with the addition of a steep ramp increase shortly after thirty milliseconds, see Figure 6.7. A reference speed of 1000rpm was given and the resulting phase current and speed curves are shown in Figure's 6.8 and 6.9 respectively.

$$T_L = \begin{cases} 0 & ,t < 0.02 \\ 0.375 + 0.375\sin(\omega(t-0.02)-\pi/2) & ,t < 0.02 \\ 0 & ,t < 0.032 \\ + \quad 750(t-03032) & ,0.0320 < t < 0.0327 \\ 0.5 & t > 0.0327 \end{cases} \tag{6.22}$$

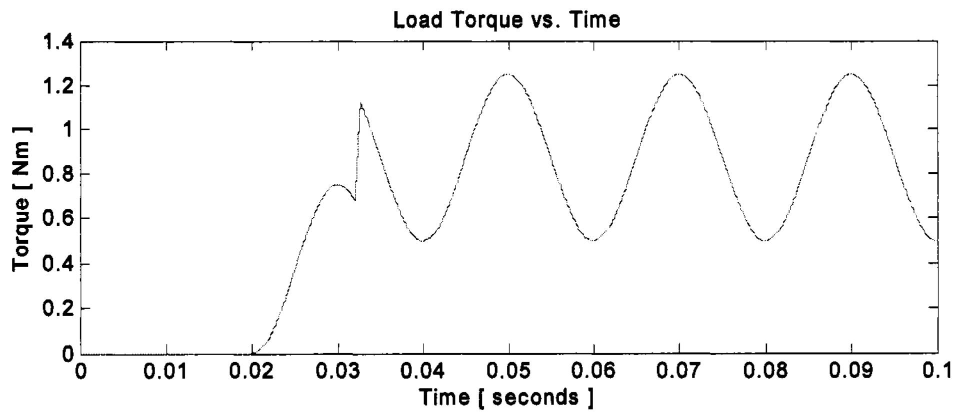


Figure 6.7 Load torque profile applied during simulations.

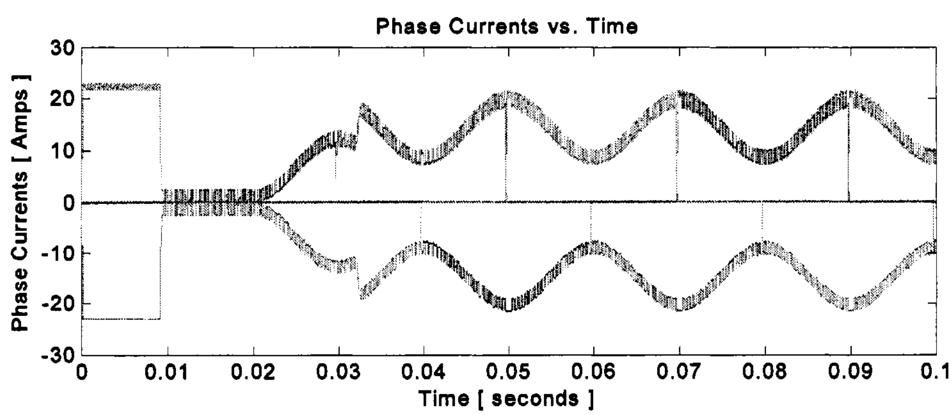


Figure 6.8 Resulting motor phase currents due to applied torque.

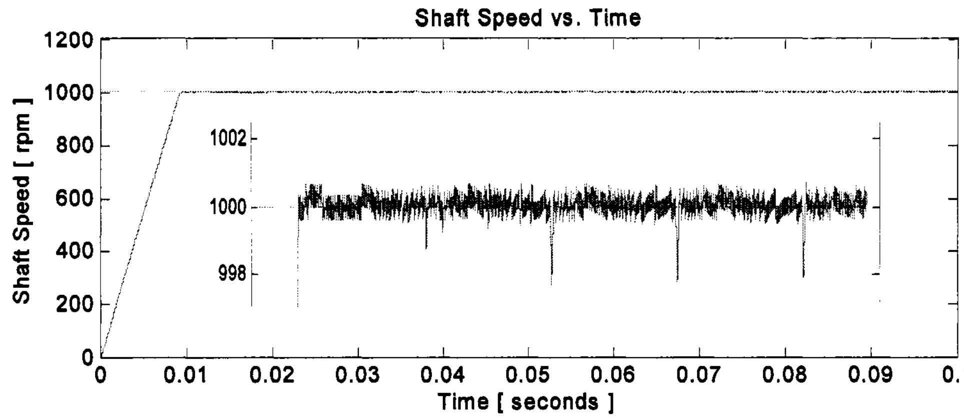


Figure 6.9 Simulated rotor speed with proposed digital control.

It is evident in Figure 6.9 that the digital control has been developed to achieve precise speed regulation under dynamic load disturbances. The high and low actuating signals (currents) are continuously updated with the load estimation provided by the state estimator, which allows for the precise speed regulation.

6.7 Experimental Results

The introduction of a closed from calculation for the digital controller sampling period was validated via the computer simulations. Also shown, was that the observer does in fact provide the disturbance rejection quality when the load torque is unknown. Further validation is sought through the use of the experimental setup descried in Chapter 3. The experimental setup load was changed from a constant load torque to a linear load. Power resistors were applied to the terminals of a dc generator as depicted in Figure 6.10. The same motor that was modeled was then used to validate the digital control introduced in this thesis.

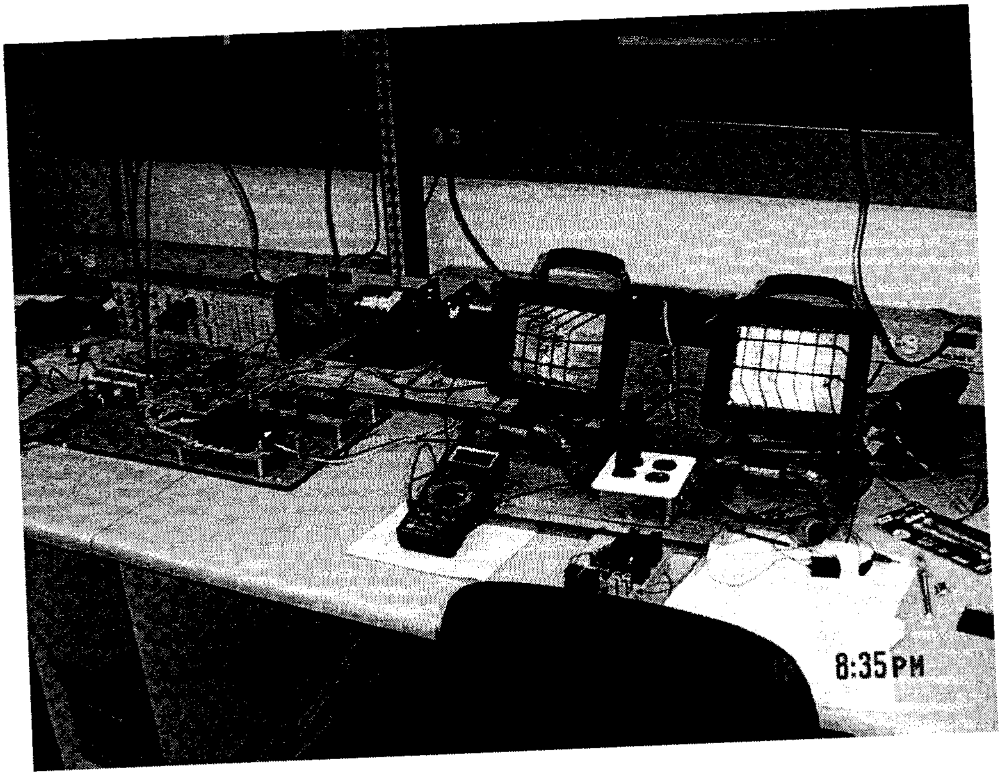


Figure 6.10 Experimental Setup with Linear Load.

The experimental set-up was used to verify the simulation results from Section 6.6. Testing was done for two unknown load conditions: two 500W and a single light utilized as loads. The load changes with speed, therefore the observer must provide accurate load information in order to achieve speed regulation between zero speed and rated speed. Figure 6.10 illustrates the first set of experimental results. Three waveforms were captured during a ten second duration. At approximately each second an increase in speed was initialized in increments of 300rpm so as to cover the entire speed range of zero to rated speed. It is evident that the motor responds to the speed command changes and is well regulated as shown in "channel 2" of the capture oscilloscope image in Figure 6.11. Also depicted in the figure is the load applied to the motor, which is changing with speed. The top portion of the figure displays the phase currents of the ten second interval.

The current envelop shows the increase in motor current needed to increase speed and to overcome that added load. Figure 6.12 shows the commanded speed and actual speed as captured by the data acquisition feature in dSpace. Appendix B provides zoomed in views of the speed regulation during each step change in speed. The speed deviation during the step changes is summarized by Figure 6.12. As expected from design equations, choosing a sampling frequency of 800Hz resulted in an average speed ripple close to twenty revolutions per minute.

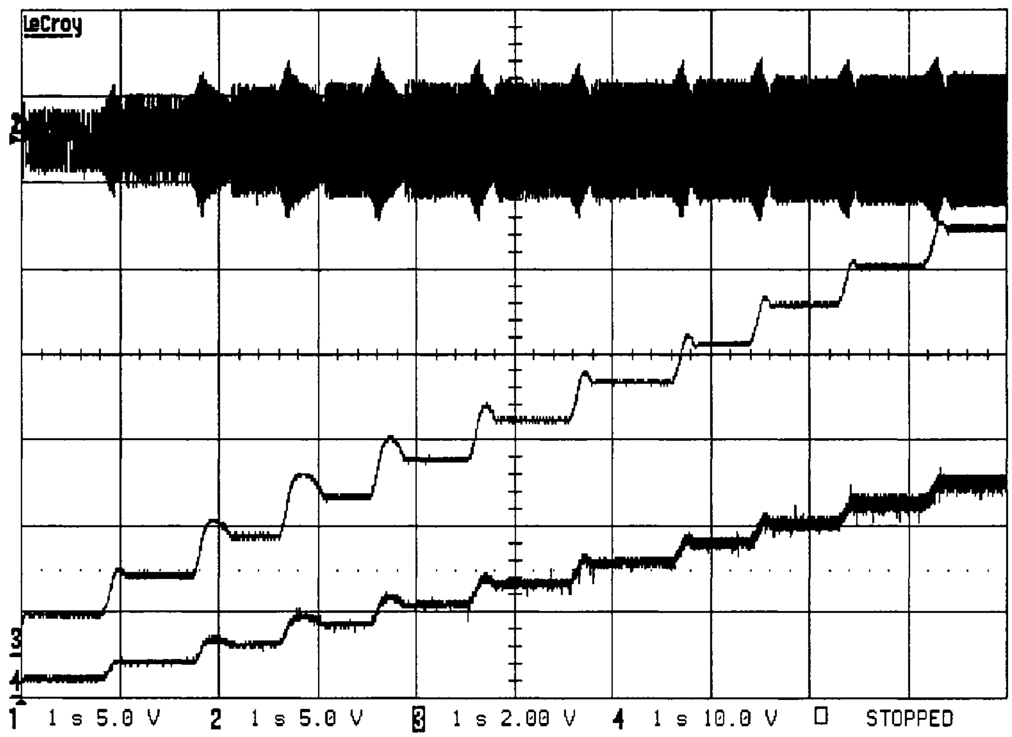


Figure 6.11 Speed regulation zero speed to rated speed with linear load.

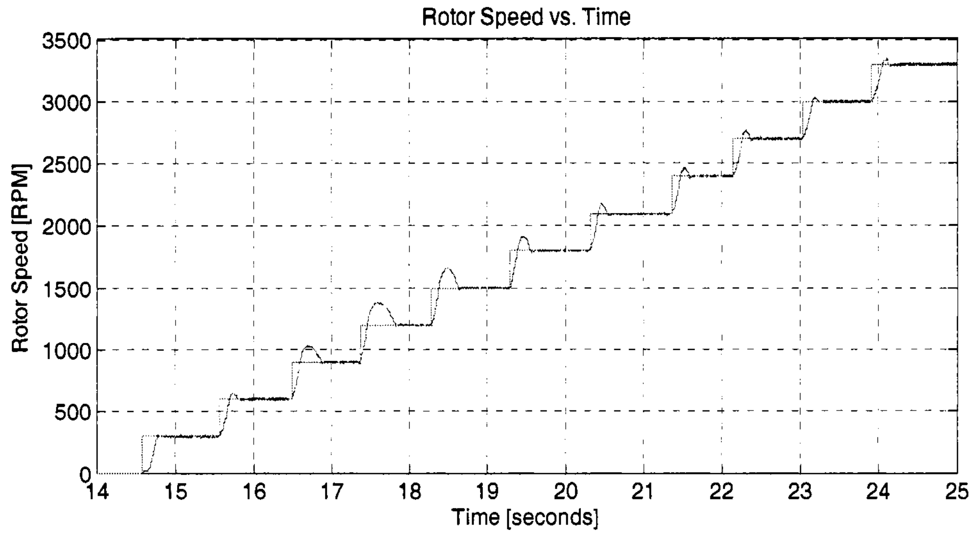


Figure 6.12 Speed and commanded speed vs. time.

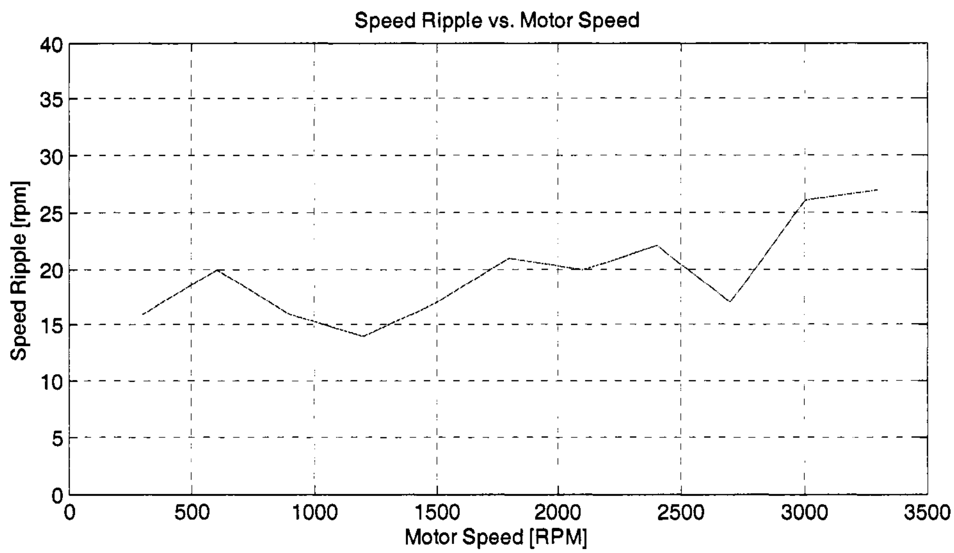


Figure 6.12 Speed ripple vs. motor speed.

A second set of simulations was conducted with have the linear load. The results are presented in the same manner with detailed speed regulation plots provided in Appendix C. In this case, it is shown that speed regulation is once again achieved under an unknown non-constant load. Figure 6.15 illustrates the speed ripple through the operating speed range. The speed ripple is roughly at twenty eight revolutions per minutes, slightly higher than the designed value but well within acceptable limits.

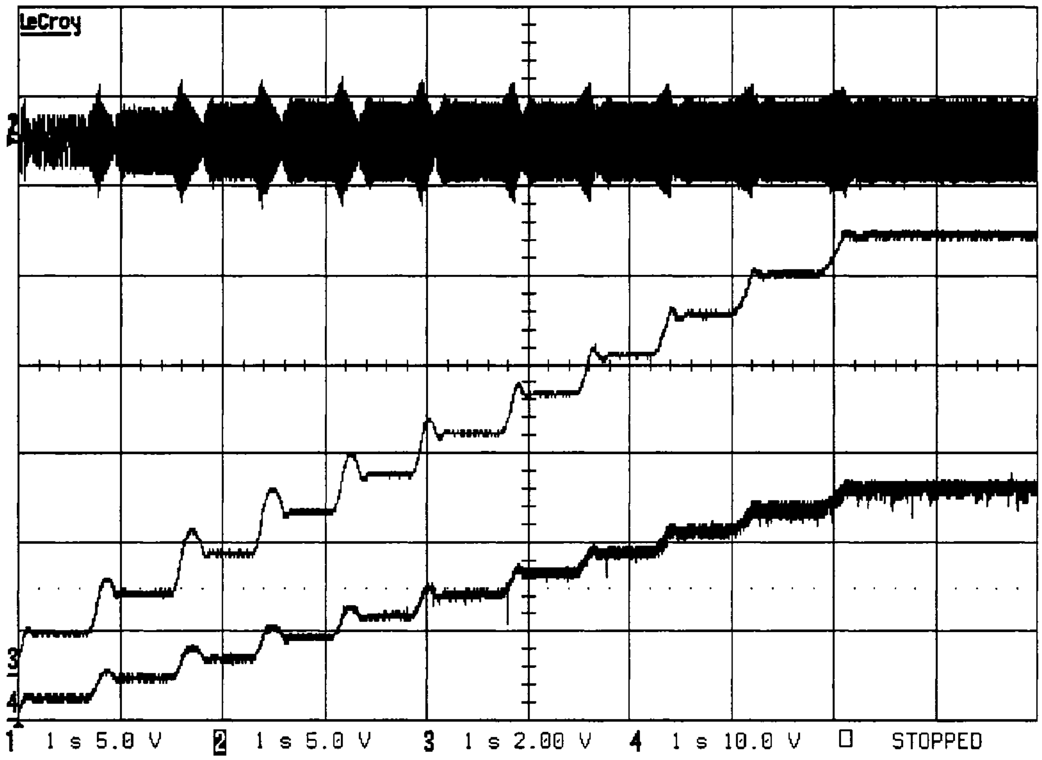


Figure 6.13 Speed regulation zero speed to rated speed with linear load.

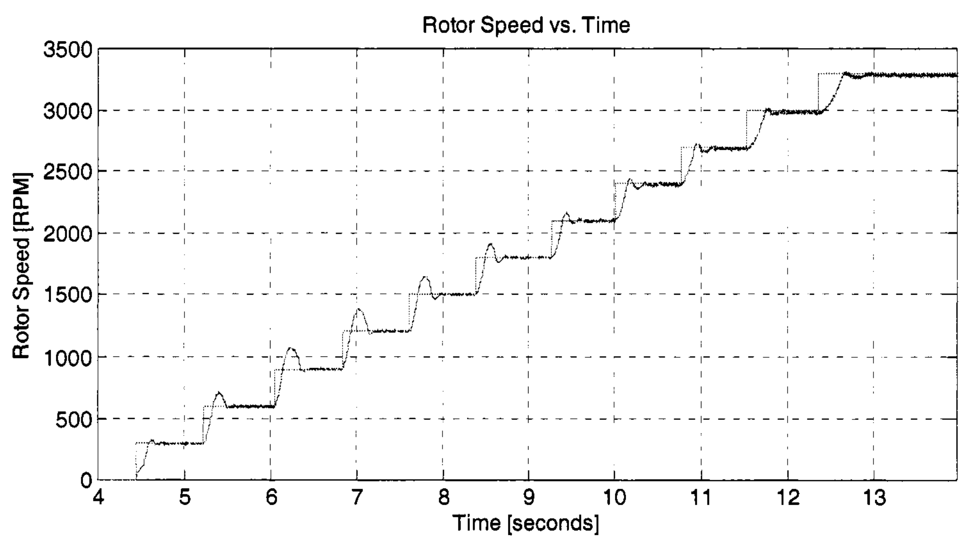


Figure 6.14 Speed and commanded speed vs. time.

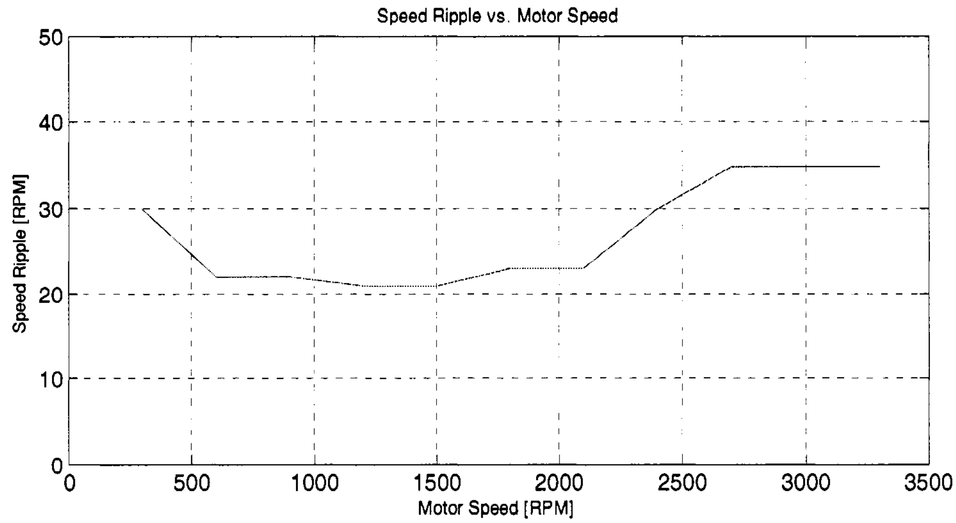


Figure 6.15 Speed ripple vs. motor speed.

6.8 Conclusions

Disturbance rejection under a fixed sample period was outlined for the digital controller. The idea behind the new concept was to exploit the known fact of electromechanical systems. That fact is that the electrical time constants, in general, are at least an order of magnitude faster than the time constants associated with the mechanical dynamics. By quickly alternating the electromagnetic torque (which is proportional to the current), an average torque is produced resulting in an average speed. The digital controller was implemented in the MATLAB/Simulink environment. Derived control design equations were used to calculate the sampling period (T_P) required to minimize the overall speed ripple to a desired maximum value. Also, the equations allowed for the real time computation of the high and low actuation signals. The design procedure for the state observer was outlined so as to achieve any desired bandwidth for the observer.

In conclusion, the design procedure was created in terms of motor parameters so as to be applicable to any BLDC motor. The simulation results determined that the proposed control accurately tracked the reference speed command despite a time varying load disturbance, which is in fact the objective of any high performance BLDC controller.

CHAPTER 7

CONCLUSION

7.1 Summary of Results

A review of brushless dc (BLDC) motor drives was presented in the second chapter. Basic BLDC machine operating principles were outlined in detail which allowed for the development of a precise simulation model. Also, a experimental test setup was constructed so as to validate simulations carried out in Chapters 5 and 6. A review of classic control theory applied to motor drives was presented followed by the introduction of the novel digital control technique.

The new concept was introduced and implement in two ways, namely Conduction Angle Control and Current Mode Control. Design equations and procedures were developed for both implementation methods. In Chapter 5, Conduction angle control was simulated and validated experimentally for known constant loads. Chapter 6 focused on Current Mode Control subject to unknown non-constant loads. It was shown that for both methods, the controller design equations accurately defined the control parameters to regulate rotor angular velocity.

7.2 Conclusions and Future Work

The new control concept was introduced and experimentally validated. A goal of this thesis was to developed a low cost controller for applications were inefficient single phase induction motors are used. Due to the simplistic nature of this control, it has the potential to be implemented in a low cost application specific integrated circuit (ASIC). Therefore, future work related to this thesis will focus on the development of a stand alone ASIC capable of delivering the benefits of BLDC motor drive at a lower cost. It was shown that for higher end applications, the observer and fixed time period combination can be implemented into a DSP. Therefore, the developed controller is well suitable for both low and high end application.

APPENDIX A

CONDUCTION ANGLE CONTROL EXPERIMENTAL RESULTS

Table A.1 Experimental Parameters for 0.1Nm Load

Given	Calculated Values	Desired Speeds
$T_L = 0.1Nm$	$I(\omega_{ss} = \omega_H) = I_H = 1.52A$	$\omega^* = 900\,rpm$
$\omega_H = 1300\,rpm$	$I(\omega_{ss} = \omega_L) = I_L = 1.17A$	$\omega^* = 1000\,rpm$
$\omega_L = 800\,rpm$	$\theta_{2,\min} = \dfrac{I_L}{I_H} = 0.77$	$\omega^* = 1100\,rpm$

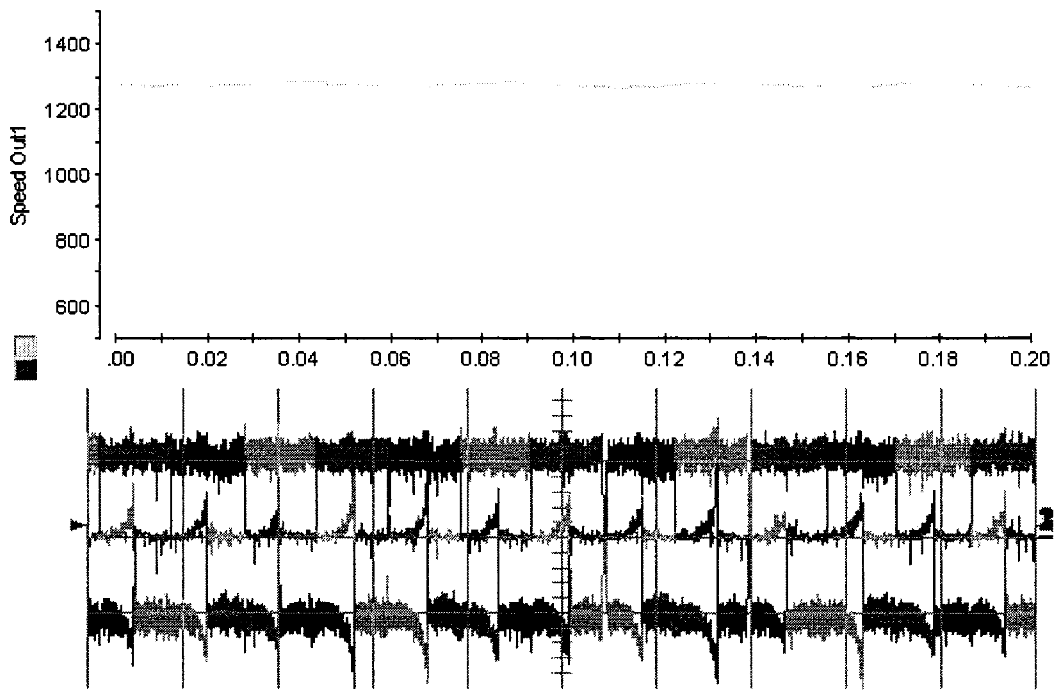


Figure A.1 Speed & Current Results for ω_H with a 0.1Nm Load

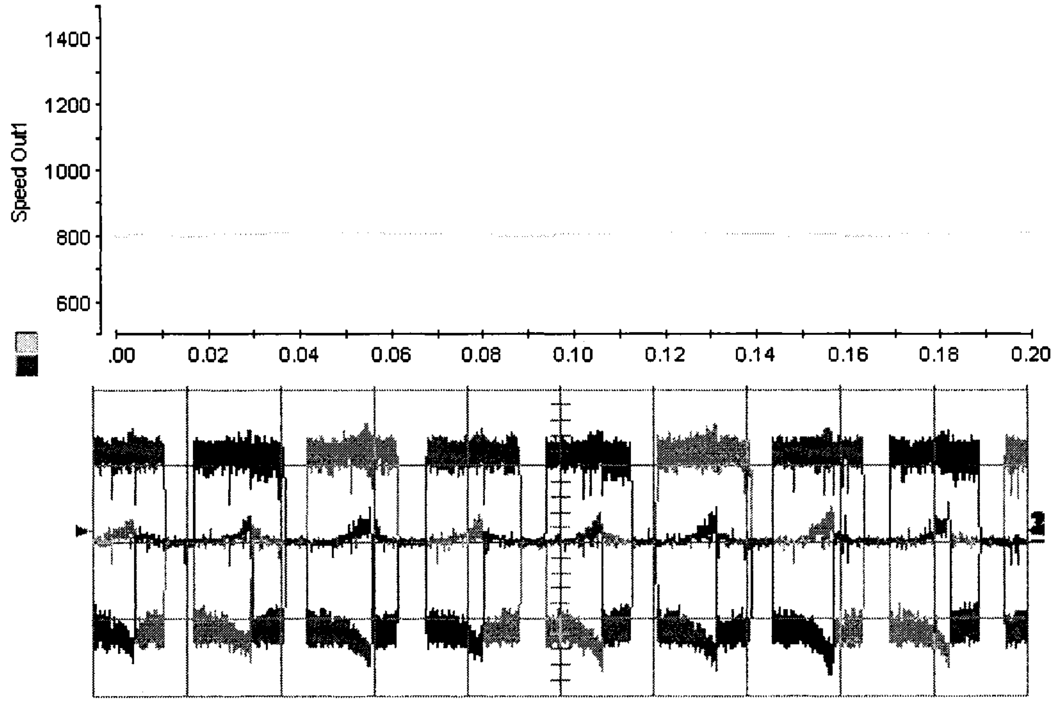


Figure A.2 Speed & Current for ω_L with a 0.1Nm Load.

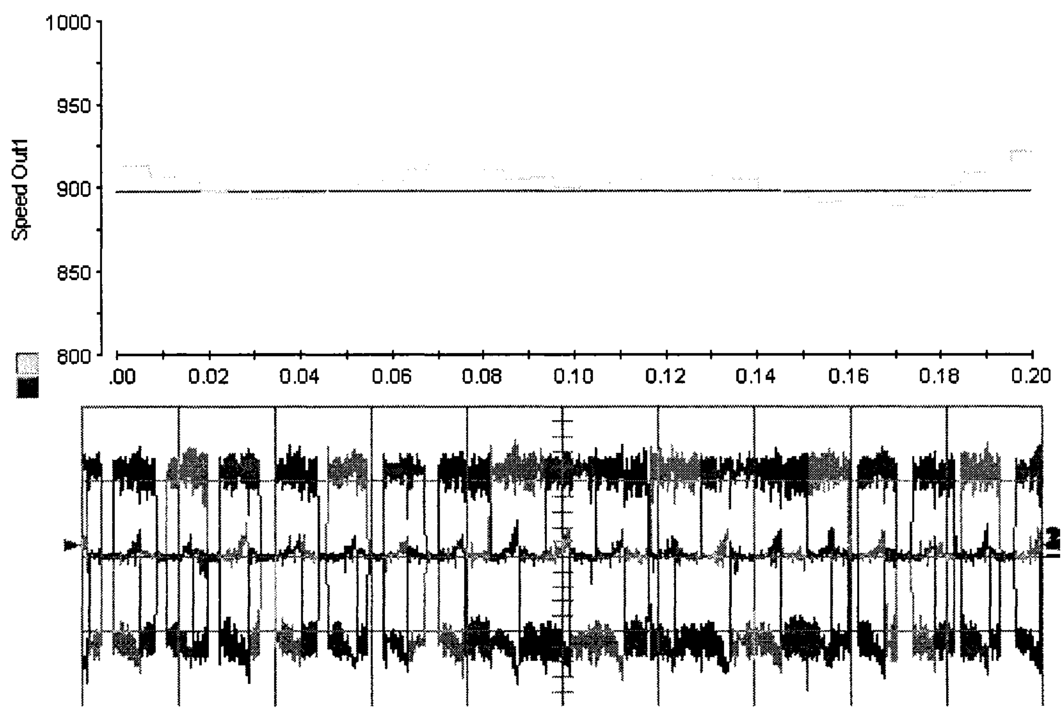


Figure A.3 Speed & Current Results for ω^* = 900 rpm with a 0.1Nm Load

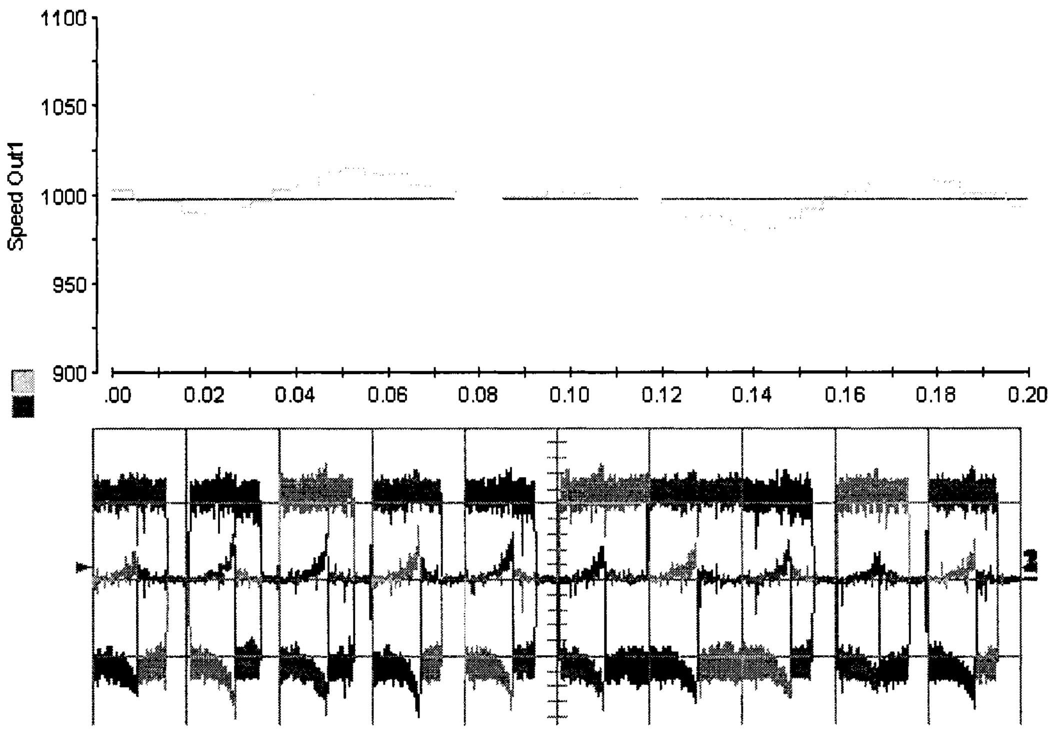


Figure A.4 Speed & Current Results for ω* = 1000 rpm with a 0.1Nm Load

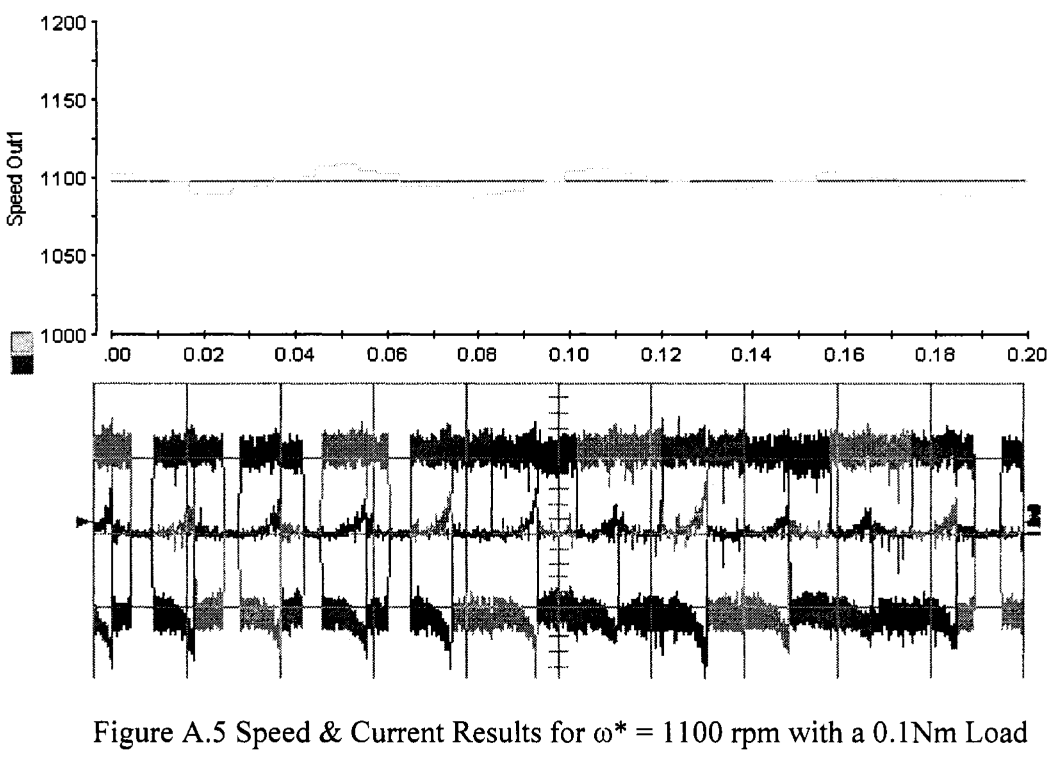


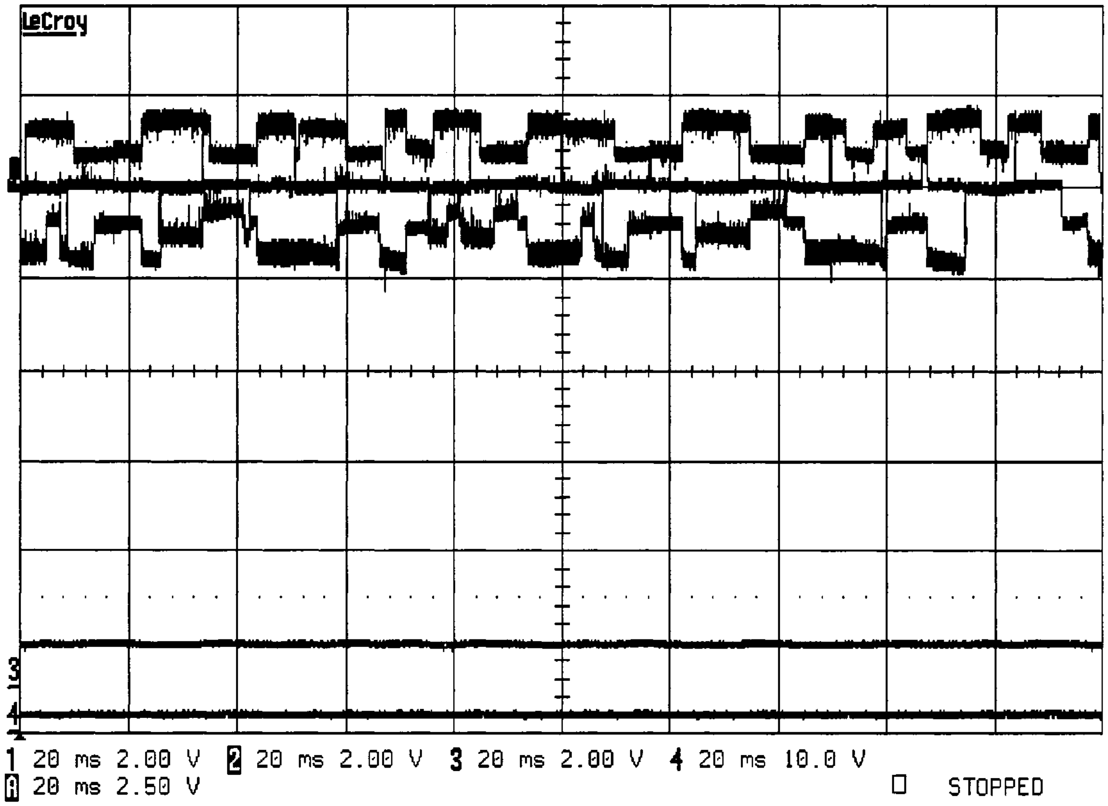
Figure A.5 Speed & Current Results for ω* = 1100 rpm with a 0.1Nm Load

Table A.2 Summary of Experimental Results for 0.1Nm Load

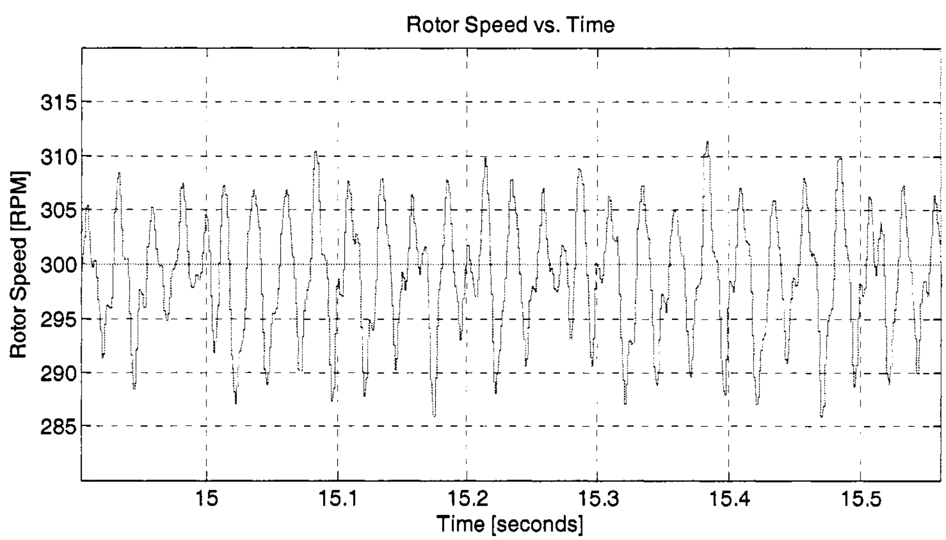
ω^*	$\omega_{average}$	$\omega_{\%Ripple}$
900 *rpm*	905 *rpm*	2.3 %
1000 *rpm*	1000 *rpm*	3.3 %
1100 *rpm*	1100 *rpm*	2.3 %

APPENDIX B

CURRENT MODE EXPERIMENTAL RESULTS: SET 1

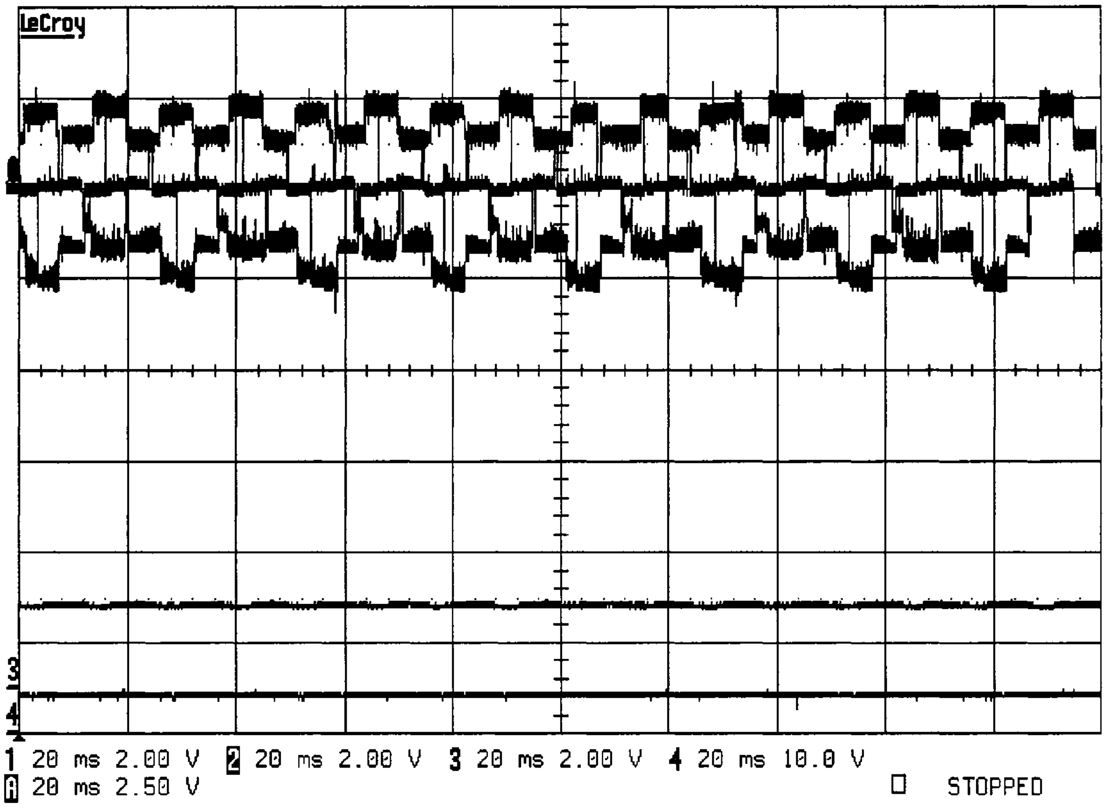


(a)

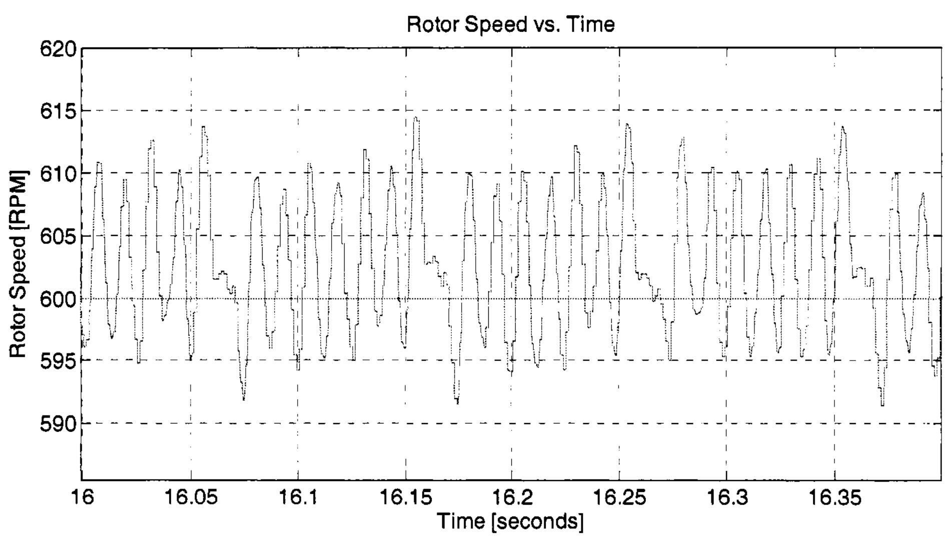


(b)

Figure B.1 Results at 300rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

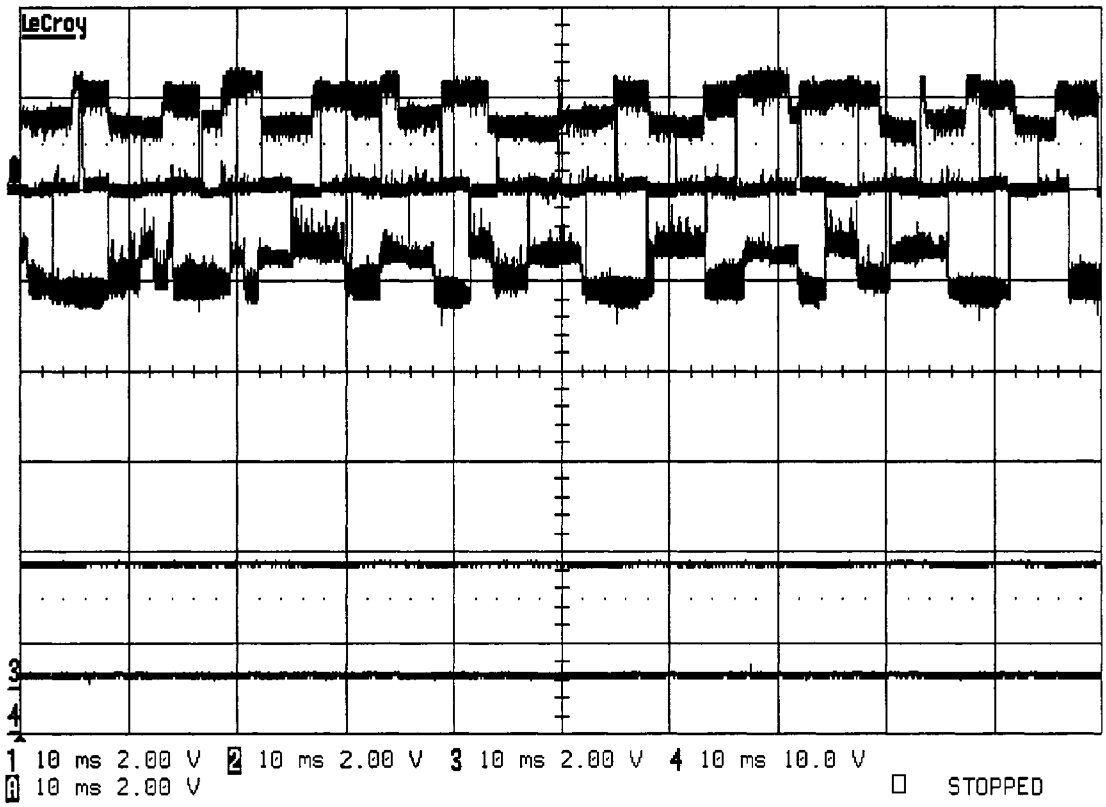


(a)

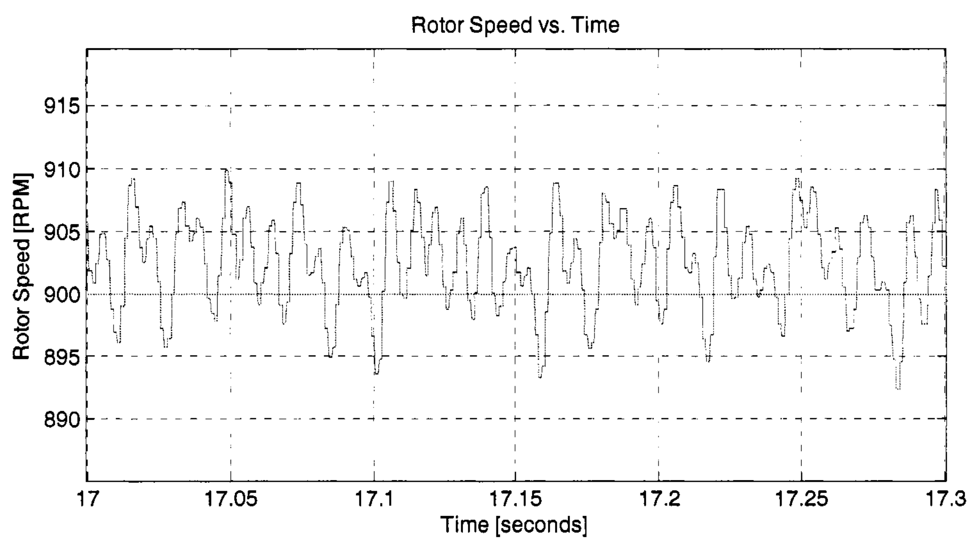


(b)

Figure B.2 Results at 600rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

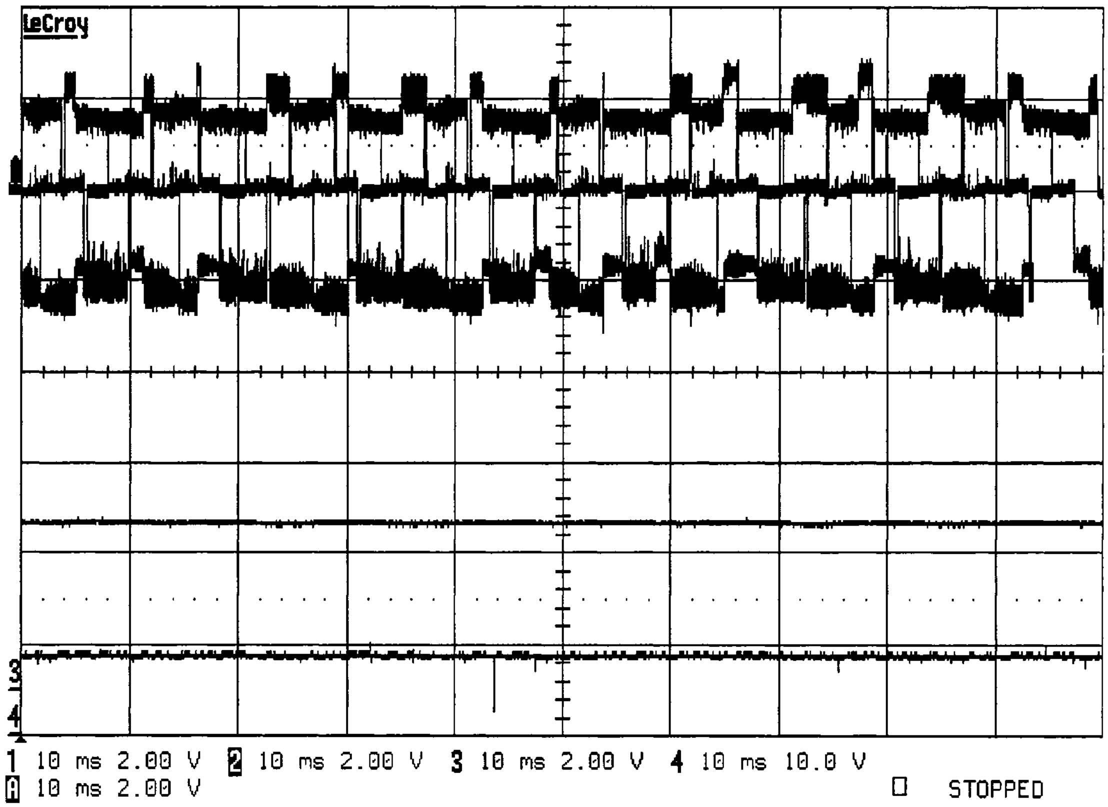


(a)

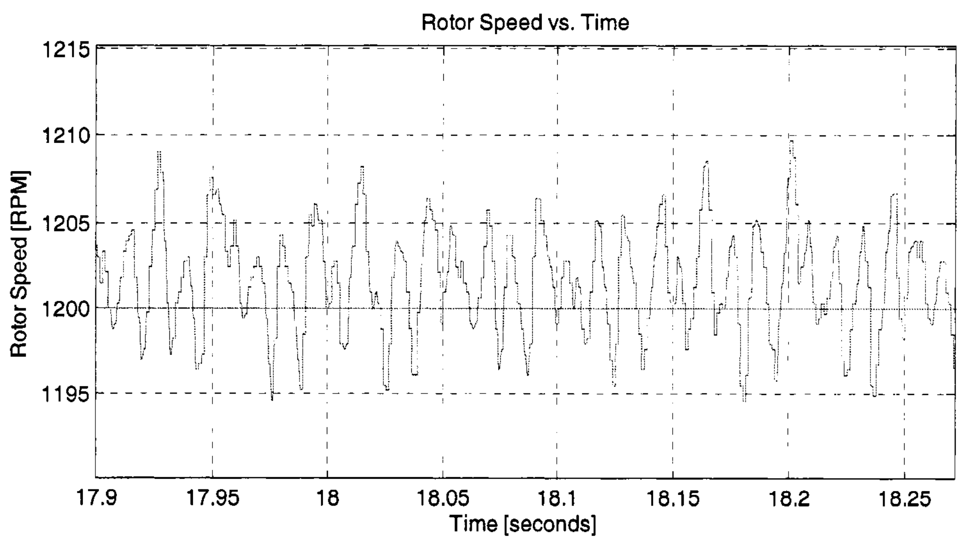


(b)

Figure B.3 Results at 900rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

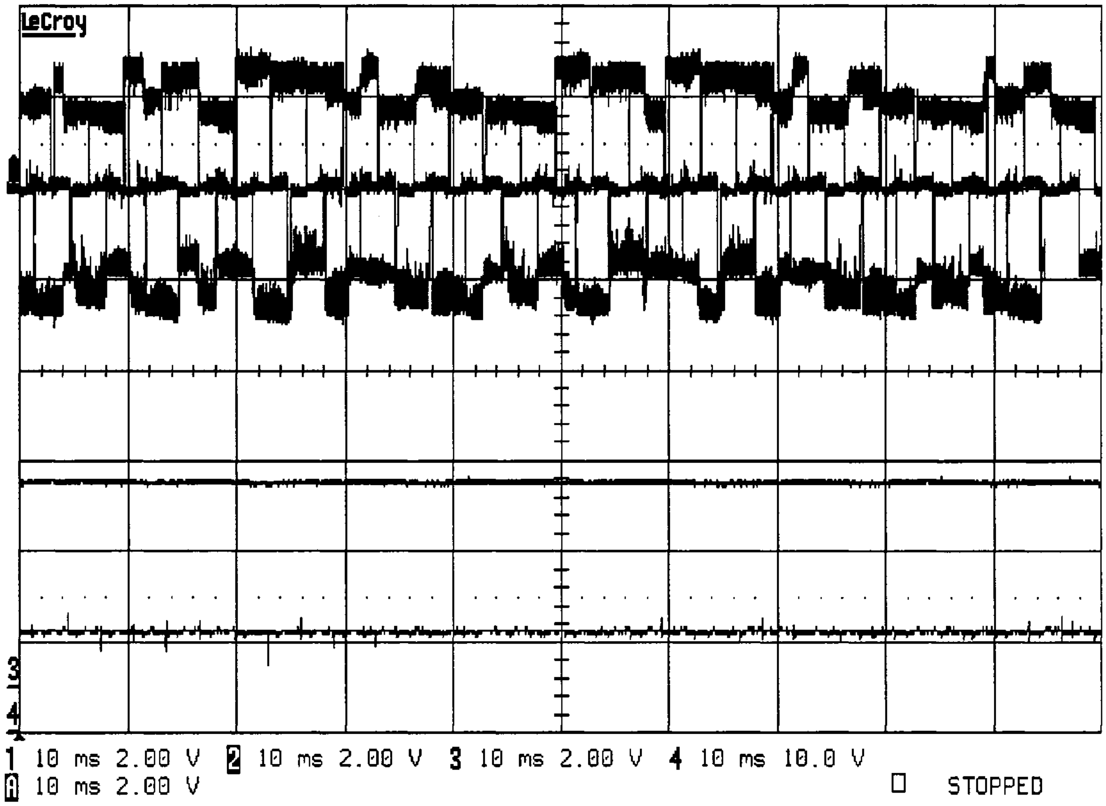


(a)

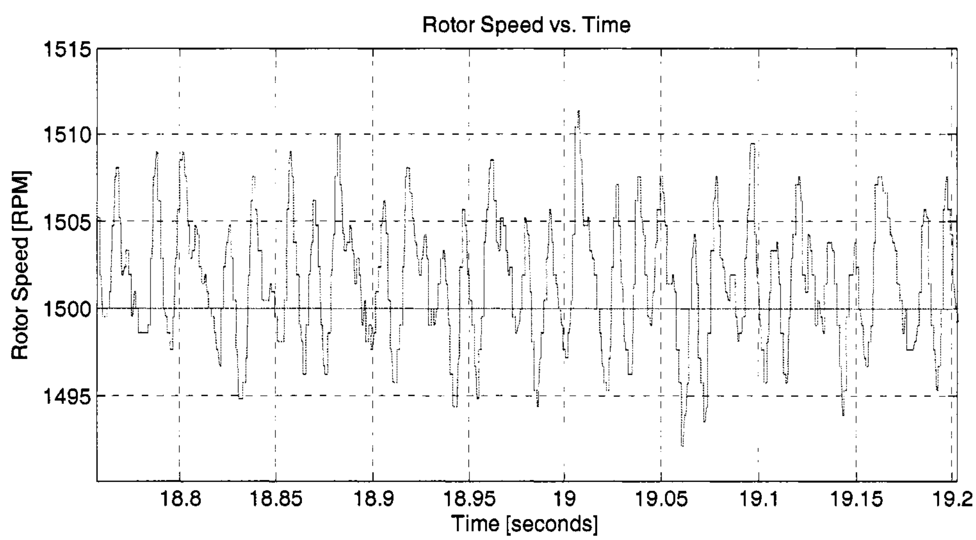


(b)

Figure B.4 Results at 1200rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

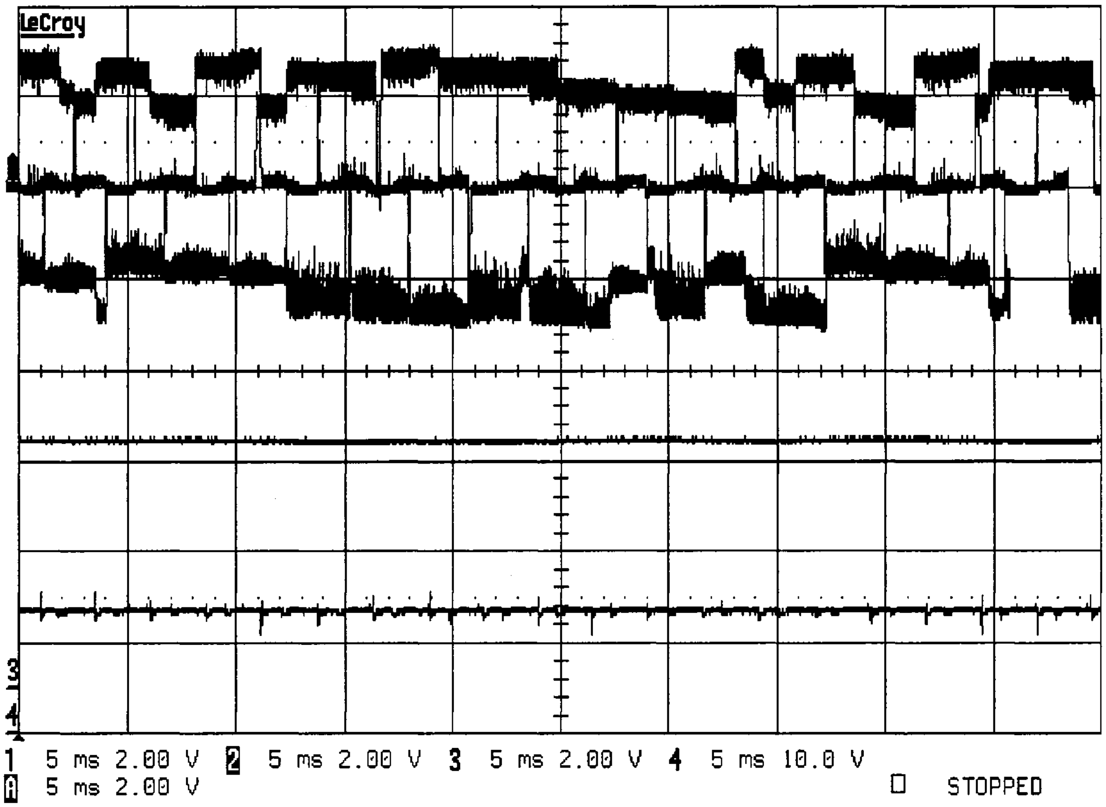


(a)

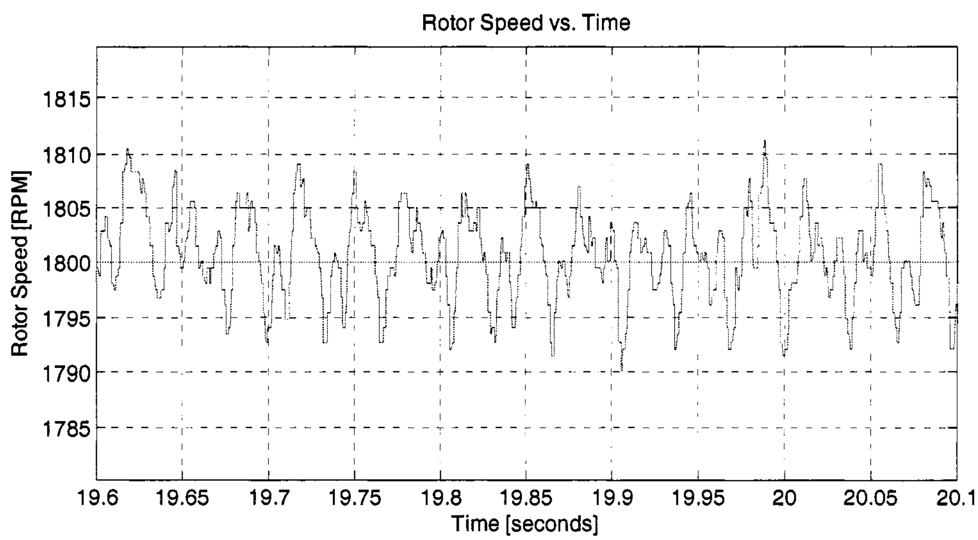


(b)

Figure B.5 Results at 1500rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

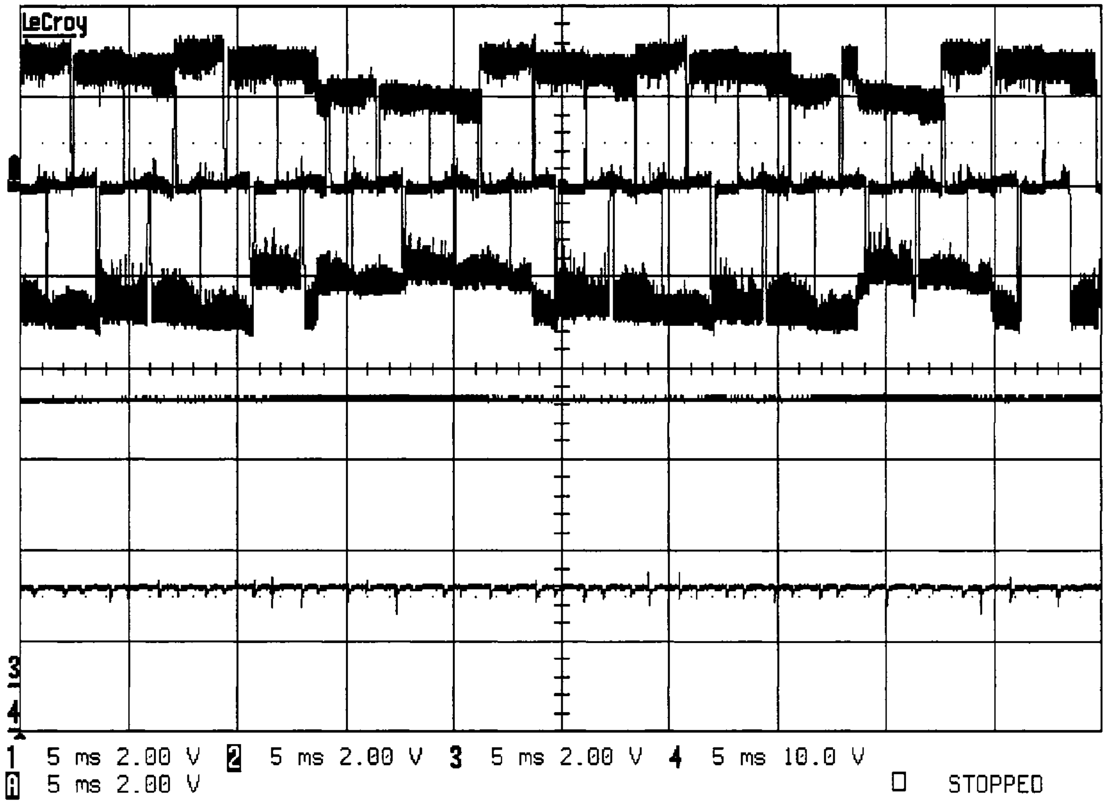


(a)

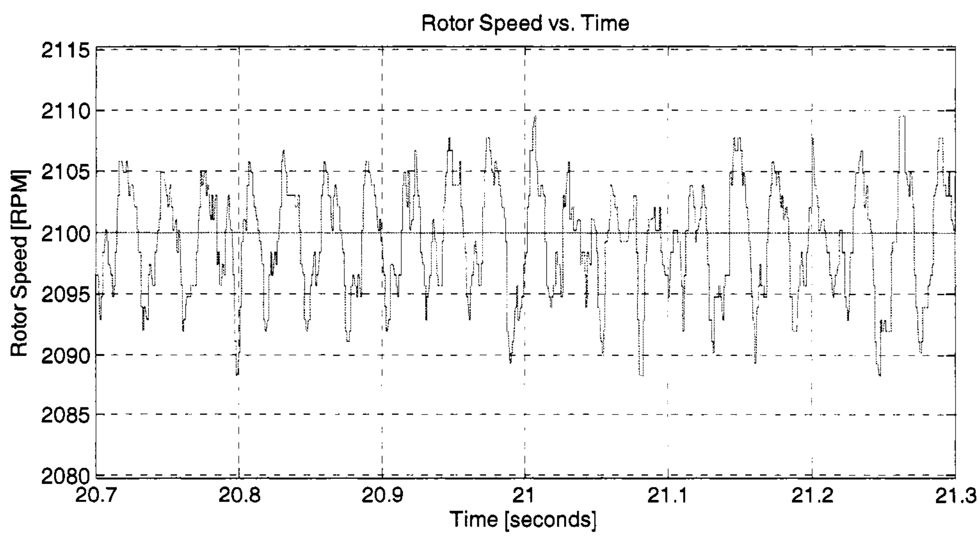


(b)

Figure B.6 Results at 1800rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

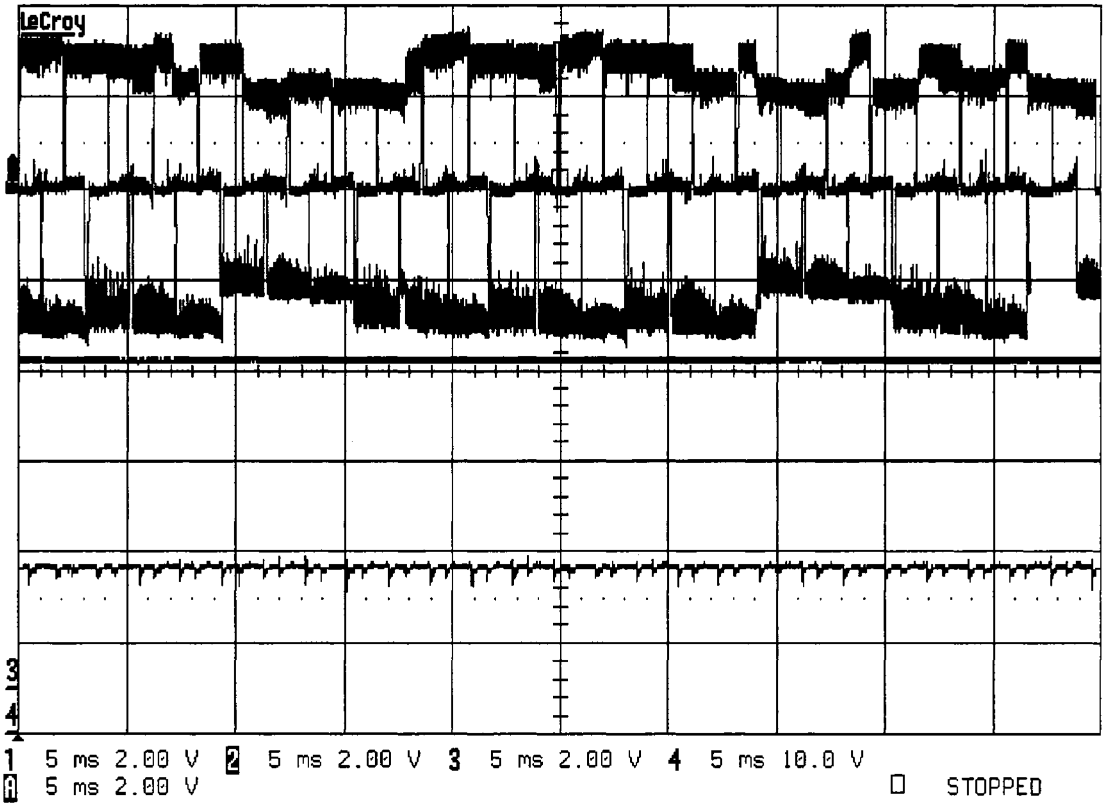


(a)

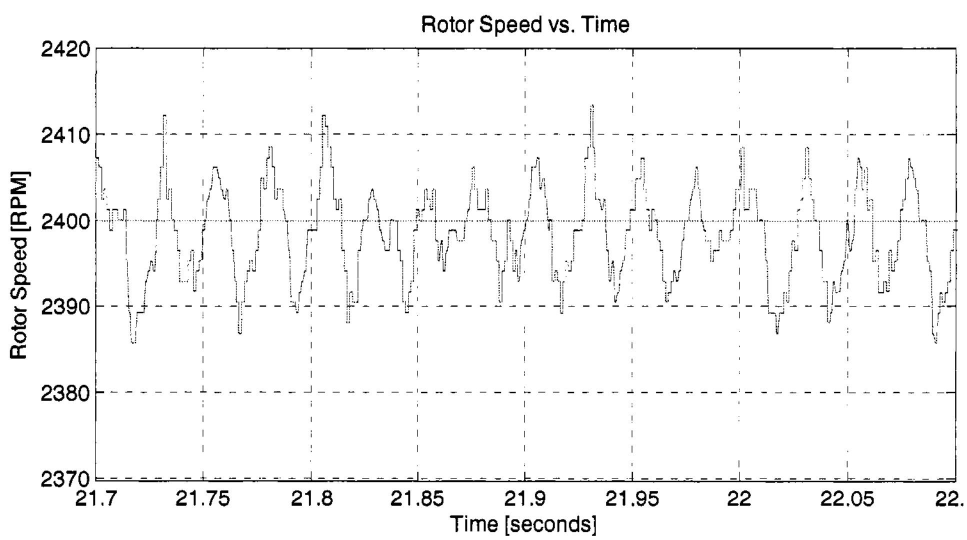


(b)

Figure B.7 Results at 2100rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

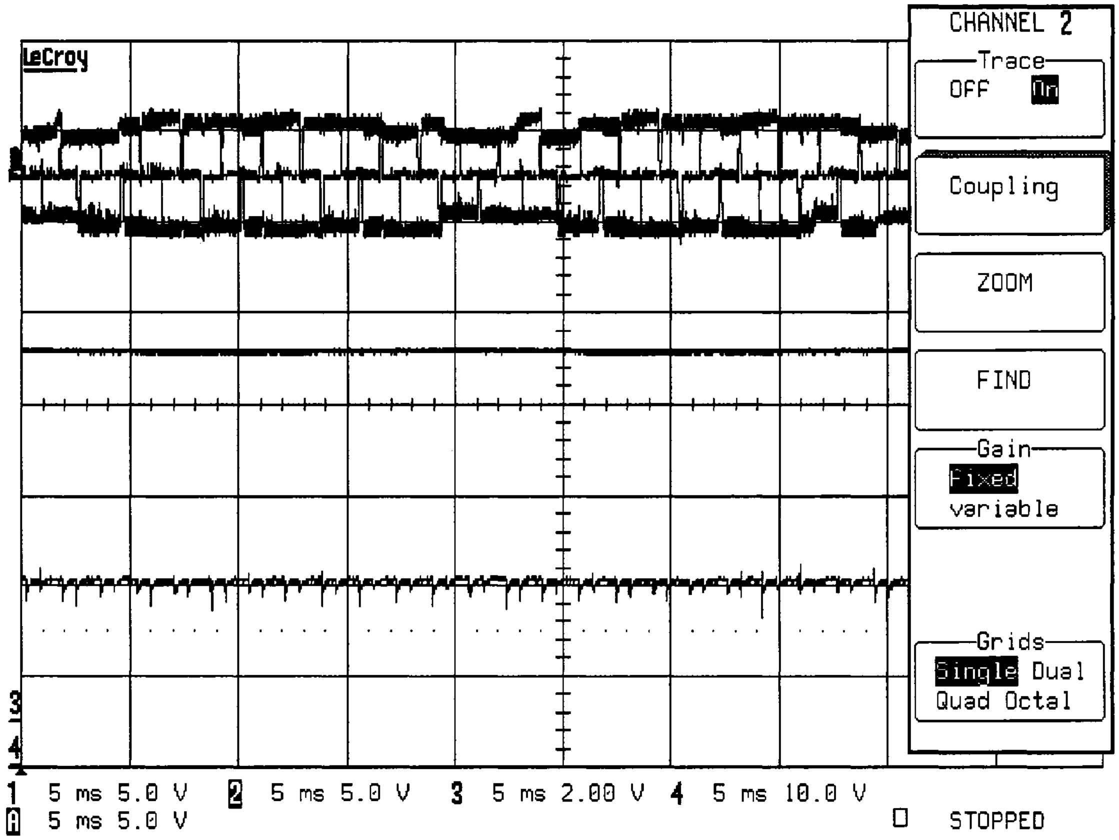


(a)

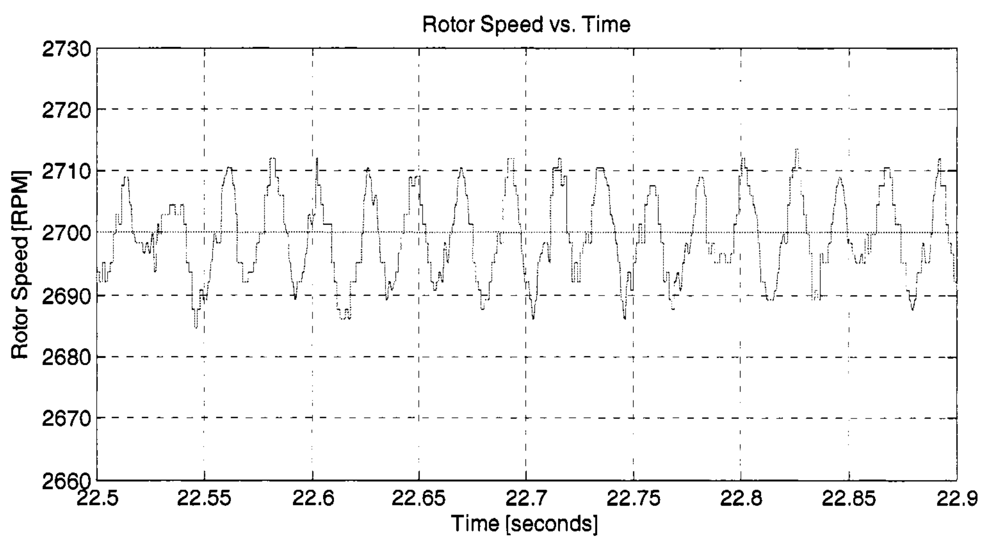


(b)

Figure B.8 Results at 2400rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

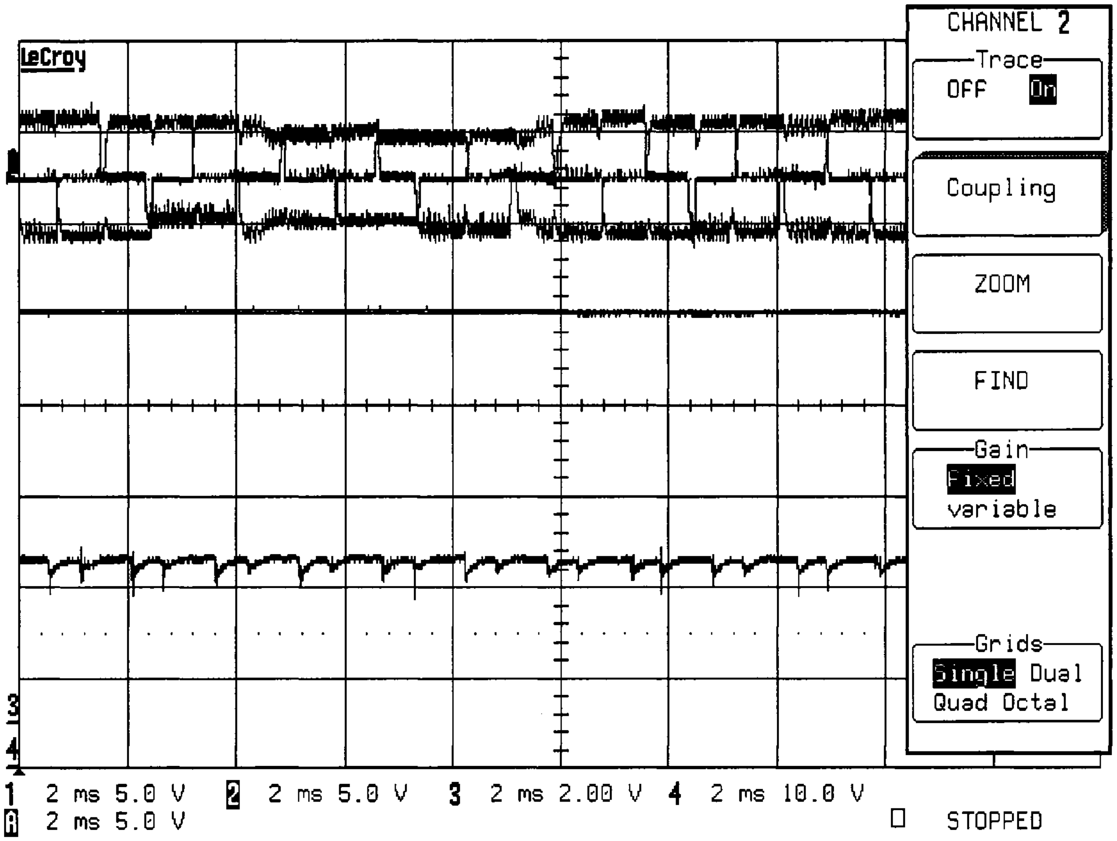


(a)

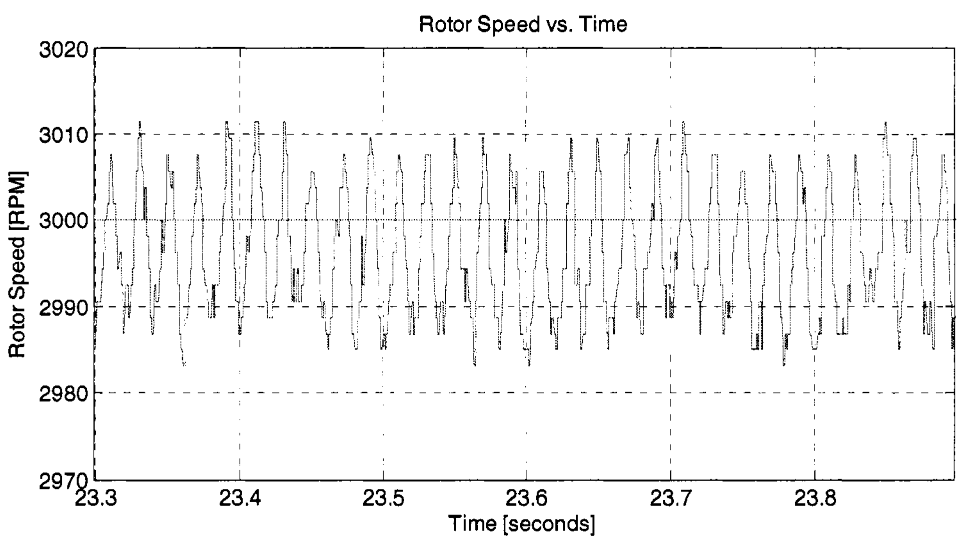


(b)

Figure B.9 Results at 2700rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

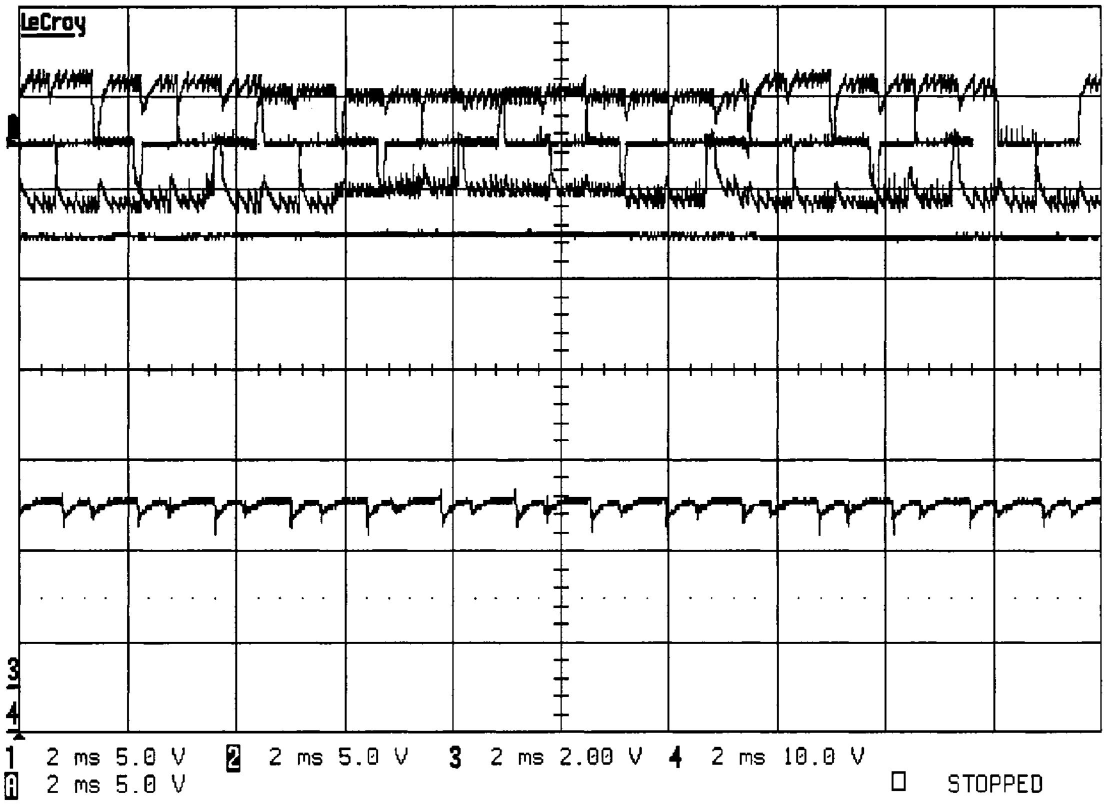


(a)

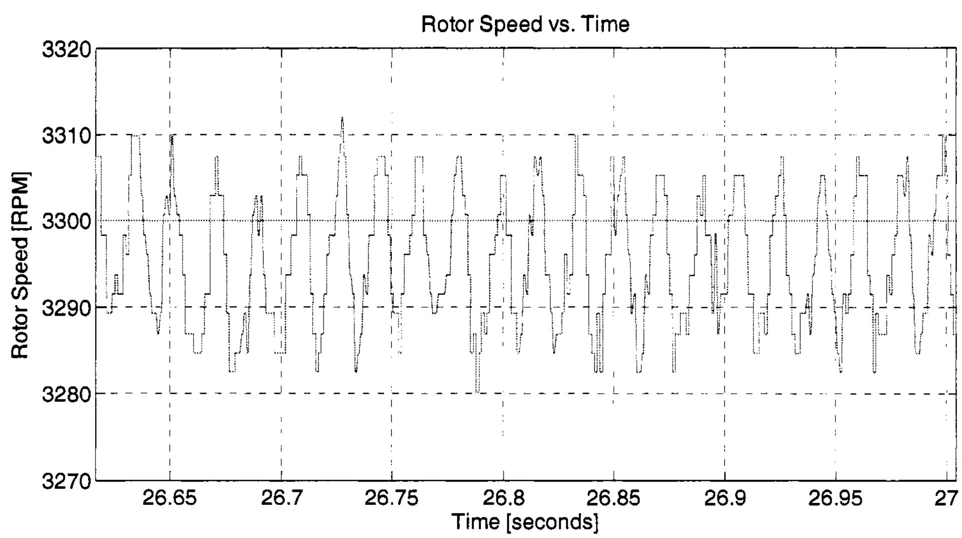


(b)

Figure B.10 Results at 3000rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.



(a)

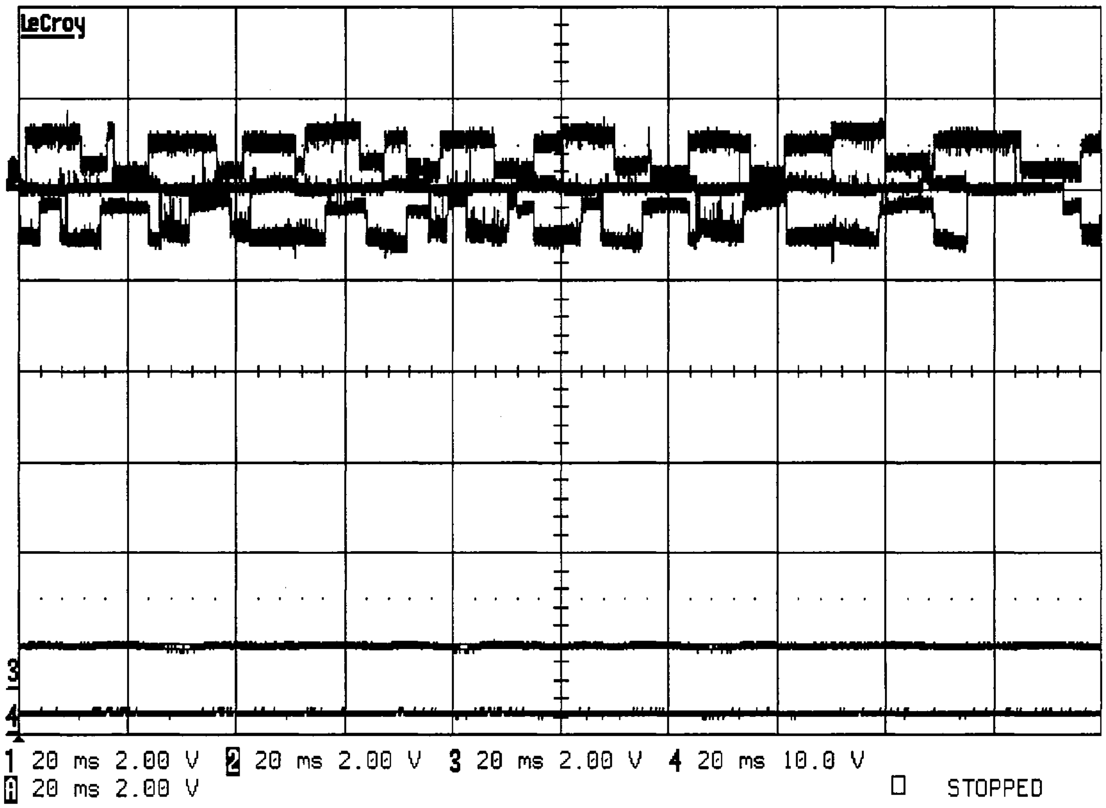


(b)

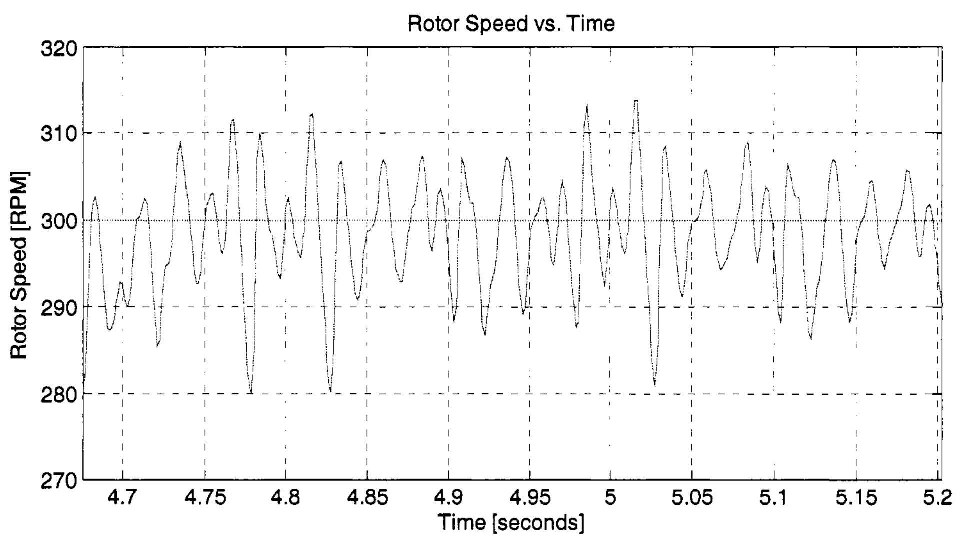
Figure B.11 Results at 3300rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

APPENDIX C

CURRENT MODE EXPERIMENTAL RESULTS: SET 2

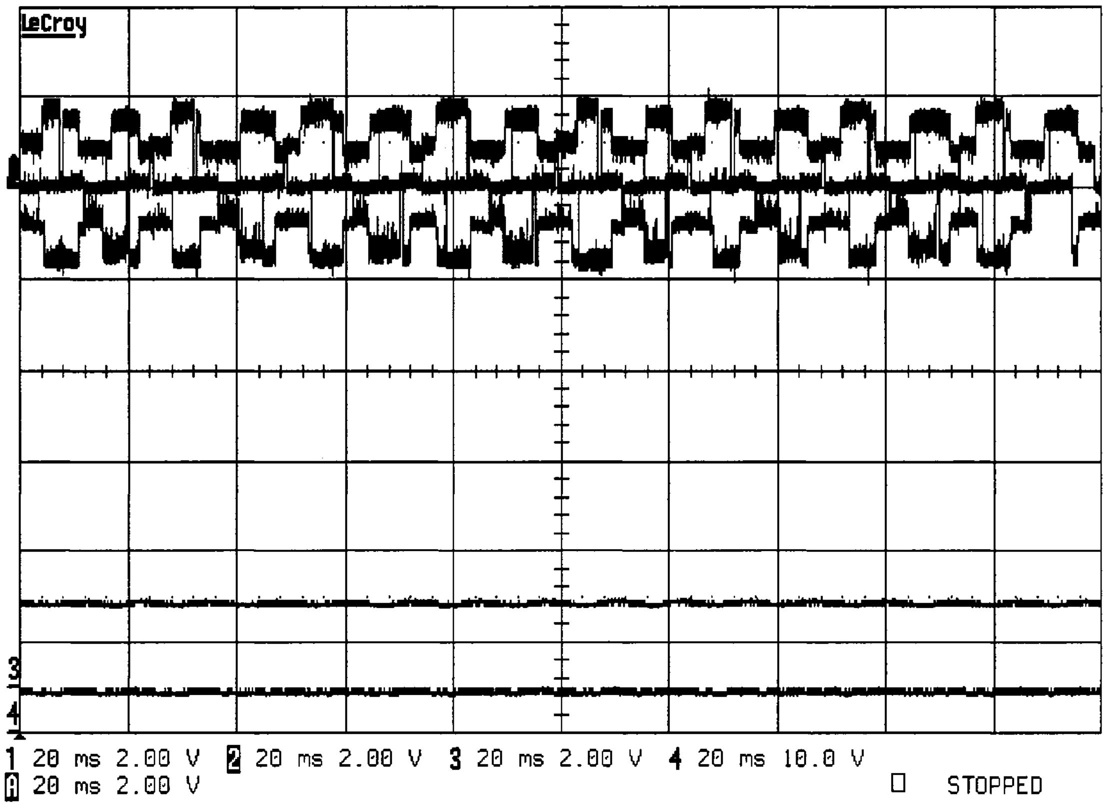


(a)

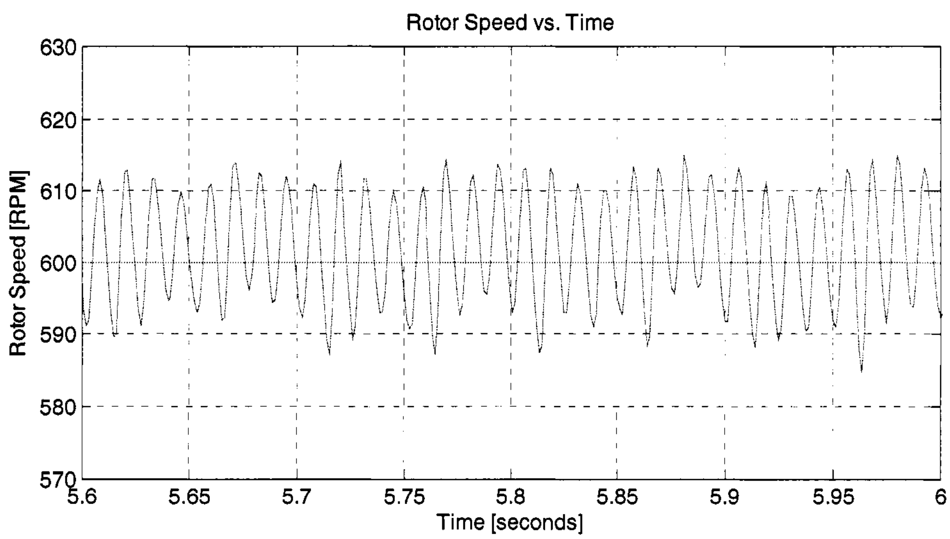


(b)

Figure C.1 Results at 300rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

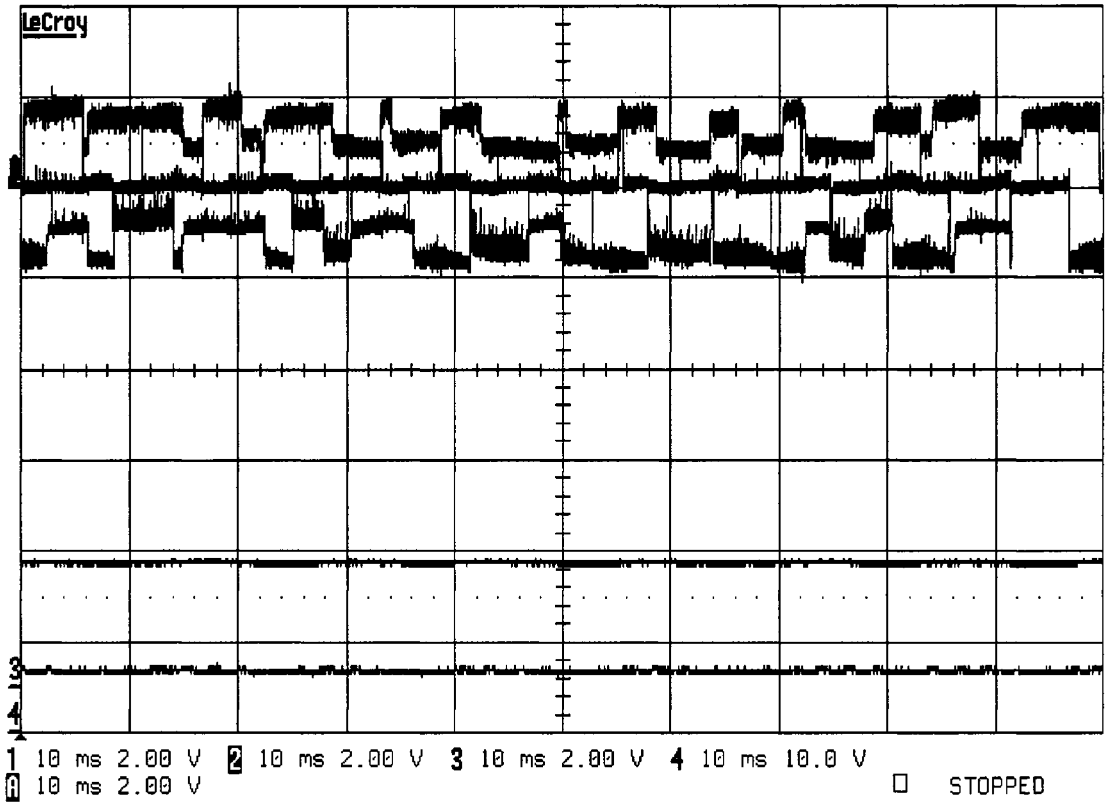


(a)

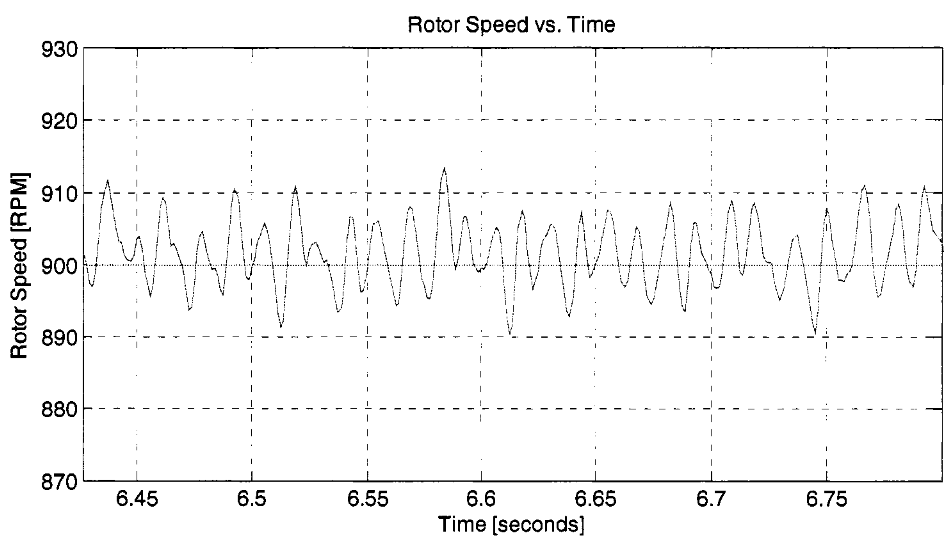


(b)

Figure C.2 Results at 600rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

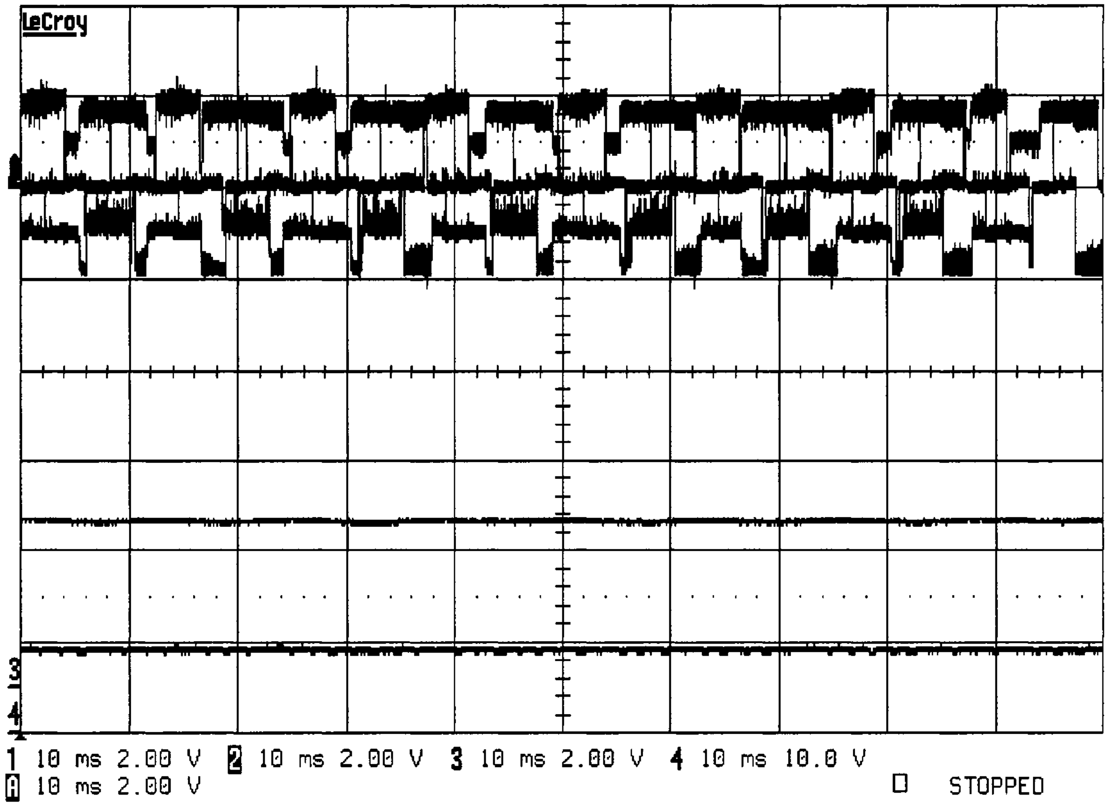


(a)

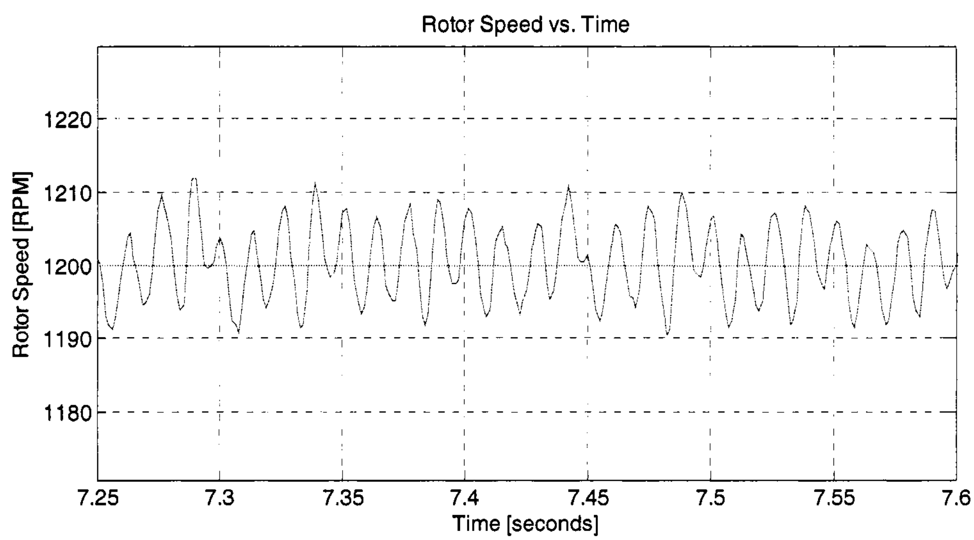


(b)

Figure C.3 Results at 900rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

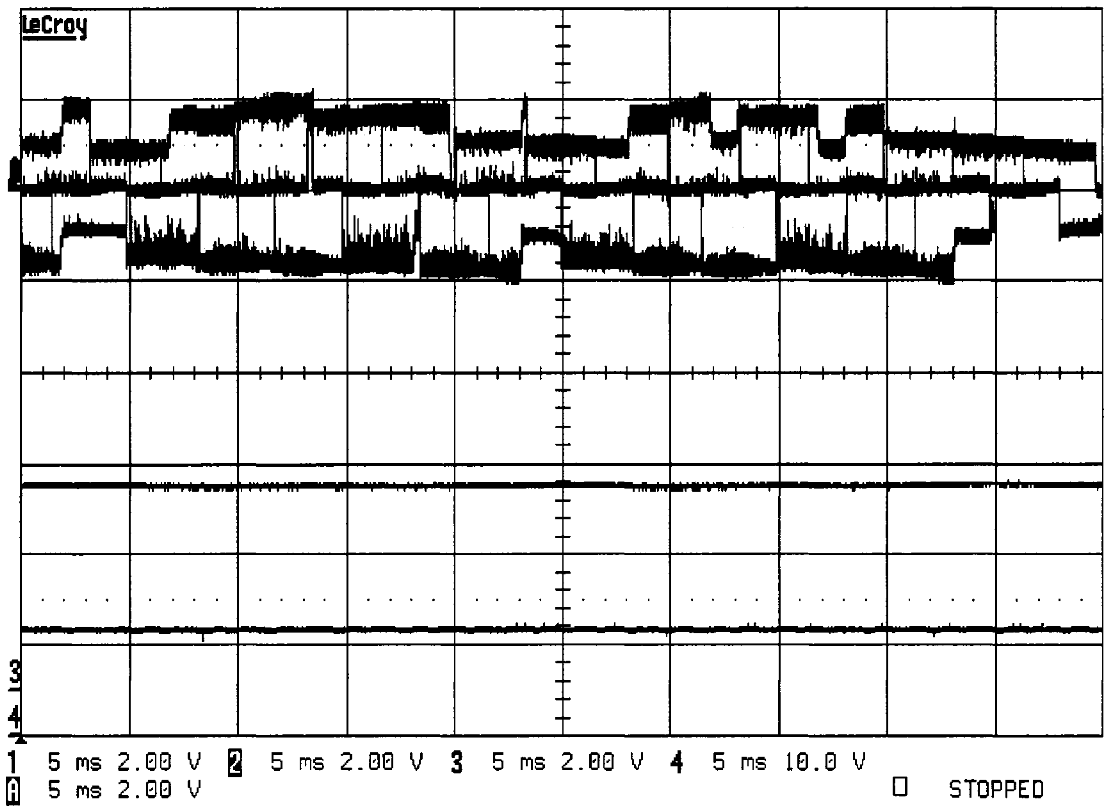


(a)

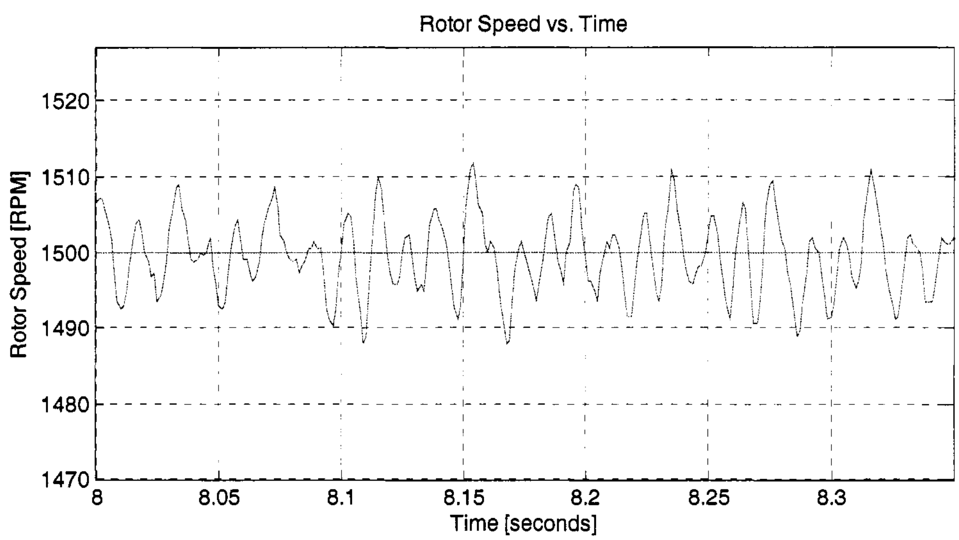


(b)

Figure C.4 Results at 1200rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

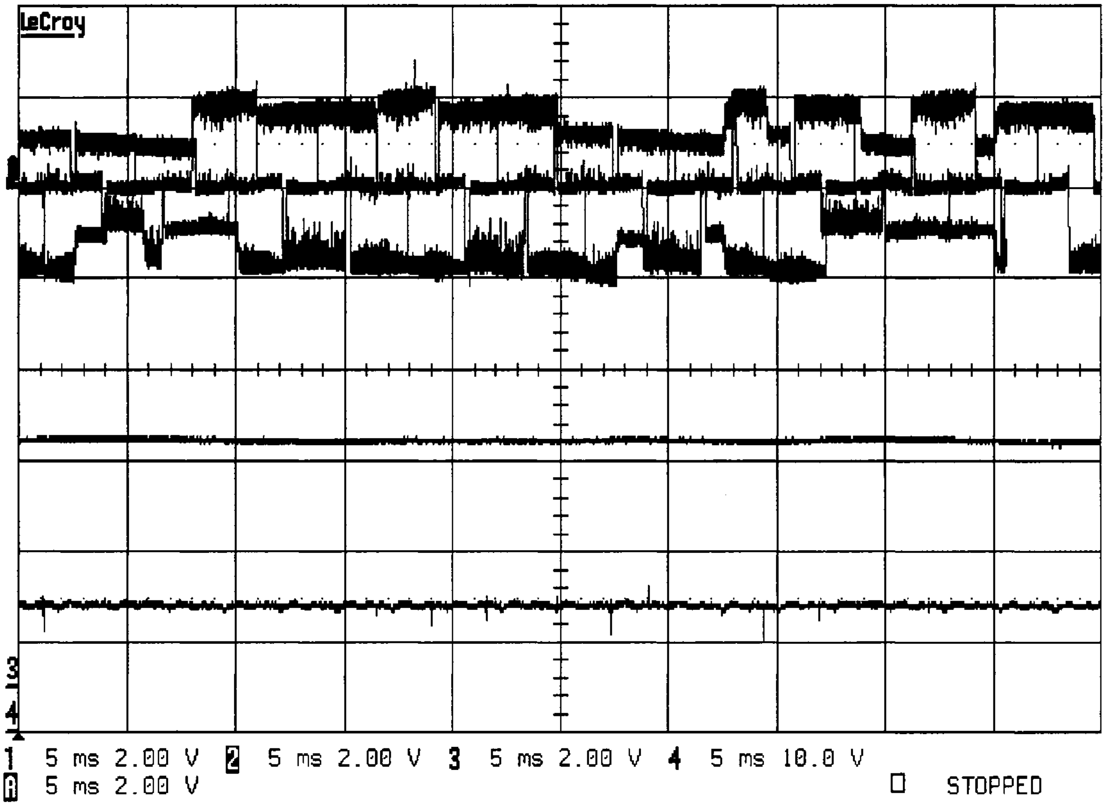


(a)

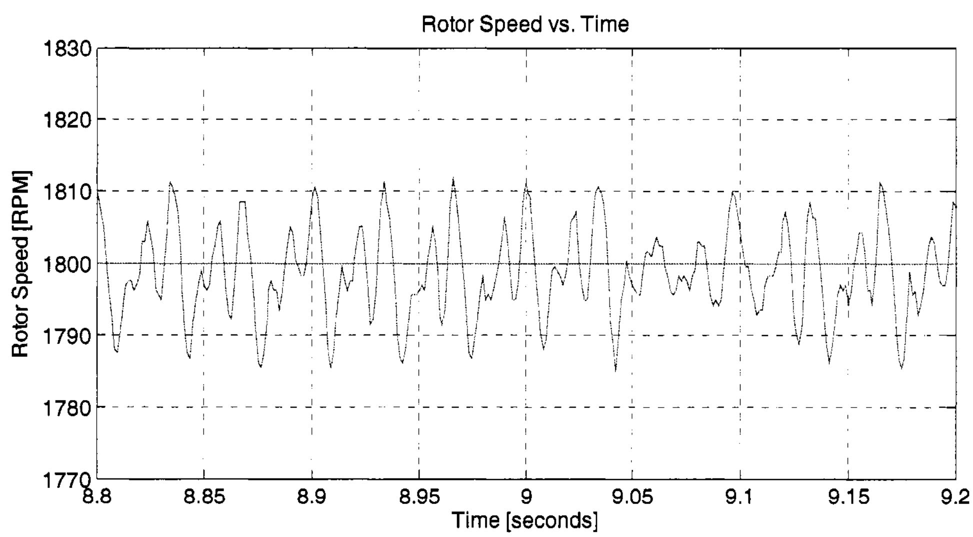


(b)

Figure C.5 Results at 1500rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

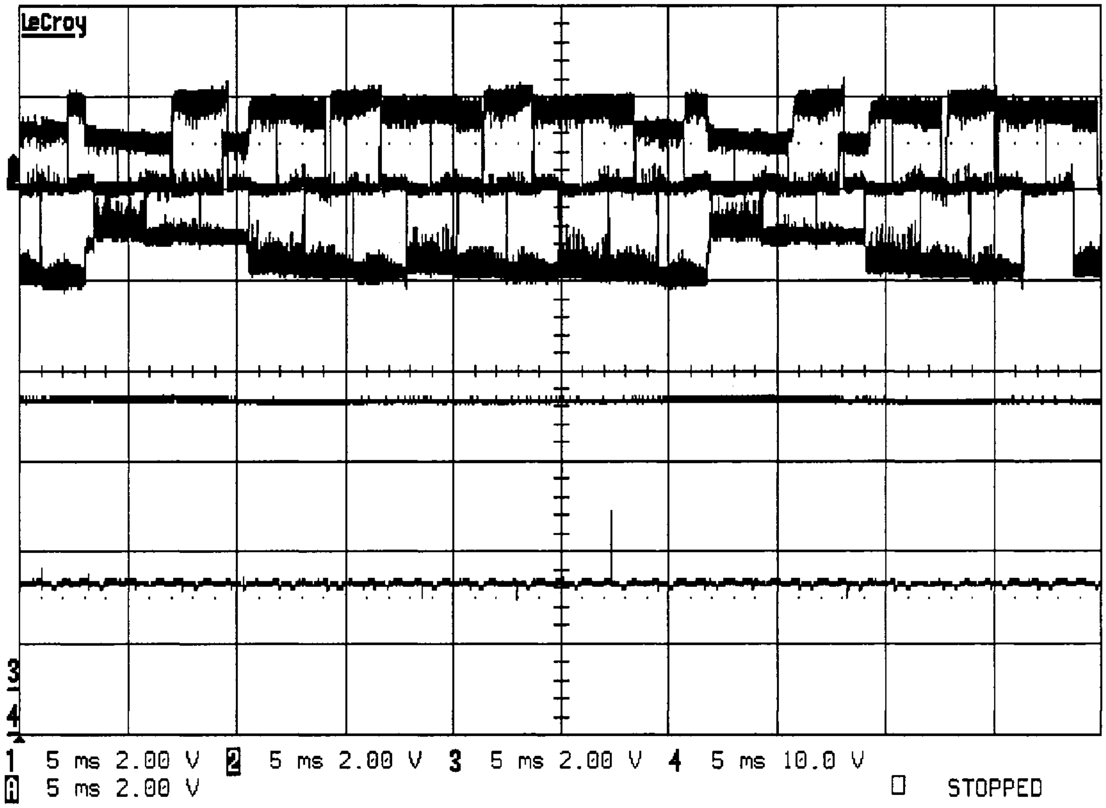


(a)

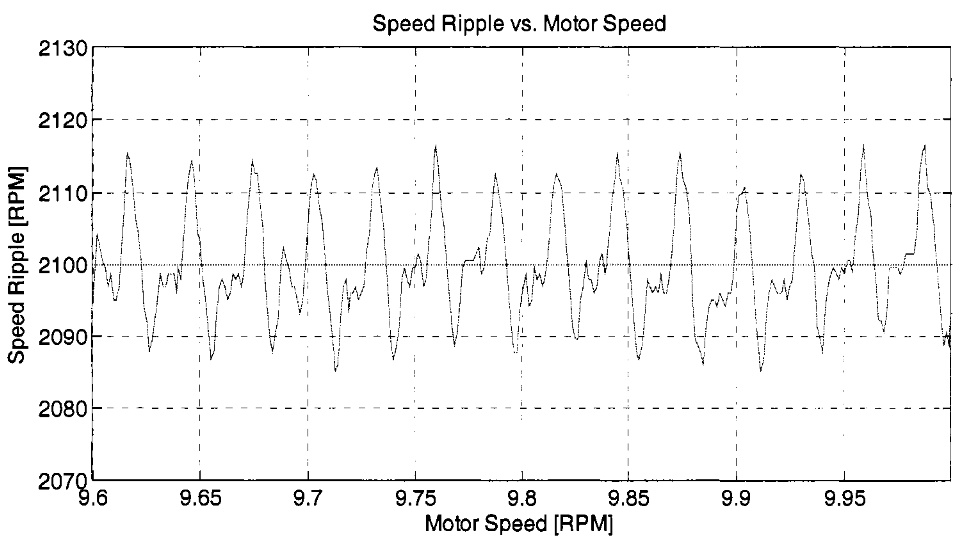


(b)

Figure C.6 Results at 1800rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

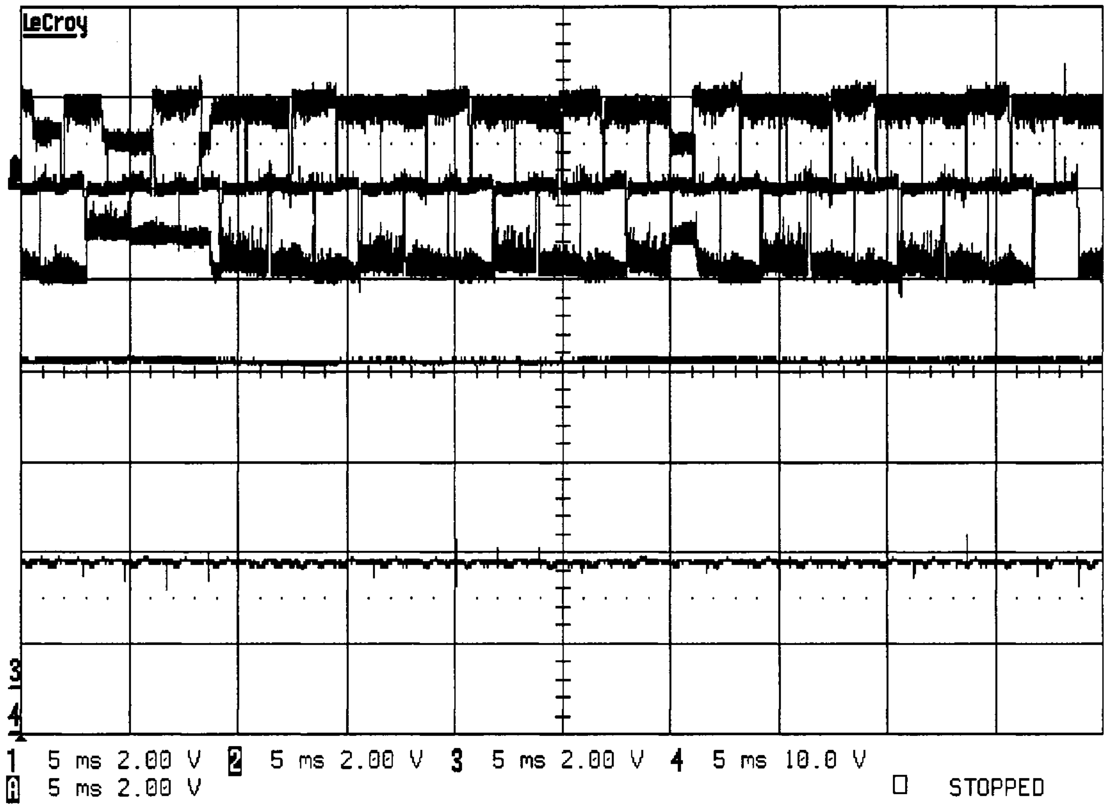


(a)

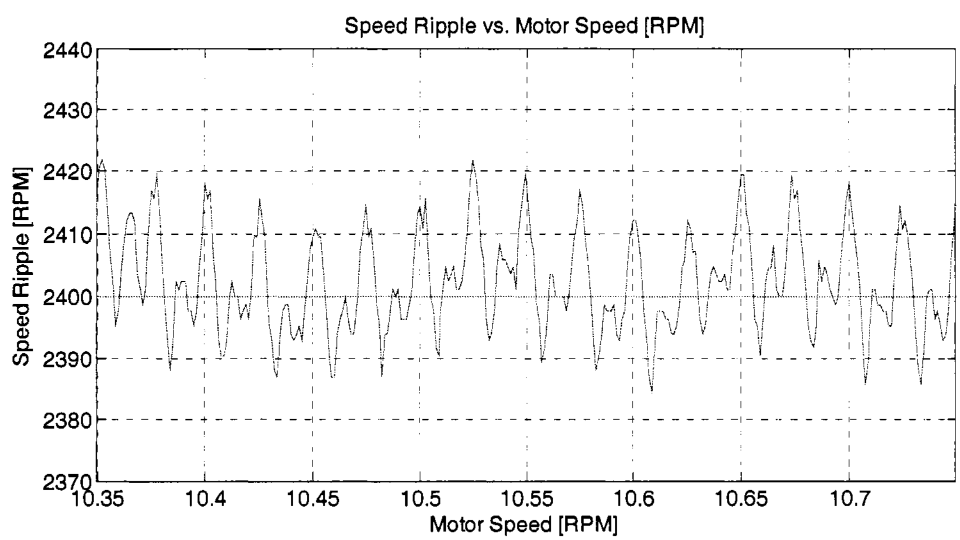


(b)

Figure C.7 Results at 2100rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

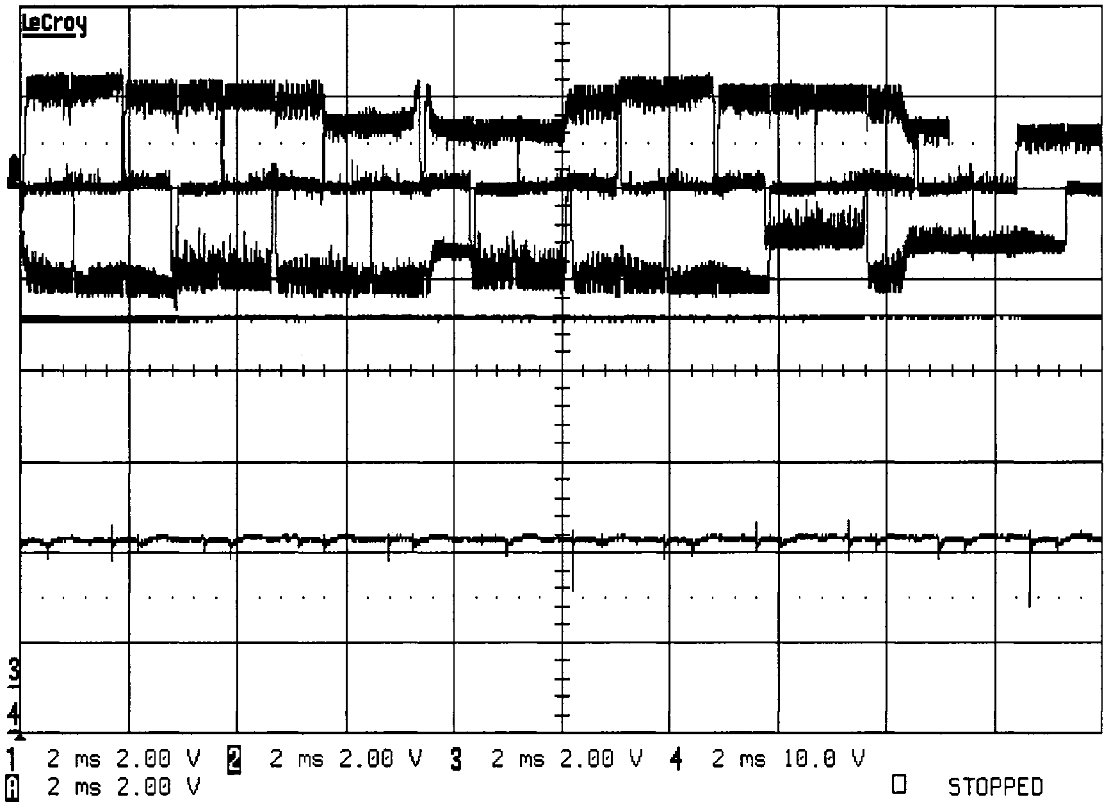


(a)

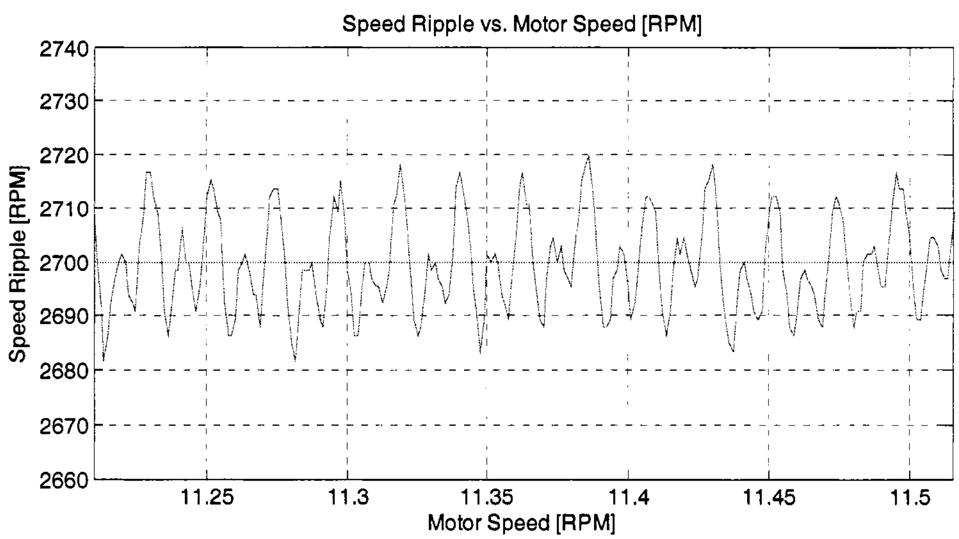


(b)

Figure C.8 Results at 2400rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

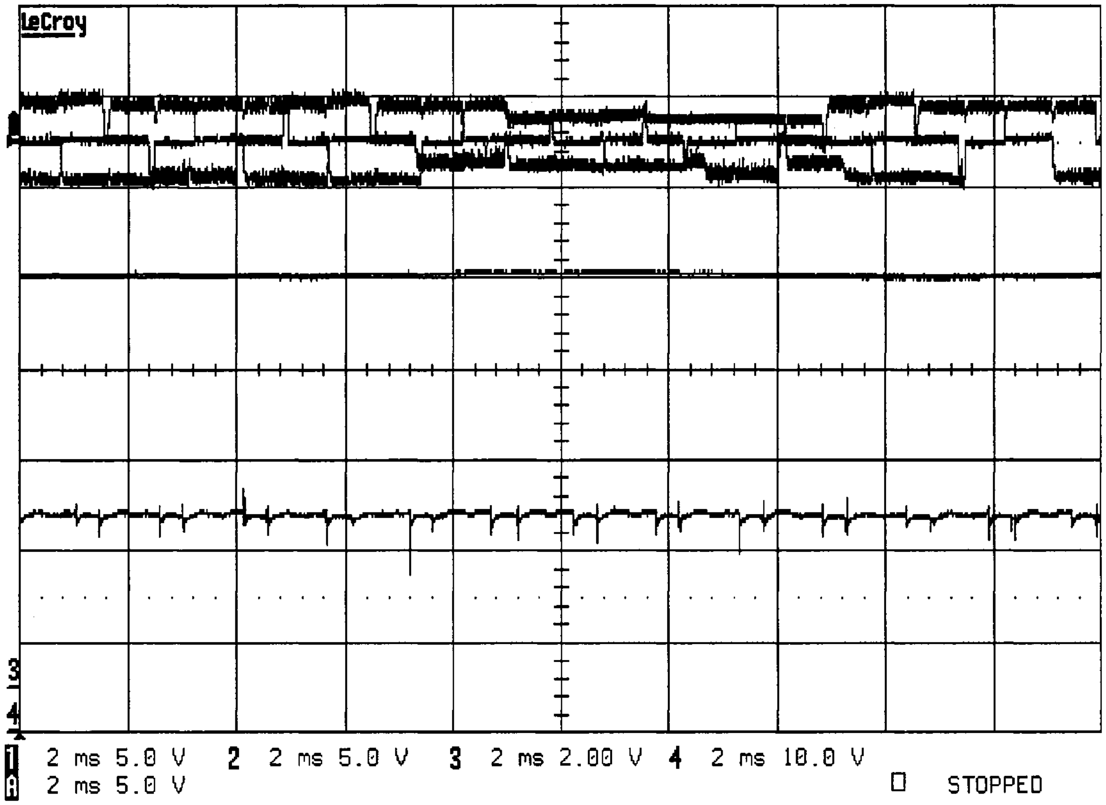


(a)

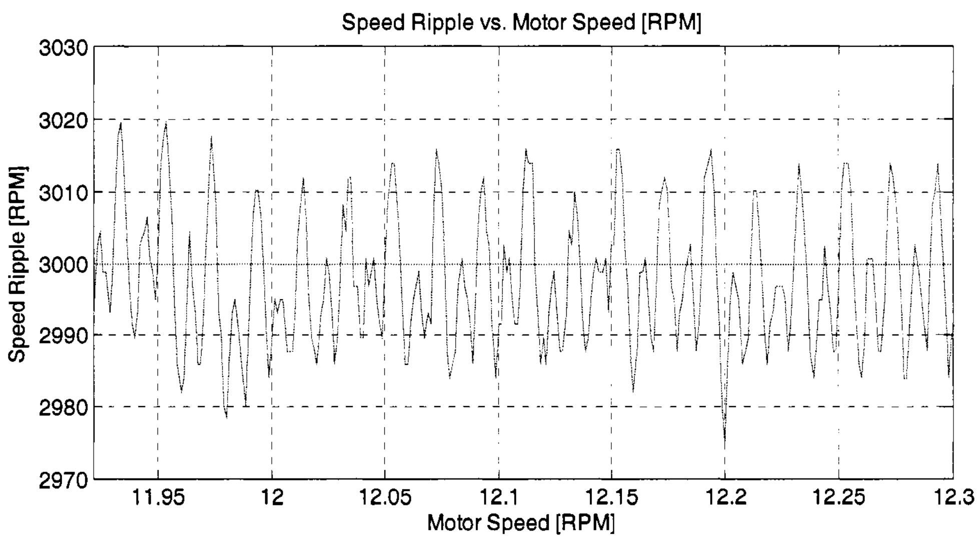


(b)

Figure C.9 Results at 2700rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

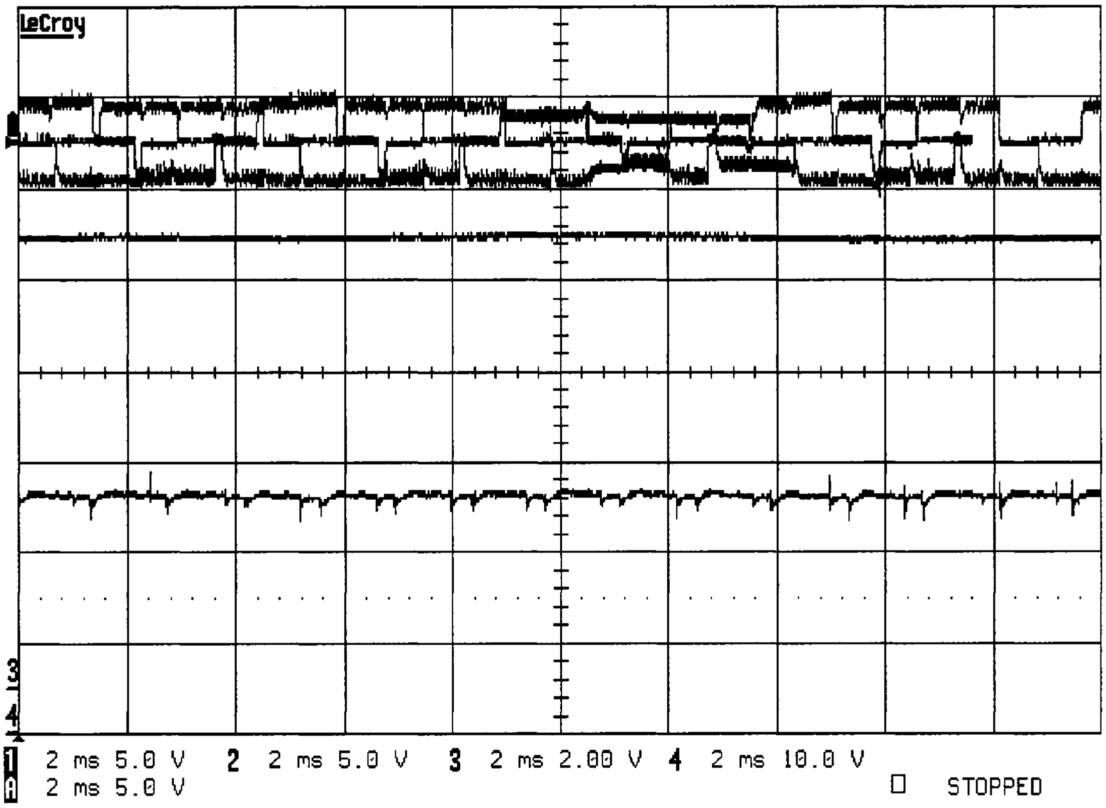


(a)

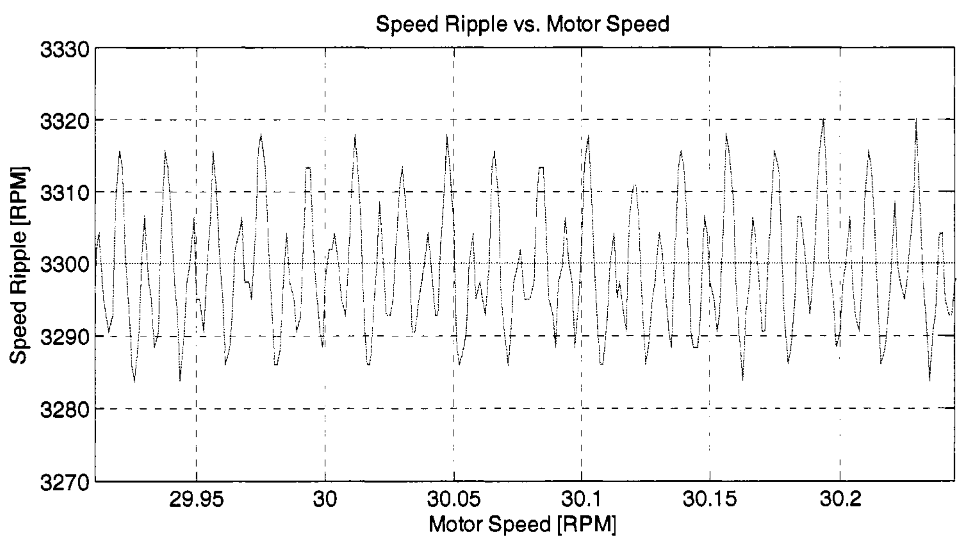


(b)

Figure C.10 Results at 3000rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.



(a)



(b)

Figure C.11 Results at 3300rpm: (a) Phase currents vs. time (b) Speed ripple vs. time.

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